Qualcomm Technologies, Inc.

PM6225 Hardware Register Description

Module Name: ADC_CORE1_USR

Base Address: 0x00003100

Date: 12/29/2025 01:40 AM

Register Quick Reference:

Register	Offset	Type
ADC_CORE1_USR_REVISION1	0x00003100	Read
ADC_CORE1_USR_REVISION2	0x00003101	Read
ADC_CORE1_USR_REVISION3	0x00003102	Read
ADC_CORE1_USR_REVISION4	0x00003103	Read
ADC_CORE1_USR_PERPH_TYPE	0x00003104	Read
ADC_CORE1_USR_PERPH_SUBTYPE	0x00003105	Read
ADC_CORE1_USR_STATUS1	0x00003108	Read
ADC_CORE1_USR_STATUS2	0x00003109	Read
ADC_CORE1_USR_IBAT_MEAS	0x0000310F	Read
ADC_CORE1_USR_INT_RT_STS	0x00003110	Read
ADC_CORE1_USR_INT_SET_TYPE	0x00003111	Read/Write
ADC_CORE1_USR_INT_POLARITY_HIGH	0x00003112	Read/Write
ADC_CORE1_USR_INT_POLARITY_LOW	0x00003113	Read/Write
ADC_CORE1_USR_INT_LATCHED_CLR	0x00003114	Write
ADC_CORE1_USR_INT_EN_SET	0x00003115	Read/Write
ADC_CORE1_USR_INT_EN_CLR	0x00003116	Read/Write
ADC_CORE1_USR_INT_LATCHED_STS	0x00003118	Read
ADC_CORE1_USR_INT_PENDING_STS	0x00003119	Read
ADC_CORE1_USR_INT_MID_SEL	0x0000311A	Read/Write
ADC_CORE1_USR_INT_PRIORITY	0x0000311B	Read/Write
ADC_CORE1_USR_ADC_DIG_PARAM	0x00003142	Read/Write
ADC_CORE1_USR_FAST_AVG_CTL	0x00003143	Read/Write
ADC_CORE1_USR_ADC_CH_SEL_CTL	0x00003144	Read/Write
ADC_CORE1_USR_DELAY_CTL	0x00003145	Read/Write
ADC_CORE1_USR_EN_CTL1	0x00003146	Read/Write
ADC_CORE1_USR_CONV_REQ	0x00003147	Write
ADC_CORE1_USR_DATA0	0x00003150	Read

Register	Offset	Type
ADC_CORE1_USR_DATA1	0x00003151	Read

ADC_CORE1_USR_REVISION1 | 0x00003100

Type: Read

Reset: 0x00000003

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

ADC_CORE1_USR_REVISION2 | 0x00003101

Type: Read

Reset: 0x00000005

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

ADC_CORE1_USR_REVISION3 | 0x00003102

Type: Read

Reset: 0x00000002

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

ADC_CORE1_USR_REVISION4 | 0x00003103

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

ADC_CORE1_USR_PERPH_TYPE | 0x00003104

Type: Read

Reset: 0x00000008

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	ADC

ADC_CORE1_USR_PERPH_SUBTYPE | 0x00003105

Type: Read

Reset: 0x00000030

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	ADC sub type

ADC_CORE1_USR_STATUS1 | 0x00003108

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
0 : 0	EOC	End of conversion status flag. Bit is de-asserted when arbiter is servicing a conversion request and asserted when conversion is completed. 0: CONV_NOT_COMPLETE 1: CONV_COMPLETE
1 : 1	REQ_STS	REQ_STS mirrors the REQ bit. When REQ is asserted the arbiter stores a descriptor in the conversion request queue. Bit is cleared when ADC conversion is completed. 0: REQ_NOT_IN_PROGRESS 1: REQ_IN_PROGRESS

ADC_CORE1_USR_STATUS2 | 0x00003109

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
7 : 5	CONV_SEQ_STATE	Conversion request FSM states. 0: IDLE_REQ 1: WAIT_HOLDOFF_RST 2: WAIT_HOLDOFF 3: RST_CH_ARB 4: GET_NEXT_CH 5: WAIT_ADC_EOC 6: WAIT_ADC_EOC2 7: GEN_IRQ

ADC_CORE1_USR_IBAT_MEAS | 0x0000310F

Type: Read

Reset: 0x00000000

Read-Only register to indicate whether the battery current measurement is supported

Bits	Name	Description
0 : 0	SUPPORTED	0 - battery current measurement is NOT supported; 1 - battery current measurement IS supported 0: IBAT_MEAS_UNSUPPORTED 1: IBAT_MEAS_SUPPORTED

ADC_CORE1_USR_INT_RT_STS | 0x00003110

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	EOC_INT_RT_STS	Secure process end of conversion interrupt. Active high signal two tcxo_clk cycles wide. 0: CONV_COMPLETE_INT_FALSE 1: CONV_COMPLETE_INT_TRUE

ADC_CORE1_USR_INT_SET_TYPE | 0x00003111

Type: Read/Write

Reset: 0x00000000

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	EOC_SET_INT_TYPE	EOC interrupt set type 0: EOC_LEVEL 1: EOC_EDGE

ADC_CORE1_USR_INT_POLARITY_HIGH | 0x00003112

Type: Read/Write

Reset: 0x00000000

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	EOC_INT_HIGH	EOC interrupt high polarity enabled 0: EOC_INT_POL_HIGH_DISABLED 1: EOC_INT_POL_HIGH_ENABLED

ADC_CORE1_USR_INT_POLARITY_LOW | 0x00003113

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	EOC_INT_LOW	EOC interrupt low polarity enabled 0: EOC_INT_POL_LOW_DISABLED 1: EOC_INT_POL_LOW_ENABLED

ADC_CORE1_USR_INT_LATCHED_CLR | 0x00003114

Type: Write

Reset: 0x00000000

Writing a '1' to a bit in this register will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits

Bits	Name	Description
0 : 0	EOC_INT_LATCHED_CLR	EOC interrupt latched clear

ADC_CORE1_USR_INT_EN_SET | 0x00003115

Type: Read/Write

Reset: 0x00000000

Writing '0' to a bit in this register has no effect. Writing a '1' to a bit in this register will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	EOC_INT_EN_SET	EOC interrupt enable set 0: EOC_INT_DISABLED 1: EOC_INT_ENBLED

ADC_CORE1_USR_INT_EN_CLR | 0x00003116

Type: Read/Write

Reset: 0x00000000

Writing a '0' to a bit in this register has no effect. Writing a '1' to a bit in this register will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	EOC_INT_EN_CLR	EOC interrupt enable clear 0: EOC_INT_DISABLED 1: EOC_INT_ENBLED

ADC_CORE1_USR_INT_LATCHED_STS | 0x00003118

Type: Read

Reset: 0x00000000

Debug: Latched (Sticky) Interrupt. '1' indiecates that the interrutp has triggered. Once the latched bit is set it can only be cleared by writing the clear bit.

Bits	Name	Description
0 : 0	EOC_INT_LATCHED_STS	EOC interrupt latched 0: EOC_INT_LATCHED_FALSE 1: EOC_INT_LATCHED_TRUE

ADC_CORE1_USR_INT_PENDING_STS | 0x00003119

Type: Read

Reset: 0x00000000

Debug: Pending is set if interrupt has been sent but not cleared.

Bits	Name	Description
0 : 0	EOC_INT_PENDING_STS	EOC interrupt pending 0: EOC_INT_PENDING_FALSE 1: EOC_INT_PENDING_TRUE

ADC_CORE1_USR_INT_MID_SEL | 0x0000311A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	Selects the MID that will receive the interrupt

ADC_CORE1_USR_INT_PRIORITY | 0x0000311B

Type: Read/Write

Reset: 0x00000000

Selects the SPMI interrupt priority

Bits	Name	Description
0 : 0	INT_PRIORITY	Selects the SPMI interrupt priority 0: SR 1: A

ADC_CORE1_USR_ADC_DIG_PARAM | 0x00003142

Type: Read/Write

Reset: 0x00000018

PMIC_STATIC

Bits	Name	Description
3 : 2	DEC_RATIO_SEL	Select the conversion decimation ratio: * DEC_RATIO_SEL = 0 ==> STEP1 = 170, STEP2 = 80 , Decimation Ratio = STEP1+STEP2 = 250 * DEC_RATIO_SEL = 1 ==> STEP1 = 340, STEP2 = 80 , Decimation Ratio = STEP1+STEP2 = 420 * DEC_RATIO_SEL = 2 ==> STEP1 = 680, STEP2 = 160, Decimation Ratio = STEP1+STEP2 = 840 The above table reflects the default ADC_CMN setting (CONV_MODE = 0 and DIG_BOOST_EN = 1). Please refer to the DIG_BOOST_EN field description in ADC_CMN peripheral if a non-default settings are used for CONV_MODE and DIG_BOOST_EN. 0: DEC_RATIO_SHORT 1: DEC_RATIO_MEDIUM 2: DEC_RATIO_LONG
5 : 4	CAL_SEL	Select calibration method 0: No_cal 1: ratio_cal 2: abs_cal
6 : 6	CAL_VAL	Select calibration values 0: timer_cal 1: new_cal

ADC_CORE1_USR_FAST_AVG_CTL | 0x00003143

Type: Read/Write

Reset: 0x00000080

Fast Average Enable PMIC_STATIC

Bits	Name	Description
2 : 0	FAST_AVG_SAMPLES	Select number of samples for use in fast average mode (i.e. 2^(value). 0: AVG_1_SAMPLE 1: AVG_2_SAMPLES 2: AVG_4_SAMPLES 3: AVG_8_SAMPLES 4: AVG_16_SAMPLES
7 : 7	FAST_AVG_EN	Enabling will select low latency for multiple averaged conversions by overlapping window samples. When this is disabled, averaged samples are run back-to-back without overlapping. 0: FAST_AVG_DISABLED 1: FAST_AVG_ENABLED

ADC_CORE1_USR_ADC_CH_SEL_CTL | 0x00003144

Type: Read/Write

Reset: 0x00000006

ADC Channel selection PMIC_STATIC

Bits	Name	Description
7 : 0	ADC_CH_SEL	ADC Channel selection, please refer to chip TDOS or sharepoint (go/adc_ch_sel) for more information

ADC_CORE1_USR_DELAY_CTL | 0x00003145

Type: Read/Write

Reset: 0x00000000

Settle Delay PMIC_STATIC

Bits	Name	Description
3 : 0	HW_SETTLE_DELAY	Time between AMUX getting configured and the ADC starting conversion. Delay = 15us for value 0, 100us*(value) for 0<value<8, and 1ms*2^(value-8) otherwise 0: hw_settle_delay_15us 1: hw_settle_delay_100us 2: hw_settle_delay_200us 3: hw_settle_delay_300us 4: hw_settle_delay_400us 5: hw_settle_delay_500us 6: hw_settle_delay_600us 7: hw_settle_delay_700us 8: hw_settle_delay_1ms 9: hw_settle_delay_2ms 10: hw_settle_delay_4ms 11: hw_settle_delay_8ms 12: hw_settle_delay_16ms 13: hw_settle_delay_32ms 14: hw_settle_delay_64ms 15: hw_settle_delay_128ms

ADC_CORE1_USR_EN_CTL1 | 0x00003146

Type: Read/Write

Reset: 0x00000000

Enables ADC module.

Bits	Name	Description
7 : 7	ADC_EN	Enables ADC module. Writing a '0' to this field will abort the ongoing conversion if there is any 0: ADC_DISABLED 1: ADC_ENABLED

ADC_CORE1_USR_CONV_REQ | 0x00003147

Type: Write

Reset: 0x00000000

Conversion Request

Bits	Name	Description
7 : 7	REQ	Conversion request strobe. Note, logic only allows writes of 0x1 to this field. 0x0 writes to this field are ignored. 0: NO_OPERATION 1: START_CONV_REQ

ADC_CORE1_USR_DATA0 | 0x00003150

Type: Read

Reset: 0x00000000

ADC Sample Byte 0

Bits	Name	Description
7 : 0	DATA_7_0	Low byte of ADC output for V-only, I-only, or VBAT of a V/I(synchronous or interleaved) conversions. DATA[15:0]={DATA1[7:0],DATA0[7:0]} forms a 16-bit 2's complement number. 1. For a V-only or VBAT of a V/I conversion, The actual voltage it represents is DATA[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, DATA[15:0]/65536*7.5 Volts. 2. For a I-only conversion, The actual current it represents is DATA[15:0]/32767*10 Amps if SUBTYPE is between 0x28 and 0x2F, or DATA[15:0]/32767*5 Amps if SUBTYPE is between 0x38 and 0x3F.

ADC_CORE1_USR_DATA1 | 0x00003151

Type: Read

Reset: 0x00000000

ADC Sample Byte 1

Bits	Name	Description
7 : 0	DATA_15_8	High byte of ADC output for V-only, I-only, or VBAT of a V/I(synchronous or interleaved) conversions. DATA[15:0]={DATA1[7:0],DATA0[7:0]} forms a 16-bit 2's complement number. 1. For a V-only or VBAT of a V/I conversion, The actual voltage it represents is DATA[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, DATA[15:0]/65536*7.5 Volts. 2. For a I-only conversion, The actual current it represents is DATA[15:0]/32767*10 Amps if SUBTYPE is between 0x28 and 0x2F, or DATA[15:0]/32767*5 Amps if SUBTYPE is between 0x38 and 0x3F.

Module Name: ADC_CORE2_MDM

Base Address: 0x00003200

Register Quick Reference:

Register	Offset	Type
ADC_CORE2_MDM_REVISION1	0x00003200	Read
ADC_CORE2_MDM_REVISION2	0x00003201	Read
ADC_CORE2_MDM_REVISION3	0x00003202	Read
ADC_CORE2_MDM_REVISION4	0x00003203	Read
ADC_CORE2_MDM_PERPH_TYPE	0x00003204	Read
ADC_CORE2_MDM_PERPH_SUBTYPE	0x00003205	Read
ADC_CORE2_MDM_STATUS1	0x00003208	Read

Register	Offset	Type
ADC_CORE2_MDM_STATUS2	0x00003209	Read
ADC_CORE2_MDM_IBAT_MEAS	0x0000320F	Read
ADC_CORE2_MDM_INT_RT_STS	0x00003210	Read
ADC_CORE2_MDM_INT_SET_TYPE	0x00003211	Read/Write
ADC_CORE2_MDM_INT_POLARITY_HIGH	0x00003212	Read/Write
ADC_CORE2_MDM_INT_POLARITY_LOW	0x00003213	Read/Write
ADC_CORE2_MDM_INT_LATCHED_CLR	0x00003214	Write
ADC_CORE2_MDM_INT_EN_SET	0x00003215	Read/Write
ADC_CORE2_MDM_INT_EN_CLR	0x00003216	Read/Write
ADC_CORE2_MDM_INT_LATCHED_STS	0x00003218	Read
ADC_CORE2_MDM_INT_PENDING_STS	0x00003219	Read
ADC_CORE2_MDM_INT_MID_SEL	0x0000321A	Read/Write
ADC_CORE2_MDM_INT_PRIORITY	0x0000321B	Read/Write
ADC_CORE2_MDM_ADC_DIG_PARAM	0x00003242	Read/Write
ADC_CORE2_MDM_FAST_AVG_CTL	0x00003243	Read/Write
ADC_CORE2_MDM_ADC_CH_SEL_CTL	0x00003244	Read/Write
ADC_CORE2_MDM_DELAY_CTL	0x00003245	Read/Write
ADC_CORE2_MDM_EN_CTL1	0x00003246	Read/Write
ADC_CORE2_MDM_CONV_REQ	0x00003247	Write
ADC_CORE2_MDM_DATA0	0x00003250	Read
ADC_CORE2_MDM_DATA1	0x00003251	Read

ADC_CORE2_MDM_REVISION1 | 0x00003200

Type: Read

Reset: 0x00000003

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

ADC_CORE2_MDM_REVISION2 | 0x00003201

Type: Read

Reset: 0x00000005

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

ADC_CORE2_MDM_REVISION3 | 0x00003202

Type: Read

Reset: 0x00000002

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

ADC_CORE2_MDM_REVISION4 | 0x00003203

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

ADC_CORE2_MDM_PERPH_TYPE | 0x00003204

Type: Read

Reset: 0x00000008

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	ADC

ADC_CORE2_MDM_PERPH_SUBTYPE | 0x00003205

Type: Read

Reset: 0x00000031

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	ADC sub type

ADC_CORE2_MDM_STATUS1 | 0x00003208

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
0 : 0	EOC	End of conversion status flag. Bit is de-asserted when arbiter is servicing a conversion request and asserted when conversion is completed. 0: CONV_NOT_COMPLETE 1: CONV_COMPLETE
1 : 1	REQ_STS	REQ_STS mirrors the REQ bit. When REQ is asserted the arbiter stores a descriptor in the conversion request queue. Bit is cleared when ADC conversion is completed. 0: REQ_NOT_IN_PROGRESS 1: REQ_IN_PROGRESS

ADC_CORE2_MDM_STATUS2 | 0x00003209

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
7 : 5	CONV_SEQ_STATE	Conversion request FSM states. 0: IDLE_REQ 1: WAIT_HOLDOFF_RST 2: WAIT_HOLDOFF 3: RST_CH_ARB 4: GET_NEXT_CH 5: WAIT_ADC_EOC 6: WAIT_ADC_EOC2 7: GEN_IRQ

ADC_CORE2_MDM_IBAT_MEAS | 0x0000320F

Type: Read

Reset: 0x00000000

Read-Only register to indicate whether the battery current measurement is supported

Bits	Name	Description
0 : 0	SUPPORTED	0 - battery current measurement is NOT supported; 1 - battery current measurement IS supported 0: IBAT_MEAS_UNSUPPORTED 1: IBAT_MEAS_SUPPORTED

ADC_CORE2_MDM_INT_RT_STS | 0x00003210

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	EOC_INT_RT_STS	Secure process end of conversion interrupt. Active high signal two tcxo_clk cycles wide. 0: CONV_COMPLETE_INT_FALSE 1: CONV_COMPLETE_INT_TRUE

ADC_CORE2_MDM_INT_SET_TYPE | 0x00003211

Type: Read/Write

Reset: 0x00000000

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	EOC_SET_INT_TYPE	EOC interrupt set type 0: EOC_LEVEL 1: EOC_EDGE

ADC_CORE2_MDM_INT_POLARITY_HIGH | 0x00003212

Type: Read/Write

Reset: 0x00000000

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	EOC_INT_HIGH	EOC interrupt high polarity enabled 0: EOC_INT_POL_HIGH_DISABLED 1: EOC_INT_POL_HIGH_ENABLED

ADC_CORE2_MDM_INT_POLARITY_LOW | 0x00003213

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	EOC_INT_LOW	EOC interrupt low polarity enabled 0: EOC_INT_POL_LOW_DISABLED 1: EOC_INT_POL_LOW_ENABLED

ADC_CORE2_MDM_INT_LATCHED_CLR | 0x00003214

Type: Write

Reset: 0x00000000

Writing a '1' to a bit in this register will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits

Bits	Name	Description
0 : 0	EOC_INT_LATCHED_CLR	EOC interrupt latched clear

ADC_CORE2_MDM_INT_EN_SET | 0x00003215

Type: Read/Write

Reset: 0x00000000

Writing '0' to a bit in this register has no effect. Writing a '1' to a bit in this register will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	EOC_INT_EN_SET	EOC interrupt enable set 0: EOC_INT_DISABLED 1: EOC_INT_ENBLED

ADC_CORE2_MDM_INT_EN_CLR | 0x00003216

Type: Read/Write

Reset: 0x00000000

Writing a '0' to a bit in this register has no effect. Writing a '1' to a bit in this register will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	EOC_INT_EN_CLR	EOC interrupt enable clear 0: EOC_INT_DISABLED 1: EOC_INT_ENBLED

ADC_CORE2_MDM_INT_LATCHED_STS | 0x00003218

Type: Read

Reset: 0x00000000

Debug: Latched (Sticky) Interrupt. '1' indiecates that the interrutp has triggered. Once the latched bit is set it can only be cleared by writing the clear bit.

Bits	Name	Description
0 : 0	EOC_INT_LATCHED_STS	EOC interrupt latched 0: EOC_INT_LATCHED_FALSE 1: EOC_INT_LATCHED_TRUE

ADC_CORE2_MDM_INT_PENDING_STS | 0x00003219

Type: Read

Reset: 0x00000000

Debug: Pending is set if interrupt has been sent but not cleared.

Bits	Name	Description
0 : 0	EOC_INT_PENDING_STS	EOC interrupt pending 0: EOC_INT_PENDING_FALSE 1: EOC_INT_PENDING_TRUE

ADC_CORE2_MDM_INT_MID_SEL | 0x0000321A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	Selects the MID that will receive the interrupt

ADC_CORE2_MDM_INT_PRIORITY | 0x0000321B

Type: Read/Write

Reset: 0x00000000

Selects the SPMI interrupt priority

Bits	Name	Description
0 : 0	INT_PRIORITY	Selects the SPMI interrupt priority 0: SR 1: A

ADC_CORE2_MDM_ADC_DIG_PARAM | 0x00003242

Type: Read/Write

Reset: 0x00000018

PMIC_STATIC

Bits	Name	Description
3 : 2	DEC_RATIO_SEL	Select the conversion decimation ratio: * DEC_RATIO_SEL = 0 ==> STEP1 = 170, STEP2 = 80 , Decimation Ratio = STEP1+STEP2 = 250 * DEC_RATIO_SEL = 1 ==> STEP1 = 340, STEP2 = 80 , Decimation Ratio = STEP1+STEP2 = 420 * DEC_RATIO_SEL = 2 ==> STEP1 = 680, STEP2 = 160, Decimation Ratio = STEP1+STEP2 = 840 The above table reflects the default ADC_CMN setting (CONV_MODE = 0 and DIG_BOOST_EN = 1). Please refer to the DIG_BOOST_EN field description in ADC_CMN peripheral if a non-default settings are used for CONV_MODE and DIG_BOOST_EN. 0: DEC_RATIO_SHORT 1: DEC_RATIO_MEDIUM 2: DEC_RATIO_LONG
5 : 4	CAL_SEL	Select calibration method 0: No_cal 1: ratio_cal 2: abs_cal
6 : 6	CAL_VAL	Select calibration values 0: timer_cal 1: new_cal

ADC_CORE2_MDM_FAST_AVG_CTL | 0x00003243

Type: Read/Write

Reset: 0x00000080

Fast Average Enable PMIC_STATIC

Bits	Name	Description
2 : 0	FAST_AVG_SAMPLES	Select number of samples for use in fast average mode (i.e. 2^(value). 0: AVG_1_SAMPLE 1: AVG_2_SAMPLES 2: AVG_4_SAMPLES 3: AVG_8_SAMPLES 4: AVG_16_SAMPLES
7 : 7	FAST_AVG_EN	Enabling will select low latency for multiple averaged conversions by overlapping window samples. When this is disabled, averaged samples are run back-to-back without overlapping. 0: FAST_AVG_DISABLED 1: FAST_AVG_ENABLED

ADC_CORE2_MDM_ADC_CH_SEL_CTL | 0x00003244

Type: Read/Write

Reset: 0x00000006

ADC Channel selection PMIC_STATIC

Bits	Name	Description
7 : 0	ADC_CH_SEL	ADC Channel selection, please refer to chip TDOS or sharepoint (go/adc_ch_sel) for more information

ADC_CORE2_MDM_DELAY_CTL | 0x00003245

Type: Read/Write

Reset: 0x00000000

Settle Delay PMIC_STATIC

Bits	Name	Description
3 : 0	HW_SETTLE_DELAY	Time between AMUX getting configured and the ADC starting conversion. Delay = 15us for value 0, 100us*(value) for 0<value<8, and 1ms*2^(value-8) otherwise 0: hw_settle_delay_15us 1: hw_settle_delay_100us 2: hw_settle_delay_200us 3: hw_settle_delay_300us 4: hw_settle_delay_400us 5: hw_settle_delay_500us 6: hw_settle_delay_600us 7: hw_settle_delay_700us 8: hw_settle_delay_1ms 9: hw_settle_delay_2ms 10: hw_settle_delay_4ms 11: hw_settle_delay_8ms 12: hw_settle_delay_16ms 13: hw_settle_delay_32ms 14: hw_settle_delay_64ms 15: hw_settle_delay_128ms

ADC_CORE2_MDM_EN_CTL1 | 0x00003246

Type: Read/Write

Reset: 0x00000000

Enables ADC module.

Bits	Name	Description
7 : 7	ADC_EN	Enables ADC module. Writing a '0' to this field will abort the ongoing conversion if there is any 0: ADC_DISABLED 1: ADC_ENABLED

ADC_CORE2_MDM_CONV_REQ | 0x00003247

Type: Write

Reset: 0x00000000

Conversion Request

Bits	Name	Description
7 : 7	REQ	Conversion request strobe. Note, logic only allows writes of 0x1 to this field. 0x0 writes to this field are ignored. 0: NO_OPERATION 1: START_CONV_REQ

ADC_CORE2_MDM_DATA0 | 0x00003250

Type: Read

Reset: 0x00000000

ADC Sample Byte 0

Bits	Name	Description
7 : 0	DATA_7_0	Low byte of ADC output for V-only, I-only, or VBAT of a V/I(synchronous or interleaved) conversions. DATA[15:0]={DATA1[7:0],DATA0[7:0]} forms a 16-bit 2's complement number. 1. For a V-only or VBAT of a V/I conversion, The actual voltage it represents is DATA[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, DATA[15:0]/65536*7.5 Volts. 2. For a I-only conversion, The actual current it represents is DATA[15:0]/32767*10 Amps if SUBTYPE is between 0x28 and 0x2F, or DATA[15:0]/32767*5 Amps if SUBTYPE is between 0x38 and 0x3F.

ADC_CORE2_MDM_DATA1 | 0x00003251

Type: Read

Reset: 0x00000000

ADC Sample Byte 1

Bits	Name	Description
7 : 0	DATA_15_8	High byte of ADC output for V-only, I-only, or VBAT of a V/I(synchronous or interleaved) conversions. DATA[15:0]={DATA1[7:0],DATA0[7:0]} forms a 16-bit 2's complement number. 1. For a V-only or VBAT of a V/I conversion, The actual voltage it represents is DATA[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, DATA[15:0]/65536*7.5 Volts. 2. For a I-only conversion, The actual current it represents is DATA[15:0]/32767*10 Amps if SUBTYPE is between 0x28 and 0x2F, or DATA[15:0]/32767*5 Amps if SUBTYPE is between 0x38 and 0x3F.

Module Name: ADC_CORE5_BTM4

Base Address: 0x00003500

Register Quick Reference:

Register	Offset	Type
ADC_CORE5_BTM4_REVISION1	0x00003500	Read
ADC_CORE5_BTM4_REVISION2	0x00003501	Read
ADC_CORE5_BTM4_REVISION3	0x00003502	Read
ADC_CORE5_BTM4_REVISION4	0x00003503	Read
ADC_CORE5_BTM4_PERPH_TYPE	0x00003504	Read
ADC_CORE5_BTM4_PERPH_SUBTYPE	0x00003505	Read
ADC_CORE5_BTM4_STATUS2	0x00003509	Read
ADC_CORE5_BTM4_STATUS_LOW	0x0000350A	Read
ADC_CORE5_BTM4_STATUS_HIGH	0x0000350B	Read
ADC_CORE5_BTM4_NUM_BTM	0x0000350F	Read
ADC_CORE5_BTM4_INT_RT_STS	0x00003510	Read
ADC_CORE5_BTM4_INT_SET_TYPE	0x00003511	Read/Write
ADC_CORE5_BTM4_INT_POLARITY_HIGH	0x00003512	Read/Write
ADC_CORE5_BTM4_INT_POLARITY_LOW	0x00003513	Read/Write
ADC_CORE5_BTM4_INT_LATCHED_CLR	0x00003514	Write
ADC_CORE5_BTM4_INT_EN_SET	0x00003515	Read/Write
ADC_CORE5_BTM4_INT_EN_CLR	0x00003516	Read/Write
ADC_CORE5_BTM4_INT_LATCHED_STS	0x00003518	Read
ADC_CORE5_BTM4_INT_PENDING_STS	0x00003519	Read
ADC_CORE5_BTM4_INT_MID_SEL	0x0000351A	Read/Write
ADC_CORE5_BTM4_INT_PRIORITY	0x0000351B	Read/Write
ADC_CORE5_BTM4_BTM_FAULT_TRIGGER_CTL	0x00003540	Read/Write
ADC_CORE5_BTM4_BTM_PBS_TRIGGER_CTL	0x00003541	Read/Write
ADC_CORE5_BTM4_ADC_DIG_PARAM	0x00003542	Read/Write
ADC_CORE5_BTM4_FAST_AVG_CTL	0x00003543	Read/Write
ADC_CORE5_BTM4_MEAS_INTERVAL_CTL	0x00003544	Read/Write
ADC_CORE5_BTM4_MEAS_INTERVAL_CTL2	0x00003545	Read/Write
ADC_CORE5_BTM4_EN_CTL1	0x00003546	Read/Write
ADC_CORE5_BTM4_CONV_REQ	0x00003547	Write

Register	Offset	Type
ADC_CORE5_BTM4_DATA_HOLD_CTL	0x00003550	Read/Write
ADC_CORE5_BTM4_BTM_ADC_CH_SEL_CTL_0	0x00003560	Read/Write
ADC_CORE5_BTM4_BTM_LOW_THR0_0	0x00003561	Read/Write
ADC_CORE5_BTM4_BTM_LOW_THR1_0	0x00003562	Read/Write
ADC_CORE5_BTM4_BTM_HIGH_THR0_0	0x00003563	Read/Write
ADC_CORE5_BTM4_BTM_HIGH_THR1_0	0x00003564	Read/Write
ADC_CORE5_BTM4_BTM_MEAS_INTERVAL_CTL_0	0x00003565	Read/Write
ADC_CORE5_BTM4_BTM_CTL_0	0x00003566	Read/Write
ADC_CORE5_BTM4_BTM_EN_0	0x00003567	Read/Write
ADC_CORE5_BTM4_BTM_ADC_CH_SEL_CTL_1	0x00003568	Read/Write
ADC_CORE5_BTM4_BTM_LOW_THR0_1	0x00003569	Read/Write
ADC_CORE5_BTM4_BTM_LOW_THR1_1	0x0000356A	Read/Write
ADC_CORE5_BTM4_BTM_HIGH_THR0_1	0x0000356B	Read/Write
ADC_CORE5_BTM4_BTM_HIGH_THR1_1	0x0000356C	Read/Write
ADC_CORE5_BTM4_BTM_MEAS_INTERVAL_CTL_1	0x0000356D	Read/Write
ADC_CORE5_BTM4_BTM_CTL_1	0x0000356E	Read/Write
ADC_CORE5_BTM4_BTM_EN_1	0x0000356F	Read/Write
ADC_CORE5_BTM4_BTM2P_ADC_CH_SEL_CTL_0	0x00003570	Read/Write
ADC_CORE5_BTM4_BTM2P_LOW_THR0_0	0x00003571	Read/Write
ADC_CORE5_BTM4_BTM2P_LOW_THR1_0	0x00003572	Read/Write
ADC_CORE5_BTM4_BTM2P_HIGH_THR0_0	0x00003573	Read/Write
ADC_CORE5_BTM4_BTM2P_HIGH_THR1_0	0x00003574	Read/Write
ADC_CORE5_BTM4_BTM2P_MEAS_INTERVAL_CTL_0	0x00003575	Read/Write
ADC_CORE5_BTM4_BTM2P_CTL_0	0x00003576	Read/Write
ADC_CORE5_BTM4_BTM2P_EN_0	0x00003577	Read/Write
ADC_CORE5_BTM4_BTM2P_ADC_CH_SEL_CTL_1	0x00003578	Read/Write
ADC_CORE5_BTM4_BTM2P_LOW_THR0_1	0x00003579	Read/Write
ADC_CORE5_BTM4_BTM2P_LOW_THR1_1	0x0000357A	Read/Write
ADC_CORE5_BTM4_BTM2P_HIGH_THR0_1	0x0000357B	Read/Write
ADC_CORE5_BTM4_BTM2P_HIGH_THR1_1	0x0000357C	Read/Write
ADC_CORE5_BTM4_BTM2P_MEAS_INTERVAL_CTL_1	0x0000357D	Read/Write
ADC_CORE5_BTM4_BTM2P_CTL_1	0x0000357E	Read/Write

Register	Offset	Type
ADC_CORE5_BTM4_BTM2P_EN_1	0x0000357F	Read/Write
ADC_CORE5_BTM4_BTM_DATA0_0	0x000035A0	Read
ADC_CORE5_BTM4_BTM_DATA1_0	0x000035A1	Read
ADC_CORE5_BTM4_BTM_DATA0_1	0x000035A2	Read
ADC_CORE5_BTM4_BTM_DATA1_1	0x000035A3	Read
ADC_CORE5_BTM4_BTM2P_DATA0_0	0x000035A4	Read
ADC_CORE5_BTM4_BTM2P_DATA1_0	0x000035A5	Read
ADC_CORE5_BTM4_BTM2P_DATA0_1	0x000035A6	Read
ADC_CORE5_BTM4_BTM2P_DATA1_1	0x000035A7	Read

ADC_CORE5_BTM4_REVISION1 | 0x00003500

Type: Read

Reset: 0x00000003

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

ADC_CORE5_BTM4_REVISION2 | 0x00003501

Type: Read

Reset: 0x00000005

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

ADC_CORE5_BTM4_REVISION3 | 0x00003502

Type: Read

Reset: 0x00000002

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

ADC_CORE5_BTM4_REVISION4 | 0x00003503

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

ADC_CORE5_BTM4_PERPH_TYPE | 0x00003504

Type: Read

Reset: 0x00000008

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	ADC

ADC_CORE5_BTM4_PERPH_SUBTYPE | 0x00003505

Type: Read

Reset: 0x00000034

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	ADC sub type

ADC_CORE5_BTM4_STATUS2 | 0x00003509

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
7 : 5	CONV_SEQ_STATE	Conversion request states selected by SEL_FSM register field. SEL_FSM Signal 0 Request FSM #1 state[3;0] 1 Request FSM #2 state[3;0] 2 Request FSM #3 state[3;0] 0: IDLE_REQ 1: WAIT_HOLDOFF_RST 2: WAIT_HOLDOFF 3: RST_CH_ARB 4: GET_NEXT_CH 5: WAIT_ADC_EOC 6: WAIT_ADC_EOC2 7: GEN_IRQ

ADC_CORE5_BTM4_STATUS_LOW | 0x0000350A

Type: Read

Reset: 0x00000000

Indicates measurement(s) where ADC read is less than low threshold.
PMIC_SCOPE=NUM_BTM__NUM_BTM_PS

Bits	Name	Description
3 : 0	LOW	Under-low-threshold status register field. Every BTM channel has one bit in this status register to indicate whether its conversion result is lower than its {LOW_THR_15_8[7:0], LOW_THR_7_0[7:0]} setting. For example, channel #0 status is at bit 0, channel #1 status is at bit 1, and so on. 0: LOW_FALSE 1: LOW_TRUE

ADC_CORE5_BTM4_STATUS_HIGH | 0x0000350B

Type: Read

Reset: 0x00000000

Indicates measurement(s) where ADC read is greater than high threshold.
PMIC_SCOPE=NUM_BTM__NUM_BTM_PS

Bits	Name	Description
3 : 0	HIGH	Above-high-threshold status register field. Every BTM channel has one bit in this status register to indicate whether its conversion result is higher than its {HIGH_THR_15_8[7:0], HIGH_THR_7_0[7:0]} setting. For example, channel #0 status is at bit 0, channel #1 status is at bit 1, and so on. 0: HIGH_FALSE 1: HIGH_TRUE

ADC_CORE5_BTM4_NUM_BTM | 0x0000350F

Type: Read

Reset: 0x00000004

Read-Only register to indicate the number of available BTM channels

Bits	Name	Description
7 : 0	NUM_BTM	Number of available BTM channels

ADC_CORE5_BTM4_INT_RT_STS | 0x00003510

Type: Read

Reset: 0x00000000

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	THR_INT_RT_STS	Threshold interrupt set type 0: THR_INT_FALSE 1: THR_INT_TRUE

ADC_CORE5_BTM4_INT_SET_TYPE | 0x00003511

Type: Read/Write

Reset: 0x00000000

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	THR_INT_SET_TYPE	Threshold interrupt high polarity enabled 0: THR_INT_POL_HIGH_DISABLED 1: THR_INT_POL_HIGH_ENABLED

ADC_CORE5_BTM4_INT_POLARITY_HIGH | 0x00003512

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	THR_INT_HIGH	Threshold interrupt high polarity enabled 0: HIGH_THR_INT_POL_HIGH_DISABLED 1: HIGH_THR_INT_POL_HIGH_ENABLED

ADC_CORE5_BTM4_INT_POLARITY_LOW | 0x00003513

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	THR_INT_LOW	Threshold interrupt low polarity enabled 0: THR_INT_POL_DISABLED 1: THR_INT_POL_ENABLED

ADC_CORE5_BTM4_INT_LATCHED_CLR | 0x00003514

Type: Write

Reset: 0x00000000

Writing a '1' to a bit in this register will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits

Bits	Name	Description
0 : 0	THR_INT_LATCHED_CLR	Threshold interrupt latched clear 0: THR_INT_FALSE 1: THR_INT_TRUE

ADC_CORE5_BTM4_INT_EN_SET | 0x00003515

Type: Read/Write

Reset: 0x00000000

Writing '0' to a bit in this register has no effect. Writing a '1' to a bit in this register will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	THR_INT_EN_SET	Threshold interrupt enable set 0: THR_INT_DISABLED 1: THR_INT_ENBLED

ADC_CORE5_BTM4_INT_EN_CLR | 0x00003516

Type: Read/Write

Reset: 0x00000000

Writing a '0' to a bit in this register has no effect. Writing a '1' to a bit in this register will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	THR_INT_EN_CLR	Threshold interrupt enable clear 0: THR_INT_DISABLED 1: THR_INT_ENBLED

ADC_CORE5_BTM4_INT_LATCHED_STS | 0x00003518

Type: Read

Reset: 0x00000000

Debug: Latched (Sticky) Interrupt. '1' indiecates that the interrutp has triggered. Once the latched bit is set it can only be cleared by writing the clear bit.

Bits	Name	Description
0 : 0	THR_INT_LATCHED_STS	Threshold interrupt latched 0: THR_INT_LATCHED_FALSE 1: THR_INT_LATCHED_TRUE

ADC_CORE5_BTM4_INT_PENDING_STS | 0x00003519

Type: Read

Reset: 0x00000000

Debug: Pending is set if interrupt has been sent but not cleared.

Bits	Name	Description
0 : 0	THR_INT_PENDING_STS	Threshold interrupt pending 0: THR_INT_PENDING_FALSE 1: THR_INT_PENDING_TRUE

ADC_CORE5_BTM4_INT_MID_SEL | 0x0000351A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	Selects the MID that will receive the interrupt

ADC_CORE5_BTM4_INT_PRIORITY | 0x0000351B

Type: Read/Write

Reset: 0x00000000

Selects the SPMI interrupt priority

Bits	Name	Description
0 : 0	INT_PRIORITY	Selects the SPMI interrupt priority 0: SR 1: A

ADC_CORE5_BTM4_BTM_FAULT_TRIGGER_CTL | 0x00003540

Type: Read/Write

Reset: 0x00000000

BTM FAULT Trigger Control Register PMIC_STATIC, PMIC_SCOPE=NUM_BTM__NUM_BTM_PS, PMIC_LOCKEDBYBIT=LOCKBIT_BTM_PERPH

Bits	Name	Description
3 : 0	CH_EN	Select which channel's threshold crossing flag should assert BTM trigger to PON. Each channel is enabled/disabled by one bit of this register. For example, channel #0 is enabled/disabled by bit 0, channel #1 is enabled/disabled by bit 1, and so on.

ADC_CORE5_BTM4_BTM_PBS_TRIGGER_CTL | 0x00003541

Type: Read/Write

Reset: 0x00000000

BTM PBS Trigger Control Register PMIC_STATIC, PMIC_SCOPE=NUM_BTM__NUM_BTM_PS, PMIC_LOCKEDBYBIT=LOCKBIT_BTM_PERPH

Bits	Name	Description
3 : 0	CH_EN	Select which channel's threshold crossing flag should assert BTM trigger to PBS. Each channel is enabled/disabled by one bit of this register. For example, channel #0 is enabled/disabled by bit 0, channel #1 is enabled/disabled by bit 1, and so on.

ADC_CORE5_BTM4_ADC_DIG_PARAM | 0x00003542

Type: Read/Write

Reset: 0x00000008

PMIC_STATIC

Bits	Name	Description
3 : 2	DEC_RATIO_SEL	Select the conversion decimation ratio: * DEC_RATIO_SEL = 0 ==> STEP1 = 170, STEP2 = 80 , Decimation Ratio = STEP1+STEP2 = 250 * DEC_RATIO_SEL = 1 ==> STEP1 = 340, STEP2 = 80 , Decimation Ratio = STEP1+STEP2 = 420 * DEC_RATIO_SEL = 2 ==> STEP1 = 680, STEP2 = 160, Decimation Ratio = STEP1+STEP2 = 840 The above table reflects the default ADC_CMN setting (CONV_MODE = 0 and DIG_BOOST_EN = 1). Please refer to the DIG_BOOST_EN field description in ADC_CMN peripheral if a non-default settings are used for CONV_MODE and DIG_BOOST_EN. 0: DEC_RATIO_SHORT 1: DEC_RATIO_MEDIUM 2: DEC_RATIO_LONG

ADC_CORE5_BTM4_FAST_AVG_CTL | 0x00003543

Type: Read/Write

Reset: 0x00000080

Fast Average Enable PMIC_STATIC

Bits	Name	Description
2 : 0	FAST_AVG_SAMPLES	Select number of samples for use in fast average mode (i.e. 2^(value). 0: AVG_1_SAMPLE 1: AVG_2_SAMPLES 2: AVG_4_SAMPLES 3: AVG_8_SAMPLES 4: AVG_16_SAMPLES
7 : 7	FAST_AVG_EN	Enabling will select low latency for multiple averaged conversions by overlapping window samples. When this is disabled, averaged samples are run back-to-back without overlapping. 0: FAST_AVG_DISABLED 1: FAST_AVG_ENABLED

ADC_CORE5_BTM4_MEAS_INTERVAL_CTL | 0x00003544

Type: Read/Write

Reset: 0x00000000

Interval Mode Control PMIC_STATIC

Bits	Name	Description
3 : 0	MEAS_INTERVAL_TIME1	Select measurement interval time (i.e., If value=0, use 0ms, else use 2^(value+4)/32768). 0: meas_interval1_0ms 1: meas_interval1_1p0ms 2: meas_interval1_2p0ms 3: meas_interval1_3p9ms 4: meas_interval1_7p8ms 5: meas_interval1_15p6ms 6: meas_interval1_31p3ms 7: meas_interval1_62p5ms 8: meas_interval1_125ms 9: meas_interval1_250ms 10: meas_interval1_500ms 11: meas_interval1_1s 12: meas_interval1_2s 13: meas_interval1_4s 14: meas_interval1_8s 15: meas_interval1_16s

ADC_CORE5_BTM4_MEAS_INTERVAL_CTL2 | 0x00003545

Type: Read/Write

Reset: 0x00000000

PMIC_STATIC

Bits	Name	Description
3 : 0	MEAS_INTERVAL_TIME3	Large timer: Select measurement interval time in seconds. 0: meas_interval3_0s 1: meas_interval3_1s 2: meas_interval3_2s 3: meas_interval3_3s 4: meas_interval3_4s 5: meas_interval3_5s 6: meas_interval3_6s 7: meas_interval3_7s 8: meas_interval3_8s 9: meas_interval3_9s 10: meas_interval3_10s 11: meas_interval3_11s 12: meas_interval3_12s 13: meas_interval3_13s 14: meas_interval3_14s 15: meas_interval3_15s
7 : 4	MEAS_INTERVAL_TIME2	Small timer: Select measurement interval time in 100ms increments. 0: meas_interval2_0ms 1: meas_interval2_100ms 2: meas_interval2_200ms 3: meas_interval2_300ms 4: meas_interval2_400ms 5: meas_interval2_500ms 6: meas_interval2_600ms 7: meas_interval2_700ms 8: meas_interval2_800ms 9: meas_interval2_900ms 10: meas_interval2_1000ms 11: meas_interval2_1100ms 12: meas_interval2_1200ms 13: meas_interval2_1300ms 14: meas_interval2_1400ms 15: meas_interval2_1500ms

ADC_CORE5_BTM4_EN_CTL1 | 0x00003546

Type: Read/Write

Reset: 0x00000000

Enables ADC module. PMIC_LOCKEDBYBIT=LOCKBIT_BTM_PERPH

Bits	Name	Description
7 : 7	ADC_EN	Enables ADC module. Writing a '0' to this field will abort all the ongoing BTM conversions if there is any 0: ADC_DISABLED 1: ADC_ENABLED

ADC_CORE5_BTM4_CONV_REQ | 0x00003547

Type: Write

Reset: 0x00000000

Conversion Request

Bits	Name	Description
7 : 7	REQ	Conversion request strobe. Note, logic only allows writes of 0x1 to this field. 0x0 writes to this field are ignored. 0: NO_OPERATION 1: START_CONV_REQ

ADC_CORE5_BTM4_DATA_HOLD_CTL | 0x00003550

Type: Read/Write

Reset: 0x00000000

Data Hold Control register

Bits	Name	Description
0 : 0	HOLD	Every time this register bit is written to a '1', the {Mx_DATA1[7:0], Mx_DATA0[7:0]} data and STATUS_HIGH/STATUS_LOW status will be loaded to a shadow register for read access. However, if this register is a '0', the free-running {Mx_DATA1[7:0], Mx_DATA0[7:0]} data and STATUS_HIGH/STATUS_LOW status will be read instead. 0: FREE_RUNNING 1: HOLD_DATA

ADC_CORE5_BTM4_BTM_ADC_CH_SEL_CTL_0 | 0x00003560

Type: Read/Write

Reset: 0x000000FF

ADC Channel selection PMIC_STATIC, PMIC_LOCKEDBYBIT=LOCKBIT_BTM0

Bits	Name	Description
7 : 0	ADC_CH_SEL	ADC Channel selection, please refer to chip TDOS or sharepoint (go/adc_ch_sel) for more information

ADC_CORE5_BTM4_BTM_LOW_THR0_0 | 0x00003561

Type: Read/Write

Reset: 0x00000001

Low Threshold Byte 0 PMIC_LOCKEDBYBIT=LOCKBIT_BTM0

Bits	Name	Description
7 : 0	LOW_THR_7_0	Low byte of low threshold detector. BTM_LOW_THR[15:0]={BTM_LOW_THR1[7:0],BTM_LOW_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_LOW_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_LOW_THR[15:0]/65536*7.5 Volts. A write to this register must be followed by a BTM_LOW_THR1 write.

ADC_CORE5_BTM4_BTM_LOW_THR1_0 | 0x00003562

Type: Read/Write

Reset: 0x00000080

Low Threshold Byte 1 PMIC_LOCKEDBYBIT=LOCKBIT_BTM0

Bits	Name	Description
7 : 0	LOW_THR_15_8	High byte of low threshold detector. BTM_LOW_THR[15:0]={BTM_LOW_THR1[7:0],BTM_LOW_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_LOW_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_LOW_THR[15:0]/65536*7.5 Volts. A write to this register must be preceded by a BTM_LOW_THR0 write.

ADC_CORE5_BTM4_BTM_HIGH_THR0_0 | 0x00003563

Type: Read/Write

Reset: 0x000000FF

High Threshold Byte 0 PMIC_LOCKEDBYBIT=LOCKBIT_BTM0

Bits	Name	Description
7 : 0	HIGH_THR_7_0	Low byte of high threshold detector. BTM_HIGH_THR[15:0]={BTM_HIGH_THR1[7:0],BTM_HIGH_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_HIGH_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_HIGH_THR[15:0]/65536*7.5 Volts. A write to this register must be followed by a BTM_HIGH_THR1 write.

ADC_CORE5_BTM4_BTM_HIGH_THR1_0 | 0x00003564

Type: Read/Write

Reset: 0x0000007F

High Threshold Byte 1 PMIC_LOCKEDBYBIT=LOCKBIT_BTM0

Bits	Name	Description
7 : 0	HIGH_THR_15_8	High byte of high threshold detector. BTM_HIGH_THR[15:0]={BTM_HIGH_THR1[7:0],BTM_HIGH_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_HIGH_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_HIGH_THR[15:0]/65536*7.5 Volts. A write to this register must be preceded by a BTM_HIGH_THR0 write.

ADC_CORE5_BTM4_BTM_MEAS_INTERVAL_CTL_0 | 0x00003565

Type: Read/Write

Reset: 0x00000000

Interval mode select PMIC_LOCKEDBYBIT=LOCKBIT_BTM0

Bits	Name	Description
1 : 0	MEAS_INTERVAL_TIME	Select which interval timer to use 0: USING_TIMER1 1: USING_TIMER2 2: USING_TIMER3

ADC_CORE5_BTM4_BTM_CTL_0 | 0x00003566

Type: Read/Write

Reset: 0x00000000

BTM control PMIC_STATIC, PMIC_LOCKEDBYBIT=LOCKBIT_BTM0

Bits	Name	Description
3 : 0	HW_SETTLE_DELAY	Time between AMUX getting configured and the ADC starting conversion. Delay = 15us for value 0, 100us*(value) for 0<value<8, and 1ms*2^(value-8) otherwise 0: hw_settle_delay_15us 1: hw_settle_delay_100us 2: hw_settle_delay_200us 3: hw_settle_delay_300us 4: hw_settle_delay_400us 5: hw_settle_delay_500us 6: hw_settle_delay_600us 7: hw_settle_delay_700us 8: hw_settle_delay_1ms 9: hw_settle_delay_2ms 10: hw_settle_delay_4ms 11: hw_settle_delay_8ms 12: hw_settle_delay_16ms 13: hw_settle_delay_32ms 14: hw_settle_delay_64ms 15: hw_settle_delay_128ms
5 : 4	CAL_SEL	Select calibration method: 0: No_cal 1: ratio_cal 2: abs_cal
6 : 6	CAL_VAL	Select calibration value 0: timer_cal 1: new_cal

ADC_CORE5_BTM4_BTM_EN_0 | 0x00003567

Type: Read/Write

Reset: 0x00000000

BTM enable PMIC_LOCKEDBYBIT=LOCKBIT_BTM0

Bits	Name	Description
0 : 0	LOW_THR_INT_EN	Enables low threshold for interrupt 0: LOW_THR_INT_DISABLED 1: LOW_THR_INT_ENABLED
1 : 1	HIGH_THR_INT_EN	Enables high threshold for interrupt 0: HIGH_THR_INT_DISABLED 1: HIGH_THR_INT_ENABLED
7 : 7	MEAS_EN	Enables measurement in auto-sequence. Writing a '0' to this field will abort the ongoing conversion if there is any 0: MEAS_DISABLE 1: MEAS_ENABLE

ADC_CORE5_BTM4_BTM_ADC_CH_SEL_CTL_1 | 0x00003568

Type: Read/Write

Reset: 0x00000000

ADC Channel selection PMIC_STATIC, PMIC_LOCKEDBYBIT=LOCKBIT_BTM1

Bits	Name	Description
7 : 0	ADC_CH_SEL	ADC Channel selection, please refer to chip TDOS or sharepoint (go/adc_ch_sel) for more information

ADC_CORE5_BTM4_BTM_LOW_THR0_1 | 0x00003569

Type: Read/Write

Reset: 0x00000001

Low Threshold Byte 0 PMIC_LOCKEDBYBIT=LOCKBIT_BTM1

Bits	Name	Description
7 : 0	LOW_THR_7_0	Low byte of low threshold detector. BTM_LOW_THR[15:0]={BTM_LOW_THR1[7:0],BTM_LOW_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_LOW_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_LOW_THR[15:0]/65536*7.5 Volts. A write to this register must be followed by a BTM_LOW_THR1 write.

ADC_CORE5_BTM4_BTM_LOW_THR1_1 | 0x0000356A

Type: Read/Write

Reset: 0x00000080

Low Threshold Byte 1 PMIC_LOCKEDBYBIT=LOCKBIT_BTM1

Bits	Name	Description
7 : 0	LOW_THR_15_8	High byte of low threshold detector. BTM_LOW_THR[15:0]={BTM_LOW_THR1[7:0],BTM_LOW_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_LOW_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_LOW_THR[15:0]/65536*7.5 Volts. A write to this register must be preceded by a BTM_LOW_THR0 write.

ADC_CORE5_BTM4_BTM_HIGH_THR0_1 | 0x0000356B

Type: Read/Write

Reset: 0x000000FF

High Threshold Byte 0 PMIC_LOCKEDBYBIT=LOCKBIT_BTM1

Bits	Name	Description
7 : 0	HIGH_THR_7_0	Low byte of high threshold detector. BTM_HIGH_THR[15:0]={BTM_HIGH_THR1[7:0],BTM_HIGH_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_HIGH_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_HIGH_THR[15:0]/65536*7.5 Volts. A write to this register must be followed by a BTM_HIGH_THR1 write.

ADC_CORE5_BTM4_BTM_HIGH_THR1_1 | 0x0000356C

Type: Read/Write

Reset: 0x0000007F

High Threshold Byte 1 PMIC_LOCKEDBYBIT=LOCKBIT_BTM1

Bits	Name	Description
7 : 0	HIGH_THR_15_8	High byte of high threshold detector. BTM_HIGH_THR[15:0]={BTM_HIGH_THR1[7:0],BTM_HIGH_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_HIGH_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_HIGH_THR[15:0]/65536*7.5 Volts. A write to this register must be preceded by a BTM_HIGH_THR0 write.

ADC_CORE5_BTM4_BTM_MEAS_INTERVAL_CTL_1 | 0x0000356D

Type: Read/Write

Reset: 0x00000000

Interval mode select PMIC_LOCKEDBYBIT=LOCKBIT_BTM1

Bits	Name	Description
1 : 0	MEAS_INTERVAL_TIME	Select which interval timer to use 0: USING_TIMER1 1: USING_TIMER2 2: USING_TIMER3

ADC_CORE5_BTM4_BTM_CTL_1 | 0x0000356E

Type: Read/Write

Reset: 0x00000000

BTM control PMIC_STATIC, PMIC_LOCKEDBYBIT=LOCKBIT_BTM1

Bits	Name	Description
3 : 0	HW_SETTLE_DELAY	Time between AMUX getting configured and the ADC starting conversion. Delay = 15us for value 0, 100us*(value) for 0<value<8, and 1ms*2^(value-8) otherwise 0: hw_settle_delay_15us 1: hw_settle_delay_100us 2: hw_settle_delay_200us 3: hw_settle_delay_300us 4: hw_settle_delay_400us 5: hw_settle_delay_500us 6: hw_settle_delay_600us 7: hw_settle_delay_700us 8: hw_settle_delay_1ms 9: hw_settle_delay_2ms 10: hw_settle_delay_4ms 11: hw_settle_delay_8ms 12: hw_settle_delay_16ms 13: hw_settle_delay_32ms 14: hw_settle_delay_64ms 15: hw_settle_delay_128ms
5 : 4	CAL_SEL	Select calibration method: 0: No_cal 1: ratio_cal 2: abs_cal
6 : 6	CAL_VAL	Select calibration value 0: timer_cal 1: new_cal

ADC_CORE5_BTM4_BTM_EN_1 | 0x0000356F

Type: Read/Write

Reset: 0x00000000

BTM enable PMIC_LOCKEDBYBIT=LOCKBIT_BTM1

Bits	Name	Description
0 : 0	LOW_THR_INT_EN	Enables low threshold for interrupt 0: LOW_THR_INT_DISABLED 1: LOW_THR_INT_ENABLED
1 : 1	HIGH_THR_INT_EN	Enables high threshold for interrupt 0: HIGH_THR_INT_DISABLED 1: HIGH_THR_INT_ENABLED

Bits	Name	Description
7 : 7	MEAS_EN	Enables measurement in auto-sequence. Writing a '0' to this field will abort the ongoing conversion if there is any 0: MEAS_DISABLE 1: MEAS_ENABLE

ADC_CORE5_BTM4_BTM2P_ADC_CH_SEL_CTL_0 | 0x00003570

Type: Read/Write

Reset: 0x00000000

ADC Channel selection PMIC_STATIC

Bits	Name	Description
7 : 0	ADC_CH_SEL	ADC Channel selection, please refer to chip TDOS or sharepoint (go/adc_ch_sel) for more information

ADC_CORE5_BTM4_BTM2P_LOW_THR0_0 | 0x00003571

Type: Read/Write

Reset: 0x00000001

Low Threshold Byte 0

Bits	Name	Description
7 : 0	LOW_THR_7_0	Low byte of low threshold detector. BTM_LOW_THR[15:0]={BTM_LOW_THR1[7:0],BTM_LOW_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_LOW_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_LOW_THR[15:0]/65536*7.5 Volts. A write to this register must be followed by a BTM_LOW_THR1 write.

ADC_CORE5_BTM4_BTM2P_LOW_THR1_0 | 0x00003572

Type: Read/Write

Reset: 0x00000080

Low Threshold Byte 1

Bits	Name	Description
7 : 0	LOW_THR_15_8	High byte of low threshold detector. BTM_LOW_THR[15:0]={BTM_LOW_THR1[7:0],BTM_LOW_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_LOW_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_LOW_THR[15:0]/65536*7.5 Volts. A write to this register must be preceded by a BTM_LOW_THR0 write.

ADC_CORE5_BTM4_BTM2P_HIGH_THR0_0 | 0x00003573

Type: Read/Write

Reset: 0x000000FF

High Threshold Byte 0

Bits	Name	Description
7 : 0	HIGH_THR_7_0	Low byte of high threshold detector. BTM_HIGH_THR[15:0]={BTM_HIGH_THR1[7:0],BTM_HIGH_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_HIGH_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_HIGH_THR[15:0]/65536*7.5 Volts. A write to this register must be followed by a BTM_HIGH_THR1 write.

ADC_CORE5_BTM4_BTM2P_HIGH_THR1_0 | 0x00003574

Type: Read/Write

Reset: 0x0000007F

High Threshold Byte 1

Bits	Name	Description
7 : 0	HIGH_THR_15_8	High byte of high threshold detector. BTM_HIGH_THR[15:0]={BTM_HIGH_THR1[7:0],BTM_HIGH_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_HIGH_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_HIGH_THR[15:0]/65536*7.5 Volts. A write to this register must be preceded by a BTM_HIGH_THR0 write.

ADC_CORE5_BTM4_BTM2P_MEAS_INTERVAL_CTL_0 | 0x00003575

Type: Read/Write

Reset: 0x00000000

Interval mode select

Bits	Name	Description
1 : 0	MEAS_INTERVAL_TIME	Select which interval timer to use 0: USING_TIMER1 1: USING_TIMER2 2: USING_TIMER3

ADC_CORE5_BTM4_BTM2P_CTL_0 | 0x00003576

Type: Read/Write

Reset: 0x00000000

BTM control PMIC_STATIC

Bits	Name	Description
3 : 0	HW_SETTLE_DELAY	Time between AMUX getting configured and the ADC starting conversion. Delay = 15us for value 0, 100us*(value) for 0<value<8, and 1ms*2^(value-8) otherwise 0: hw_settle_delay_15us 1: hw_settle_delay_100us 2: hw_settle_delay_200us 3: hw_settle_delay_300us 4: hw_settle_delay_400us 5: hw_settle_delay_500us 6: hw_settle_delay_600us 7: hw_settle_delay_700us 8: hw_settle_delay_1ms 9: hw_settle_delay_2ms 10: hw_settle_delay_4ms 11: hw_settle_delay_8ms 12: hw_settle_delay_16ms 13: hw_settle_delay_32ms 14: hw_settle_delay_64ms 15: hw_settle_delay_128ms
5 : 4	CAL_SEL	Select calibration method: 0: No_cal 1: ratio_cal 2: abs_cal

Bits	Name	Description
6 : 6	CAL_VAL	Select calibration value 0: timer_cal 1: new_cal

ADC_CORE5_BTM4_BTM2P_EN_0 | 0x00003577

Type: Read/Write

Reset: 0x00000000

BTM enable

Bits	Name	Description
0 : 0	LOW_THR_INT_EN	Enables low threshold for interrupt 0: LOW_THR_INT_DISABLED 1: LOW_THR_INT_ENABLED
1 : 1	HIGH_THR_INT_EN	Enables high threshold for interrupt 0: HIGH_THR_INT_DISABLED 1: HIGH_THR_INT_ENABLED
7 : 7	MEAS_EN	Enables measurement in auto-sequence. Writing a '0' to this field will abort the ongoing conversion if there is any 0: MEAS_DISABLE 1: MEAS_ENABLE

ADC_CORE5_BTM4_BTM2P_ADC_CH_SEL_CTL_1 | 0x00003578

Type: Read/Write

Reset: 0x00000000

ADC Channel selection PMIC_STATIC

Bits	Name	Description
7 : 0	ADC_CH_SEL	ADC Channel selection, please refer to chip TDOS or sharepoint (go/adc_ch_sel) for more information

ADC_CORE5_BTM4_BTM2P_LOW_THR0_1 | 0x00003579

Type: Read/Write

Reset: 0x00000001

Low Threshold Byte 0

Bits	Name	Description
7 : 0	LOW_THR_7_0	Low byte of low threshold detector. BTM_LOW_THR[15:0]={BTM_LOW_THR1[7:0],BTM_LOW_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_LOW_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_LOW_THR[15:0]/65536*7.5 Volts. A write to this register must be followed by a BTM_LOW_THR1 write.

ADC_CORE5_BTM4_BTM2P_LOW_THR1_1 | 0x0000357A

Type: Read/Write

Reset: 0x00000080

Low Threshold Byte 1

Bits	Name	Description
7 : 0	LOW_THR_15_8	High byte of low threshold detector. BTM_LOW_THR[15:0]={BTM_LOW_THR1[7:0],BTM_LOW_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_LOW_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_LOW_THR[15:0]/65536*7.5 Volts. A write to this register must be preceded by a BTM_LOW_THR0 write.

ADC_CORE5_BTM4_BTM2P_HIGH_THR0_1 | 0x0000357B

Type: Read/Write

Reset: 0x000000FF

High Threshold Byte 0

Bits	Name	Description
7 : 0	HIGH_THR_7_0	Low byte of high threshold detector. BTM_HIGH_THR[15:0]={BTM_HIGH_THR1[7:0],BTM_HIGH_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_HIGH_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_HIGH_THR[15:0]/65536*7.5 Volts. A write to this register must be followed by a BTM_HIGH_THR1 write.

ADC_CORE5_BTM4_BTM2P_HIGH_THR1_1 | 0x0000357C

Type: Read/Write

Reset: 0x0000007F

High Threshold Byte 1

Bits	Name	Description
7 : 0	HIGH_THR_15_8	High byte of high threshold detector. BTM_HIGH_THR[15:0]={BTM_HIGH_THR1[7:0],BTM_HIGH_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_HIGH_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_HIGH_THR[15:0]/65536*7.5 Volts. A write to this register must be preceded by a BTM_HIGH_THR0 write.

ADC_CORE5_BTM4_BTM2P_MEAS_INTERVAL_CTL_1 | 0x0000357D

Type: Read/Write

Reset: 0x00000000

Interval mode select

Bits	Name	Description
1 : 0	MEAS_INTERVAL_TIME	Select which interval timer to use 0: USING_TIMER1 1: USING_TIMER2 2: USING_TIMER3

ADC_CORE5_BTM4_BTM2P_CTL_1 | 0x0000357E

Type: Read/Write

Reset: 0x00000000

BTM control PMIC_STATIC

Bits	Name	Description
3 : 0	HW_SETTLE_DELAY	Time between AMUX getting configured and the ADC starting conversion. Delay = 15us for value 0, 100us*(value) for 0<value<8, and 1ms*2^(value-8) otherwise 0: hw_settle_delay_15us 1: hw_settle_delay_100us 2: hw_settle_delay_200us 3: hw_settle_delay_300us 4: hw_settle_delay_400us 5: hw_settle_delay_500us 6: hw_settle_delay_600us 7: hw_settle_delay_700us 8: hw_settle_delay_1ms 9: hw_settle_delay_2ms 10: hw_settle_delay_4ms 11: hw_settle_delay_8ms 12: hw_settle_delay_16ms 13: hw_settle_delay_32ms 14: hw_settle_delay_64ms 15: hw_settle_delay_128ms
5 : 4	CAL_SEL	Select calibration method: 0: No_cal 1: ratio_cal 2: abs_cal
6 : 6	CAL_VAL	Select calibration value 0: timer_cal 1: new_cal

ADC_CORE5_BTM4_BTM2P_EN_1 | 0x0000357F

Type: Read/Write

Reset: 0x00000000

BTM enable

Bits	Name	Description
0 : 0	LOW_THR_INT_EN	Enables low threshold for interrupt 0: LOW_THR_INT_DISABLED 1: LOW_THR_INT_ENABLED
1 : 1	HIGH_THR_INT_EN	Enables high threshold for interrupt 0: HIGH_THR_INT_DISABLED 1: HIGH_THR_INT_ENABLED

Bits	Name	Description
7 : 7	MEAS_EN	Enables measurement in auto-sequence. Writing a '0' to this field will abort the ongoing conversion if there is any 0: MEAS_DISABLE 1: MEAS_ENABLE

ADC_CORE5_BTM4_BTM_DATA0_0 | 0x000035A0

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA_7_0	Low byte of ADC output for V-only, I-only, or VBAT of a V/I(synchronous or interleaved) conversions. BTM_DATA[15:0]={BTM_DATA1[7:0],BTM_DATA0[7:0]} forms a 16-bit 2's complement number. 1. For a V-only or VBAT of a V/I conversion, The actual voltage it represents is BTM_DATA[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_DATA[15:0]/65536*7.5 Volts. 2. For a I-only conversion, The actual current it represents is BTM_DATA[15:0]/32767*10 Amps if SUBTYPE is between 0x28 and 0x2F, or BTM_DATA[15:0]/32767*5 Amps if SUBTYPE is between 0x38 and 0x3F.

ADC_CORE5_BTM4_BTM_DATA1_0 | 0x000035A1

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA_15_8	High byte of ADC output for V-only, I-only, or VBAT of a V/I(synchronous or interleaved) conversions. BTM_DATA[15:0]={BTM_DATA1[7:0],BTM_DATA0[7:0]} forms a 16-bit 2's complement number. 1. For a V-only or VBAT of a V/I conversion, The actual voltage it represents is BTM_DATA[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_DATA[15:0]/65536*7.5 Volts. 2. For a I-only conversion, The actual current it represents is BTM_DATA[15:0]/32767*10 Amps if SUBTYPE is between 0x28 and 0x2F, or BTM_DATA[15:0]/32767*5 Amps if SUBTYPE is between 0x38 and 0x3F.

ADC_CORE5_BTM4_BTM_DATA0_1 | 0x000035A2

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA_7_0	Low byte of ADC output for V-only, I-only, or VBAT of a V/I(synchronous or interleaved) conversions. BTM_DATA[15:0]={BTM_DATA1[7:0],BTM_DATA0[7:0]} forms a 16-bit 2's complement number. 1. For a V-only or VBAT of a V/I conversion, The actual voltage it represents is BTM_DATA[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_DATA[15:0]/65536*7.5 Volts. 2. For a I-only conversion, The actual current it represents is BTM_DATA[15:0]/32767*10 Amps if SUBTYPE is between 0x28 and 0x2F, or BTM_DATA[15:0]/32767*5 Amps if SUBTYPE is between 0x38 and 0x3F.

ADC_CORE5_BTM4_BTM_DATA1_1 | 0x000035A3

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA_15_8	High byte of ADC output for V-only, I-only, or VBAT of a V/I(synchronous or interleaved) conversions. BTM_DATA[15:0]={BTM_DATA1[7:0],BTM_DATA0[7:0]} forms a 16-bit 2's complement number. 1. For a V-only or VBAT of a V/I conversion, The actual voltage it represents is BTM_DATA[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_DATA[15:0]/65536*7.5 Volts. 2. For a I-only conversion, The actual current it represents is BTM_DATA[15:0]/32767*10 Amps if SUBTYPE is between 0x28 and 0x2F, or BTM_DATA[15:0]/32767*5 Amps if SUBTYPE is between 0x38 and 0x3F.

ADC_CORE5_BTM4_BTM2P_DATA0_0 | 0x000035A4

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA_7_0	Low byte of ADC output for V-only, I-only, or VBAT of a V/I(synchronous or interleaved) conversions. BTM_DATA[15:0]={BTM_DATA1[7:0],BTM_DATA0[7:0]} forms a 16-bit 2's complement number. 1. For a V-only or VBAT of a V/I conversion, The actual voltage it represents is BTM_DATA[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_DATA[15:0]/65536*7.5 Volts. 2. For a I-only conversion, The actual current it represents is BTM_DATA[15:0]/32767*10 Amps if SUBTYPE is between 0x28 and 0x2F, or BTM_DATA[15:0]/32767*5 Amps if SUBTYPE is between 0x38 and 0x3F.

ADC_CORE5_BTM4_BTM2P_DATA1_0 | 0x000035A5

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA_15_8	High byte of ADC output for V-only, I-only, or VBAT of a V/I(synchronous or interleaved) conversions. BTM_DATA[15:0]={BTM_DATA1[7:0],BTM_DATA0[7:0]} forms a 16-bit 2's complement number. 1. For a V-only or VBAT of a V/I conversion, The actual voltage it represents is BTM_DATA[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_DATA[15:0]/65536*7.5 Volts. 2. For a I-only conversion, The actual current it represents is BTM_DATA[15:0]/32767*10 Amps if SUBTYPE is between 0x28 and 0x2F, or BTM_DATA[15:0]/32767*5 Amps if SUBTYPE is between 0x38 and 0x3F.

ADC_CORE5_BTM4_BTM2P_DATA0_1 | 0x000035A6

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA_7_0	Low byte of ADC output for V-only, I-only, or VBAT of a V/I(synchronous or interleaved) conversions. BTM_DATA[15:0]={BTM_DATA1[7:0],BTM_DATA0[7:0]} forms a 16-bit 2's complement number. 1. For a V-only or VBAT of a V/I conversion, The actual voltage it represents is BTM_DATA[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_DATA[15:0]/65536*7.5 Volts. 2. For a I-only conversion, The actual current it represents is BTM_DATA[15:0]/32767*10 Amps if SUBTYPE is between 0x28 and 0x2F, or BTM_DATA[15:0]/32767*5 Amps if SUBTYPE is between 0x38 and 0x3F.

ADC_CORE5_BTM4_BTM2P_DATA1_1 | 0x000035A7

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA_15_8	High byte of ADC output for V-only, I-only, or VBAT of a V/I(synchronous or interleaved) conversions. BTM_DATA[15:0]={BTM_DATA1[7:0],BTM_DATA0[7:0]} forms a 16-bit 2's complement number. 1. For a V-only or VBAT of a V/I conversion, The actual voltage it represents is BTM_DATA[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_DATA[15:0]/65536*7.5 Volts. 2. For a I-only conversion, The actual current it represents is BTM_DATA[15:0]/32767*10 Amps if SUBTYPE is between 0x28 and 0x2F, or BTM_DATA[15:0]/32767*5 Amps if SUBTYPE is between 0x38 and 0x3F.

Module Name: ADC_CORE5_BTM2

Base Address: 0x00003800

Register Quick Reference:

Register	Offset	Type
ADC_CORE5_BTM2_REVISION1	0x00003800	Read
ADC_CORE5_BTM2_REVISION2	0x00003801	Read
ADC_CORE5_BTM2_REVISION3	0x00003802	Read
ADC_CORE5_BTM2_REVISION4	0x00003803	Read
ADC_CORE5_BTM2_PERPH_TYPE	0x00003804	Read
ADC_CORE5_BTM2_PERPH_SUBTYPE	0x00003805	Read
ADC_CORE5_BTM2_STATUS2	0x00003809	Read
ADC_CORE5_BTM2_STATUS_LOW	0x0000380A	Read
ADC_CORE5_BTM2_STATUS_HIGH	0x0000380B	Read
ADC_CORE5_BTM2_NUM_BTM	0x0000380F	Read
ADC_CORE5_BTM2_INT_RT_STS	0x00003810	Read
ADC_CORE5_BTM2_INT_SET_TYPE	0x00003811	Read/Write
ADC_CORE5_BTM2_INT_POLARITY_HIGH	0x00003812	Read/Write
ADC_CORE5_BTM2_INT_POLARITY_LOW	0x00003813	Read/Write
ADC_CORE5_BTM2_INT_LATCHED_CLR	0x00003814	Write
ADC_CORE5_BTM2_INT_EN_SET	0x00003815	Read/Write
ADC_CORE5_BTM2_INT_EN_CLR	0x00003816	Read/Write
ADC_CORE5_BTM2_INT_LATCHED_STS	0x00003818	Read
ADC_CORE5_BTM2_INT_PENDING_STS	0x00003819	Read
ADC_CORE5_BTM2_INT_MID_SEL	0x0000381A	Read/Write
ADC_CORE5_BTM2_INT_PRIORITY	0x0000381B	Read/Write

Register	Offset	Type
ADC_CORE5_BTM2_BTM_FAULT_TRIGGER_CTL	0x00003840	Read/Write
ADC_CORE5_BTM2_BTM_PBS_TRIGGER_CTL	0x00003841	Read/Write
ADC_CORE5_BTM2_ADC_DIG_PARAM	0x00003842	Read/Write
ADC_CORE5_BTM2_FAST_AVG_CTL	0x00003843	Read/Write
ADC_CORE5_BTM2_MEAS_INTERVAL_CTL	0x00003844	Read/Write
ADC_CORE5_BTM2_MEAS_INTERVAL_CTL2	0x00003845	Read/Write
ADC_CORE5_BTM2_EN_CTL1	0x00003846	Read/Write
ADC_CORE5_BTM2_CONV_REQ	0x00003847	Write
ADC_CORE5_BTM2_DATA_HOLD_CTL	0x00003850	Read/Write
ADC_CORE5_BTM2_BTM_ADC_CH_SEL_CTL_0	0x00003860	Read/Write
ADC_CORE5_BTM2_BTM_LOW_THR0_0	0x00003861	Read/Write
ADC_CORE5_BTM2_BTM_LOW_THR1_0	0x00003862	Read/Write
ADC_CORE5_BTM2_BTM_HIGH_THR0_0	0x00003863	Read/Write
ADC_CORE5_BTM2_BTM_HIGH_THR1_0	0x00003864	Read/Write
ADC_CORE5_BTM2_BTM_MEAS_INTERVAL_CTL_0	0x00003865	Read/Write
ADC_CORE5_BTM2_BTM_CTL_0	0x00003866	Read/Write
ADC_CORE5_BTM2_BTM_EN_0	0x00003867	Read/Write
ADC_CORE5_BTM2_BTM_ADC_CH_SEL_CTL_1	0x00003868	Read/Write
ADC_CORE5_BTM2_BTM_LOW_THR0_1	0x00003869	Read/Write
ADC_CORE5_BTM2_BTM_LOW_THR1_1	0x0000386A	Read/Write
ADC_CORE5_BTM2_BTM_HIGH_THR0_1	0x0000386B	Read/Write
ADC_CORE5_BTM2_BTM_HIGH_THR1_1	0x0000386C	Read/Write
ADC_CORE5_BTM2_BTM_MEAS_INTERVAL_CTL_1	0x0000386D	Read/Write
ADC_CORE5_BTM2_BTM_CTL_1	0x0000386E	Read/Write
ADC_CORE5_BTM2_BTM_EN_1	0x0000386F	Read/Write
ADC_CORE5_BTM2_BTM_DATA0_0	0x000038A0	Read
ADC_CORE5_BTM2_BTM_DATA1_0	0x000038A1	Read
ADC_CORE5_BTM2_BTM_DATA0_1	0x000038A2	Read
ADC_CORE5_BTM2_BTM_DATA1_1	0x000038A3	Read

ADC_CORE5_BTM2_REVISION1 | 0x00003800

Type: Read

Reset: 0x00000003

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

ADC_CORE5_BTM2_REVISION2 | 0x00003801

Type: Read

Reset: 0x00000005

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

ADC_CORE5_BTM2_REVISION3 | 0x00003802

Type: Read

Reset: 0x00000002

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

ADC_CORE5_BTM2_REVISION4 | 0x00003803

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

ADC_CORE5_BTM2_PERPH_TYPE | 0x00003804

Type: Read

Reset: 0x00000008

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	ADC

ADC_CORE5_BTM2_PERPH_SUBTYPE | 0x00003805

Type: Read

Reset: 0x00000034

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	ADC sub type

ADC_CORE5_BTM2_STATUS2 | 0x00003809

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
7 : 5	CONV_SEQ_STATE	Conversion request states selected by SEL_FSM register field. SEL_FSM Signal 0 Request FSM #1 state[3;0] 1 Request FSM #2 state[3;0] 2 Request FSM #3 state[3;0] 0: IDLE_REQ 1: WAIT_HOLDOFF_RST 2: WAIT_HOLDOFF 3: RST_CH_ARB 4: GET_NEXT_CH 5: WAIT_ADC_EOC 6: WAIT_ADC_EOC2 7: GEN_IRQ

ADC_CORE5_BTM2_STATUS_LOW | 0x0000380A

Type: Read

Reset: 0x00000000

Indicates measurement(s) where ADC read is less than low threshold.
PMIC_SCOPE=NUM_BTM__NUM_BTM_PS

Bits	Name	Description
1 : 0	LOW	Under-low-threshold status register field. Every BTM channel has one bit in this status register to indicate whether its conversion result is lower than its {LOW_THR_15_8[7:0], LOW_THR_7_0[7:0]} setting. For example, channel #0 status is at bit 0, channel #1 status is at bit 1, and so on. 0: LOW_FALSE 1: LOW_TRUE

ADC_CORE5_BTM2_STATUS_HIGH | 0x0000380B

Type: Read

Reset: 0x00000000

Indicates measurement(s) where ADC read is greater than high threshold.
PMIC_SCOPE=NUM_BTM__NUM_BTM_PS

Bits	Name	Description
1 : 0	HIGH	Above-high-threshold status register field. Every BTM channel has one bit in this status register to indicate whether its conversion result is higher than its {HIGH_THR_15_8[7:0], HIGH_THR_7_0[7:0]} setting. For example, channel #0 status is at bit 0, channel #1 status is at bit 1, and so on. 0: HIGH_FALSE 1: HIGH_TRUE

ADC_CORE5_BTM2_NUM_BTM | 0x0000380F

Type: Read

Reset: 0x00000002

Read-Only register to indicate the number of available BTM channels

Bits	Name	Description
7 : 0	NUM_BTM	Number of available BTM channels

ADC_CORE5_BTM2_INT_RT_STS | 0x00003810

Type: Read

Reset: 0x00000000

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	THR_INT_RT_STS	Threshold interrupt set type 0: THR_INT_FALSE 1: THR_INT_TRUE

ADC_CORE5_BTM2_INT_SET_TYPE | 0x00003811

Type: Read/Write

Reset: 0x00000000

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	THR_INT_SET_TYPE	Threshold interrupt high polarity enabled 0: THR_INT_POL_HIGH_DISABLED 1: THR_INT_POL_HIGH_ENABLED

ADC_CORE5_BTM2_INT_POLARITY_HIGH | 0x00003812

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	THR_INT_HIGH	Threshold interrupt high polarity enabled 0: HIGH_THR_INT_POL_HIGH_DISABLED 1: HIGH_THR_INT_POL_HIGH_ENABLED

ADC_CORE5_BTM2_INT_POLARITY_LOW | 0x00003813

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	THR_INT_LOW	Threshold interrupt low polarity enabled 0: THR_INT_POL_DISABLED 1: THR_INT_POL_ENABLED

ADC_CORE5_BTM2_INT_LATCHED_CLR | 0x00003814

Type: Write

Reset: 0x00000000

Writing a '1' to a bit in this register will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits

Bits	Name	Description
0 : 0	THR_INT_LATCHED_CLR	Threshold interrupt latched clear 0: THR_INT_FALSE 1: THR_INT_TRUE

ADC_CORE5_BTM2_INT_EN_SET | 0x00003815

Type: Read/Write

Reset: 0x00000000

Writing '0' to a bit in this register has no effect. Writing a '1' to a bit in this register will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	THR_INT_EN_SET	Threshold interrupt enable set 0: THR_INT_DISABLED 1: THR_INT_ENBLED

ADC_CORE5_BTM2_INT_EN_CLR | 0x00003816

Type: Read/Write

Reset: 0x00000000

Writing a '0' to a bit in this register has no effect. Writing a '1' to a bit in this register will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	THR_INT_EN_CLR	Threshold interrupt enable clear 0: THR_INT_DISABLED 1: THR_INT_ENBLED

ADC_CORE5_BTM2_INT_LATCHED_STS | 0x00003818

Type: Read

Reset: 0x00000000

Debug: Latched (Sticky) Interrupt. '1' indiecates that the interrutp has triggered. Once the latched bit is set it can only be cleared by writing the clear bit.

Bits	Name	Description
0 : 0	THR_INT_LATCHED_STS	Threshold interrupt latched 0: THR_INT_LATCHED_FALSE 1: THR_INT_LATCHED_TRUE

ADC_CORE5_BTM2_INT_PENDING_STS | 0x00003819

Type: Read

Reset: 0x00000000

Debug: Pending is set if interrupt has been sent but not cleared.

Bits	Name	Description
0 : 0	THR_INT_PENDING_STS	Threshold interrupt pending 0: THR_INT_PENDING_FALSE 1: THR_INT_PENDING_TRUE

ADC_CORE5_BTM2_INT_MID_SEL | 0x0000381A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	Selects the MID that will receive the interrupt

ADC_CORE5_BTM2_INT_PRIORITY | 0x0000381B

Type: Read/Write

Reset: 0x00000000

Selects the SPMI interrupt priority

Bits	Name	Description
0 : 0	INT_PRIORITY	Selects the SPMI interrupt priority 0: SR 1: A

ADC_CORE5_BTM2_BTM_FAULT_TRIGGER_CTL | 0x00003840

Type: Read/Write

Reset: 0x00000000

BTM FAULT Trigger Control Register PMIC_STATIC, PMIC_SCOPE=NUM_BTM__NUM_BTM_PS, PMIC_LOCKEDBYBIT=LOCKBIT_BTM_PERPH

Bits	Name	Description
1 : 0	CH_EN	Select which channel's threshold crossing flag should assert BTM trigger to PON. Each channel is enabled/disabled by one bit of this register. For example, channel #0 is enabled/disabled by bit 0, channel #1 is enabled/disabled by bit 1, and so on.

ADC_CORE5_BTM2_BTM_PBS_TRIGGER_CTL | 0x00003841

Type: Read/Write

Reset: 0x00000000

BTM PBS Trigger Control Register PMIC_STATIC, PMIC_SCOPE=NUM_BTM__NUM_BTM_PS, PMIC_LOCKEDBYBIT=LOCKBIT_BTM_PERPH

Bits	Name	Description
1 : 0	CH_EN	Select which channel's threshold crossing flag should assert BTM trigger to PBS. Each channel is enabled/disabled by one bit of this register. For example, channel #0 is enabled/disabled by bit 0, channel #1 is enabled/disabled by bit 1, and so on.

ADC_CORE5_BTM2_ADC_DIG_PARAM | 0x00003842

Type: Read/Write

Reset: 0x00000008

PMIC_STATIC

Bits	Name	Description
3 : 2	DEC_RATIO_SEL	Select the conversion decimation ratio: * DEC_RATIO_SEL = 0 ==> STEP1 = 170, STEP2 = 80 , Decimation Ratio = STEP1+STEP2 = 250 * DEC_RATIO_SEL = 1 ==> STEP1 = 340, STEP2 = 80 , Decimation Ratio = STEP1+STEP2 = 420 * DEC_RATIO_SEL = 2 ==> STEP1 = 680, STEP2 = 160, Decimation Ratio = STEP1+STEP2 = 840 The above table reflects the default ADC_CMN setting (CONV_MODE = 0 and DIG_BOOST_EN = 1). Please refer to the DIG_BOOST_EN field description in ADC_CMN peripheral if a non-default settings are used for CONV_MODE and DIG_BOOST_EN. 0: DEC_RATIO_SHORT 1: DEC_RATIO_MEDIUM 2: DEC_RATIO_LONG

ADC_CORE5_BTM2_FAST_AVG_CTL | 0x00003843

Type: Read/Write

Reset: 0x00000080

Fast Average Enable PMIC_STATIC

Bits	Name	Description
2 : 0	FAST_AVG_SAMPLES	Select number of samples for use in fast average mode (i.e. 2^(value). 0: AVG_1_SAMPLE 1: AVG_2_SAMPLES 2: AVG_4_SAMPLES 3: AVG_8_SAMPLES 4: AVG_16_SAMPLES
7 : 7	FAST_AVG_EN	Enabling will select low latency for multiple averaged conversions by overlapping window samples. When this is disabled, averaged samples are run back-to-back without overlapping. 0: FAST_AVG_DISABLED 1: FAST_AVG_ENABLED

ADC_CORE5_BTM2_MEAS_INTERVAL_CTL | 0x00003844

Type: Read/Write

Reset: 0x00000000

Interval Mode Control PMIC_STATIC

Bits	Name	Description
3 : 0	MEAS_INTERVAL_TIME1	Select measurement interval time (i.e., If value=0, use 0ms, else use 2^(value+4)/32768). 0: meas_interval1_0ms 1: meas_interval1_1p0ms 2: meas_interval1_2p0ms 3: meas_interval1_3p9ms 4: meas_interval1_7p8ms 5: meas_interval1_15p6ms 6: meas_interval1_31p3ms 7: meas_interval1_62p5ms 8: meas_interval1_125ms 9: meas_interval1_250ms 10: meas_interval1_500ms 11: meas_interval1_1s 12: meas_interval1_2s 13: meas_interval1_4s 14: meas_interval1_8s 15: meas_interval1_16s

ADC_CORE5_BTM2_MEAS_INTERVAL_CTL2 | 0x00003845

Type: Read/Write

Reset: 0x00000000

PMIC_STATIC

Bits	Name	Description
3 : 0	MEAS_INTERVAL_TIME3	Large timer: Select measurement interval time in seconds. 0: meas_interval3_0s 1: meas_interval3_1s 2: meas_interval3_2s 3: meas_interval3_3s 4: meas_interval3_4s 5: meas_interval3_5s 6: meas_interval3_6s 7: meas_interval3_7s 8: meas_interval3_8s 9: meas_interval3_9s 10: meas_interval3_10s 11: meas_interval3_11s 12: meas_interval3_12s 13: meas_interval3_13s 14: meas_interval3_14s 15: meas_interval3_15s
7 : 4	MEAS_INTERVAL_TIME2	Small timer: Select measurement interval time in 100ms increments. 0: meas_interval2_0ms 1: meas_interval2_100ms 2: meas_interval2_200ms 3: meas_interval2_300ms 4: meas_interval2_400ms 5: meas_interval2_500ms 6: meas_interval2_600ms 7: meas_interval2_700ms 8: meas_interval2_800ms 9: meas_interval2_900ms 10: meas_interval2_1000ms 11: meas_interval2_1100ms 12: meas_interval2_1200ms 13: meas_interval2_1300ms 14: meas_interval2_1400ms 15: meas_interval2_1500ms

ADC_CORE5_BTM2_EN_CTL1 | 0x00003846

Type: Read/Write

Reset: 0x00000000

Enables ADC module. PMIC_LOCKEDBYBIT=LOCKBIT_BTM_PERPH

Bits	Name	Description
7 : 7	ADC_EN	Enables ADC module. Writing a '0' to this field will abort all the ongoing BTM conversions if there is any 0: ADC_DISABLED 1: ADC_ENABLED

ADC_CORE5_BTM2_CONV_REQ | 0x00003847

Type: Write

Reset: 0x00000000

Conversion Request

Bits	Name	Description
7 : 7	REQ	Conversion request strobe. Note, logic only allows writes of 0x1 to this field. 0x0 writes to this field are ignored. 0: NO_OPERATION 1: START_CONV_REQ

ADC_CORE5_BTM2_DATA_HOLD_CTL | 0x00003850

Type: Read/Write

Reset: 0x00000000

Data Hold Control register

Bits	Name	Description
0 : 0	HOLD	Every time this register bit is written to a '1', the {Mx_DATA1[7:0], Mx_DATA0[7:0]} data and STATUS_HIGH/STATUS_LOW status will be loaded to a shadow register for read access. However, if this register is a '0', the free-running {Mx_DATA1[7:0], Mx_DATA0[7:0]} data and STATUS_HIGH/STATUS_LOW status will be read instead. 0: FREE_RUNNING 1: HOLD_DATA

ADC_CORE5_BTM2_BTM_ADC_CH_SEL_CTL_0 | 0x00003860

Type: Read/Write

Reset: 0x000000FF

ADC Channel selection PMIC_STATIC, PMIC_LOCKEDBYBIT=LOCKBIT_BTM0

Bits	Name	Description
7 : 0	ADC_CH_SEL	ADC Channel selection, please refer to chip TDOS or sharepoint (go/adc_ch_sel) for more information

ADC_CORE5_BTM2_BTM_LOW_THR0_0 | 0x00003861

Type: Read/Write

Reset: 0x00000001

Low Threshold Byte 0 PMIC_LOCKEDBYBIT=LOCKBIT_BTM0

Bits	Name	Description
7 : 0	LOW_THR_7_0	Low byte of low threshold detector. BTM_LOW_THR[15:0]={BTM_LOW_THR1[7:0],BTM_LOW_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_LOW_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_LOW_THR[15:0]/65536*7.5 Volts. A write to this register must be followed by a BTM_LOW_THR1 write.

ADC_CORE5_BTM2_BTM_LOW_THR1_0 | 0x00003862

Type: Read/Write

Reset: 0x00000080

Low Threshold Byte 1 PMIC_LOCKEDBYBIT=LOCKBIT_BTM0

Bits	Name	Description
7 : 0	LOW_THR_15_8	High byte of low threshold detector. BTM_LOW_THR[15:0]={BTM_LOW_THR1[7:0],BTM_LOW_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_LOW_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_LOW_THR[15:0]/65536*7.5 Volts. A write to this register must be preceded by a BTM_LOW_THR0 write.

ADC_CORE5_BTM2_BTM_HIGH_THR0_0 | 0x00003863

Type: Read/Write

Reset: 0x000000FF

High Threshold Byte 0 PMIC_LOCKEDBYBIT=LOCKBIT_BTM0

Bits	Name	Description
7 : 0	HIGH_THR_7_0	Low byte of high threshold detector. BTM_HIGH_THR[15:0]={BTM_HIGH_THR1[7:0],BTM_HIGH_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_HIGH_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_HIGH_THR[15:0]/65536*7.5 Volts. A write to this register must be followed by a BTM_HIGH_THR1 write.

ADC_CORE5_BTM2_BTM_HIGH_THR1_0 | 0x00003864

Type: Read/Write

Reset: 0x0000007F

High Threshold Byte 1 PMIC_LOCKEDBYBIT=LOCKBIT_BTM0

Bits	Name	Description
7 : 0	HIGH_THR_15_8	High byte of high threshold detector. BTM_HIGH_THR[15:0]={BTM_HIGH_THR1[7:0],BTM_HIGH_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_HIGH_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_HIGH_THR[15:0]/65536*7.5 Volts. A write to this register must be preceded by a BTM_HIGH_THR0 write.

ADC_CORE5_BTM2_BTM_MEAS_INTERVAL_CTL_0 | 0x00003865

Type: Read/Write

Reset: 0x00000000

Interval mode select PMIC_LOCKEDBYBIT=LOCKBIT_BTM0

Bits	Name	Description
1 : 0	MEAS_INTERVAL_TIME	Select which interval timer to use 0: USING_TIMER1 1: USING_TIMER2 2: USING_TIMER3

ADC_CORE5_BTM2_BTM_CTL_0 | 0x00003866

Type: Read/Write

Reset: 0x00000000

BTM control PMIC_STATIC, PMIC_LOCKEDBYBIT=LOCKBIT_BTM0

Bits	Name	Description
3 : 0	HW_SETTLE_DELAY	Time between AMUX getting configured and the ADC starting conversion. Delay = 15us for value 0, 100us*(value) for 0<value<8, and 1ms*2^(value-8) otherwise 0: hw_settle_delay_15us 1: hw_settle_delay_100us 2: hw_settle_delay_200us 3: hw_settle_delay_300us 4: hw_settle_delay_400us 5: hw_settle_delay_500us 6: hw_settle_delay_600us 7: hw_settle_delay_700us 8: hw_settle_delay_1ms 9: hw_settle_delay_2ms 10: hw_settle_delay_4ms 11: hw_settle_delay_8ms 12: hw_settle_delay_16ms 13: hw_settle_delay_32ms 14: hw_settle_delay_64ms 15: hw_settle_delay_128ms
5 : 4	CAL_SEL	Select calibration method: 0: No_cal 1: ratio_cal 2: abs_cal
6 : 6	CAL_VAL	Select calibration value 0: timer_cal 1: new_cal

ADC_CORE5_BTM2_BTM_EN_0 | 0x00003867

Type: Read/Write

Reset: 0x00000000

BTM enable PMIC_LOCKEDBYBIT=LOCKBIT_BTM0

Bits	Name	Description
0 : 0	LOW_THR_INT_EN	Enables low threshold for interrupt 0: LOW_THR_INT_DISABLED 1: LOW_THR_INT_ENABLED
1 : 1	HIGH_THR_INT_EN	Enables high threshold for interrupt 0: HIGH_THR_INT_DISABLED 1: HIGH_THR_INT_ENABLED
7 : 7	MEAS_EN	Enables measurement in auto-sequence. Writing a '0' to this field will abort the ongoing conversion if there is any 0: MEAS_DISABLE 1: MEAS_ENABLE

ADC_CORE5_BTM2_BTM_ADC_CH_SEL_CTL_1 | 0x00003868

Type: Read/Write

Reset: 0x00000000

ADC Channel selection PMIC_STATIC, PMIC_LOCKEDBYBIT=LOCKBIT_BTM1

Bits	Name	Description
7 : 0	ADC_CH_SEL	ADC Channel selection, please refer to chip TDOS or sharepoint (go/adc_ch_sel) for more information

ADC_CORE5_BTM2_BTM_LOW_THR0_1 | 0x00003869

Type: Read/Write

Reset: 0x00000001

Low Threshold Byte 0 PMIC_LOCKEDBYBIT=LOCKBIT_BTM1

Bits	Name	Description
7 : 0	LOW_THR_7_0	Low byte of low threshold detector. BTM_LOW_THR[15:0]={BTM_LOW_THR1[7:0],BTM_LOW_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_LOW_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_LOW_THR[15:0]/65536*7.5 Volts. A write to this register must be followed by a BTM_LOW_THR1 write.

ADC_CORE5_BTM2_BTM_LOW_THR1_1 | 0x0000386A

Type: Read/Write

Reset: 0x00000080

Low Threshold Byte 1 PMIC_LOCKEDBYBIT=LOCKBIT_BTM1

Bits	Name	Description
7 : 0	LOW_THR_15_8	High byte of low threshold detector. BTM_LOW_THR[15:0]={BTM_LOW_THR1[7:0],BTM_LOW_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_LOW_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_LOW_THR[15:0]/65536*7.5 Volts. A write to this register must be preceded by a BTM_LOW_THR0 write.

ADC_CORE5_BTM2_BTM_HIGH_THR0_1 | 0x0000386B

Type: Read/Write

Reset: 0x000000FF

High Threshold Byte 0 PMIC_LOCKEDBYBIT=LOCKBIT_BTM1

Bits	Name	Description
7 : 0	HIGH_THR_7_0	Low byte of high threshold detector. BTM_HIGH_THR[15:0]={BTM_HIGH_THR1[7:0],BTM_HIGH_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_HIGH_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_HIGH_THR[15:0]/65536*7.5 Volts. A write to this register must be followed by a BTM_HIGH_THR1 write.

ADC_CORE5_BTM2_BTM_HIGH_THR1_1 | 0x0000386C

Type: Read/Write

Reset: 0x0000007F

High Threshold Byte 1 PMIC_LOCKEDBYBIT=LOCKBIT_BTM1

Bits	Name	Description
7 : 0	HIGH_THR_15_8	High byte of high threshold detector. BTM_HIGH_THR[15:0]={BTM_HIGH_THR1[7:0],BTM_HIGH_THR0[7:0]} forms a 16-bit 2's complement number. The actual voltage it represents is BTM_HIGH_THR[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_HIGH_THR[15:0]/65536*7.5 Volts. A write to this register must be preceded by a BTM_HIGH_THR0 write.

ADC_CORE5_BTM2_BTM_MEAS_INTERVAL_CTL_1 | 0x0000386D

Type: Read/Write

Reset: 0x00000000

Interval mode select PMIC_LOCKEDBYBIT=LOCKBIT_BTM1

Bits	Name	Description
1 : 0	MEAS_INTERVAL_TIME	Select which interval timer to use 0: USING_TIMER1 1: USING_TIMER2 2: USING_TIMER3

ADC_CORE5_BTM2_BTM_CTL_1 | 0x0000386E

Type: Read/Write

Reset: 0x00000000

BTM control PMIC_STATIC, PMIC_LOCKEDBYBIT=LOCKBIT_BTM1

Bits	Name	Description
3 : 0	HW_SETTLE_DELAY	Time between AMUX getting configured and the ADC starting conversion. Delay = 15us for value 0, 100us*(value) for 0<value<8, and 1ms*2^(value-8) otherwise 0: hw_settle_delay_15us 1: hw_settle_delay_100us 2: hw_settle_delay_200us 3: hw_settle_delay_300us 4: hw_settle_delay_400us 5: hw_settle_delay_500us 6: hw_settle_delay_600us 7: hw_settle_delay_700us 8: hw_settle_delay_1ms 9: hw_settle_delay_2ms 10: hw_settle_delay_4ms 11: hw_settle_delay_8ms 12: hw_settle_delay_16ms 13: hw_settle_delay_32ms 14: hw_settle_delay_64ms 15: hw_settle_delay_128ms
5 : 4	CAL_SEL	Select calibration method: 0: No_cal 1: ratio_cal 2: abs_cal
6 : 6	CAL_VAL	Select calibration value 0: timer_cal 1: new_cal

ADC_CORE5_BTM2_BTM_EN_1 | 0x0000386F

Type: Read/Write

Reset: 0x00000000

BTM enable PMIC_LOCKEDBYBIT=LOCKBIT_BTM1

Bits	Name	Description
0 : 0	LOW_THR_INT_EN	Enables low threshold for interrupt 0: LOW_THR_INT_DISABLED 1: LOW_THR_INT_ENABLED
1 : 1	HIGH_THR_INT_EN	Enables high threshold for interrupt 0: HIGH_THR_INT_DISABLED 1: HIGH_THR_INT_ENABLED

Bits	Name	Description
7 : 7	MEAS_EN	Enables measurement in auto-sequence. Writing a '0' to this field will abort the ongoing conversion if there is any 0: MEAS_DISABLE 1: MEAS_ENABLE

ADC_CORE5_BTM2_BTM_DATA0_0 | 0x000038A0

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA_7_0	Low byte of ADC output for V-only, I-only, or VBAT of a V/I(synchronous or interleaved) conversions. BTM_DATA[15:0]={BTM_DATA1[7:0],BTM_DATA0[7:0]} forms a 16-bit 2's complement number. 1. For a V-only or VBAT of a V/I conversion, The actual voltage it represents is BTM_DATA[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_DATA[15:0]/65536*7.5 Volts. 2. For a I-only conversion, The actual current it represents is BTM_DATA[15:0]/32767*10 Amps if SUBTYPE is between 0x28 and 0x2F, or BTM_DATA[15:0]/32767*5 Amps if SUBTYPE is between 0x38 and 0x3F.

ADC_CORE5_BTM2_BTM_DATA1_0 | 0x000038A1

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA_15_8	High byte of ADC output for V-only, I-only, or VBAT of a V/I(synchronous or interleaved) conversions. BTM_DATA[15:0]={BTM_DATA1[7:0],BTM_DATA0[7:0]} forms a 16-bit 2's complement number. 1. For a V-only or VBAT of a V/I conversion, The actual voltage it represents is BTM_DATA[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_DATA[15:0]/65536*7.5 Volts. 2. For a I-only conversion, The actual current it represents is BTM_DATA[15:0]/32767*10 Amps if SUBTYPE is between 0x28 and 0x2F, or BTM_DATA[15:0]/32767*5 Amps if SUBTYPE is between 0x38 and 0x3F.

ADC_CORE5_BTM2_BTM_DATA0_1 | 0x000038A2

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA_7_0	Low byte of ADC output for V-only, I-only, or VBAT of a V/I(synchronous or interleaved) conversions. BTM_DATA[15:0]={BTM_DATA1[7:0],BTM_DATA0[7:0]} forms a 16-bit 2's complement number. 1. For a V-only or VBAT of a V/I conversion, The actual voltage it represents is BTM_DATA[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_DATA[15:0]/65536*7.5 Volts. 2. For a I-only conversion, The actual current it represents is BTM_DATA[15:0]/32767*10 Amps if SUBTYPE is between 0x28 and 0x2F, or BTM_DATA[15:0]/32767*5 Amps if SUBTYPE is between 0x38 and 0x3F.

ADC_CORE5_BTM2_BTM_DATA1_1 | 0x000038A3

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA_15_8	High byte of ADC output for V-only, I-only, or VBAT of a V/I(synchronous or interleaved) conversions. BTM_DATA[15:0]={BTM_DATA1[7:0],BTM_DATA0[7:0]} forms a 16-bit 2's complement number. 1. For a V-only or VBAT of a V/I conversion, The actual voltage it represents is BTM_DATA[15:0]/115600*7.5 Volts if DIG_BOOST_EN=1, otherwise, BTM_DATA[15:0]/65536*7.5 Volts. 2. For a I-only conversion, The actual current it represents is BTM_DATA[15:0]/32767*10 Amps if SUBTYPE is between 0x28 and 0x2F, or BTM_DATA[15:0]/32767*5 Amps if SUBTYPE is between 0x38 and 0x3F.

Module Name: SPMI

Base Address: 0x00000600

Register Quick Reference:

Register	Offset	Type
SPMI_SPMI_BUF_CFG	0x00000640	Read/Write

SPMI_SPMI_BUF_CFG | 0x00000640

Type: Read/Write

Reset: 0x00000001

No description provided for this register.

Bits	Name	Description
1 : 0	BUFFER_STRENGTH	SPMI Buffer Drive Strength Configuration 0: LOW10PF 1: MID20PF 2: HIGH40PF 3: VERYHIGH50PF
4 : 4	RFFE_LOW_DRIVE_EN	Enable RFFE Low Drive 0: RFFE_LOW_DRIVE_EN_false 1: RFFE_LOW_DRIVE_EN_true

Module Name: PON

Base Address: 0x00000800

Register Quick Reference:

Register	Offset	Type
PON_REVISION1	0x00000800	Read

Register	Offset	Type
PON_REVISION2	0x00000801	Read
PON_REVISION3	0x00000802	Read
PON_REVISION4	0x00000803	Read
PON_PERPH_TYPE	0x00000804	Read
PON_PERPH_SUBTYPE	0x00000805	Read
PON_PON_PBL_STATUS	0x00000807	Read
PON_INT_RT_STS	0x00000810	Read
PON_INT_SET_TYPE	0x00000811	Read/Write
PON_INT_POLARITY_HIGH	0x00000812	Read/Write
PON_INT_POLARITY_LOW	0x00000813	Read/Write
PON_INT_LATCHED_CLR	0x00000814	Write
PON_INT_EN_SET	0x00000815	Read/Write
PON_INT_EN_CLR	0x00000816	Read/Write
PON_INT_LATCHED_STS	0x00000818	Read
PON_INT_PENDING_STS	0x00000819	Read
PON_INT_MID_SEL	0x0000081A	Read/Write
PON_INT_PRIORITY	0x0000081B	Read/Write
PON_KPDPWR_N_RESET_S1_TIMER	0x00000840	Read/Write
PON_KPDPWR_N_RESET_S2_TIMER	0x00000841	Read/Write
PON_KPDPWR_N_RESET_S2_CTL	0x00000842	Read/Write
PON_KPDPWR_N_RESET_S2_CTL2	0x00000843	Read/Write
PON_RESIN_N_RESET_S1_TIMER	0x00000844	Read/Write
PON_RESIN_N_RESET_S2_TIMER	0x00000845	Read/Write
PON_RESIN_N_RESET_S2_CTL	0x00000846	Read/Write
PON_RESIN_N_RESET_S2_CTL2	0x00000847	Read/Write
PON_RESIN_AND_KPDPWR_RESET_S1_TIMER	0x00000848	Read/Write
PON_RESIN_AND_KPDPWR_RESET_S2_TIMER	0x00000849	Read/Write
PON_RESIN_AND_KPDPWR_RESET_S2_CTL	0x0000084A	Read/Write
PON_RESIN_AND_KPDPWR_RESET_S2_CTL2	0x0000084B	Read/Write
PON_GP2_RESET_S1_TIMER	0x0000084C	Read/Write
PON_GP2_RESET_S2_TIMER	0x0000084D	Read/Write
PON_GP2_RESET_S2_CTL	0x0000084E	Read/Write

Register	Offset	Type
PON_GP2_RESET_S2_CTL2	0x0000084F	Read/Write
PON_GP1_RESET_S1_TIMER	0x00000850	Read/Write
PON_GP1_RESET_S2_TIMER	0x00000851	Read/Write
PON_GP1_RESET_S2_CTL	0x00000852	Read/Write
PON_GP1_RESET_S2_CTL2	0x00000853	Read/Write
PON_PMIC_WD_RESET_S1_TIMER	0x00000854	Read/Write
PON_PMIC_WD_RESET_S2_TIMER	0x00000855	Read/Write
PON_PMIC_WD_RESET_S2_CTL	0x00000856	Read/Write
PON_PMIC_WD_RESET_S2_CTL2	0x00000857	Read/Write
PON_PMIC_WD_RESET_PET	0x00000858	Write
PON_PS_HOLD_RESET_CTL	0x0000085A	Read/Write
PON_PS_HOLD_RESET_CTL2	0x0000085B	Read/Write
PON_PULL_CTL	0x00000870	Read/Write
PON_DEBOUNCE_CTL	0x00000871	Read/Write
PON_RESET_S3_SRC	0x00000874	Read/Write
PON_RESET_S3_TIMER	0x00000875	Read/Write
PON_SMPL_CTL	0x0000087F	Read/Write
PON_PON_TRIGGER_EN	0x00000880	Read/Write
PON_UVLO_RB_STATUS	0x00000885	Read
PON_OVLO_RB_STATUS	0x00000887	Read
PON_PS_HOLD_STATUS	0x0000089A	Read
PON_PON_REASON1	0x000008C0	Read
PON_WARM_RESET_REASON1	0x000008C2	Read
PON_ON_REASON	0x000008C4	Read
PON_POFF_REASON1	0x000008C5	Read
PON_OFF_REASON	0x000008C7	Read
PON_FAULT_REASON1	0x000008C8	Read
PON_FAULT_REASON2	0x000008C9	Read
PON_S3_RESET_REASON	0x000008CA	Read
PON_SOFT_RESET_REASON1	0x000008CB	Read

PON_REVISION1 | 0x00000800

Type: Read

Reset: 0x00000001

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

PON_REVISION2 | 0x00000801

Type: Read

Reset: 0x00000001

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

PON_REVISION3 | 0x00000802

Type: Read

Reset: 0x00000011

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

PON_REVISION4 | 0x00000803

Type: Read

Reset: 0x00000000

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

PON_PERPH_TYPE | 0x00000804

Type: Read

Reset: 0x00000001

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	No description provided for this bit field. 1: PON

PON_PERPH_SUBTYPE | 0x00000805

Type: Read

Reset: 0x00000004

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	No description provided for this bit field. 1: PON_PRIMARY 2: PON_SECONDARY 3: PON_1REG 4: PON_GEN2_PRIMARY 5: PON_GEN2_SECONDARY 7: PON_GEN2_NO_XVDD_RB

PON_PON_PBL_STATUS | 0x00000807

Type: Read

Reset: 0x00000000

Stage 2 reset generation and register access error status.

Bits	Name	Description
6 : 6	XVDD_RB_OCCURRED	XVDD_RB was asserted druing the last power cycle 0: NO_RESET 1: RESET_OCCURRED
7 : 7	DVDD_RB_OCCURRED	DVDD_RB was asserted during the last power cycle 0: NO_RESET 1: RESET_OCCURRED

PON_INT_RT_STS | 0x00000810

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	KPDPWR_ON	KPDPWR_N has been asserted for longer than his debounce timer 0: INT_RT_STATUS_LOW 1: INT_RT_STATUS_HIGH
1 : 1	RESIN_ON	RESIN_N has been asserted for longer than his debounce timer 0: INT_RT_STATUS_LOW 1: INT_RT_STATUS_HIGH
2 : 2	CBLPWR_ON	CBLPWR_N has been asserted for longer than his debounce timer 0: INT_RT_STATUS_LOW 1: INT_RT_STATUS_HIGH
3 : 3	KPDPWR_BARK	warning that a reset event has been triggered by KPDPWR_N 0: INT_RT_STATUS_LOW 1: INT_RT_STATUS_HIGH
4 : 4	RESIN_BARK	warning that a reset event has been triggered by RESIN_N 0: INT_RT_STATUS_LOW 1: INT_RT_STATUS_HIGH

Bits	Name	Description
5 : 5	K_R_BARK	warning that a reset event has been triggered by asserting RESIN_N and KPDPWR_N simultaneously 0: INT_RT_STATUS_LOW 1: INT_RT_STATUS_HIGH
6 : 6	PMIC_WD_BARK	warning that a reset event has been triggered by the PMIC Watchdog timer 0: INT_RT_STATUS_LOW 1: INT_RT_STATUS_HIGH
7 : 7	SOFT_RESET_OCCURED	warning that a reset event has been triggered by the PMIC Watchdog timer 0: INT_RT_STATUS_LOW 1: INT_RT_STATUS_HIGH

PON_INT_SET_TYPE | 0x00000811

Type: Read/Write

Reset: 0x00000000

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	KPDPWR_ON	No description provided for this bit field. 0: LEVEL 1: EDGE
1 : 1	RESIN_ON	No description provided for this bit field. 0: LEVEL 1: EDGE
2 : 2	CBLPWR_ON	No description provided for this bit field. 0: LEVEL 1: EDGE
3 : 3	KPDPWR_BARK	No description provided for this bit field. 0: LEVEL 1: EDGE
4 : 4	RESIN_BARK	No description provided for this bit field. 0: LEVEL 1: EDGE
5 : 5	K_R_BARK	No description provided for this bit field. 0: LEVEL 1: EDGE

Bits	Name	Description
6 : 6	PMIC_WD_BARK	No description provided for this bit field. 0: LEVEL 1: EDGE
7 : 7	SOFT_RESET_OCCURED	No description provided for this bit field. 0: LEVEL 1: EDGE

PON_INT_POLARITY_HIGH | 0x00000812

Type: Read/Write

Reset: 0x00000000

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	KPDPWR_ON	No description provided for this bit field. 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	RESIN_ON	No description provided for this bit field. 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
2 : 2	CBLPWR_ON	No description provided for this bit field. 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
3 : 3	KPDPWR_BARK	No description provided for this bit field. 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
4 : 4	RESIN_BARK	No description provided for this bit field. 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
5 : 5	K_R_BARK	No description provided for this bit field. 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
6 : 6	PMIC_WD_BARK	No description provided for this bit field. 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
7 : 7	SOFT_RESET_OCCURED	No description provided for this bit field. 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

PON_INT_POLARITY_LOW | 0x00000813

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	KPDPWR_ON	No description provided for this bit field. 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	RESIN_ON	No description provided for this bit field. 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
2 : 2	CBLPWR_ON	No description provided for this bit field. 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
3 : 3	KPDPWR_BARK	No description provided for this bit field. 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
4 : 4	RESIN_BARK	No description provided for this bit field. 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
5 : 5	K_R_BARK	No description provided for this bit field. 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
6 : 6	PMIC_WD_BARK	No description provided for this bit field. 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
7 : 7	SOFT_RESET_OCCURED	No description provided for this bit field. 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

PON_INT_LATCHED_CLR | 0x00000814

Type: Write

Reset: 0x00000000

writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits

Bits	Name	Description
0 : 0	KPDPWR_ON	No description provided for this bit field.
1 : 1	RESIN_ON	No description provided for this bit field.
2 : 2	CBLPWR_ON	No description provided for this bit field.
3 : 3	KPDPWR_BARK	No description provided for this bit field.
4 : 4	RESIN_BARK	No description provided for this bit field.
5 : 5	K_R_BARK	No description provided for this bit field.
6 : 6	PMIC_WD_BARK	No description provided for this bit field.
7 : 7	SOFT_RESET_OCCURED	No description provided for this bit field.

PON_INT_EN_SET | 0x00000815

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	KPDPWR_ON	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	RESIN_ON	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
2 : 2	CBLPWR_ON	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
3 : 3	KPDPWR_BARK	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
4 : 4	RESIN_BARK	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
5 : 5	K_R_BARK	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

Bits	Name	Description
6 : 6	PMIC_WD_BARK	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
7 : 7	SOFT_RESET_OCCURED	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

PON_INT_EN_CLR | 0x00000816

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	KPDPWR_ON	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	RESIN_ON	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
2 : 2	CBLPWR_ON	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
3 : 3	KPDPWR_BARK	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
4 : 4	RESIN_BARK	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
5 : 5	K_R_BARK	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
6 : 6	PMIC_WD_BARK	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
7 : 7	SOFT_RESET_OCCURED	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

PON_INT_LATCHED_STS | 0x00000818

Type: Read

Reset: 0x00000000

Debug: Latched (Sticky) Interrupt. '1' indiecates that the interrutp has triggered. Once the latched bit is set it can only be cleared by writing the clear bit.

Bits	Name	Description
0 : 0	KPDPWR_ON	No description provided for this bit field. 0: NO_INT_RECEIVED 1: INTERRUPT_RECEIVED
1 : 1	RESIN_ON	No description provided for this bit field. 0: NO_INT_RECEIVED 1: INTERRUPT_RECEIVED
2 : 2	CBLPWR_ON	No description provided for this bit field. 0: NO_INT_RECEIVED 1: INTERRUPT_RECEIVED
3 : 3	KPDPWR_BARK	No description provided for this bit field. 0: NO_INT_RECEIVED 1: INTERRUPT_RECEIVED
4 : 4	RESIN_BARK	No description provided for this bit field. 0: NO_INT_RECEIVED 1: INTERRUPT_RECEIVED
5 : 5	K_R_BARK	No description provided for this bit field. 0: NO_INT_RECEIVED 1: INTERRUPT_RECEIVED
6 : 6	PMIC_WD_BARK	No description provided for this bit field. 0: NO_INT_RECEIVED 1: INTERRUPT_RECEIVED
7 : 7	SOFT_RESET_OCCURED	No description provided for this bit field. 0: NO_INT_RECEIVED 1: INTERRUPT_RECEIVED

PON_INT_PENDING_STS | 0x00000819

Type: Read

Reset: 0x00000000

Debug: Pending is set if interrupt has been sent but not cleared.

Bits	Name	Description
0 : 0	KPDPWR_ON	No description provided for this bit field. 0: NO_INT_PENDING 1: INTERRUPT_PENDING
1 : 1	RESIN_ON	No description provided for this bit field. 0: NO_INT_PENDING 1: INTERRUPT_PENDING
2 : 2	CBLPWR_ON	No description provided for this bit field. 0: NO_INT_PENDING 1: INTERRUPT_PENDING
3 : 3	KPDPWR_BARK	No description provided for this bit field. 0: NO_INT_PENDING 1: INTERRUPT_PENDING
4 : 4	RESIN_BARK	No description provided for this bit field. 0: NO_INT_PENDING 1: INTERRUPT_PENDING
5 : 5	K_R_BARK	No description provided for this bit field. 0: NO_INT_PENDING 1: INTERRUPT_PENDING
6 : 6	PMIC_WD_BARK	No description provided for this bit field. 0: NO_INT_PENDING 1: INTERRUPT_PENDING
7 : 7	SOFT_RESET_OCCURED	No description provided for this bit field. 0: NO_INT_PENDING 1: INTERRUPT_PENDING

PON_INT_MID_SEL | 0x0000081A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: MID0 1: MID1 2: MID2 3: MID3

PON_INT_PRIORITY **| 0x0000081B**

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: SR 1: A

PON_KPDPWR_N_RESET_S1_TIMER | 0x00000840

Type: Read/Write

Reset: 0x0000000F

Stage 1 (Bark) Timer. Bark cannot be disabled, but interrupt can be disabled if necessary

Bits	Name	Description
3 : 0	S1_TIMER	Time that the debounced trigger must be held before bark is sent to MSM This field can only be updated when block is disabled (i.e. 8 32k always-on(aon) clock cycles after writing 0 to S2_RESET_EN and PON_TRIGGER_EN:KPDPWR_N fields). 0: ms_0 1: ms_32 2: ms_56 3: ms_80 4: ms_128 5: ms_184 6: ms_272 7: ms_408 8: ms_608 9: ms_904 10: ms_1352 11: ms_2048 12: ms_3072 13: ms_4480 14: ms_6720 15: ms_10256

PON_KPDPWR_N_RESET_S2_TIMER | 0x00000841

Type: Read/Write

Reset: 0x00000007

Stage 2 (bite) configuration

Bits	Name	Description
2 : 0	S2_TIMER	Time that debounced trigger must be held before S2 reset occurs {0ms, 10ms, 50ms, 100ms, 250ms, 500ms, 1s, 2s} This field can only be updated when block is disabled (i.e. 8 32k always-on(aon) clock cycles after writing 0 to S2_RESET_EN field). 0: ms_0 1: ms_10 2: ms_50 3: ms_100 4: ms_250 5: ms_500 6: s_1 7: s_2

PON_KPDPWR_N_RESET_S2_CTL | 0x00000842

Type: Read/Write

Reset: 0x00000004

Stage 2 (bite) configuration

Bits	Name	Description
3 : 0	RESET_TYPE	This field can only be updated when block is disabled (i.e. 8 32k always-on(aon) clock cycles after writing 0 to S2_RESET_EN field). 0: RESERVED0 1: WARM_RESET 2: IMMEDIATE_XVDD_SHUTDOWN 3: RESERVED3 4: SHUTDOWN 5: DVDD_SHUTDOWN 6: XVDD_SHUTDOWN 7: HARD_RESET 8: DVDD_HARD_RESET 9: XVDD_HARD_RESET 10: WARM_RESET_AND_DVDD_SHUTDOWN 11: WARM_RESET_AND_XVDD_SHUTDOWN 12: WARM_RESET_AND_SHUTDOWN 13: WARM_RESET_THEN_HARD_RESET 14: WARM_RESET_THEN_DVDD_HARD_RESET 15: WARM_RESET_THEN_XVDD_HARD_RESET

PON_KPDPWR_N_RESET_S2_CTL2 | 0x00000843

Type: Read/Write

Reset: 0x00000000

Stage 2 (bite) configuration

Bits	Name	Description
7 : 7	S2_RESET_EN	Enable Stage 2 reset Field is synchronized by a 2-stage shift register so, for reliable hardware operation, the mininum time allowed between write operations is 3 32k always-on(aon) clock cycles. 0: Disabled 1: Enabled

PON_RESIN_N_RESET_S1_TIMER | 0x00000844

Type: Read/Write

Reset: 0x0000000F

Stage 1 (Bark) Timer. Bark cannot be disabled, but interrupt can be disabled if necessary

Bits	Name	Description
3 : 0	S1_TIMER	Time that the debounced trigger must be held before bark is sent to MSM -- This field can only be updated when block is disabled (i.e. 8 32k always-on(aon) clock cycles after writing 0 to S2_RESET_EN field). 0: ms_0 1: ms_32 2: ms_56 3: ms_80 4: ms_128 5: ms_184 6: ms_272 7: ms_408 8: ms_608 9: ms_904 10: ms_1352 11: ms_2048 12: ms_3072 13: ms_4480 14: ms_6720 15: ms_10256

PON_RESIN_N_RESET_S2_TIMER | 0x00000845

Type: Read/Write

Reset: 0x00000007

Stage 2 (bite) configuration

Bits	Name	Description
2 : 0	S2_TIMER	Time that debounced trigger must be held before S2 reset occurs {0ms, 10ms, 50ms, 100ms, 250ms, 500ms, 1s, 2s} This field can only be updated when block is disabled (i.e. 8 32k always-on(aon) clock cycles after writing 0 to S2_RESET_EN field). 0: ms_0 1: ms_10 2: ms_50 3: ms_100 4: ms_250 5: ms_500 6: s_1 7: s_2

PON_RESIN_N_RESET_S2_CTL | 0x00000846

Type: Read/Write

Reset: 0x00000004

Stage 2 (bite) configuration

Bits	Name	Description
3 : 0	RESET_TYPE	This field can only be updated when block is disabled (i.e. 8 32k always-on(aon) clock cycles after writing 0 to S2_RESET_EN field). 0: RESERVED0 1: WARM_RESET 2: IMMEDIATE_XVDD_SHUTDOWN 3: RESERVED3 4: SHUTDOWN 5: DVDD_SHUTDOWN 6: XVDD_SHUTDOWN 7: HARD_RESET 8: DVDD_HARD_RESET 9: XVDD_HARD_RESET 10: WARM_RESET_AND_DVDD_SHUTDOWN 11: WARM_RESET_AND_XVDD_SHUTDOWN 12: WARM_RESET_AND_SHUTDOWN 13: WARM_RESET_THEN_HARD_RESET 14: WARM_RESET_THEN_DVDD_HARD_RESET 15: WARM_RESET_THEN_XVDD_HARD_RESET

PON_RESIN_N_RESET_S2_CTL2 | 0x00000847

Type: Read/Write

Reset: 0x00000000

Stage 2 (bite) configuration

Bits	Name	Description
7 : 7	S2_RESET_EN	Enable Stage 2 reset Field is synchronized by a 2-stage shift register so, for reliable hardware operation, the mininum time allowed between write operations is 3 32k always-on(aon) clock cycles. 0: Disabled 1: Enabled

PON_RESIN_AND_KPDPWR_RESET_S1_TIMER | 0x00000848

Type: Read/Write

Reset: 0x0000000F

Stage 1 (Bark) Timer. Bark cannot be disabled, but interrupt can be disabled if necessary

Bits	Name	Description
3 : 0	S1_TIMER	Time that the debounced trigger must be held before bark is sent to MSM -- This field can only be updated when block is disabled (i.e. 8 32k always-on(aon) clock cycles after writing 0 to S2_RESET_EN field). 0: ms_0 1: ms_32 2: ms_56 3: ms_80 4: ms_128 5: ms_184 6: ms_272 7: ms_408 8: ms_608 9: ms_904 10: ms_1352 11: ms_2048 12: ms_3072 13: ms_4480 14: ms_6720 15: ms_10256

PON_RESIN_AND_KPDPWR_RESET_S2_TIMER | 0x00000849

Type: Read/Write

Reset: 0x00000007

Stage 2 (bite) configuration

Bits	Name	Description
2 : 0	S2_TIMER	Time that debounced trigger must be held before S2 reset occurs {0ms, 10ms, 50ms, 100ms, 250ms, 500ms, 1s, 2s} This field can only be updated when block is disabled (i.e. 8 32k always-on(aon) clock cycles after writing 0 to S2_RESET_EN field). 0: ms_0 1: ms_10 2: ms_50 3: ms_100 4: ms_250 5: ms_500 6: s_1 7: s_2

PON_RESIN_AND_KPDPWR_RESET_S2_CTL | 0x0000084A

Type: Read/Write

Reset: 0x00000004

Stage 2 (bite) configuration

Bits	Name	Description
3 : 0	RESET_TYPE	This field can only be updated when block is disabled (i.e. 8 32k always-on(aon) clock cycles after writing 0 to S2_RESET_EN field). 0: RESERVED0 1: WARM_RESET 2: IMMEDIATE_XVDD_SHUTDOWN 3: RESERVED3 4: SHUTDOWN 5: DVDD_SHUTDOWN 6: XVDD_SHUTDOWN 7: HARD_RESET 8: DVDD_HARD_RESET 9: XVDD_HARD_RESET 10: WARM_RESET_AND_DVDD_SHUTDOWN 11: WARM_RESET_AND_XVDD_SHUTDOWN 12: WARM_RESET_AND_SHUTDOWN 13: WARM_RESET_THEN_HARD_RESET 14: WARM_RESET_THEN_DVDD_HARD_RESET 15: WARM_RESET_THEN_XVDD_HARD_RESET

PON_RESIN_AND_KPDPWR_RESET_S2_CTL2 | 0x0000084B

Type: Read/Write

Reset: 0x00000000

Stage 2 (bite) configuration

Bits	Name	Description
7 : 7	S2_RESET_EN	Enable Stage 2 reset Field is synchronized by a 2-stage shift register so, for reliable hardware operation, the mininum time allowed between write operations is 3 32k always-on(aon) clock cycles. 0: Disabled 1: Enabled

PON_GP2_RESET_S1_TIMER | 0x0000084C

Type: Read/Write

Reset: 0x0000000F

Stage 1 (Bark) Timer. Bark cannot be disabled, but interrupt can be disabled if necessary

Bits	Name	Description
3 : 0	S1_TIMER	Time that the debounced trigger must be held before bark is sent to MSM. This field can only be updated when block is disabled (i.e. 8 32k always-on(aon) clock cycles after writing 0 to S2_RESET_EN field). 0: ms_0 1: ms_32 2: ms_56 3: ms_80 4: ms_128 5: ms_184 6: ms_272 7: ms_408 8: ms_608 9: ms_904 10: ms_1352 11: ms_2048 12: ms_3072 13: ms_4480 14: ms_6720 15: ms_10256

PON_GP2_RESET_S2_TIMER **| 0x0000084D**

Type: Read/Write

Reset: 0x00000007

Stage 2 (bite) configuration

Bits	Name	Description
2 : 0	S2_TIMER	Time that debounced trigger must be held before S2 reset occurs {0ms, 10ms, 50ms, 100ms, 250ms, 500ms, 1s, 2s} This field can only be updated when block is disabled (i.e. 8 32k always-on(aon) clock cycles after writing 0 to S2_RESET_EN field). 0: ms_0 1: ms_10 2: ms_50 3: ms_100 4: ms_250 5: ms_500 6: s_1 7: s_2

PON_GP2_RESET_S2_CTL | 0x0000084E

Type: Read/Write

Reset: 0x00000004

Stage 2 (bite) configuration

Bits	Name	Description
3 : 0	RESET_TYPE	This field can only be updated when block is disabled (i.e. 8 32k always-on(aon) clock cycles after writing 0 to S2_RESET_EN field). 0: RESERVED0 1: WARM_RESET 2: IMMEDIATE_XVDD_SHUTDOWN 3: RESERVED3 4: SHUTDOWN 5: DVDD_SHUTDOWN 6: XVDD_SHUTDOWN 7: HARD_RESET 8: DVDD_HARD_RESET 9: XVDD_HARD_RESET 10: WARM_RESET_AND_DVDD_SHUTDOWN 11: WARM_RESET_AND_XVDD_SHUTDOWN 12: WARM_RESET_AND_SHUTDOWN 13: WARM_RESET_THEN_HARD_RESET 14: WARM_RESET_THEN_DVDD_HARD_RESET 15: WARM_RESET_THEN_XVDD_HARD_RESET

PON_GP2_RESET_S2_CTL2 | 0x0000084F

Type: Read/Write

Reset: 0x00000000

Stage 2 (bite) configuration

Bits	Name	Description
7 : 7	S2_RESET_EN	Enable Stage 2 reset Field is synchronized by a 2-stage shift register so, for reliable hardware operation, the mininum time allowed between write operations is 3 32k always-on(aon) clock cycles. 0: Disabled 1: Enabled

PON_GP1_RESET_S1_TIMER | 0x00000850

Type: Read/Write

Reset: 0x0000000F

Stage 1 (Bark) Timer. Bark cannot be disabled, but interrupt can be disabled if necessary

Bits	Name	Description
3 : 0	S1_TIMER	Time that the debounced trigger must be held before bark is sent to MSM -- This field can only be updated when block is disabled (i.e. 8 32k always-on(aon) clock cycles after writing 0 to S2_RESET_EN field). 0: ms_0 1: ms_32 2: ms_56 3: ms_80 4: ms_128 5: ms_184 6: ms_272 7: ms_408 8: ms_608 9: ms_904 10: ms_1352 11: ms_2048 12: ms_3072 13: ms_4480 14: ms_6720 15: ms_10256

PON_GP1_RESET_S2_TIMER | 0x00000851

Type: Read/Write

Reset: 0x00000007

Stage 2 (bite) configuration

Bits	Name	Description
2 : 0	S2_TIMER	Time that debounced trigger must be held before S2 reset occurs {0ms, 10ms, 50ms, 100ms, 250ms, 500ms, 1s, 2s} This field can only be updated when block is disabled (i.e. 8 32k always-on(aon) clock cycles after writing 0 to S2_RESET_EN field). 0: ms_0 1: ms_10 2: ms_50 3: ms_100 4: ms_250 5: ms_500 6: s_1 7: s_2

PON_GP1_RESET_S2_CTL | 0x00000852

Type: Read/Write

Reset: 0x00000004

Stage 2 (bite) configuration

Bits	Name	Description
3 : 0	RESET_TYPE	This field can only be updated when block is disabled (i.e. 8 32k always-on(aon) clock cycles after writing 0 to S2_RESET_EN field). 0: RESERVED0 1: WARM_RESET 2: IMMEDIATE_XVDD_SHUTDOWN 3: RESERVED3 4: SHUTDOWN 5: DVDD_SHUTDOWN 6: XVDD_SHUTDOWN 7: HARD_RESET 8: DVDD_HARD_RESET 9: XVDD_HARD_RESET 10: WARM_RESET_AND_DVDD_SHUTDOWN 11: WARM_RESET_AND_XVDD_SHUTDOWN 12: WARM_RESET_AND_SHUTDOWN 13: WARM_RESET_THEN_HARD_RESET 14: WARM_RESET_THEN_DVDD_HARD_RESET 15: WARM_RESET_THEN_XVDD_HARD_RESET

PON_GP1_RESET_S2_CTL2 | 0x00000853

Type: Read/Write

Reset: 0x00000000

Stage 2 (bite) configuration

Bits	Name	Description
7 : 7	S2_RESET_EN	Enable Stage 2 reset Field is synchronized by a 2-stage shift register so, for reliable hardware operation, the mininum time allowed between write operations is 3 32k always-on(aon) clock cycles. 0: Disabled 1: Enabled

PON_PMIC_WD_RESET_S1_TIMER | 0x00000854

Type: Read/Write

Reset: 0x0000001F

Stage 1 (Bark) Timer. Bark cannot be disabled, but interrupt can be disabled if necessary

Bits	Name	Description
6 : 0	S1_TIMER	Time that the debounced trigger must be held before bark is sent to MSM (seconds) -- 0 - 127 seconds, default 31 seconds. Program hex value of decimal count desired (not binary coded). This is a shadowed field so, for reliable hardware operation, the mininum time allowed between write operations is 8 32k always-on(aon) clock cycles. 0: Sec_0 1: Sec_1 2: Sec_2 3: Sec_3 4: Sec_4 5: Sec_5 6: Sec_6 7: Sec_7 8: Sec_8 9: Sec_9 10: Sec_10 11: Sec_11 12: Sec_12 13: Sec_13 14: Sec_14 15: Sec_15 16: Sec_16 17: Sec_17 18: Sec_18 19: Sec_19 20: Sec_20 21: Sec_21 22: Sec_22 23: Sec_23 24: Sec_24 25: Sec_25 26: Sec_26 27: Sec_27 28: Sec_28 29: Sec_29 30: Sec_30 31: Sec_31 32: Sec_32 33: Sec_33 34: Sec_34 35: Sec_35 36: Sec_36 37: Sec_37 38: Sec_38 39: Sec_39 40: Sec_40 41: Sec_41 42: Sec_42 43: Sec_43 44: Sec_44 45: Sec_45 46: Sec_46 47: Sec_47 48: Sec_48 49: Sec_49 50: Sec_50 51: Sec_51 52: Sec_52 53: Sec_53 54: Sec_54

Bits	Name	Description
6 : 0	S1_TIMER	55: Sec_55 56: Sec_56 57: Sec_57 58: Sec_58 59: Sec_59 60: Sec_60 61: Sec_61 62: Sec_62 63: Sec_63 64: Sec_64 65: Sec_65 66: Sec_66 67: Sec_67 68: Sec_68 69: Sec_69 70: Sec_70 71: Sec_71 72: Sec_72 73: Sec_73 74: Sec_74 75: Sec_75 76: Sec_76 77: Sec_77 78: Sec_78 79: Sec_79 80: Sec_80 81: Sec_81 82: Sec_82 83: Sec_83 84: Sec_84 85: Sec_85 86: Sec_86 87: Sec_87 88: Sec_88 89: Sec_89 90: Sec_90 91: Sec_91 92: Sec_92 93: Sec_93 94: Sec_94 95: Sec_95 96: Sec_96 97: Sec_97 98: Sec_98 99: Sec_99 100: Sec_100 101: Sec_101 102: Sec_102 103: Sec_103 104: Sec_104 105: Sec_105 106: Sec_106 107: Sec_107 108: Sec_108 109: Sec_109

Bits	Name	Description
6 : 0	S1_TIMER	110: Sec_110 111: Sec_111 112: Sec_112 113: Sec_113 114: Sec_114 115: Sec_115 116: Sec_116 117: Sec_117 118: Sec_118 119: Sec_119 120: Sec_120 121: Sec_121 122: Sec_122 123: Sec_123 124: Sec_124 125: Sec_125 126: Sec_126

PON_PMIC_WD_RESET_S2_TIMER | 0x00000855

Type: Read/Write

Reset: 0x00000001

Stage 2 (bite) configuration

Bits	Name	Description
6 : 0	S2_TIMER	Time that debounced trigger must be held before S2 reset occurs -- 0 - 127 seconds (default = 32 seconds). Program hex value of decimal count desired (Not binary coded). Timer starts after WD bark expires This is a shadowed field so, for reliable hardware operation, the mininum time allowed between write operations is 8 32k always-on(aon) clock cycles. 0: Sec_0 1: Sec_1 2: Sec_2 3: Sec_3 4: Sec_4 5: Sec_5 6: Sec_6 7: Sec_7 8: Sec_8 9: Sec_9 10: Sec_10 11: Sec_11 12: Sec_12 13: Sec_13 14: Sec_14 15: Sec_15 16: Sec_16 17: Sec_17 18: Sec_18 19: Sec_19 20: Sec_20 21: Sec_21 22: Sec_22 23: Sec_23 24: Sec_24 25: Sec_25 26: Sec_26 27: Sec_27 28: Sec_28 29: Sec_29 30: Sec_30 31: Sec_31 32: Sec_32 33: Sec_33 34: Sec_34 35: Sec_35 36: Sec_36 37: Sec_37 38: Sec_38 39: Sec_39 40: Sec_40 41: Sec_41 42: Sec_42 43: Sec_43 44: Sec_44 45: Sec_45 46: Sec_46 47: Sec_47 48: Sec_48 49: Sec_49 50: Sec_50 51: Sec_51 52: Sec_52 53: Sec_53 54: Sec_54

Bits	Name	Description
6 : 0	S2_TIMER	55: Sec_55 56: Sec_56 57: Sec_57 58: Sec_58 59: Sec_59 60: Sec_60 61: Sec_61 62: Sec_62 63: Sec_63 64: Sec_64 65: Sec_65 66: Sec_66 67: Sec_67 68: Sec_68 69: Sec_69 70: Sec_70 71: Sec_71 72: Sec_72 73: Sec_73 74: Sec_74 75: Sec_75 76: Sec_76 77: Sec_77 78: Sec_78 79: Sec_79 80: Sec_80 81: Sec_81 82: Sec_82 83: Sec_83 84: Sec_84 85: Sec_85 86: Sec_86 87: Sec_87 88: Sec_88 89: Sec_89 90: Sec_90 91: Sec_91 92: Sec_92 93: Sec_93 94: Sec_94 95: Sec_95 96: Sec_96 97: Sec_97 98: Sec_98 99: Sec_99 100: Sec_100 101: Sec_101 102: Sec_102 103: Sec_103 104: Sec_104 105: Sec_105 106: Sec_106 107: Sec_107 108: Sec_108 109: Sec_109

Bits	Name	Description
6 : 0	S2_TIMER	110: Sec_110 111: Sec_111 112: Sec_112 113: Sec_113 114: Sec_114 115: Sec_115 116: Sec_116 117: Sec_117 118: Sec_118 119: Sec_119 120: Sec_120 121: Sec_121 122: Sec_122 123: Sec_123 124: Sec_124 125: Sec_125 126: Sec_126

PON_PMIC_WD_RESET_S2_CTL | 0x00000856

Type: Read/Write

Reset: 0x00000006

Stage 2 (bite) configuration.

Bits	Name	Description
3 : 0	RESET_TYPE	This field can only be updated when block is disabled (i.e. 8 32k always-on(aon) clock cycles after writing 0 to S2_RESET_EN field). 0: RESERVED0 1: WARM_RESET 2: IMMEDIATE_XVDD_SHUTDOWN 3: RESERVED3 4: SHUTDOWN 5: DVDD_SHUTDOWN 6: XVDD_SHUTDOWN 7: HARD_RESET 8: DVDD_HARD_RESET 9: XVDD_HARD_RESET 10: RESERVED10 11: RESERVED11 12: RESERVED12 13: RESERVED13 14: RESERVED14 15: RESERVED15

PON_PMIC_WD_RESET_S2_CTL2 | 0x00000857

Type: Read/Write

Reset: 0x00000000

Stage 2 (bite) configuration.

Bits	Name	Description
7 : 7	S2_RESET_EN	Enable Stage 2 reset Field is synchronized by a 2-stage shift register so, for reliable hardware operation, the mininum time allowed between write operations is 3 32k always-on(aon) clock cycles. 0: Disabled 1: Enabled

PON_PMIC_WD_RESET_PET | 0x00000858

Type: Write

Reset: 0x00000000

Stage 2 (bite) configuration

Bits	Name	Description
0 : 0	WATCHDOG_PET	Writing '1' to this bit will clear the PMIC WD timer. Writing '0' has no effect. This is a synchronized field so, for reliable hardware operation, the mininum time allowed between write operations is 5 32k always-on(aon) clock cycles. 0: NOP_WD 1: PET_WD

PON_PS_HOLD_RESET_CTL | 0x0000085A

Type: Read/Write

Reset: 0x00000004

Stage 2 (bite) configuration

Bits	Name	Description
3 : 0	RESET_TYPE	This is a shadowed field so, for reliable hardware operation, the mininum time allowed between write operations is 8 32k always-on(aon) clock cycles. 0: RESERVED0 1: WARM_RESET 2: IMMEDIATE_XVDD_SHUTDOWN 3: RESERVED3 4: SHUTDOWN 5: DVDD_SHUTDOWN 6: XVDD_SHUTDOWN 7: HARD_RESET 8: DVDD_HARD_RESET 9: XVDD_HARD_RESET 10: RESERVED10 11: RESERVED11 12: RESERVED12 13: RESERVED13 14: RESERVED14 15: RESERVED15

PON_PS_HOLD_RESET_CTL2 | 0x0000085B

Type: Read/Write

Reset: 0x00000080

Stage 2 (bite) configuration

Bits	Name	Description
7 : 7	S2_RESET_EN	Enable reset Field is synchronized by a 2-stage shift register so, for reliable hardware operation, the mininum time allowed between write operations is 3 32k always-on(aon) clock cycles. 0: Disabled 1: Enabled

PON_PULL_CTL | 0x00000870

Type: Read/Write

Reset: 0x00000007

No description provided for this register.

Bits	Name	Description
0 : 0	RESIN_N_PU_EN	No description provided for this bit field. 0: PU_DISABLED 1: PU_ENABLED
1 : 1	KPDPWR_N_PU_EN	No description provided for this bit field. 0: PU_DISABLED 1: PU_ENABLED
2 : 2	CBLPWR_N_PU_EN	No description provided for this bit field. 0: PU_DISABLED 1: PU_ENABLED
3 : 3	PON1_PD_EN	No description provided for this bit field. 0: PD_DISABLED 1: PD_ENABLED

PON_DEBOUNCE_CTL | 0x00000871

Type: Read/Write

Reset: 0x00000005

No description provided for this register.

Bits	Name	Description
3 : 0	DEBOUNCE	KPD/CBL/RESIN/RESIN_AND_KPD: Time delay for keypad input, cable, reset input, reset input && keypad input input de-bouncing. Only signal assertion is de-bounced. Delay = (1/1024)* 2^ (x-4) The supported setting range for this field is [0, 12]. This is a shadowed field so, for reliable hardware operation, the minimum time allowed between write operations is 5 32k always-on(aon) clock cycles. 0: ms_0p06 1: ms_0p12 2: ms_0p24 3: ms_0p49 4: ms_0p98 5: ms_2p0 6: ms_3p9 7: ms_7p8 8: mS_15p6 9: mS_31p2 10: mS_62p5 11: mS_125 12: mS_250 13: RESERVED1 14: RESERVED2 15: RESERVED3

PON_RESET_S3_SRC | 0x00000874

Type: Read/Write

Reset: 0x00000003

Choose source for stage 3 (Full Complete Shutdown). This is a write once register. PMIC_WRITE_ONCE

Bits	Name	Description
1 : 0	RESET_S3_SOURCE	00=KPDPWR_N, 01=RESIN_N, 10=(KPDPWR_N and RESIN_N both need to be asserted, 11=either KPDPWR_N or RESIN_N) This is a shadowed field so, for reliable hardware operation, the mininum time allowed between write operations is 5 32k always-on(aon) clock cycles. 0: KPDPWR_N 1: RESIN_N 2: KPDPWR_AND_RESIN 3: KPDPWR_OR_RESIN

PON_RESET_S3_TIMER | 0x00000875

Type: Read/Write

Reset: 0x00000004

Time trigger must be held before S3 reset occurs (seconds) PMIC_LOCKEDBYBIT=LOCKBIT_D1

Bits	Name	Description
2 : 0	S3_TIMER	Time trigger must be held before S3 reset occurs. 000 = Instant, else 2^(x) seconds (2 to 128) This is a shadowed field so, for reliable hardware operation, the mininum time allowed between write operations is 5 32k always-on(aon) clock cycles. 0: Immediate 1: sec_2 2: sec_4 3: sec_8 4: sec_16 5: sec_32 6: sec_64 7: sec_128

PON_SMPL_CTL | 0x0000087F

Type: Read/Write

Reset: 0x00000000

SMPL control register

Bits	Name	Description
7 : 7	SMPL_EN	Enable SMPL timer 0: SMPL_DISABLE 1: SMPL_ENABLED

PON_PON_TRIGGER_EN | 0x00000880

Type: Read/Write

Reset: 0x000000E4

Power on trigger enables. Each field is synchronized by a 2-stage shift register so, for reliable hardware operation, the mininum time allowed between write operations is 3 32k always-on(aon) clock cycles.

Bits	Name	Description
2 : 2	RTC	Enable PON trigger for RTC 0: Disabled 1: Enabled
3 : 3	DC_CHG	Enable PON trigger for DC CHG 0: Disabled 1: Enabled
4 : 4	USB_CHG	Enable PON trigger for USB CHG 0: Disabled 1: Enabled
5 : 5	PON1	Enable PON trigger for PON1 0: Disabled 1: Enabled
6 : 6	CBLPWR_N	Enable PON trigger for CBL_PWR_N 0: Disabled 1: Enabled
7 : 7	KPDPWR_N	Enable PON trigger for new KPDPWR press 0: Disabled 1: Enabled

PON_UVLO_RB_STATUS | 0x00000885

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	UVLO_RB	Status of debounced UVLO_RB signal 0: DEASSERTED 1: ASSERTED

PON_OVLO_RB_STATUS | 0x00000887

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	OVLO_RB	Status of deboucned OVLO_RB signal 0: DEASSERTED 1: ASSERTED

PON_PS_HOLD_STATUS | 0x0000089A

Type: Read

Reset: 0x00000000

Status of the PS_HOLD input to the PMIC

Bits	Name	Description
7 : 7	PS_HOLD	Status of the PS_HOLD input to the PMIC 0: ASSERTED 1: DEASSERTED

PON_PON_REASON1 | 0x000008C0

Type: Read

Reset: 0x00000000

Reasons that the PMIC entered the PON state. All zeros mean that no trigger received

Bits	Name	Description
0 : 0	HARD_RESET	Triggered from a Hard Reset event (check POFF reason for the trigger) 1: TRIGGER_RECEIVED
1 : 1	SMPL	Triggered from SMPL 1: TRIGGER_RECEIVED
2 : 2	RTC	Triggered from RTC 1: TRIGGER_RECEIVED
3 : 3	DC_CHG	Triggered from DC charger 1: TRIGGER_RECEIVED
4 : 4	USB_CHG	Triggered from USB charger 1: TRIGGER_RECEIVED
5 : 5	PON1	Triggered from PON1 1: TRIGGER_RECEIVED

Bits	Name	Description
6 : 6	CBLPWR_N	Triggered from CBL_PWR1_N 1: TRIGGER_RECEIVED
7 : 7	KPDPWR_N	Triggered from new KPDPWR press 1: TRIGGER_RECEIVED

PON_WARM_RESET_REASON1 | 0x000008C2

Type: Read

Reset: 0x00000000

Reasons that PMIC entered the Warm Reset state. This register is automatically reset when the PMIC turns off itself or experiences a system fault.

Bits	Name	Description
0 : 0	SOFT	Triggered by Software 1: TRIGGER_RECEIVED
1 : 1	PS_HOLD	Triggered by PS_HOLD 1: TRIGGER_RECEIVED
2 : 2	PMIC_WD	Triggered by PMIC Watchdog 1: TRIGGER_RECEIVED
3 : 3	GP1	Triggered by Keypad_Reset1 1: TRIGGER_RECEIVED
4 : 4	GP2	Triggered by Keypad_Reset2 1: TRIGGER_RECEIVED
5 : 5	KPDPWR_AND_RESIN	Triggered by simultaneous KPDPWR_N + RESIN_N 1: TRIGGER_RECEIVED
6 : 6	RESIN_N	Triggred by RESIN_N 1: TRIGGER_RECEIVED
7 : 7	KPDPWR_N	Triggered by KPDPWR_N 1: TRIGGER_RECEIVED

PON_ON_REASON | 0x000008C4

Type: Read

Reset: 0x00000000

Reasons that the PMIC entered the ON state. All zeros mean that no trigger received

Bits	Name	Description
6 : 6	WARM_SEQ	PMIC entered ON state because of a warm-reset sequence 1: TRIGGER_RECEIVED
7 : 7	PON_SEQ	PMIC entered ON state because of a normal powering-on sequence 1: TRIGGER_RECEIVED

PON_POFF_REASON1 | 0x000008C5

Type: Read

Reset: 0x00000000

Reasons that the PMIC entered POFF state left the on state and commenced a shutdown sequence. All zeros mean that no trigger received.

Bits	Name	Description
0 : 0	SOFT	Triggered by Software 1: TRIGGER_RECEIVED
1 : 1	PS_HOLD	Triggered by PS_HOLD 1: TRIGGER_RECEIVED
2 : 2	PMIC_WD	Triggered by PMIC Watchdog 1: TRIGGER_RECEIVED
3 : 3	GP1	Triggered by Keypad_Reset1 1: TRIGGER_RECEIVED
4 : 4	GP2	Triggered by Keypad_Reset2 1: TRIGGER_RECEIVED
5 : 5	KPDPWR_AND_RESIN	Triggered by simultaneous KPDPWR_N + RESIN_N 1: TRIGGER_RECEIVED
6 : 6	RESIN_N	Triggred by RESIN_N 1: TRIGGER_RECEIVED
7 : 7	KPDPWR_N	Triggered by KPDPWR_N 1: TRIGGER_RECEIVED

PON_OFF_REASON | 0x000008C7

Type: Read

Reset: 0x00000000

Reasons that the PMIC entered the OFF state, All zeros mean that no trigger received

Bits	Name	Description
2 : 2	RAW_XVDD_RB_OCCURED	The assertion of this field indicates that a raw_xvdd_rb event occured and the PMIC hasn't experienced another OFF state after this event.,TRIGGER_RECEIVED=1,,,'
3 : 3	RAW_DVDD_RB_OCCURED	The assertion of this field indicates that a raw_dvdd_rb event occured and the PMIC hasn't experienced another OFF state after this event.,TRIGGER_RECEIVED=1,,,'
4 : 4	IMMEDIATE_XVDD_SHUTDOWN	PMIC entered OFF state because of an IMMEDIATE_XVDD_SHUTDOWN S2 Reset 1: TRIGGER_RECEIVED
5 : 5	S3_RESET	PMIC entered OFF state because of a S3 Reset 1: TRIGGER_RECEIVED
6 : 6	FAULT_SEQ	PMIC entered OFF state because of a fault-handling sequence 1: TRIGGER_RECEIVED
7 : 7	POFF_SEQ	PMIC entered OFF state because of a normal powering-off sequence 1: TRIGGER_RECEIVED

PON_FAULT_REASON1 | 0x000008C8

Type: Read

Reset: 0x00000000

Reasons that the PMIC entered the FAULT state. All zeros mean that no faults received

Bits	Name	Description
0 : 0	GP_FAULT0	Triggered by general-purpose fault #0 1: TRIGGER_RECEIVED
1 : 1	GP_FAULT1	Triggered by general-purpose fault #1 1: TRIGGER_RECEIVED
2 : 2	GP_FAULT2	Triggered by general-purpose fault #2 1: TRIGGER_RECEIVED
3 : 3	GP_FAULT3	Triggered by general-purpose fault #3 1: TRIGGER_RECEIVED
4 : 4	MBG_FAULT	Triggered by MBG fault event 1: TRIGGER_RECEIVED
5 : 5	OVLO	Triggered by OVLO event 1: TRIGGER_RECEIVED
6 : 6	UVLO	Triggered by UVLO event 1: TRIGGER_RECEIVED

Bits	Name	Description
7 : 7	AVDD_RB	Triggered by AVDD_RB event 1: TRIGGER_RECEIVED

PON_FAULT_REASON2 | 0x000008C9

Type: Read

Reset: 0x00000000

Reasons that the PMIC entered the FAULT state. All zeros mean that no trigger received

Bits	Name	Description
3 : 3	FAULT_N	Triggered by FAULT_N bus 1: TRIGGER_RECEIVED
4 : 4	PBS_WATCHDOG_TO	Triggered by PBS WATCHDOG 1: TRIGGER_RECEIVED
5 : 5	PBS_NACK	Triggered by PBS NACK event 1: TRIGGER_RECEIVED
6 : 6	RESTART_PON	Triggered by RESTART PON 1: TRIGGER_RECEIVED
7 : 7	OTST3	Triggered by OTST3 1: TRIGGER_RECEIVED

PON_S3_RESET_REASON | 0x000008CA

Type: Read

Reset: 0x00000000

Reasons that the PMIC experienced S3 reset, All zeros mean that no trigger received. Clear this register by writing to this register. This is a synchronized address so, for reliable hardware operation, the mininum time allowed between write operations is 5 32k always-on(aon) clock cycles.

Bits	Name	Description
4 : 4	FAULT_N	Triggered by FAULT_N being low more than 10 clock cycles (32k clock) 1: TRIGGER_RECEIVED
5 : 5	PBS_WATCHDOG_TO	Triggered by PBS_WATCHDOG_TO in FAULT state 1: TRIGGER_RECEIVED

Bits	Name	Description
6 : 6	PBS_NACK	Triggered by PBS_NACK in FAULT state 1: TRIGGER_RECEIVED
7 : 7	KPDPWR_ANDOR_RESIN	Triggered by KPDPWR_N, or RESIN_N, or KPDPWR_AND_RESIN, or KPDPWR_OR_RESIN 1: TRIGGER_RECEIVED

PON_SOFT_RESET_REASON1 | 0x000008CB

Type: Read

Reset: 0x00000000

Reasons that the global_soft_rb was asserted PMIC registers were reset. All zeros mean that no trigger received.

Bits	Name	Description
0 : 0	SOFT	Triggered by Software 1: TRIGGER_RECEIVED

Module Name: RF_CLK1

Base Address: 0x00005400

Register Quick Reference:

Register	Offset	Type
RF_CLK1_REVISION1	0x00005400	Read
RF_CLK1_REVISION2	0x00005401	Read
RF_CLK1_REVISION3	0x00005402	Read
RF_CLK1_REVISION4	0x00005403	Read
RF_CLK1_PERPH_TYPE	0x00005404	Read
RF_CLK1_PERPH_SUBTYPE	0x00005405	Read
RF_CLK1_STATUS1	0x00005408	Read
RF_CLK1_INT_SELF_CLEARING	0x0000541D	Read
RF_CLK1_MODE_CTL	0x0000543F	Read/Write
RF_CLK1_EDGE_CTL1	0x00005443	Read/Write
RF_CLK1_DRV_CTL1	0x00005444	Read/Write
RF_CLK1_CLKBUFF_HOLD_CTL	0x00005445	Read/Write
RF_CLK1_EN_CTL	0x00005446	Read/Write

Register	Offset	Type
RF_CLK1_PC_EN_CTL	0x0000544A	Read/Write
RF_CLK1_CLKBUFF_FOLLOW_HOLD_REQ	0x0000544B	Read/Write

RF_CLK1_REVISION1 | 0x00005400

Type: Read

Reset: 0x00000000

HW Version Register [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

RF_CLK1_REVISION2 | 0x00005401

Type: Read

Reset: 0x00000005

HW Version Register [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

RF_CLK1_REVISION3 | 0x00005402

Type: Read

Reset: 0x00000001

HW Version Register [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

RF_CLK1_REVISION4 | 0x00005403

Type: Read

Reset: 0x00000004

HW Version Register [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

RF_CLK1_PERPH_TYPE | 0x00005404

Type: Read

Reset: 0x00000006

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	Clock 6: CLOCK

RF_CLK1_PERPH_SUBTYPE | 0x00005405

Type: Read

Reset: 0x00000009

Peripheral SubType PMIC_CONSTANT

Bits	Name	Description
7 : 0	SUBTYPE	RF clock 9: RF_CLK

RF_CLK1_STATUS1 | 0x00005408

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
6 : 6	CLK_ENA_HOLD	Clock enable is latched from another clock peripheral 0: CLK_UNLATCH 1: CLK_LATCH
7 : 7	CLK_OK	Indicates Hardware or Software enable and includes warmup delay 0: CLK_OFF 1: CLK_ON

RF_CLK1_INT_SELF_CLEARING | 0x0000541D

Type: Read

Reset: 0x00000001

Selects the SPMI interrupt priority

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	Self_clearing or regular interrupt 0: regular_interrupt 1: self_clearing_interrupt

RF_CLK1_MODE_CTL | 0x0000543F

Type: Read/Write

Reset: 0x00000007

No description provided for this register.

Bits	Name	Description
2 : 0	MODE	GPIO Mode: 0: Digital Input 1: Digital Output 2: Digital Input and Output 3: Analog Pass Through 4-7: Mission 0: DIG_IN 1: DIG_OUT 2: DIG_INOUT 3: ANA_PASS_THRU 4: CLK_MODE

RF_CLK1_EDGE_CTL1 | 0x00005443

Type: Read/Write

Reset: 0x00000002

No description provided for this register.

Bits	Name	Description
1 : 0	OUT_EDGE	Edge Rate Control: 0 - 1X Slowest 1 - 2X 2 - 3X 3 - 4X Fastest 4-7 Disable 0: ONE_X 1: TWO_X 2: THREE_X 3: FOUR_X

RF_CLK1_DRV_CTL1 | 0x00005444

Type: Read/Write

Reset: 0x00000002

No description provided for this register.

Bits	Name	Description
1 : 0	OUT_DRV	Drive Strength Control: 0 - 1X Lowest Drive, Highest Impedance 1 - 2X 2 - 3X 3 - 4X Highest Drive, Lowest impedance 0: ONE_X 1: TWO_X 2: THREE_X 3: FOUR_X

RF_CLK1_CLKBUFF_HOLD_CTL | 0x00005445

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 6	CLKBUFF_HOLD	Each bit reserved to different EE, set anybit (plus pin buff ena) to enable holding clkbuff feature of other peripherals

RF_CLK1_EN_CTL | 0x00005446

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	CLK_EN	Software Enable 0 - not enable 1 - Enable 0: CLK_NOT_FORCE 1: CLK_FORCE_EN

RF_CLK1_PC_EN_CTL | 0x0000544A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	FOLLOW_PC_EN	Pin Control Enable 0 - do not follow pin control 1 - follow pin control 0: NOT_FOLLOW_PIN 1: FOLLOW_PIN
1 : 1	PC_POLARITY	Pin Control Polarity 0 - logic high enables the clock 1 - logic low enables the clock 0: POS_PINCONTROL_POLARITY 1: NEG_PINCONTROL_POLARITY
2 : 2	FOLLOW_GPIO_IN_EN	Allow GPIO enable buffer 0 - disallow GPIO enables the clock 1 - allow GPIO enables the clock 0: NOT_FOLLOW_GPIO_ENA 1: FOLLOW_GPIO_ENA

RF_CLK1_CLKBUFF_FOLLOW_HOLD_REQ | 0x0000544B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 6	FOLLOW_HOLD_REQ	Each bit reserved to different EE, set anybit to allow holding buffer enabled by other peripherals

Module Name: SLEEP_CLK1

Base Address: 0x00005A00

Register Quick Reference:

Register	Offset	Type
SLEEP_CLK1_REVISION1	0x00005A00	Read
SLEEP_CLK1_REVISION2	0x00005A01	Read
SLEEP_CLK1_PERPH_TYPE	0x00005A04	Read
SLEEP_CLK1_PERPH_SUBTYPE	0x00005A05	Read
SLEEP_CLK1_EN_CTL	0x00005A46	Read/Write

Register	Offset	Type
SLEEP_CLK1_CAL_RC3	0x00005A5A	Read/Write
SLEEP_CLK1_CAL_RC4	0x00005A5B	Read/Write

SLEEP_CLK1_REVISION1 | 0x00005A00

Type: Read

Reset: 0x00000000

HW Version Register [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

SLEEP_CLK1_REVISION2 | 0x00005A01

Type: Read

Reset: 0x00000005

HW Version Register [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

SLEEP_CLK1_PERPH_TYPE | 0x00005A04

Type: Read

Reset: 0x00000006

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	Clock 6: CLOCK

SLEEP_CLK1_PERPH_SUBTYPE | 0x00005A05

Type: Read

Reset: 0x0000000C

Peripheral SubType PMIC_CONSTANT

Bits	Name	Description
7 : 0	SUBTYPE	Sleep Clock 12: SLP_CLK

SLEEP_CLK1_EN_CTL | 0x00005A46

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	FOLLOW_HW_SLPCLK_PAD_EN	Follow Hardward Enable (VREG_SLEEP_PAD_IO_OK) Sleep Clock Driver 0: NOT_FOLLOW_HW_SLPCLK_PAD_EN 1: FOLLOW_HW_SLPCLK_PAD_EN
7 : 7	FORCE_SLPCLK_PAD_EN	Force Enable Sleep Clock Driver 0: NOT_FORCE_SLPCLK_PAD_EN 1: FORCE_SLPCLK_PAD_EN

SLEEP_CLK1_CAL_RC3 | 0x00005A5A

Type: Read/Write

Reset: 0x00000001

No description provided for this register.

Bits	Name	Description
0 : 0	LFRC_DRIFT_DET_EN_BATT	lfrc dirft detector enabled when battery is present 0: DRIFT_DET_DISABLED 1: DRIFT_DET_ENABLED

SLEEP_CLK1_CAL_RC4 | 0x00005A5B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
4 : 4	LFRC_DRIFT_DET_EN_COIN	lfrc dirft detector enabled when coin cell/cap is present 0: DRIFT_DET_DISABLED 1: DRIFT_DET_ENABLED
6 : 6	COINCELL_GOOD	COINCELL_GOOD Indicate whether a qualified coin cell is installed 0: WEAK_COINCAP 1: STRONG_COINCAP
7 : 7	CALRC_EN	CalRC enable 0: CALRC_DISABLED 1: CALRC_ENABLED

Module Name: RTC_RW
Base Address: 0x00006000
Register Quick Reference:

Register	Offset	Type
RTC_RW_REVISION1	0x00006000	Read
RTC_RW_REVISION2	0x00006001	Read
RTC_RW_PERPH_TYPE	0x00006004	Read
RTC_RW_PERPH_SUBTYPE	0x00006005	Read
RTC_RW_STATUS1	0x00006008	Read
RTC_RW_STATUS2	0x00006009	Read/Write
RTC_RW_EN_CTL1	0x00006046	Read/Write
RTC_RW_RDATA0	0x00006048	Read
RTC_RW_RDATA1	0x00006049	Read
RTC_RW_RDATA2	0x0000604A	Read

Register	Offset	Type
RTC_RW_RDATA3	0x0000604B	Read

RTC_RW_REVISION1 | 0x00006000

Type: Read

Reset: 0x00000000

HW Version Register [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

RTC_RW_REVISION2 | 0x00006001

Type: Read

Reset: 0x00000003

HW Version Register [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

RTC_RW_PERPH_TYPE | 0x00006004

Type: Read

Reset: 0x00000007

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	RTC 7: RTC

RTC_RW_PERPH_SUBTYPE | 0x00006005

Type: Read

Reset: 0x00000001

Peripheral SubType PMIC_CONSTANT

Bits	Name	Description
7 : 0	SUBTYPE	RTC RW 1: RTC_RW

RTC_RW_STATUS1 | 0x00006008

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
7 : 7	RTC_OK	0 = RTC is disabled 1 = RTC is enabled 0: RTC_OFF 1: RTC_ON

RTC_RW_STATUS2 | 0x00006009

Type: Read/Write

Reset: 0x00000080

Status Registers

Bits	Name	Description
7 : 7	RTC_RST	0 = RTC is good 1 = RTC has reset 0: RTC_GOOD 1: RTC_RESETED

RTC_RW_EN_CTL1 | 0x00006046

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	RTC_EN	RTC_EN - enables the real-time clock 0: RTC_COUNTER_DIS 1: RTC_COUNTER_EN

RTC_RW_RDATA0 | 0x00006048

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	RTC_RDATA0	RTC 32-bit counter [7:0] value

RTC_RW_RDATA1 | 0x00006049

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	RTC_RDATA1	RTC 32-bit counter [15:8] value

RTC_RW_RDATA2 | 0x0000604A

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	RTC_RDATA2	RTC 32-bit counter [23:16] value

RTC_RW_RDATA3 | 0x0000604B

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	RTC_RDATA3	RTC 32-bit counter [31:24] value

Module Name: RTC_ALARM

Base Address: 0x00006100

Register Quick Reference:

Register	Offset	Type
RTC_ALARM_REVISION1	0x00006100	Read
RTC_ALARM_REVISION2	0x00006101	Read
RTC_ALARM_PERPH_TYPE	0x00006104	Read
RTC_ALARM_PERPH_SUBTYPE	0x00006105	Read
RTC_ALARM_STATUS1	0x00006108	Read
RTC_ALARM_INT_RT_STS	0x00006110	Read
RTC_ALARM_INT_SET_TYPE	0x00006111	Read/Write
RTC_ALARM_INT_POLARITY_HIGH	0x00006112	Read/Write
RTC_ALARM_INT_POLARITY_LOW	0x00006113	Read/Write
RTC_ALARM_INT_LATCHED_CLR	0x00006114	Write
RTC_ALARM_INT_EN_SET	0x00006115	Read/Write
RTC_ALARM_INT_EN_CLR	0x00006116	Read/Write
RTC_ALARM_INT_LATCHED_STS	0x00006118	Read
RTC_ALARM_INT_PENDING_STS	0x00006119	Read
RTC_ALARM_INT_MID_SEL	0x0000611A	Read/Write

Register	Offset	Type
RTC_ALARM_INT_PRIORITY	0x0000611B	Read/Write
RTC_ALARM_ALARM_DATA0	0x00006140	Read/Write
RTC_ALARM_ALARM_DATA1	0x00006141	Read/Write
RTC_ALARM_ALARM_DATA2	0x00006142	Read/Write
RTC_ALARM_ALARM_DATA3	0x00006143	Read/Write
RTC_ALARM_EN_CTL1	0x00006146	Read/Write
RTC_ALARM_ALARM_CTL	0x00006147	Read/Write
RTC_ALARM_ALARM_CLR	0x00006148	Write

RTC_ALARM_REVISION1 | 0x00006100

Type: Read

Reset: 0x00000000

HW Version Register [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

RTC_ALARM_REVISION2 | 0x00006101

Type: Read

Reset: 0x00000003

HW Version Register [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

RTC_ALARM_PERPH_TYPE | 0x00006104

Type: Read

Reset: 0x00000007

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	RTC 7: RTC

RTC_ALARM_PERPH_SUBTYPE | 0x00006105

Type: Read

Reset: 0x00000003

Peripheral SubType PMIC_CONSTANT

Bits	Name	Description
7 : 0	SUBTYPE	RTC_ALARM 3: RTC_ALARM

RTC_ALARM_STATUS1 | 0x00006108

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
7 : 7	RTC_ALARM_OK	0 = ALARM is not enabled 1 = ALARM is enabled 0: RTC_ALARM_DIS 1: RTC_ALARM_EN

RTC_ALARM_INT_RT_STS | 0x00006110

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
1 : 1	RTC_ALARM	No description provided for this bit field. 0: RTC_ALARM_NOT_EXPIRED 1: RTC_ALARM_EXPIRED

RTC_ALARM_INT_SET_TYPE | 0x00006111

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 1	RTC_ALARM	No description provided for this bit field. 0: LEVEL 1: EDGE

RTC_ALARM_INT_POLARITY_HIGH | 0x00006112

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 1	RTC_ALARM	No description provided for this bit field. 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

RTC_ALARM_INT_POLARITY_LOW | 0x00006113

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 1	RTC_ALARM	No description provided for this bit field. 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

RTC_ALARM_INT_LATCHED_CLR | 0x00006114

Type: Write

Reset: 0x00000000

writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits

Bits	Name	Description
1 : 1	RTC_ALARM	No description provided for this bit field.

RTC_ALARM_INT_EN_SET | 0x00006115

Type: Read/Write

Reset: 0x00000000

PMIC_SET_MASK

Bits	Name	Description
1 : 1	RTC_ALARM	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

RTC_ALARM_INT_EN_CLR | 0x00006116

Type: Read/Write

Reset: 0x00000000

PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
1 : 1	RTC_ALARM	No description provided for this bit field.

RTC_ALARM_INT_LATCHED_STS | 0x00006118

Type: Read

Reset: 0x00000000

Debug: Latched (Sticky) Interrupt. '1' indicates that the interrupt has triggered. Once the latched bit is set it can only be cleared by writing the clear bit.

Bits	Name	Description
1 : 1	RTC_ALARM	No description provided for this bit field. 0: NO_INT_LATCHED 1: INTERRUPT_LATCHED

RTC_ALARM_INT_PENDING_STS | 0x00006119

Type: Read

Reset: 0x00000000

Debug: Pending is set if interrupt has been sent but not cleared.

Bits	Name	Description
1 : 1	RTC_ALARM	No description provided for this bit field. 0: NO_INT_PENDING 1: INTERRUPT_PENDING

RTC_ALARM_INT_MID_SEL | 0x0000611A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: MID0 1: MID1 2: MID2 3: MID3

RTC_ALARM_INT_PRIORITY | 0x0000611B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: SR 1: A

RTC_ALARM_ALARM_DATA0 | 0x00006140

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	RTC_ALARM_DATA0	RTC_ALARM_DATA0 - Real time alarm value [7:0].

RTC_ALARM_ALARM_DATA1 | 0x00006141

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	RTC_ALARM_DATA1	RTC_ALARM_DATA1 - Real time alarm value [15:8].

RTC_ALARM_ALARM_DATA2 | 0x00006142

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	RTC_ALARM_DATA2	RTC_ALARM_DATA2 - Real time alarm value [23:16].

RTC_ALARM_ALARM_DATA3 | 0x00006143

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	RTC_ALARM_DATA3	RTC_ALARM_DATA3 - Real time alarm value [31:24].

RTC_ALARM_EN_CTL1 | 0x00006146

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	ABORT_EN	ABORT_EN - Enable the abort on PERPH_RB feature. If the PMIC fails to power up within 4 seconds, the alarm will be masked to stop repeated power cycling. 0: RTC_STARTUP_DOESNT_ABORT 1: RTC_STARTUP_ABORT_EN
7 : 7	ALARM_EN	ALARM_EN - enables the real-time clock alarm 0: RTC_ALARM_DIS 1: RTC_ALARM_EN

RTC_ALARM_ALARM_CTL | 0x00006147

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	ALARM_CTL	Control RTC alarmed when (greater or equal) or (equal) to timeset 0: ALARMED_GREATER 1: ALARMED_EQUAL

RTC_ALARM_ALARM_CLR | 0x00006148

Type: Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	ALARM_CLR	RTC alarm cleared by writing 1

Module Name: REVID

Base Address: 0x00000100

Register Quick Reference:

Register	Offset	Type
REVID_REVISION3	0x00000102	Read
REVID_REVISION4	0x00000103	Read

REVID_REVISION3 | 0x00000102

Type: Read

Reset: 0x00000001

HW Version Regster [23:16] PMIC_REALPIN

Bits	Name	Description
7 : 0	METAL	This number is incremented on a metal-only revision of the chip

REVID_REVISION4 | 0x00000103

Type: Read

Reset: 0x00000001

HW Version Regster [31:24] PMIC_REALPIN

Bits	Name	Description
7 : 0	ALL_LAYER	This number is incremented everytime there is an all layer revision of the chip

Module Name: SDAM1

Base Address: 0x0000B000

Register Quick Reference:

Register	Offset	Type
SDAM1_INT_RT_STS	0x0000B010	Read
SDAM1_INT_SET_TYPE	0x0000B011	Read
SDAM1_INT_POLARITY_HIGH	0x0000B012	Read
SDAM1_INT_POLARITY_LOW	0x0000B013	Read
SDAM1_INT_LATCHED_CLR	0x0000B014	Write
SDAM1_INT_EN_SET	0x0000B015	Read/Write
SDAM1_INT_EN_CLR	0x0000B016	Read/Write
SDAM1_INT_LATCHED_STS	0x0000B018	Read
SDAM1_INT_PENDING_STS	0x0000B019	Read
SDAM1_INT_MID_SEL	0x0000B01A	Read
SDAM1_INT_PRIORITY	0x0000B01B	Read
SDAM1_MEM_000	0x0000B040	Read/Write
SDAM1_MEM_001	0x0000B041	Read/Write
SDAM1_MEM_002	0x0000B042	Read/Write
SDAM1_MEM_003	0x0000B043	Read/Write
SDAM1_MEM_004	0x0000B044	Read/Write
SDAM1_MEM_005	0x0000B045	Read/Write
SDAM1_MEM_006	0x0000B046	Read/Write
SDAM1_MEM_007	0x0000B047	Read/Write
SDAM1_MEM_008	0x0000B048	Read/Write
SDAM1_MEM_009	0x0000B049	Read/Write
SDAM1_MEM_010	0x0000B04A	Read/Write
SDAM1_MEM_011	0x0000B04B	Read/Write
SDAM1_MEM_012	0x0000B04C	Read/Write

Register	Offset	Type
SDAM1_MEM_013	0x0000B04D	Read/Write
SDAM1_MEM_014	0x0000B04E	Read/Write
SDAM1_MEM_015	0x0000B04F	Read/Write
SDAM1_MEM_016	0x0000B050	Read/Write
SDAM1_MEM_017	0x0000B051	Read/Write
SDAM1_MEM_018	0x0000B052	Read/Write
SDAM1_MEM_019	0x0000B053	Read/Write
SDAM1_MEM_020	0x0000B054	Read/Write
SDAM1_MEM_021	0x0000B055	Read/Write
SDAM1_MEM_022	0x0000B056	Read/Write
SDAM1_MEM_023	0x0000B057	Read/Write
SDAM1_MEM_024	0x0000B058	Read/Write
SDAM1_MEM_025	0x0000B059	Read/Write
SDAM1_MEM_026	0x0000B05A	Read/Write
SDAM1_MEM_027	0x0000B05B	Read/Write
SDAM1_MEM_028	0x0000B05C	Read/Write
SDAM1_MEM_029	0x0000B05D	Read/Write
SDAM1_MEM_030	0x0000B05E	Read/Write
SDAM1_MEM_031	0x0000B05F	Read/Write
SDAM1_MEM_032	0x0000B060	Read/Write
SDAM1_MEM_033	0x0000B061	Read/Write
SDAM1_MEM_034	0x0000B062	Read/Write
SDAM1_MEM_035	0x0000B063	Read/Write
SDAM1_MEM_036	0x0000B064	Read/Write
SDAM1_MEM_037	0x0000B065	Read/Write
SDAM1_MEM_038	0x0000B066	Read/Write
SDAM1_MEM_039	0x0000B067	Read/Write
SDAM1_MEM_040	0x0000B068	Read/Write
SDAM1_MEM_041	0x0000B069	Read/Write
SDAM1_MEM_042	0x0000B06A	Read/Write
SDAM1_MEM_043	0x0000B06B	Read/Write
SDAM1_MEM_044	0x0000B06C	Read/Write

Register	Offset	Type
SDAM1_MEM_045	0x0000B06D	Read/Write
SDAM1_MEM_046	0x0000B06E	Read/Write
SDAM1_MEM_047	0x0000B06F	Read/Write
SDAM1_MEM_048	0x0000B070	Read/Write
SDAM1_MEM_049	0x0000B071	Read/Write
SDAM1_MEM_050	0x0000B072	Read/Write
SDAM1_MEM_051	0x0000B073	Read/Write
SDAM1_MEM_052	0x0000B074	Read/Write
SDAM1_MEM_053	0x0000B075	Read/Write
SDAM1_MEM_054	0x0000B076	Read/Write
SDAM1_MEM_055	0x0000B077	Read/Write
SDAM1_MEM_056	0x0000B078	Read/Write
SDAM1_MEM_057	0x0000B079	Read/Write
SDAM1_MEM_058	0x0000B07A	Read/Write
SDAM1_MEM_059	0x0000B07B	Read/Write
SDAM1_MEM_060	0x0000B07C	Read/Write
SDAM1_MEM_061	0x0000B07D	Read/Write
SDAM1_MEM_062	0x0000B07E	Read/Write
SDAM1_MEM_063	0x0000B07F	Read/Write
SDAM1_MEM_064	0x0000B080	Read/Write
SDAM1_MEM_065	0x0000B081	Read/Write
SDAM1_MEM_066	0x0000B082	Read/Write
SDAM1_MEM_067	0x0000B083	Read/Write
SDAM1_MEM_068	0x0000B084	Read/Write
SDAM1_MEM_069	0x0000B085	Read/Write
SDAM1_MEM_070	0x0000B086	Read/Write
SDAM1_MEM_071	0x0000B087	Read/Write
SDAM1_MEM_072	0x0000B088	Read/Write
SDAM1_MEM_073	0x0000B089	Read/Write
SDAM1_MEM_074	0x0000B08A	Read/Write
SDAM1_MEM_075	0x0000B08B	Read/Write
SDAM1_MEM_076	0x0000B08C	Read/Write

Register	Offset	Type
SDAM1_MEM_077	0x0000B08D	Read/Write
SDAM1_MEM_078	0x0000B08E	Read/Write
SDAM1_MEM_079	0x0000B08F	Read/Write
SDAM1_MEM_080	0x0000B090	Read/Write
SDAM1_MEM_081	0x0000B091	Read/Write
SDAM1_MEM_082	0x0000B092	Read/Write
SDAM1_MEM_083	0x0000B093	Read/Write
SDAM1_MEM_084	0x0000B094	Read/Write
SDAM1_MEM_085	0x0000B095	Read/Write
SDAM1_MEM_086	0x0000B096	Read/Write
SDAM1_MEM_087	0x0000B097	Read/Write
SDAM1_MEM_088	0x0000B098	Read/Write
SDAM1_MEM_089	0x0000B099	Read/Write
SDAM1_MEM_090	0x0000B09A	Read/Write
SDAM1_MEM_091	0x0000B09B	Read/Write
SDAM1_MEM_092	0x0000B09C	Read/Write
SDAM1_MEM_093	0x0000B09D	Read/Write
SDAM1_MEM_094	0x0000B09E	Read/Write
SDAM1_MEM_095	0x0000B09F	Read/Write
SDAM1_MEM_096	0x0000B0A0	Read/Write
SDAM1_MEM_097	0x0000B0A1	Read/Write
SDAM1_MEM_098	0x0000B0A2	Read/Write
SDAM1_MEM_099	0x0000B0A3	Read/Write
SDAM1_MEM_100	0x0000B0A4	Read/Write
SDAM1_MEM_101	0x0000B0A5	Read/Write
SDAM1_MEM_102	0x0000B0A6	Read/Write
SDAM1_MEM_103	0x0000B0A7	Read/Write
SDAM1_MEM_104	0x0000B0A8	Read/Write
SDAM1_MEM_105	0x0000B0A9	Read/Write
SDAM1_MEM_106	0x0000B0AA	Read/Write
SDAM1_MEM_107	0x0000B0AB	Read/Write
SDAM1_MEM_108	0x0000B0AC	Read/Write

Register	Offset	Type
SDAM1_MEM_109	0x0000B0AD	Read/Write
SDAM1_MEM_110	0x0000B0AE	Read/Write
SDAM1_MEM_111	0x0000B0AF	Read/Write
SDAM1_MEM_112	0x0000B0B0	Read/Write
SDAM1_MEM_113	0x0000B0B1	Read/Write
SDAM1_MEM_114	0x0000B0B2	Read/Write
SDAM1_MEM_115	0x0000B0B3	Read/Write
SDAM1_MEM_116	0x0000B0B4	Read/Write
SDAM1_MEM_117	0x0000B0B5	Read/Write
SDAM1_MEM_118	0x0000B0B6	Read/Write
SDAM1_MEM_119	0x0000B0B7	Read/Write
SDAM1_MEM_120	0x0000B0B8	Read/Write
SDAM1_MEM_121	0x0000B0B9	Read/Write
SDAM1_MEM_122	0x0000B0BA	Read/Write
SDAM1_MEM_123	0x0000B0BB	Read/Write
SDAM1_MEM_124	0x0000B0BC	Read/Write
SDAM1_MEM_125	0x0000B0BD	Read/Write
SDAM1_MEM_126	0x0000B0BE	Read/Write
SDAM1_MEM_127	0x0000B0BF	Read/Write

SDAM1_INT_RT_STS | 0x0000B010

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 1	STS	Interrupt Real Time Status Bits 0: INT_RT_STATUS_LOW 1: INT_RT_STATUS_HIGH

SDAM1_INT_SET_TYPE | 0x0000B011

Type: Read

Reset: 0x00000002

No description provided for this register.

Bits	Name	Description
1 : 1	TYPE	0 = use level trigger interrupts, 1 = use edge trigger interrupts 0: LEVEL 1: EDGE

SDAM1_INT_POLARITY_HIGH | 0x0000B012

Type: Read

Reset: 0x00000002

No description provided for this register.

Bits	Name	Description
1 : 1	POLARITY	1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

SDAM1_INT_POLARITY_LOW | 0x0000B013

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 1	POLARITY	1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

SDAM1_INT_LATCHED_CLR | 0x0000B014

Type: Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 1	INT_LATCHED_CLR	writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits

SDAM1_INT_EN_SET | 0x0000B015

Type: Read/Write

Reset: 0x00000000

PMIC_SET_MASK

Bits	Name	Description
1 : 1	INT_EN	writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status 0: INT_DISABLED 1: INT_ENABLED

SDAM1_INT_EN_CLR | 0x0000B016

Type: Read/Write

Reset: 0x00000000

PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
1 : 1	INT_EN	writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status 0: INT_DISABLED 1: INT_ENABLED

SDAM1_INT_LATCHED_STS | 0x0000B018

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 1	INT_LATCHED_STS	Debug: Latched (Sticky) Interrupt. '1' indiecates that the interrutp has triggered. Once the latched bit is set it can only be cleared by writing the clear bit. 0: NO_INT_LATCHED 1: INTERRUPT_LATCHED

SDAM1_INT_PENDING_STS | 0x0000B019

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 1	INT_PENDING_STS	Debug: Pending is set if interrupt has been sent but not cleared. 0: NO_INT_PENDING 1: INTERRUPT_PENDING

SDAM1_INT_MID_SEL | 0x0000B01A

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 0	INT_MID_SEL	Selects the MID that will receive the interrupt 0: MID0 1: MID1 2: MID2 3: MID3

SDAM1_INT_PRIORITY | 0x0000B01B

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	INT_PRIORITY	SR=0 A=1 0: SR 1: A

SDAM1_MEM_000 | 0x0000B040

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_001 | 0x0000B041

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_002 | 0x0000B042

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_003 | 0x0000B043

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_004 | 0x0000B044

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_005 | 0x0000B045

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_006 | 0x0000B046

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_007 | 0x0000B047

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_008 | 0x0000B048

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_009 | 0x0000B049

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_010 | 0x0000B04A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_011 | 0x0000B04B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_012 | 0x0000B04C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_013 | 0x0000B04D

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_014 | 0x0000B04E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_015 | 0x0000B04F

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_016 | 0x0000B050

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_017 | 0x0000B051

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_018 | 0x0000B052

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_019 | 0x0000B053

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_020 | 0x0000B054

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_021 | 0x0000B055

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_022 | 0x0000B056

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_023 | 0x0000B057

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_024 | 0x0000B058

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_025 | 0x0000B059

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_026 | 0x0000B05A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_027 | 0x0000B05B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_028 | 0x0000B05C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_029 | 0x0000B05D

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_030 | 0x0000B05E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_031 | 0x0000B05F

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_032 | 0x0000B060

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_033 | 0x0000B061

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_034 | 0x0000B062

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_035 | 0x0000B063

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_036 | 0x0000B064

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_037 | 0x0000B065

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_038 | 0x0000B066

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_039 | **0x0000B067**

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_040 | 0x0000B068

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_041 | 0x0000B069

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_042 | 0x0000B06A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_043 | 0x0000B06B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_044 | 0x0000B06C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_045 | 0x0000B06D

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_046 | 0x0000B06E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_047 | 0x0000B06F

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_048 | 0x0000B070

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_049 | 0x0000B071

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_050 | 0x0000B072

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_051 | 0x0000B073

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_052 | 0x0000B074

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_053 | 0x0000B075

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_054 | 0x0000B076

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_055 | 0x0000B077

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_056 | 0x0000B078

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_057 | 0x0000B079

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_058 | 0x0000B07A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_059 | 0x0000B07B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_060 | 0x0000B07C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_061 | 0x0000B07D

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_062 | 0x0000B07E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_063 | 0x0000B07F

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_064 | 0x0000B080

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_065 | 0x0000B081

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_066 | 0x0000B082

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_067 | 0x0000B083

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_068 | 0x0000B084

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_069 | 0x0000B085

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_070 | 0x0000B086

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_071 | 0x0000B087

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_072 | 0x0000B088

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_073 | 0x0000B089

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_074 | 0x0000B08A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_075 | 0x0000B08B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_076 | 0x0000B08C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_077 | 0x0000B08D

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_078 | 0x0000B08E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_079 | 0x0000B08F

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_080 | 0x0000B090

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_081 | 0x0000B091

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_082 | 0x0000B092

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_083 | 0x0000B093

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_084 | 0x0000B094

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_085 | 0x0000B095

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_086 | 0x0000B096

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_087 | 0x0000B097

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_088 | 0x0000B098

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_089 | 0x0000B099

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_090 | 0x0000B09A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_091 | 0x0000B09B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_092 | 0x0000B09C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_093 | **0x0000B09D**

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_094 | 0x0000B09E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_095 | 0x0000B09F

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_096 | 0x0000B0A0

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_097 | 0x0000B0A1

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_098 | 0x0000B0A2

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_099 | 0x0000B0A3

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_100 | 0x0000B0A4

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_101 | 0x0000B0A5

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_102 | 0x0000B0A6

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_103 | 0x0000B0A7

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_104 | 0x0000B0A8

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_105 | 0x0000B0A9

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_106 | 0x0000B0AA

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_107 | 0x0000B0AB

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_108 | 0x0000B0AC

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_109 | 0x0000B0AD

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_110 | 0x0000B0AE

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_111 | 0x0000B0AF

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_112 | 0x0000B0B0

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_113 | 0x0000B0B1

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_114 | 0x0000B0B2

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_115 | 0x0000B0B3

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_116 | 0x0000B0B4

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_117 | 0x0000B0B5

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_118 | 0x0000B0B6

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_119 | 0x0000B0B7

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_120 | 0x0000B0B8

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_121 | 0x0000B0B9

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_122 | 0x0000B0BA

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_123 | 0x0000B0BB

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_124 | 0x0000B0BC

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_125 | 0x0000B0BD

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_126 | 0x0000B0BE

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM1_MEM_127 | 0x0000B0BF

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

Module Name: SDAM2

Base Address: 0x0000B100

Register Quick Reference:

Register	Offset	Type
SDAM2_INT_RT_STS	0x0000B110	Read
SDAM2_INT_SET_TYPE	0x0000B111	Read
SDAM2_INT_POLARITY_HIGH	0x0000B112	Read
SDAM2_INT_POLARITY_LOW	0x0000B113	Read
SDAM2_INT_LATCHED_CLR	0x0000B114	Write
SDAM2_INT_EN_SET	0x0000B115	Read/Write
SDAM2_INT_EN_CLR	0x0000B116	Read/Write

Register	Offset	Type
SDAM2_INT_LATCHED_STS	0x0000B118	Read
SDAM2_INT_PENDING_STS	0x0000B119	Read
SDAM2_INT_MID_SEL	0x0000B11A	Read
SDAM2_INT_PRIORITY	0x0000B11B	Read
SDAM2_MEM_000	0x0000B140	Read/Write
SDAM2_MEM_001	0x0000B141	Read/Write
SDAM2_MEM_002	0x0000B142	Read/Write
SDAM2_MEM_003	0x0000B143	Read/Write
SDAM2_MEM_004	0x0000B144	Read/Write
SDAM2_MEM_005	0x0000B145	Read/Write
SDAM2_MEM_006	0x0000B146	Read/Write
SDAM2_MEM_007	0x0000B147	Read/Write
SDAM2_MEM_008	0x0000B148	Read/Write
SDAM2_MEM_009	0x0000B149	Read/Write
SDAM2_MEM_010	0x0000B14A	Read/Write
SDAM2_MEM_011	0x0000B14B	Read/Write
SDAM2_MEM_012	0x0000B14C	Read/Write
SDAM2_MEM_013	0x0000B14D	Read/Write
SDAM2_MEM_014	0x0000B14E	Read/Write
SDAM2_MEM_015	0x0000B14F	Read/Write
SDAM2_MEM_016	0x0000B150	Read/Write
SDAM2_MEM_017	0x0000B151	Read/Write
SDAM2_MEM_018	0x0000B152	Read/Write
SDAM2_MEM_019	0x0000B153	Read/Write
SDAM2_MEM_020	0x0000B154	Read/Write
SDAM2_MEM_021	0x0000B155	Read/Write
SDAM2_MEM_022	0x0000B156	Read/Write
SDAM2_MEM_023	0x0000B157	Read/Write
SDAM2_MEM_024	0x0000B158	Read/Write
SDAM2_MEM_025	0x0000B159	Read/Write
SDAM2_MEM_026	0x0000B15A	Read/Write
SDAM2_MEM_027	0x0000B15B	Read/Write

Register	Offset	Type
SDAM2_MEM_028	0x0000B15C	Read/Write
SDAM2_MEM_029	0x0000B15D	Read/Write
SDAM2_MEM_030	0x0000B15E	Read/Write
SDAM2_MEM_031	0x0000B15F	Read/Write
SDAM2_MEM_032	0x0000B160	Read/Write
SDAM2_MEM_033	0x0000B161	Read/Write
SDAM2_MEM_034	0x0000B162	Read/Write
SDAM2_MEM_035	0x0000B163	Read/Write
SDAM2_MEM_036	0x0000B164	Read/Write
SDAM2_MEM_037	0x0000B165	Read/Write
SDAM2_MEM_038	0x0000B166	Read/Write
SDAM2_MEM_039	0x0000B167	Read/Write
SDAM2_MEM_040	0x0000B168	Read/Write
SDAM2_MEM_041	0x0000B169	Read/Write
SDAM2_MEM_042	0x0000B16A	Read/Write
SDAM2_MEM_043	0x0000B16B	Read/Write
SDAM2_MEM_044	0x0000B16C	Read/Write
SDAM2_MEM_045	0x0000B16D	Read/Write
SDAM2_MEM_046	0x0000B16E	Read/Write
SDAM2_MEM_047	0x0000B16F	Read/Write
SDAM2_MEM_048	0x0000B170	Read/Write
SDAM2_MEM_049	0x0000B171	Read/Write
SDAM2_MEM_050	0x0000B172	Read/Write
SDAM2_MEM_051	0x0000B173	Read/Write
SDAM2_MEM_052	0x0000B174	Read/Write
SDAM2_MEM_053	0x0000B175	Read/Write
SDAM2_MEM_054	0x0000B176	Read/Write
SDAM2_MEM_055	0x0000B177	Read/Write
SDAM2_MEM_056	0x0000B178	Read/Write
SDAM2_MEM_057	0x0000B179	Read/Write
SDAM2_MEM_058	0x0000B17A	Read/Write
SDAM2_MEM_059	0x0000B17B	Read/Write

Register	Offset	Type
SDAM2_MEM_060	0x0000B17C	Read/Write
SDAM2_MEM_061	0x0000B17D	Read/Write
SDAM2_MEM_062	0x0000B17E	Read/Write
SDAM2_MEM_063	0x0000B17F	Read/Write
SDAM2_MEM_064	0x0000B180	Read/Write
SDAM2_MEM_065	0x0000B181	Read/Write
SDAM2_MEM_066	0x0000B182	Read/Write
SDAM2_MEM_067	0x0000B183	Read/Write
SDAM2_MEM_068	0x0000B184	Read/Write
SDAM2_MEM_069	0x0000B185	Read/Write
SDAM2_MEM_070	0x0000B186	Read/Write
SDAM2_MEM_071	0x0000B187	Read/Write
SDAM2_MEM_072	0x0000B188	Read/Write
SDAM2_MEM_073	0x0000B189	Read/Write
SDAM2_MEM_074	0x0000B18A	Read/Write
SDAM2_MEM_075	0x0000B18B	Read/Write
SDAM2_MEM_076	0x0000B18C	Read/Write
SDAM2_MEM_077	0x0000B18D	Read/Write
SDAM2_MEM_078	0x0000B18E	Read/Write
SDAM2_MEM_079	0x0000B18F	Read/Write
SDAM2_MEM_080	0x0000B190	Read/Write
SDAM2_MEM_081	0x0000B191	Read/Write
SDAM2_MEM_082	0x0000B192	Read/Write
SDAM2_MEM_083	0x0000B193	Read/Write
SDAM2_MEM_084	0x0000B194	Read/Write
SDAM2_MEM_085	0x0000B195	Read/Write
SDAM2_MEM_086	0x0000B196	Read/Write
SDAM2_MEM_087	0x0000B197	Read/Write
SDAM2_MEM_088	0x0000B198	Read/Write
SDAM2_MEM_089	0x0000B199	Read/Write
SDAM2_MEM_090	0x0000B19A	Read/Write
SDAM2_MEM_091	0x0000B19B	Read/Write

Register	Offset	Type
SDAM2_MEM_092	0x0000B19C	Read/Write
SDAM2_MEM_093	0x0000B19D	Read/Write
SDAM2_MEM_094	0x0000B19E	Read/Write
SDAM2_MEM_095	0x0000B19F	Read/Write
SDAM2_MEM_096	0x0000B1A0	Read/Write
SDAM2_MEM_097	0x0000B1A1	Read/Write
SDAM2_MEM_098	0x0000B1A2	Read/Write
SDAM2_MEM_099	0x0000B1A3	Read/Write
SDAM2_MEM_100	0x0000B1A4	Read/Write
SDAM2_MEM_101	0x0000B1A5	Read/Write
SDAM2_MEM_102	0x0000B1A6	Read/Write
SDAM2_MEM_103	0x0000B1A7	Read/Write
SDAM2_MEM_104	0x0000B1A8	Read/Write
SDAM2_MEM_105	0x0000B1A9	Read/Write
SDAM2_MEM_106	0x0000B1AA	Read/Write
SDAM2_MEM_107	0x0000B1AB	Read/Write
SDAM2_MEM_108	0x0000B1AC	Read/Write
SDAM2_MEM_109	0x0000B1AD	Read/Write
SDAM2_MEM_110	0x0000B1AE	Read/Write
SDAM2_MEM_111	0x0000B1AF	Read/Write
SDAM2_MEM_112	0x0000B1B0	Read/Write
SDAM2_MEM_113	0x0000B1B1	Read/Write
SDAM2_MEM_114	0x0000B1B2	Read/Write
SDAM2_MEM_115	0x0000B1B3	Read/Write
SDAM2_MEM_116	0x0000B1B4	Read/Write
SDAM2_MEM_117	0x0000B1B5	Read/Write
SDAM2_MEM_118	0x0000B1B6	Read/Write
SDAM2_MEM_119	0x0000B1B7	Read/Write
SDAM2_MEM_120	0x0000B1B8	Read/Write
SDAM2_MEM_121	0x0000B1B9	Read/Write
SDAM2_MEM_122	0x0000B1BA	Read/Write
SDAM2_MEM_123	0x0000B1BB	Read/Write

Register	Offset	Type
SDAM2_MEM_124	0x0000B1BC	Read/Write
SDAM2_MEM_125	0x0000B1BD	Read/Write
SDAM2_MEM_126	0x0000B1BE	Read/Write
SDAM2_MEM_127	0x0000B1BF	Read/Write

SDAM2_INT_RT_STS | 0x0000B110

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 1	STS	Interrupt Real Time Status Bits 0: INT_RT_STATUS_LOW 1: INT_RT_STATUS_HIGH

SDAM2_INT_SET_TYPE | 0x0000B111

Type: Read

Reset: 0x00000002

No description provided for this register.

Bits	Name	Description
1 : 1	TYPE	0 = use level trigger interrupts, 1 = use edge trigger interrupts 0: LEVEL 1: EDGE

SDAM2_INT_POLARITY_HIGH | 0x0000B112

Type: Read

Reset: 0x00000002

No description provided for this register.

Bits	Name	Description
1 : 1	POLARITY	1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

SDAM2_INT_POLARITY_LOW | 0x0000B113

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 1	POLARITY	1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

SDAM2_INT_LATCHED_CLR | 0x0000B114

Type: Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 1	INT_LATCHED_CLR	writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits

SDAM2_INT_EN_SET | 0x0000B115

Type: Read/Write

Reset: 0x00000000

PMIC_SET_MASK

Bits	Name	Description
1 : 1	INT_EN	writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status 0: INT_DISABLED 1: INT_ENABLED

SDAM2_INT_EN_CLR | 0x0000B116

Type: Read/Write

Reset: 0x00000000

PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
1 : 1	INT_EN	writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status 0: INT_DISABLED 1: INT_ENABLED

SDAM2_INT_LATCHED_STS | 0x0000B118

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 1	INT_LATCHED_STS	Debug: Latched (Sticky) Interrupt. '1' indiecates that the interrutp has triggered. Once the latched bit is set it can only be cleared by writing the clear bit. 0: NO_INT_LATCHED 1: INTERRUPT_LATCHED

SDAM2_INT_PENDING_STS | 0x0000B119

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 1	INT_PENDING_STS	Debug: Pending is set if interrupt has been sent but not cleared. 0: NO_INT_PENDING 1: INTERRUPT_PENDING

SDAM2_INT_MID_SEL | 0x0000B11A

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 0	INT_MID_SEL	Selects the MID that will receive the interrupt 0: MID0 1: MID1 2: MID2 3: MID3

SDAM2_INT_PRIORITY | 0x0000B11B

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	INT_PRIORITY	SR=0 A=1 0: SR 1: A

SDAM2_MEM_000 | 0x0000B140

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_001 | 0x0000B141

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_002 | 0x0000B142

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_003 | 0x0000B143

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_004 | 0x0000B144

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_005 | 0x0000B145

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_006 | 0x0000B146

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_007 | 0x0000B147

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_008 | 0x0000B148

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_009 | 0x0000B149

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_010 | 0x0000B14A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_011 | 0x0000B14B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_012 | 0x0000B14C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_013 | 0x0000B14D

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_014 | 0x0000B14E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_015 | 0x0000B14F

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_016 | 0x0000B150

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_017 | 0x0000B151

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_018 | 0x0000B152

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_019 | 0x0000B153

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_020 | 0x0000B154

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_021 | 0x0000B155

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_022 | 0x0000B156

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_023 | 0x0000B157

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_024 | 0x0000B158

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_025 | 0x0000B159

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_026 | 0x0000B15A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_027 | 0x0000B15B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_028 | 0x0000B15C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_029 | 0x0000B15D

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_030 | 0x0000B15E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_031 | 0x0000B15F

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_032 | 0x0000B160

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_033 | **0x0000B161**

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_034 | 0x0000B162

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_035 | 0x0000B163

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_036 | 0x0000B164

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_037 | 0x0000B165

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_038 | 0x0000B166

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_039 | 0x0000B167

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_040 | 0x0000B168

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_041 | 0x0000B169

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_042 | 0x0000B16A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_043 | 0x0000B16B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_044 | 0x0000B16C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_045 | 0x0000B16D

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_046 | 0x0000B16E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_047 | 0x0000B16F

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_048 | 0x0000B170

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_049 | 0x0000B171

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_050 | 0x0000B172

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_051 | **0x0000B173**

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_052 | 0x0000B174

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_053 | 0x0000B175

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_054 | 0x0000B176

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_055 | 0x0000B177

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_056 | 0x0000B178

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_057 | 0x0000B179

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_058 | 0x0000B17A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_059 | 0x0000B17B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_060 | 0x0000B17C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_061 | 0x0000B17D

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_062 | 0x0000B17E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_063 | 0x0000B17F

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_064 | 0x0000B180

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_065 | 0x0000B181

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_066 | 0x0000B182

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_067 | 0x0000B183

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_068 | 0x0000B184

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_069 | 0x0000B185

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_070 | 0x0000B186

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_071 | 0x0000B187

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_072 | 0x0000B188

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_073 | 0x0000B189

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_074 | 0x0000B18A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_075 | 0x0000B18B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_076 | 0x0000B18C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_077 | 0x0000B18D

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_078 | 0x0000B18E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_079 | 0x0000B18F

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_080 | 0x0000B190

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_081 | 0x0000B191

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_082 | 0x0000B192

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_083 | 0x0000B193

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_084 | 0x0000B194

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_085 | 0x0000B195

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_086 | 0x0000B196

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_087 | 0x0000B197

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_088 | 0x0000B198

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_089 | 0x0000B199

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_090 | 0x0000B19A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_091 | 0x0000B19B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_092 | 0x0000B19C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_093 | 0x0000B19D

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_094 | 0x0000B19E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_095 | 0x0000B19F

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_096 | 0x0000B1A0

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_097 | 0x0000B1A1

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_098 | 0x0000B1A2

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_099 | 0x0000B1A3

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_100 | 0x0000B1A4

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_101 | 0x0000B1A5

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_102 | 0x0000B1A6

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_103 | 0x0000B1A7

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_104 | 0x0000B1A8

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_105 | 0x0000B1A9

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_106 | 0x0000B1AA

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_107 | 0x0000B1AB

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_108 | 0x0000B1AC

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_109 | 0x0000B1AD

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_110 | 0x0000B1AE

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_111 | 0x0000B1AF

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_112 | 0x0000B1B0

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_113 | 0x0000B1B1

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_114 | 0x0000B1B2

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_115 | 0x0000B1B3

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_116 | 0x0000B1B4

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_117 | 0x0000B1B5

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_118 | 0x0000B1B6

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_119 | 0x0000B1B7

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_120 | 0x0000B1B8

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_121 | 0x0000B1B9

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_122 | 0x0000B1BA

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_123 | **0x0000B1BB**

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_124 | 0x0000B1BC

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_125 | 0x0000B1BD

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_126 | 0x0000B1BE

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM2_MEM_127 | 0x0000B1BF

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

Module Name: SDAM3

Base Address: 0x0000B200

Register Quick Reference:

Register	Offset	Type
SDAM3_INT_RT_STS	0x0000B210	Read
SDAM3_INT_SET_TYPE	0x0000B211	Read
SDAM3_INT_POLARITY_HIGH	0x0000B212	Read
SDAM3_INT_POLARITY_LOW	0x0000B213	Read
SDAM3_INT_LATCHED_CLR	0x0000B214	Write
SDAM3_INT_EN_SET	0x0000B215	Read/Write
SDAM3_INT_EN_CLR	0x0000B216	Read/Write
SDAM3_INT_LATCHED_STS	0x0000B218	Read
SDAM3_INT_PENDING_STS	0x0000B219	Read
SDAM3_INT_MID_SEL	0x0000B21A	Read
SDAM3_INT_PRIORITY	0x0000B21B	Read
SDAM3_MEM_000	0x0000B240	Read/Write
SDAM3_MEM_001	0x0000B241	Read/Write
SDAM3_MEM_002	0x0000B242	Read/Write
SDAM3_MEM_003	0x0000B243	Read/Write
SDAM3_MEM_004	0x0000B244	Read/Write
SDAM3_MEM_005	0x0000B245	Read/Write

Register	Offset	Type
SDAM3_MEM_006	0x0000B246	Read/Write
SDAM3_MEM_007	0x0000B247	Read/Write
SDAM3_MEM_008	0x0000B248	Read/Write
SDAM3_MEM_009	0x0000B249	Read/Write
SDAM3_MEM_010	0x0000B24A	Read/Write
SDAM3_MEM_011	0x0000B24B	Read/Write
SDAM3_MEM_012	0x0000B24C	Read/Write
SDAM3_MEM_013	0x0000B24D	Read/Write
SDAM3_MEM_014	0x0000B24E	Read/Write
SDAM3_MEM_015	0x0000B24F	Read/Write
SDAM3_MEM_016	0x0000B250	Read/Write
SDAM3_MEM_017	0x0000B251	Read/Write
SDAM3_MEM_018	0x0000B252	Read/Write
SDAM3_MEM_019	0x0000B253	Read/Write
SDAM3_MEM_020	0x0000B254	Read/Write
SDAM3_MEM_021	0x0000B255	Read/Write
SDAM3_MEM_022	0x0000B256	Read/Write
SDAM3_MEM_023	0x0000B257	Read/Write
SDAM3_MEM_024	0x0000B258	Read/Write
SDAM3_MEM_025	0x0000B259	Read/Write
SDAM3_MEM_026	0x0000B25A	Read/Write
SDAM3_MEM_027	0x0000B25B	Read/Write
SDAM3_MEM_028	0x0000B25C	Read/Write
SDAM3_MEM_029	0x0000B25D	Read/Write
SDAM3_MEM_030	0x0000B25E	Read/Write
SDAM3_MEM_031	0x0000B25F	Read/Write
SDAM3_MEM_032	0x0000B260	Read/Write
SDAM3_MEM_033	0x0000B261	Read/Write
SDAM3_MEM_034	0x0000B262	Read/Write
SDAM3_MEM_035	0x0000B263	Read/Write
SDAM3_MEM_036	0x0000B264	Read/Write
SDAM3_MEM_037	0x0000B265	Read/Write

Register	Offset	Type
SDAM3_MEM_038	0x0000B266	Read/Write
SDAM3_MEM_039	0x0000B267	Read/Write
SDAM3_MEM_040	0x0000B268	Read/Write
SDAM3_MEM_041	0x0000B269	Read/Write
SDAM3_MEM_042	0x0000B26A	Read/Write
SDAM3_MEM_043	0x0000B26B	Read/Write
SDAM3_MEM_044	0x0000B26C	Read/Write
SDAM3_MEM_045	0x0000B26D	Read/Write
SDAM3_MEM_046	0x0000B26E	Read/Write
SDAM3_MEM_047	0x0000B26F	Read/Write
SDAM3_MEM_048	0x0000B270	Read/Write
SDAM3_MEM_049	0x0000B271	Read/Write
SDAM3_MEM_050	0x0000B272	Read/Write
SDAM3_MEM_051	0x0000B273	Read/Write
SDAM3_MEM_052	0x0000B274	Read/Write
SDAM3_MEM_053	0x0000B275	Read/Write
SDAM3_MEM_054	0x0000B276	Read/Write
SDAM3_MEM_055	0x0000B277	Read/Write
SDAM3_MEM_056	0x0000B278	Read/Write
SDAM3_MEM_057	0x0000B279	Read/Write
SDAM3_MEM_058	0x0000B27A	Read/Write
SDAM3_MEM_059	0x0000B27B	Read/Write
SDAM3_MEM_060	0x0000B27C	Read/Write
SDAM3_MEM_061	0x0000B27D	Read/Write
SDAM3_MEM_062	0x0000B27E	Read/Write
SDAM3_MEM_063	0x0000B27F	Read/Write
SDAM3_MEM_064	0x0000B280	Read/Write
SDAM3_MEM_065	0x0000B281	Read/Write
SDAM3_MEM_066	0x0000B282	Read/Write
SDAM3_MEM_067	0x0000B283	Read/Write
SDAM3_MEM_068	0x0000B284	Read/Write
SDAM3_MEM_069	0x0000B285	Read/Write

Register	Offset	Type
SDAM3_MEM_070	0x0000B286	Read/Write
SDAM3_MEM_071	0x0000B287	Read/Write
SDAM3_MEM_072	0x0000B288	Read/Write
SDAM3_MEM_073	0x0000B289	Read/Write
SDAM3_MEM_074	0x0000B28A	Read/Write
SDAM3_MEM_075	0x0000B28B	Read/Write
SDAM3_MEM_076	0x0000B28C	Read/Write
SDAM3_MEM_077	0x0000B28D	Read/Write
SDAM3_MEM_078	0x0000B28E	Read/Write
SDAM3_MEM_079	0x0000B28F	Read/Write
SDAM3_MEM_080	0x0000B290	Read/Write
SDAM3_MEM_081	0x0000B291	Read/Write
SDAM3_MEM_082	0x0000B292	Read/Write
SDAM3_MEM_083	0x0000B293	Read/Write
SDAM3_MEM_084	0x0000B294	Read/Write
SDAM3_MEM_085	0x0000B295	Read/Write
SDAM3_MEM_086	0x0000B296	Read/Write
SDAM3_MEM_087	0x0000B297	Read/Write
SDAM3_MEM_088	0x0000B298	Read/Write
SDAM3_MEM_089	0x0000B299	Read/Write
SDAM3_MEM_090	0x0000B29A	Read/Write
SDAM3_MEM_091	0x0000B29B	Read/Write
SDAM3_MEM_092	0x0000B29C	Read/Write
SDAM3_MEM_093	0x0000B29D	Read/Write
SDAM3_MEM_094	0x0000B29E	Read/Write
SDAM3_MEM_095	0x0000B29F	Read/Write
SDAM3_MEM_096	0x0000B2A0	Read/Write
SDAM3_MEM_097	0x0000B2A1	Read/Write
SDAM3_MEM_098	0x0000B2A2	Read/Write
SDAM3_MEM_099	0x0000B2A3	Read/Write
SDAM3_MEM_100	0x0000B2A4	Read/Write
SDAM3_MEM_101	0x0000B2A5	Read/Write

Register	Offset	Type
SDAM3_MEM_102	0x0000B2A6	Read/Write
SDAM3_MEM_103	0x0000B2A7	Read/Write
SDAM3_MEM_104	0x0000B2A8	Read/Write
SDAM3_MEM_105	0x0000B2A9	Read/Write
SDAM3_MEM_106	0x0000B2AA	Read/Write
SDAM3_MEM_107	0x0000B2AB	Read/Write
SDAM3_MEM_108	0x0000B2AC	Read/Write
SDAM3_MEM_109	0x0000B2AD	Read/Write
SDAM3_MEM_110	0x0000B2AE	Read/Write
SDAM3_MEM_111	0x0000B2AF	Read/Write
SDAM3_MEM_112	0x0000B2B0	Read/Write
SDAM3_MEM_113	0x0000B2B1	Read/Write
SDAM3_MEM_114	0x0000B2B2	Read/Write
SDAM3_MEM_115	0x0000B2B3	Read/Write
SDAM3_MEM_116	0x0000B2B4	Read/Write
SDAM3_MEM_117	0x0000B2B5	Read/Write
SDAM3_MEM_118	0x0000B2B6	Read/Write
SDAM3_MEM_119	0x0000B2B7	Read/Write
SDAM3_MEM_120	0x0000B2B8	Read/Write
SDAM3_MEM_121	0x0000B2B9	Read/Write
SDAM3_MEM_122	0x0000B2BA	Read/Write
SDAM3_MEM_123	0x0000B2BB	Read/Write
SDAM3_MEM_124	0x0000B2BC	Read/Write
SDAM3_MEM_125	0x0000B2BD	Read/Write
SDAM3_MEM_126	0x0000B2BE	Read/Write
SDAM3_MEM_127	0x0000B2BF	Read/Write

SDAM3_INT_RT_STS | 0x0000B210

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 1	STS	Interrupt Real Time Status Bits 0: INT_RT_STATUS_LOW 1: INT_RT_STATUS_HIGH

SDAM3_INT_SET_TYPE | 0x0000B211

Type: Read

Reset: 0x00000002

No description provided for this register.

Bits	Name	Description
1 : 1	TYPE	0 = use level trigger interrupts, 1 = use edge trigger interrupts 0: LEVEL 1: EDGE

SDAM3_INT_POLARITY_HIGH | 0x0000B212

Type: Read

Reset: 0x00000002

No description provided for this register.

Bits	Name	Description
1 : 1	POLARITY	1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

SDAM3_INT_POLARITY_LOW | 0x0000B213

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 1	POLARITY	1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

SDAM3_INT_LATCHED_CLR | 0x0000B214

Type: Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 1	INT_LATCHED_CLR	writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits

SDAM3_INT_EN_SET | 0x0000B215

Type: Read/Write

Reset: 0x00000000

PMIC_SET_MASK

Bits	Name	Description
1 : 1	INT_EN	writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status 0: INT_DISABLED 1: INT_ENABLED

SDAM3_INT_EN_CLR | 0x0000B216

Type: Read/Write

Reset: 0x00000000

PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
1 : 1	INT_EN	writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status 0: INT_DISABLED 1: INT_ENABLED

SDAM3_INT_LATCHED_STS | 0x0000B218

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 1	INT_LATCHED_STS	Debug: Latched (Sticky) Interrupt. '1' indiecates that the interrutp has triggered. Once the latched bit is set it can only be cleared by writing the clear bit. 0: NO_INT_LATCHED 1: INTERRUPT_LATCHED

SDAM3_INT_PENDING_STS | 0x0000B219

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 1	INT_PENDING_STS	Debug: Pending is set if interrupt has been sent but not cleared. 0: NO_INT_PENDING 1: INTERRUPT_PENDING

SDAM3_INT_MID_SEL | 0x0000B21A

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 0	INT_MID_SEL	Selects the MID that will receive the interrupt 0: MID0 1: MID1 2: MID2 3: MID3

SDAM3_INT_PRIORITY | 0x0000B21B

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	INT_PRIORITY	SR=0 A=1 0: SR 1: A

SDAM3_MEM_000 | 0x0000B240

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_001 | 0x0000B241

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_002 | 0x0000B242

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_003 | 0x0000B243

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_004 | 0x0000B244

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_005 | 0x0000B245

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_006 | 0x0000B246

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_007 | 0x0000B247

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_008 | 0x0000B248

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_009 | 0x0000B249

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_010 | 0x0000B24A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_011 | 0x0000B24B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_012 | 0x0000B24C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_013 | 0x0000B24D

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_014 | 0x0000B24E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_015 | 0x0000B24F

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_016 | 0x0000B250

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_017 | 0x0000B251

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_018 | 0x0000B252

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_019 | 0x0000B253

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_020 | 0x0000B254

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_021 | 0x0000B255

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_022 | 0x0000B256

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_023 | 0x0000B257

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_024 | 0x0000B258

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_025 | 0x0000B259

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_026 | 0x0000B25A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_027 | **0x0000B25B**

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_028 | 0x0000B25C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_029 | 0x0000B25D

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_030 | 0x0000B25E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_031 | 0x0000B25F

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_032 | 0x0000B260

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_033 | 0x0000B261

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_034 | 0x0000B262

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_035 | 0x0000B263

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_036 | 0x0000B264

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_037 | 0x0000B265

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_038 | 0x0000B266

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_039 | 0x0000B267

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_040 | 0x0000B268

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_041 | 0x0000B269

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_042 | 0x0000B26A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_043 | 0x0000B26B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_044 | 0x0000B26C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_045 | 0x0000B26D

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_046 | 0x0000B26E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_047 | 0x0000B26F

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_048 | 0x0000B270

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_049 | 0x0000B271

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_050 | 0x0000B272

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_051 | 0x0000B273

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_052 | 0x0000B274

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_053 | 0x0000B275

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_054 | 0x0000B276

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_055 | 0x0000B277

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_056 | 0x0000B278

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_057 | 0x0000B279

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_058 | 0x0000B27A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_059 | 0x0000B27B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_060 | 0x0000B27C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_061 | 0x0000B27D

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_062 | 0x0000B27E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_063 | **0x0000B27F**

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_064 | 0x0000B280

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_065 | 0x0000B281

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_066 | 0x0000B282

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_067 | 0x0000B283

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_068 | 0x0000B284

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_069 | 0x0000B285

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_070 | 0x0000B286

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_071 | 0x0000B287

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_072 | 0x0000B288

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_073 | 0x0000B289

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_074 | 0x0000B28A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_075 | 0x0000B28B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_076 | 0x0000B28C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_077 | 0x0000B28D

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_078 | 0x0000B28E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_079 | 0x0000B28F

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_080 | 0x0000B290

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_081 | 0x0000B291

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_082 | 0x0000B292

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_083 | 0x0000B293

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_084 | 0x0000B294

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_085 | 0x0000B295

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_086 | 0x0000B296

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_087 | 0x0000B297

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_088 | 0x0000B298

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_089 | 0x0000B299

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_090 | 0x0000B29A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_091 | 0x0000B29B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_092 | 0x0000B29C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_093 | 0x0000B29D

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_094 | 0x0000B29E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_095 | 0x0000B29F

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_096 | 0x0000B2A0

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_097 | 0x0000B2A1

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_098 | 0x0000B2A2

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_099 | 0x0000B2A3

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_100 | 0x0000B2A4

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_101 | 0x0000B2A5

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_102 | 0x0000B2A6

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_103 | 0x0000B2A7

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_104 | 0x0000B2A8

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_105 | 0x0000B2A9

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_106 | 0x0000B2AA

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_107 | 0x0000B2AB

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_108 | 0x0000B2AC

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_109 | 0x0000B2AD

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_110 | 0x0000B2AE

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_111 | 0x0000B2AF

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_112 | 0x0000B2B0

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_113 | 0x0000B2B1

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_114 | 0x0000B2B2

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_115 | 0x0000B2B3

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_116 | 0x0000B2B4

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_117 | 0x0000B2B5

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_118 | 0x0000B2B6

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_119 | 0x0000B2B7

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_120 | 0x0000B2B8

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_121 | 0x0000B2B9

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_122 | 0x0000B2BA

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_123 | 0x0000B2BB

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_124 | 0x0000B2BC

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_125 | 0x0000B2BD

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_126 | 0x0000B2BE

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM3_MEM_127 | 0x0000B2BF

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

Module Name: SDAM4

Base Address: 0x0000B300

Register Quick Reference:

Register	Offset	Type
SDAM4_INT_RT_STS	0x0000B310	Read
SDAM4_INT_SET_TYPE	0x0000B311	Read
SDAM4_INT_POLARITY_HIGH	0x0000B312	Read
SDAM4_INT_POLARITY_LOW	0x0000B313	Read
SDAM4_INT_LATCHED_CLR	0x0000B314	Write
SDAM4_INT_EN_SET	0x0000B315	Read/Write
SDAM4_INT_EN_CLR	0x0000B316	Read/Write
SDAM4_INT_LATCHED_STS	0x0000B318	Read
SDAM4_INT_PENDING_STS	0x0000B319	Read
SDAM4_INT_MID_SEL	0x0000B31A	Read
SDAM4_INT_PRIORITY	0x0000B31B	Read
SDAM4_MEM_000	0x0000B340	Read/Write
SDAM4_MEM_001	0x0000B341	Read/Write
SDAM4_MEM_002	0x0000B342	Read/Write
SDAM4_MEM_003	0x0000B343	Read/Write
SDAM4_MEM_004	0x0000B344	Read/Write
SDAM4_MEM_005	0x0000B345	Read/Write
SDAM4_MEM_006	0x0000B346	Read/Write
SDAM4_MEM_007	0x0000B347	Read/Write
SDAM4_MEM_008	0x0000B348	Read/Write
SDAM4_MEM_009	0x0000B349	Read/Write
SDAM4_MEM_010	0x0000B34A	Read/Write
SDAM4_MEM_011	0x0000B34B	Read/Write
SDAM4_MEM_012	0x0000B34C	Read/Write
SDAM4_MEM_013	0x0000B34D	Read/Write
SDAM4_MEM_014	0x0000B34E	Read/Write
SDAM4_MEM_015	0x0000B34F	Read/Write
SDAM4_MEM_016	0x0000B350	Read/Write
SDAM4_MEM_017	0x0000B351	Read/Write

Register	Offset	Type
SDAM4_MEM_018	0x0000B352	Read/Write
SDAM4_MEM_019	0x0000B353	Read/Write
SDAM4_MEM_020	0x0000B354	Read/Write
SDAM4_MEM_021	0x0000B355	Read/Write
SDAM4_MEM_022	0x0000B356	Read/Write
SDAM4_MEM_023	0x0000B357	Read/Write
SDAM4_MEM_024	0x0000B358	Read/Write
SDAM4_MEM_025	0x0000B359	Read/Write
SDAM4_MEM_026	0x0000B35A	Read/Write
SDAM4_MEM_027	0x0000B35B	Read/Write
SDAM4_MEM_028	0x0000B35C	Read/Write
SDAM4_MEM_029	0x0000B35D	Read/Write
SDAM4_MEM_030	0x0000B35E	Read/Write
SDAM4_MEM_031	0x0000B35F	Read/Write
SDAM4_MEM_032	0x0000B360	Read/Write
SDAM4_MEM_033	0x0000B361	Read/Write
SDAM4_MEM_034	0x0000B362	Read/Write
SDAM4_MEM_035	0x0000B363	Read/Write
SDAM4_MEM_036	0x0000B364	Read/Write
SDAM4_MEM_037	0x0000B365	Read/Write
SDAM4_MEM_038	0x0000B366	Read/Write
SDAM4_MEM_039	0x0000B367	Read/Write
SDAM4_MEM_040	0x0000B368	Read/Write
SDAM4_MEM_041	0x0000B369	Read/Write
SDAM4_MEM_042	0x0000B36A	Read/Write
SDAM4_MEM_043	0x0000B36B	Read/Write
SDAM4_MEM_044	0x0000B36C	Read/Write
SDAM4_MEM_045	0x0000B36D	Read/Write
SDAM4_MEM_046	0x0000B36E	Read/Write
SDAM4_MEM_047	0x0000B36F	Read/Write
SDAM4_MEM_048	0x0000B370	Read/Write
SDAM4_MEM_049	0x0000B371	Read/Write

Register	Offset	Type
SDAM4_MEM_050	0x0000B372	Read/Write
SDAM4_MEM_051	0x0000B373	Read/Write
SDAM4_MEM_052	0x0000B374	Read/Write
SDAM4_MEM_053	0x0000B375	Read/Write
SDAM4_MEM_054	0x0000B376	Read/Write
SDAM4_MEM_055	0x0000B377	Read/Write
SDAM4_MEM_056	0x0000B378	Read/Write
SDAM4_MEM_057	0x0000B379	Read/Write
SDAM4_MEM_058	0x0000B37A	Read/Write
SDAM4_MEM_059	0x0000B37B	Read/Write
SDAM4_MEM_060	0x0000B37C	Read/Write
SDAM4_MEM_061	0x0000B37D	Read/Write
SDAM4_MEM_062	0x0000B37E	Read/Write
SDAM4_MEM_063	0x0000B37F	Read/Write
SDAM4_MEM_064	0x0000B380	Read/Write
SDAM4_MEM_065	0x0000B381	Read/Write
SDAM4_MEM_066	0x0000B382	Read/Write
SDAM4_MEM_067	0x0000B383	Read/Write
SDAM4_MEM_068	0x0000B384	Read/Write
SDAM4_MEM_069	0x0000B385	Read/Write
SDAM4_MEM_070	0x0000B386	Read/Write
SDAM4_MEM_071	0x0000B387	Read/Write
SDAM4_MEM_072	0x0000B388	Read/Write
SDAM4_MEM_073	0x0000B389	Read/Write
SDAM4_MEM_074	0x0000B38A	Read/Write
SDAM4_MEM_075	0x0000B38B	Read/Write
SDAM4_MEM_076	0x0000B38C	Read/Write
SDAM4_MEM_077	0x0000B38D	Read/Write
SDAM4_MEM_078	0x0000B38E	Read/Write
SDAM4_MEM_079	0x0000B38F	Read/Write
SDAM4_MEM_080	0x0000B390	Read/Write
SDAM4_MEM_081	0x0000B391	Read/Write

Register	Offset	Type
SDAM4_MEM_082	0x0000B392	Read/Write
SDAM4_MEM_083	0x0000B393	Read/Write
SDAM4_MEM_084	0x0000B394	Read/Write
SDAM4_MEM_085	0x0000B395	Read/Write
SDAM4_MEM_086	0x0000B396	Read/Write
SDAM4_MEM_087	0x0000B397	Read/Write
SDAM4_MEM_088	0x0000B398	Read/Write
SDAM4_MEM_089	0x0000B399	Read/Write
SDAM4_MEM_090	0x0000B39A	Read/Write
SDAM4_MEM_091	0x0000B39B	Read/Write
SDAM4_MEM_092	0x0000B39C	Read/Write
SDAM4_MEM_093	0x0000B39D	Read/Write
SDAM4_MEM_094	0x0000B39E	Read/Write
SDAM4_MEM_095	0x0000B39F	Read/Write
SDAM4_MEM_096	0x0000B3A0	Read/Write
SDAM4_MEM_097	0x0000B3A1	Read/Write
SDAM4_MEM_098	0x0000B3A2	Read/Write
SDAM4_MEM_099	0x0000B3A3	Read/Write
SDAM4_MEM_100	0x0000B3A4	Read/Write
SDAM4_MEM_101	0x0000B3A5	Read/Write
SDAM4_MEM_102	0x0000B3A6	Read/Write
SDAM4_MEM_103	0x0000B3A7	Read/Write
SDAM4_MEM_104	0x0000B3A8	Read/Write
SDAM4_MEM_105	0x0000B3A9	Read/Write
SDAM4_MEM_106	0x0000B3AA	Read/Write
SDAM4_MEM_107	0x0000B3AB	Read/Write
SDAM4_MEM_108	0x0000B3AC	Read/Write
SDAM4_MEM_109	0x0000B3AD	Read/Write
SDAM4_MEM_110	0x0000B3AE	Read/Write
SDAM4_MEM_111	0x0000B3AF	Read/Write
SDAM4_MEM_112	0x0000B3B0	Read/Write
SDAM4_MEM_113	0x0000B3B1	Read/Write

Register	Offset	Type
SDAM4_MEM_114	0x0000B3B2	Read/Write
SDAM4_MEM_115	0x0000B3B3	Read/Write
SDAM4_MEM_116	0x0000B3B4	Read/Write
SDAM4_MEM_117	0x0000B3B5	Read/Write
SDAM4_MEM_118	0x0000B3B6	Read/Write
SDAM4_MEM_119	0x0000B3B7	Read/Write
SDAM4_MEM_120	0x0000B3B8	Read/Write
SDAM4_MEM_121	0x0000B3B9	Read/Write
SDAM4_MEM_122	0x0000B3BA	Read/Write
SDAM4_MEM_123	0x0000B3BB	Read/Write
SDAM4_MEM_124	0x0000B3BC	Read/Write
SDAM4_MEM_125	0x0000B3BD	Read/Write
SDAM4_MEM_126	0x0000B3BE	Read/Write
SDAM4_MEM_127	0x0000B3BF	Read/Write

SDAM4_INT_RT_STS | 0x0000B310

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 1	STS	Interrupt Real Time Status Bits 0: INT_RT_STATUS_LOW 1: INT_RT_STATUS_HIGH

SDAM4_INT_SET_TYPE | 0x0000B311

Type: Read

Reset: 0x00000002

No description provided for this register.

Bits	Name	Description
1 : 1	TYPE	0 = use level trigger interrupts, 1 = use edge trigger interrupts 0: LEVEL 1: EDGE

SDAM4_INT_POLARITY_HIGH | 0x0000B312

Type: Read

Reset: 0x00000002

No description provided for this register.

Bits	Name	Description
1 : 1	POLARITY	1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

SDAM4_INT_POLARITY_LOW | 0x0000B313

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 1	POLARITY	1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

SDAM4_INT_LATCHED_CLR | 0x0000B314

Type: Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 1	INT_LATCHED_CLR	writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits

SDAM4_INT_EN_SET | 0x0000B315

Type: Read/Write

Reset: 0x00000000

PMIC_SET_MASK

Bits	Name	Description
1 : 1	INT_EN	writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status 0: INT_DISABLED 1: INT_ENABLED

SDAM4_INT_EN_CLR | 0x0000B316

Type: Read/Write

Reset: 0x00000000

PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
1 : 1	INT_EN	writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status 0: INT_DISABLED 1: INT_ENABLED

SDAM4_INT_LATCHED_STS | 0x0000B318

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 1	INT_LATCHED_STS	Debug: Latched (Sticky) Interrupt. '1' indiecates that the interrutp has triggered. Once the latched bit is set it can only be cleared by writing the clear bit. 0: NO_INT_LATCHED 1: INTERRUPT_LATCHED

SDAM4_INT_PENDING_STS | 0x0000B319

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 1	INT_PENDING_STS	Debug: Pending is set if interrupt has been sent but not cleared. 0: NO_INT_PENDING 1: INTERRUPT_PENDING

SDAM4_INT_MID_SEL | 0x0000B31A

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 0	INT_MID_SEL	Selects the MID that will receive the interrupt 0: MID0 1: MID1 2: MID2 3: MID3

SDAM4_INT_PRIORITY | 0x0000B31B

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	INT_PRIORITY	SR=0 A=1 0: SR 1: A

SDAM4_MEM_000 | 0x0000B340

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_001 | 0x0000B341

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_002 | 0x0000B342

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_003 | 0x0000B343

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_004 | 0x0000B344

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_005 | 0x0000B345

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_006 | 0x0000B346

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_007 | 0x0000B347

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_008 | 0x0000B348

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_009 | 0x0000B349

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_010 | 0x0000B34A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_011 | 0x0000B34B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_012 | 0x0000B34C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_013 | 0x0000B34D

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_014 | 0x0000B34E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_015 | 0x0000B34F

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_016 | 0x0000B350

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_017 | 0x0000B351

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_018 | 0x0000B352

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_019 | 0x0000B353

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_020 | 0x0000B354

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_021 | 0x0000B355

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_022 | 0x0000B356

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_023 | 0x0000B357

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_024 | 0x0000B358

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_025 | 0x0000B359

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_026 | 0x0000B35A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_027 | 0x0000B35B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_028 | 0x0000B35C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_029 | 0x0000B35D

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_030 | 0x0000B35E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_031 | 0x0000B35F

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_032 | 0x0000B360

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_033 | 0x0000B361

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_034 | 0x0000B362

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_035 | 0x0000B363

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_036 | 0x0000B364

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_037 | 0x0000B365

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_038 | 0x0000B366

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_039 | 0x0000B367

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_040 | 0x0000B368

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_041 | 0x0000B369

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_042 | 0x0000B36A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_043 | 0x0000B36B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_044 | 0x0000B36C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_045 | 0x0000B36D

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_046 | 0x0000B36E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_047 | 0x0000B36F

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_048 | 0x0000B370

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_049 | 0x0000B371

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_050 | 0x0000B372

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_051 | 0x0000B373

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_052 | 0x0000B374

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_053 | 0x0000B375

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_054 | 0x0000B376

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_055 | 0x0000B377

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_056 | 0x0000B378

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_057 | 0x0000B379

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_058 | 0x0000B37A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_059 | 0x0000B37B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_060 | 0x0000B37C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_061 | 0x0000B37D

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_062 | 0x0000B37E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_063 | 0x0000B37F

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_064 | 0x0000B380

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_065 | 0x0000B381

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_066 | 0x0000B382

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_067 | 0x0000B383

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_068 | 0x0000B384

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_069 | 0x0000B385

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_070 | 0x0000B386

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_071 | 0x0000B387

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_072 | 0x0000B388

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_073 | 0x0000B389

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_074 | 0x0000B38A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_075 | **0x0000B38B**

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_076 | 0x0000B38C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_077 | 0x0000B38D

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_078 | 0x0000B38E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_079 | 0x0000B38F

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_080 | 0x0000B390

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_081 | 0x0000B391

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_082 | 0x0000B392

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_083 | 0x0000B393

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_084 | 0x0000B394

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_085 | 0x0000B395

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_086 | 0x0000B396

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_087 | 0x0000B397

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_088 | 0x0000B398

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_089 | 0x0000B399

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_090 | 0x0000B39A

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_091 | 0x0000B39B

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_092 | 0x0000B39C

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_093 | 0x0000B39D

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_094 | 0x0000B39E

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_095 | 0x0000B39F

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_096 | 0x0000B3A0

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_097 | 0x0000B3A1

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_098 | 0x0000B3A2

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_099 | 0x0000B3A3

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_100 | 0x0000B3A4

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_101 | 0x0000B3A5

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_102 | 0x0000B3A6

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_103 | 0x0000B3A7

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_104 | 0x0000B3A8

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_105 | 0x0000B3A9

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_106 | 0x0000B3AA

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_107 | 0x0000B3AB

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_108 | 0x0000B3AC

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_109 | 0x0000B3AD

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_110 | 0x0000B3AE

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_111 | 0x0000B3AF

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_112 | 0x0000B3B0

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_113 | 0x0000B3B1

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_114 | 0x0000B3B2

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_115 | 0x0000B3B3

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_116 | 0x0000B3B4

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_117 | 0x0000B3B5

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_118 | 0x0000B3B6

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_119 | 0x0000B3B7

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_120 | 0x0000B3B8

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_121 | 0x0000B3B9

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_122 | 0x0000B3BA

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_123 | 0x0000B3BB

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_124 | 0x0000B3BC

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_125 | 0x0000B3BD

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_126 | 0x0000B3BE

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

SDAM4_MEM_127 | 0x0000B3BF

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	DATA	No description provided for this bit field.

Module Name: GPIO01

Base Address: 0x0000C000

Register Quick Reference:

Register	Offset	Type
GPIO01_REVISION1	0x0000C000	Read
GPIO01_REVISION2	0x0000C001	Read
GPIO01_REVISION3	0x0000C002	Read
GPIO01_REVISION4	0x0000C003	Read
GPIO01_PERPH_TYPE	0x0000C004	Read
GPIO01_PERPH_SUBTYPE	0x0000C005	Read
GPIO01_STATUS1	0x0000C008	Read

Register	Offset	Type
GPIO01_INT_RT_STS	0x0000C010	Read
GPIO01_INT_SET_TYPE	0x0000C011	Read/Write
GPIO01_INT_POLARITY_HIGH	0x0000C012	Read/Write
GPIO01_INT_POLARITY_LOW	0x0000C013	Read/Write
GPIO01_INT_LATCHED_CLR	0x0000C014	Write
GPIO01_INT_EN_SET	0x0000C015	Read/Write
GPIO01_INT_EN_CLR	0x0000C016	Read/Write
GPIO01_INT_LATCHED_STS	0x0000C018	Read
GPIO01_INT_PENDING_STS	0x0000C019	Read
GPIO01_INT_MID_SEL	0x0000C01A	Read/Write
GPIO01_INT_PRIORITY	0x0000C01B	Read/Write
GPIO01_MODE_CTL	0x0000C040	Read/Write
GPIO01_DIG_VIN_CTL	0x0000C041	Read/Write
GPIO01_DIG_PULL_CTL	0x0000C042	Read/Write
GPIO01_DIG_OUT_DRV_CTL	0x0000C045	Read/Write
GPIO01_EN_CTL	0x0000C046	Read/Write

GPIO01_REVISION1 | 0x0000C000

Type: Read

Reset: 0x00000002

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

GPIO01_REVISION2 | 0x0000C001

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

GPIO01_REVISION3 | 0x0000C002

Type: Read

Reset: 0x00000000

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

GPIO01_REVISION4 | 0x0000C003

Type: Read

Reset: 0x00000002

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

GPIO01_PERPH_TYPE | 0x0000C004

Type: Read

Reset: 0x00000010

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	No description provided for this bit field. 16: GPIO

GPIO01_PERPH_SUBTYPE | 0x0000C005

Type: Read

Reset: 0x00000010

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	No description provided for this bit field. 16: GPIO_LV 17: GPIO_MV

GPIO01_STATUS1 | 0x0000C008

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
0 : 0	GPIO_VAL	Value read by the input buffer, if enabled 0: GPIO_INPUT_LOW 1: GPIO_INPUT_HIGH
7 : 7	GPIO_OK	No description provided for this bit field. 0: GPIO_DISABLED 1: GPIO_ENABLED

GPIO01_INT_RT_STS | 0x0000C010

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	GPIO_IN_STS	No description provided for this bit field. 0: INT_RT_STATUS_LOW 1: INT_RT_STATUS_HIGH

GPIO01_INT_SET_TYPE | 0x0000C011

Type: Read/Write

Reset: 0x00000000

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	GPIO_IN_TYPE	No description provided for this bit field. 0: LEVEL 1: EDGE

GPIO01_INT_POLARITY_HIGH | 0x0000C012

Type: Read/Write

Reset: 0x00000000

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	GPIO_IN_HIGH	No description provided for this bit field. 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

GPIO01_INT_POLARITY_LOW | 0x0000C013

Type: Read/Write

Reset: 0x00000000

1 = Interrupt will trigger on a level low (falling edge) event, 0 = level low triggering is disabled

Bits	Name	Description
0 : 0	GPIO_IN_LOW	No description provided for this bit field. 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

GPIO01_INT_LATCHED_CLR | 0x0000C014

Type: Write

Reset: 0x00000000

writing a 1 to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits

Bits	Name	Description
0 : 0	GPIO_IN_LATCHED_CLR	No description provided for this bit field.

GPIO01_INT_EN_SET | 0x0000C015

Type: Read/Write

Reset: 0x00000000

writing 0 to this register has no effect. Writing a 1 will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	GPIO_IN_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

GPIO01_INT_EN_CLR | 0x0000C016

Type: Read/Write

Reset: 0x00000000

writing 0 to this register has no effect. Writing a 1 will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	GPIO_IN_EN_CLR	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

GPIO01_INT_LATCHED_STS | 0x0000C018

Type: Read

Reset: 0x00000000

Debug: Latched (Sticky) Interrupt. 1 indiecates that the interrutp has triggered. Once the latched bit is set it can only be cleared by writing the clear bit.

Bits	Name	Description
0 : 0	GPIO_IN_LATCHED_STS	No description provided for this bit field. 0: NO_INT_LATCHED 1: INTERRUPT_LATCHED

GPIO01_INT_PENDING_STS | 0x0000C019

Type: Read

Reset: 0x00000000

Debug: Pending is set if interrupt has been sent but not cleared.

Bits	Name	Description
0 : 0	GPIO_IN_PENDING_STS	No description provided for this bit field. 0: NO_INT_PENDING 1: INTERRUPT_PENDING

GPIO01_INT_MID_SEL | 0x0000C01A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: MID0 1: MID1 2: MID2 3: MID3

GPIO01_INT_PRIORITY | 0x0000C01B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: SR 1: A

GPIO01_MODE_CTL | 0x0000C040

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 0	MODE	GPIO Mode: 0: Digital Input 1: Digital Output 2: Digital Input and Output 3: Analog Pass Through 0: DIG_IN 1: DIG_OUT 2: DIG_INOUT 3: ANA_PASS_THRU

GPIO01_DIG_VIN_CTL | 0x0000C041

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
2 : 0	VOLTAGE_SEL	Select Voltage source: 000 = VIN0 (refer to the objective spec.) 001 = VIN1 (refer to the objective spec.) 010 = Reserved 011 = Reserved 100 = Reserved 101 = Reserved 110 = Reserved 111 = Reserved (refer to the objective spec for the definition of VINx) (Voltage supply is for GPIO_MV only, GPIO_LV only has one voltage supply) 0: VIN0 1: VIN1 2: RESERVED2 3: RESERVED3 4: RESERVED4 5: RESERVED5 6: RESERVED6 7: RESERVED7

GPIO01_DIG_PULL_CTL | 0x0000C042

Type: Read/Write

Reset: 0x00000004

No description provided for this register.

Bits	Name	Description
2 : 0	PULLUP_SEL	Current source pulls: - 000 = pull-up 30uA constant - 001 = pull-up 1.5uA - 010 = pull-up 31.5uA constant - 011 = pull-up 1.5uA + 30uA boost - 100 = pull-down 10uA - 101 = no pull - 110 = Reserved - 111 = Reserved Pull up or pull down can only be enabled when master PU/PD is enabled. Pull is disabled when master enable is 0 regardless of this register's setting (see GPIO MDOS for more information of pull enable control) 0: PULLUP_30UA 1: PULLUP_1P5UA 2: PULLUP_31P5UA 3: PULLUP_1P5UA_30UA_BOOST 4: PULLDOWN_10UA 5: NO_PULL 6: RESERVED6 7: RESERVED7

GPIO01_DIG_OUT_DRV_CTL | 0x0000C045

Type: Read/Write

Reset: 0x00000001

No description provided for this register.

Bits	Name	Description
1 : 0	OUTPUT_DRV_SEL	Digital output buffer drive strength select: 00 = Reserved 01 = Low strength 10 = Medium strength 11 = High strength 0: RESERVED 1: LOW_STR 2: MED_STR 3: HIGH_STR

Bits	Name	Description
5 : 4	OUTPUT_TYPE	Digital output buffer type configuration 00= CMOS (drive high and low) 01= Open drain NMOS (only drive low, i.e. I2C, PMOS disabled) 10= Open drain PMOS (only drive high, NMOS disabled) 11= No Drive. Both PMOS and NMOS are disabled 0: CMOS 1: OPEN_DRAIN_NMOS 2: OPEN_DRAIN_PMOS 3: NO_DRIVE

GPIO01_EN_CTL | 0x0000C046

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
7 : 7	PERPH_EN	GPIO Master Enable 0 = puts GPIO_PAD at high Z and disables the block 1 = GPIO is enabled 0: GPIO_DISABLED 1: GPIO_ENABLED

Module Name: GPIO02

Base Address: 0x0000C100

Register Quick Reference:

Register	Offset	Type
GPIO02_REVISION1	0x0000C100	Read
GPIO02_REVISION2	0x0000C101	Read
GPIO02_REVISION3	0x0000C102	Read
GPIO02_REVISION4	0x0000C103	Read
GPIO02_PERPH_TYPE	0x0000C104	Read
GPIO02_PERPH_SUBTYPE	0x0000C105	Read
GPIO02_STATUS1	0x0000C108	Read
GPIO02_INT_RT_STS	0x0000C110	Read
GPIO02_INT_SET_TYPE	0x0000C111	Read/Write
GPIO02_INT_POLARITY_HIGH	0x0000C112	Read/Write

Register	Offset	Type
GPIO02_INT_POLARITY_LOW	0x0000C113	Read/Write
GPIO02_INT_LATCHED_CLR	0x0000C114	Write
GPIO02_INT_EN_SET	0x0000C115	Read/Write
GPIO02_INT_EN_CLR	0x0000C116	Read/Write
GPIO02_INT_LATCHED_STS	0x0000C118	Read
GPIO02_INT_PENDING_STS	0x0000C119	Read
GPIO02_INT_MID_SEL	0x0000C11A	Read/Write
GPIO02_INT_PRIORITY	0x0000C11B	Read/Write
GPIO02_MODE_CTL	0x0000C140	Read/Write
GPIO02_DIG_VIN_CTL	0x0000C141	Read/Write
GPIO02_DIG_PULL_CTL	0x0000C142	Read/Write
GPIO02_DIG_OUT_DRV_CTL	0x0000C145	Read/Write
GPIO02_EN_CTL	0x0000C146	Read/Write

GPIO02_REVISION1 | 0x0000C100

Type: Read

Reset: 0x00000002

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

GPIO02_REVISION2 | 0x0000C101

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

GPIO02_REVISION3 | 0x0000C102

Type: Read

Reset: 0x00000000

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

GPIO02_REVISION4 | 0x0000C103

Type: Read

Reset: 0x00000002

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

GPIO02_PERPH_TYPE | 0x0000C104

Type: Read

Reset: 0x00000010

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	No description provided for this bit field. 16: GPIO

GPIO02_PERPH_SUBTYPE | 0x0000C105

Type: Read

Reset: 0x00000010

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	No description provided for this bit field. 16: GPIO_LV 17: GPIO_MV

GPIO02_STATUS1 | 0x0000C108

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
0 : 0	GPIO_VAL	Value read by the input buffer, if enabled 0: GPIO_INPUT_LOW 1: GPIO_INPUT_HIGH
7 : 7	GPIO_OK	No description provided for this bit field. 0: GPIO_DISABLED 1: GPIO_ENABLED

GPIO02_INT_RT_STS | 0x0000C110

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	GPIO_IN_STS	No description provided for this bit field. 0: INT_RT_STATUS_LOW 1: INT_RT_STATUS_HIGH

GPIO02_INT_SET_TYPE | 0x0000C111

Type: Read/Write

Reset: 0x00000000

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	GPIO_IN_TYPE	No description provided for this bit field. 0: LEVEL 1: EDGE

GPIO02_INT_POLARITY_HIGH | 0x0000C112

Type: Read/Write

Reset: 0x00000000

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	GPIO_IN_HIGH	No description provided for this bit field. 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

GPIO02_INT_POLARITY_LOW | 0x0000C113

Type: Read/Write

Reset: 0x00000000

1 = Interrupt will trigger on a level low (falling edge) event, 0 = level low triggering is disabled

Bits	Name	Description
0 : 0	GPIO_IN_LOW	No description provided for this bit field. 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

GPIO02_INT_LATCHED_CLR | 0x0000C114

Type: Write

Reset: 0x00000000

writing a 1 to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits

Bits	Name	Description
0 : 0	GPIO_IN_LATCHED_CLR	No description provided for this bit field.

GPIO02_INT_EN_SET | 0x0000C115

Type: Read/Write

Reset: 0x00000000

writing 0 to this register has no effect. Writing a 1 will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	GPIO_IN_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

GPIO02_INT_EN_CLR | 0x0000C116

Type: Read/Write

Reset: 0x00000000

writing 0 to this register has no effect. Writing a 1 will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	GPIO_IN_EN_CLR	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

GPIO02_INT_LATCHED_STS | 0x0000C118

Type: Read

Reset: 0x00000000

Debug: Latched (Sticky) Interrupt. 1 indiecates that the interrutp has triggered. Once the latched bit is set it can only be cleared by writing the clear bit.

Bits	Name	Description
0 : 0	GPIO_IN_LATCHED_STS	No description provided for this bit field. 0: NO_INT_LATCHED 1: INTERRUPT_LATCHED

GPIO02_INT_PENDING_STS | 0x0000C119

Type: Read

Reset: 0x00000000

Debug: Pending is set if interrupt has been sent but not cleared.

Bits	Name	Description
0 : 0	GPIO_IN_PENDING_STS	No description provided for this bit field. 0: NO_INT_PENDING 1: INTERRUPT_PENDING

GPIO02_INT_MID_SEL | 0x0000C11A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: MID0 1: MID1 2: MID2 3: MID3

GPIO02_INT_PRIORITY | 0x0000C11B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: SR 1: A

GPIO02_MODE_CTL | 0x0000C140

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 0	MODE	GPIO Mode: 0: Digital Input 1: Digital Output 2: Digital Input and Output 3: Analog Pass Through 0: DIG_IN 1: DIG_OUT 2: DIG_INOUT 3: ANA_PASS_THRU

GPIO02_DIG_VIN_CTL | 0x0000C141

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
2 : 0	VOLTAGE_SEL	Select Voltage source: 000 = VIN0 (refer to the objective spec.) 001 = VIN1 (refer to the objective spec.) 010 = Reserved 011 = Reserved 100 = Reserved 101 = Reserved 110 = Reserved 111 = Reserved (refer to the objective spec for the definition of VINx) (Voltage supply is for GPIO_MV only, GPIO_LV only has one voltage supply) 0: VIN0 1: VIN1 2: RESERVED2 3: RESERVED3 4: RESERVED4 5: RESERVED5 6: RESERVED6 7: RESERVED7

GPIO02_DIG_PULL_CTL | 0x0000C142

Type: Read/Write

Reset: 0x00000004

No description provided for this register.

Bits	Name	Description
2 : 0	PULLUP_SEL	Current source pulls: - 000 = pull-up 30uA constant - 001 = pull-up 1.5uA - 010 = pull-up 31.5uA constant - 011 = pull-up 1.5uA + 30uA boost - 100 = pull-down 10uA - 101 = no pull - 110 = Reserved - 111 = Reserved Pull up or pull down can only be enabled when master PU/PD is enabled. Pull is disabled when master enable is 0 regardless of this register's setting (see GPIO MDOS for more information of pull enable control) 0: PULLUP_30UA 1: PULLUP_1P5UA 2: PULLUP_31P5UA 3: PULLUP_1P5UA_30UA_BOOST 4: PULLDOWN_10UA 5: NO_PULL 6: RESERVED6 7: RESERVED7

GPIO02_DIG_OUT_DRV_CTL | 0x0000C145

Type: Read/Write

Reset: 0x00000001

No description provided for this register.

Bits	Name	Description
1 : 0	OUTPUT_DRV_SEL	Digital output buffer drive strength select: 00 = Reserved 01 = Low strength 10 = Medium strength 11 = High strength 0: RESERVED 1: LOW_STR 2: MED_STR 3: HIGH_STR

Bits	Name	Description
5 : 4	OUTPUT_TYPE	Digital output buffer type configuration 00= CMOS (drive high and low) 01= Open drain NMOS (only drive low, i.e. I2C, PMOS disabled) 10= Open drain PMOS (only drive high, NMOS disabled) 11= No Drive. Both PMOS and NMOS are disabled 0: CMOS 1: OPEN_DRAIN_NMOS 2: OPEN_DRAIN_PMOS 3: NO_DRIVE

GPIO02_EN_CTL | 0x0000C146

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
7 : 7	PERPH_EN	GPIO Master Enable 0 = puts GPIO_PAD at high Z and disables the block 1 = GPIO is enabled 0: GPIO_DISABLED 1: GPIO_ENABLED

Module Name: GPIO03

Base Address: 0x0000C200

Register Quick Reference:

Register	Offset	Type
GPIO03_REVISION1	0x0000C200	Read
GPIO03_REVISION2	0x0000C201	Read
GPIO03_REVISION3	0x0000C202	Read
GPIO03_REVISION4	0x0000C203	Read
GPIO03_PERPH_TYPE	0x0000C204	Read
GPIO03_PERPH_SUBTYPE	0x0000C205	Read
GPIO03_STATUS1	0x0000C208	Read
GPIO03_INT_RT_STS	0x0000C210	Read
GPIO03_INT_SET_TYPE	0x0000C211	Read/Write
GPIO03_INT_POLARITY_HIGH	0x0000C212	Read/Write

Register	Offset	Type
GPIO03_INT_POLARITY_LOW	0x0000C213	Read/Write
GPIO03_INT_LATCHED_CLR	0x0000C214	Write
GPIO03_INT_EN_SET	0x0000C215	Read/Write
GPIO03_INT_EN_CLR	0x0000C216	Read/Write
GPIO03_INT_LATCHED_STS	0x0000C218	Read
GPIO03_INT_PENDING_STS	0x0000C219	Read
GPIO03_INT_MID_SEL	0x0000C21A	Read/Write
GPIO03_INT_PRIORITY	0x0000C21B	Read/Write
GPIO03_MODE_CTL	0x0000C240	Read/Write
GPIO03_DIG_VIN_CTL	0x0000C241	Read/Write
GPIO03_DIG_PULL_CTL	0x0000C242	Read/Write
GPIO03_DIG_OUT_DRV_CTL	0x0000C245	Read/Write
GPIO03_EN_CTL	0x0000C246	Read/Write

GPIO03_REVISION1 | 0x0000C200

Type: Read

Reset: 0x00000002

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

GPIO03_REVISION2 | 0x0000C201

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

GPIO03_REVISION3 | 0x0000C202

Type: Read

Reset: 0x00000000

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

GPIO03_REVISION4 | 0x0000C203

Type: Read

Reset: 0x00000002

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

GPIO03_PERPH_TYPE | 0x0000C204

Type: Read

Reset: 0x00000010

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	No description provided for this bit field. 16: GPIO

GPIO03_PERPH_SUBTYPE | 0x0000C205

Type: Read

Reset: 0x00000010

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	No description provided for this bit field. 16: GPIO_LV 17: GPIO_MV

GPIO03_STATUS1 | 0x0000C208

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
0 : 0	GPIO_VAL	Value read by the input buffer, if enabled 0: GPIO_INPUT_LOW 1: GPIO_INPUT_HIGH
7 : 7	GPIO_OK	No description provided for this bit field. 0: GPIO_DISABLED 1: GPIO_ENABLED

GPIO03_INT_RT_STS | 0x0000C210

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	GPIO_IN_STS	No description provided for this bit field. 0: INT_RT_STATUS_LOW 1: INT_RT_STATUS_HIGH

GPIO03_INT_SET_TYPE | 0x0000C211

Type: Read/Write

Reset: 0x00000000

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	GPIO_IN_TYPE	No description provided for this bit field. 0: LEVEL 1: EDGE

GPIO03_INT_POLARITY_HIGH | 0x0000C212

Type: Read/Write

Reset: 0x00000000

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	GPIO_IN_HIGH	No description provided for this bit field. 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

GPIO03_INT_POLARITY_LOW | 0x0000C213

Type: Read/Write

Reset: 0x00000000

1 = Interrupt will trigger on a level low (falling edge) event, 0 = level low triggering is disabled

Bits	Name	Description
0 : 0	GPIO_IN_LOW	No description provided for this bit field. 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

GPIO03_INT_LATCHED_CLR | **0x0000C214**

Type: Write

Reset: 0x00000000

writing a 1 to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits

Bits	Name	Description
0 : 0	GPIO_IN_LATCHED_CLR	No description provided for this bit field.

GPIO03_INT_EN_SET | 0x0000C215

Type: Read/Write

Reset: 0x00000000

writing 0 to this register has no effect. Writing a 1 will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	GPIO_IN_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

GPIO03_INT_EN_CLR | 0x0000C216

Type: Read/Write

Reset: 0x00000000

writing 0 to this register has no effect. Writing a 1 will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	GPIO_IN_EN_CLR	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

GPIO03_INT_LATCHED_STS | 0x0000C218

Type: Read

Reset: 0x00000000

Debug: Latched (Sticky) Interrupt. 1 indiecates that the interrutp has triggered. Once the latched bit is set it can only be cleared by writing the clear bit.

Bits	Name	Description
0 : 0	GPIO_IN_LATCHED_STS	No description provided for this bit field. 0: NO_INT_LATCHED 1: INTERRUPT_LATCHED

GPIO03_INT_PENDING_STS | 0x0000C219

Type: Read

Reset: 0x00000000

Debug: Pending is set if interrupt has been sent but not cleared.

Bits	Name	Description
0 : 0	GPIO_IN_PENDING_STS	No description provided for this bit field. 0: NO_INT_PENDING 1: INTERRUPT_PENDING

GPIO03_INT_MID_SEL | 0x0000C21A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: MID0 1: MID1 2: MID2 3: MID3

GPIO03_INT_PRIORITY | 0x0000C21B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: SR 1: A

GPIO03_MODE_CTL | 0x0000C240

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 0	MODE	GPIO Mode: 0: Digital Input 1: Digital Output 2: Digital Input and Output 3: Analog Pass Through 0: DIG_IN 1: DIG_OUT 2: DIG_INOUT 3: ANA_PASS_THRU

GPIO03_DIG_VIN_CTL | 0x0000C241

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
2 : 0	VOLTAGE_SEL	Select Voltage source: 000 = VIN0 (refer to the objective spec.) 001 = VIN1 (refer to the objective spec.) 010 = Reserved 011 = Reserved 100 = Reserved 101 = Reserved 110 = Reserved 111 = Reserved (refer to the objective spec for the definition of VINx) (Voltage supply is for GPIO_MV only, GPIO_LV only has one voltage supply) 0: VIN0 1: VIN1 2: RESERVED2 3: RESERVED3 4: RESERVED4 5: RESERVED5 6: RESERVED6 7: RESERVED7

GPIO03_DIG_PULL_CTL | 0x0000C242

Type: Read/Write

Reset: 0x00000004

No description provided for this register.

Bits	Name	Description
2 : 0	PULLUP_SEL	Current source pulls: - 000 = pull-up 30uA constant - 001 = pull-up 1.5uA - 010 = pull-up 31.5uA constant - 011 = pull-up 1.5uA + 30uA boost - 100 = pull-down 10uA - 101 = no pull - 110 = Reserved - 111 = Reserved Pull up or pull down can only be enabled when master PU/PD is enabled. Pull is disabled when master enable is 0 regardless of this register's setting (see GPIO MDOS for more information of pull enable control) 0: PULLUP_30UA 1: PULLUP_1P5UA 2: PULLUP_31P5UA 3: PULLUP_1P5UA_30UA_BOOST 4: PULLDOWN_10UA 5: NO_PULL 6: RESERVED6 7: RESERVED7

GPIO03_DIG_OUT_DRV_CTL | 0x0000C245

Type: Read/Write

Reset: 0x00000001

No description provided for this register.

Bits	Name	Description
1 : 0	OUTPUT_DRV_SEL	Digital output buffer drive strength select: 00 = Reserved 01 = Low strength 10 = Medium strength 11 = High strength 0: RESERVED 1: LOW_STR 2: MED_STR 3: HIGH_STR

Bits	Name	Description
5 : 4	OUTPUT_TYPE	Digital output buffer type configuration 00= CMOS (drive high and low) 01= Open drain NMOS (only drive low, i.e. I2C, PMOS disabled) 10= Open drain PMOS (only drive high, NMOS disabled) 11= No Drive. Both PMOS and NMOS are disabled 0: CMOS 1: OPEN_DRAIN_NMOS 2: OPEN_DRAIN_PMOS 3: NO_DRIVE

GPIO03_EN_CTL | 0x0000C246

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
7 : 7	PERPH_EN	GPIO Master Enable 0 = puts GPIO_PAD at high Z and disables the block 1 = GPIO is enabled 0: GPIO_DISABLED 1: GPIO_ENABLED

Module Name: GPIO04

Base Address: 0x0000C300

Register Quick Reference:

Register	Offset	Type
GPIO04_REVISION1	0x0000C300	Read
GPIO04_REVISION2	0x0000C301	Read
GPIO04_REVISION3	0x0000C302	Read
GPIO04_REVISION4	0x0000C303	Read
GPIO04_PERPH_TYPE	0x0000C304	Read
GPIO04_PERPH_SUBTYPE	0x0000C305	Read
GPIO04_STATUS1	0x0000C308	Read
GPIO04_INT_RT_STS	0x0000C310	Read
GPIO04_INT_SET_TYPE	0x0000C311	Read/Write
GPIO04_INT_POLARITY_HIGH	0x0000C312	Read/Write

Register	Offset	Type
GPIO04_INT_POLARITY_LOW	0x0000C313	Read/Write
GPIO04_INT_LATCHED_CLR	0x0000C314	Write
GPIO04_INT_EN_SET	0x0000C315	Read/Write
GPIO04_INT_EN_CLR	0x0000C316	Read/Write
GPIO04_INT_LATCHED_STS	0x0000C318	Read
GPIO04_INT_PENDING_STS	0x0000C319	Read
GPIO04_INT_MID_SEL	0x0000C31A	Read/Write
GPIO04_INT_PRIORITY	0x0000C31B	Read/Write
GPIO04_MODE_CTL	0x0000C340	Read/Write
GPIO04_DIG_VIN_CTL	0x0000C341	Read/Write
GPIO04_DIG_PULL_CTL	0x0000C342	Read/Write
GPIO04_DIG_OUT_DRV_CTL	0x0000C345	Read/Write
GPIO04_EN_CTL	0x0000C346	Read/Write

GPIO04_REVISION1 | 0x0000C300

Type: Read

Reset: 0x00000002

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

GPIO04_REVISION2 | 0x0000C301

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

GPIO04_REVISION3 | 0x0000C302

Type: Read

Reset: 0x00000000

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

GPIO04_REVISION4 | 0x0000C303

Type: Read

Reset: 0x00000002

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

GPIO04_PERPH_TYPE | 0x0000C304

Type: Read

Reset: 0x00000010

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	No description provided for this bit field. 16: GPIO

GPIO04_PERPH_SUBTYPE | 0x0000C305

Type: Read

Reset: 0x00000010

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	No description provided for this bit field. 16: GPIO_LV 17: GPIO_MV

GPIO04_STATUS1 | 0x0000C308

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
0 : 0	GPIO_VAL	Value read by the input buffer, if enabled 0: GPIO_INPUT_LOW 1: GPIO_INPUT_HIGH
7 : 7	GPIO_OK	No description provided for this bit field. 0: GPIO_DISABLED 1: GPIO_ENABLED

GPIO04_INT_RT_STS | 0x0000C310

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	GPIO_IN_STS	No description provided for this bit field. 0: INT_RT_STATUS_LOW 1: INT_RT_STATUS_HIGH

GPIO04_INT_SET_TYPE | 0x0000C311

Type: Read/Write

Reset: 0x00000000

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	GPIO_IN_TYPE	No description provided for this bit field. 0: LEVEL 1: EDGE

GPIO04_INT_POLARITY_HIGH | 0x0000C312

Type: Read/Write

Reset: 0x00000000

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	GPIO_IN_HIGH	No description provided for this bit field. 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

GPIO04_INT_POLARITY_LOW | 0x0000C313

Type: Read/Write

Reset: 0x00000000

1 = Interrupt will trigger on a level low (falling edge) event, 0 = level low triggering is disabled

Bits	Name	Description
0 : 0	GPIO_IN_LOW	No description provided for this bit field. 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

GPIO04_INT_LATCHED_CLR | 0x0000C314

Type: Write

Reset: 0x00000000

writing a 1 to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits

Bits	Name	Description
0 : 0	GPIO_IN_LATCHED_CLR	No description provided for this bit field.

GPIO04_INT_EN_SET | 0x0000C315

Type: Read/Write

Reset: 0x00000000

writing 0 to this register has no effect. Writing a 1 will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	GPIO_IN_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

GPIO04_INT_EN_CLR | 0x0000C316

Type: Read/Write

Reset: 0x00000000

writing 0 to this register has no effect. Writing a 1 will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	GPIO_IN_EN_CLR	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

GPIO04_INT_LATCHED_STS | 0x0000C318

Type: Read

Reset: 0x00000000

Debug: Latched (Sticky) Interrupt. 1 indiecates that the interrutp has triggered. Once the latched bit is set it can only be cleared by writing the clear bit.

Bits	Name	Description
0 : 0	GPIO_IN_LATCHED_STS	No description provided for this bit field. 0: NO_INT_LATCHED 1: INTERRUPT_LATCHED

GPIO04_INT_PENDING_STS | 0x0000C319

Type: Read

Reset: 0x00000000

Debug: Pending is set if interrupt has been sent but not cleared.

Bits	Name	Description
0 : 0	GPIO_IN_PENDING_STS	No description provided for this bit field. 0: NO_INT_PENDING 1: INTERRUPT_PENDING

GPIO04_INT_MID_SEL | 0x0000C31A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: MID0 1: MID1 2: MID2 3: MID3

GPIO04_INT_PRIORITY | 0x0000C31B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: SR 1: A

GPIO04_MODE_CTL | 0x0000C340

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 0	MODE	GPIO Mode: 0: Digital Input 1: Digital Output 2: Digital Input and Output 3: Analog Pass Through 0: DIG_IN 1: DIG_OUT 2: DIG_INOUT 3: ANA_PASS_THRU

GPIO04_DIG_VIN_CTL | 0x0000C341

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
2 : 0	VOLTAGE_SEL	Select Voltage source: 000 = VIN0 (refer to the objective spec.) 001 = VIN1 (refer to the objective spec.) 010 = Reserved 011 = Reserved 100 = Reserved 101 = Reserved 110 = Reserved 111 = Reserved (refer to the objective spec for the definition of VINx) (Voltage supply is for GPIO_MV only, GPIO_LV only has one voltage supply) 0: VIN0 1: VIN1 2: RESERVED2 3: RESERVED3 4: RESERVED4 5: RESERVED5 6: RESERVED6 7: RESERVED7

GPIO04_DIG_PULL_CTL | 0x0000C342

Type: Read/Write

Reset: 0x00000004

No description provided for this register.

Bits	Name	Description
2 : 0	PULLUP_SEL	Current source pulls: - 000 = pull-up 30uA constant - 001 = pull-up 1.5uA - 010 = pull-up 31.5uA constant - 011 = pull-up 1.5uA + 30uA boost - 100 = pull-down 10uA - 101 = no pull - 110 = Reserved - 111 = Reserved Pull up or pull down can only be enabled when master PU/PD is enabled. Pull is disabled when master enable is 0 regardless of this register's setting (see GPIO MDOS for more information of pull enable control) 0: PULLUP_30UA 1: PULLUP_1P5UA 2: PULLUP_31P5UA 3: PULLUP_1P5UA_30UA_BOOST 4: PULLDOWN_10UA 5: NO_PULL 6: RESERVED6 7: RESERVED7

GPIO04_DIG_OUT_DRV_CTL | 0x0000C345

Type: Read/Write

Reset: 0x00000001

No description provided for this register.

Bits	Name	Description
1 : 0	OUTPUT_DRV_SEL	Digital output buffer drive strength select: 00 = Reserved 01 = Low strength 10 = Medium strength 11 = High strength 0: RESERVED 1: LOW_STR 2: MED_STR 3: HIGH_STR

Bits	Name	Description
5 : 4	OUTPUT_TYPE	Digital output buffer type configuration 00= CMOS (drive high and low) 01= Open drain NMOS (only drive low, i.e. I2C, PMOS disabled) 10= Open drain PMOS (only drive high, NMOS disabled) 11= No Drive. Both PMOS and NMOS are disabled 0: CMOS 1: OPEN_DRAIN_NMOS 2: OPEN_DRAIN_PMOS 3: NO_DRIVE

GPIO04_EN_CTL | 0x0000C346

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
7 : 7	PERPH_EN	GPIO Master Enable 0 = puts GPIO_PAD at high Z and disables the block 1 = GPIO is enabled 0: GPIO_DISABLED 1: GPIO_ENABLED

Module Name: GPIO05

Base Address: 0x0000C400

Register Quick Reference:

Register	Offset	Type
GPIO05_REVISION1	0x0000C400	Read
GPIO05_REVISION2	0x0000C401	Read
GPIO05_REVISION3	0x0000C402	Read
GPIO05_REVISION4	0x0000C403	Read
GPIO05_PERPH_TYPE	0x0000C404	Read
GPIO05_PERPH_SUBTYPE	0x0000C405	Read
GPIO05_STATUS1	0x0000C408	Read
GPIO05_INT_RT_STS	0x0000C410	Read
GPIO05_INT_SET_TYPE	0x0000C411	Read/Write
GPIO05_INT_POLARITY_HIGH	0x0000C412	Read/Write

Register	Offset	Type
GPIO05_INT_POLARITY_LOW	0x0000C413	Read/Write
GPIO05_INT_LATCHED_CLR	0x0000C414	Write
GPIO05_INT_EN_SET	0x0000C415	Read/Write
GPIO05_INT_EN_CLR	0x0000C416	Read/Write
GPIO05_INT_LATCHED_STS	0x0000C418	Read
GPIO05_INT_PENDING_STS	0x0000C419	Read
GPIO05_INT_MID_SEL	0x0000C41A	Read/Write
GPIO05_INT_PRIORITY	0x0000C41B	Read/Write
GPIO05_MODE_CTL	0x0000C440	Read/Write
GPIO05_DIG_VIN_CTL	0x0000C441	Read/Write
GPIO05_DIG_PULL_CTL	0x0000C442	Read/Write
GPIO05_DIG_OUT_DRV_CTL	0x0000C445	Read/Write
GPIO05_EN_CTL	0x0000C446	Read/Write

GPIO05_REVISION1 | 0x0000C400

Type: Read

Reset: 0x00000002

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

GPIO05_REVISION2 | 0x0000C401

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

GPIO05_REVISION3 | 0x0000C402

Type: Read

Reset: 0x00000000

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

GPIO05_REVISION4 | 0x0000C403

Type: Read

Reset: 0x00000002

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

GPIO05_PERPH_TYPE | 0x0000C404

Type: Read

Reset: 0x00000010

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	No description provided for this bit field. 16: GPIO

GPIO05_PERPH_SUBTYPE | 0x0000C405

Type: Read

Reset: 0x00000010

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	No description provided for this bit field. 16: GPIO_LV 17: GPIO_MV

GPIO05_STATUS1 | 0x0000C408

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
0 : 0	GPIO_VAL	Value read by the input buffer, if enabled 0: GPIO_INPUT_LOW 1: GPIO_INPUT_HIGH
7 : 7	GPIO_OK	No description provided for this bit field. 0: GPIO_DISABLED 1: GPIO_ENABLED

GPIO05_INT_RT_STS | 0x0000C410

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	GPIO_IN_STS	No description provided for this bit field. 0: INT_RT_STATUS_LOW 1: INT_RT_STATUS_HIGH

GPIO05_INT_SET_TYPE | 0x0000C411

Type: Read/Write

Reset: 0x00000000

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	GPIO_IN_TYPE	No description provided for this bit field. 0: LEVEL 1: EDGE

GPIO05_INT_POLARITY_HIGH | 0x0000C412

Type: Read/Write

Reset: 0x00000000

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	GPIO_IN_HIGH	No description provided for this bit field. 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

GPIO05_INT_POLARITY_LOW | 0x0000C413

Type: Read/Write

Reset: 0x00000000

1 = Interrupt will trigger on a level low (falling edge) event, 0 = level low triggering is disabled

Bits	Name	Description
0 : 0	GPIO_IN_LOW	No description provided for this bit field. 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

GPIO05_INT_LATCHED_CLR | 0x0000C414

Type: Write

Reset: 0x00000000

writing a 1 to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits

Bits	Name	Description
0 : 0	GPIO_IN_LATCHED_CLR	No description provided for this bit field.

GPIO05_INT_EN_SET | 0x0000C415

Type: Read/Write

Reset: 0x00000000

writing 0 to this register has no effect. Writing a 1 will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	GPIO_IN_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

GPIO05_INT_EN_CLR | 0x0000C416

Type: Read/Write

Reset: 0x00000000

writing 0 to this register has no effect. Writing a 1 will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	GPIO_IN_EN_CLR	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

GPIO05_INT_LATCHED_STS | 0x0000C418

Type: Read

Reset: 0x00000000

Debug: Latched (Sticky) Interrupt. 1 indiecates that the interrutp has triggered. Once the latched bit is set it can only be cleared by writing the clear bit.

Bits	Name	Description
0 : 0	GPIO_IN_LATCHED_STS	No description provided for this bit field. 0: NO_INT_LATCHED 1: INTERRUPT_LATCHED

GPIO05_INT_PENDING_STS | 0x0000C419

Type: Read

Reset: 0x00000000

Debug: Pending is set if interrupt has been sent but not cleared.

Bits	Name	Description
0 : 0	GPIO_IN_PENDING_STS	No description provided for this bit field. 0: NO_INT_PENDING 1: INTERRUPT_PENDING

GPIO05_INT_MID_SEL | 0x0000C41A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: MID0 1: MID1 2: MID2 3: MID3

GPIO05_INT_PRIORITY | 0x0000C41B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: SR 1: A

GPIO05_MODE_CTL | 0x0000C440

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 0	MODE	GPIO Mode: 0: Digital Input 1: Digital Output 2: Digital Input and Output 3: Analog Pass Through 0: DIG_IN 1: DIG_OUT 2: DIG_INOUT 3: ANA_PASS_THRU

GPIO05_DIG_VIN_CTL | 0x0000C441

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
2 : 0	VOLTAGE_SEL	Select Voltage source: 000 = VIN0 (refer to the objective spec.) 001 = VIN1 (refer to the objective spec.) 010 = Reserved 011 = Reserved 100 = Reserved 101 = Reserved 110 = Reserved 111 = Reserved (refer to the objective spec for the definition of VINx) (Voltage supply is for GPIO_MV only, GPIO_LV only has one voltage supply) 0: VIN0 1: VIN1 2: RESERVED2 3: RESERVED3 4: RESERVED4 5: RESERVED5 6: RESERVED6 7: RESERVED7

GPIO05_DIG_PULL_CTL | 0x0000C442

Type: Read/Write

Reset: 0x00000004

No description provided for this register.

Bits	Name	Description
2 : 0	PULLUP_SEL	Current source pulls: - 000 = pull-up 30uA constant - 001 = pull-up 1.5uA - 010 = pull-up 31.5uA constant - 011 = pull-up 1.5uA + 30uA boost - 100 = pull-down 10uA - 101 = no pull - 110 = Reserved - 111 = Reserved Pull up or pull down can only be enabled when master PU/PD is enabled. Pull is disabled when master enable is 0 regardless of this register's setting (see GPIO MDOS for more information of pull enable control) 0: PULLUP_30UA 1: PULLUP_1P5UA 2: PULLUP_31P5UA 3: PULLUP_1P5UA_30UA_BOOST 4: PULLDOWN_10UA 5: NO_PULL 6: RESERVED6 7: RESERVED7

GPIO05_DIG_OUT_DRV_CTL | 0x0000C445

Type: Read/Write

Reset: 0x00000001

No description provided for this register.

Bits	Name	Description
1 : 0	OUTPUT_DRV_SEL	Digital output buffer drive strength select: 00 = Reserved 01 = Low strength 10 = Medium strength 11 = High strength 0: RESERVED 1: LOW_STR 2: MED_STR 3: HIGH_STR

Bits	Name	Description
5 : 4	OUTPUT_TYPE	Digital output buffer type configuration 00= CMOS (drive high and low) 01= Open drain NMOS (only drive low, i.e. I2C, PMOS disabled) 10= Open drain PMOS (only drive high, NMOS disabled) 11= No Drive. Both PMOS and NMOS are disabled 0: CMOS 1: OPEN_DRAIN_NMOS 2: OPEN_DRAIN_PMOS 3: NO_DRIVE

GPIO05_EN_CTL | 0x0000C446

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
7 : 7	PERPH_EN	GPIO Master Enable 0 = puts GPIO_PAD at high Z and disables the block 1 = GPIO is enabled 0: GPIO_DISABLED 1: GPIO_ENABLED

Module Name: GPIO06

Base Address: 0x0000C500

Register Quick Reference:

Register	Offset	Type
GPIO06_REVISION1	0x0000C500	Read
GPIO06_REVISION2	0x0000C501	Read
GPIO06_REVISION3	0x0000C502	Read
GPIO06_REVISION4	0x0000C503	Read
GPIO06_PERPH_TYPE	0x0000C504	Read
GPIO06_PERPH_SUBTYPE	0x0000C505	Read
GPIO06_STATUS1	0x0000C508	Read
GPIO06_INT_RT_STS	0x0000C510	Read
GPIO06_INT_SET_TYPE	0x0000C511	Read/Write
GPIO06_INT_POLARITY_HIGH	0x0000C512	Read/Write

Register	Offset	Type
GPIO06_INT_POLARITY_LOW	0x0000C513	Read/Write
GPIO06_INT_LATCHED_CLR	0x0000C514	Write
GPIO06_INT_EN_SET	0x0000C515	Read/Write
GPIO06_INT_EN_CLR	0x0000C516	Read/Write
GPIO06_INT_LATCHED_STS	0x0000C518	Read
GPIO06_INT_PENDING_STS	0x0000C519	Read
GPIO06_INT_MID_SEL	0x0000C51A	Read/Write
GPIO06_INT_PRIORITY	0x0000C51B	Read/Write
GPIO06_MODE_CTL	0x0000C540	Read/Write
GPIO06_DIG_VIN_CTL	0x0000C541	Read/Write
GPIO06_DIG_PULL_CTL	0x0000C542	Read/Write
GPIO06_DIG_OUT_DRV_CTL	0x0000C545	Read/Write
GPIO06_EN_CTL	0x0000C546	Read/Write

GPIO06_REVISION1 | 0x0000C500

Type: Read

Reset: 0x00000002

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

GPIO06_REVISION2 | 0x0000C501

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

GPIO06_REVISION3 | 0x0000C502

Type: Read

Reset: 0x00000000

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

GPIO06_REVISION4 | 0x0000C503

Type: Read

Reset: 0x00000002

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

GPIO06_PERPH_TYPE | 0x0000C504

Type: Read

Reset: 0x00000010

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	No description provided for this bit field. 16: GPIO

GPIO06_PERPH_SUBTYPE | 0x0000C505

Type: Read

Reset: 0x00000011

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	No description provided for this bit field. 16: GPIO_LV 17: GPIO_MV

GPIO06_STATUS1 | 0x0000C508

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
0 : 0	GPIO_VAL	Value read by the input buffer, if enabled 0: GPIO_INPUT_LOW 1: GPIO_INPUT_HIGH
7 : 7	GPIO_OK	No description provided for this bit field. 0: GPIO_DISABLED 1: GPIO_ENABLED

GPIO06_INT_RT_STS | 0x0000C510

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	GPIO_IN_STS	No description provided for this bit field. 0: INT_RT_STATUS_LOW 1: INT_RT_STATUS_HIGH

GPIO06_INT_SET_TYPE | 0x0000C511

Type: Read/Write

Reset: 0x00000000

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	GPIO_IN_TYPE	No description provided for this bit field. 0: LEVEL 1: EDGE

GPIO06_INT_POLARITY_HIGH | 0x0000C512

Type: Read/Write

Reset: 0x00000000

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	GPIO_IN_HIGH	No description provided for this bit field. 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

GPIO06_INT_POLARITY_LOW | 0x0000C513

Type: Read/Write

Reset: 0x00000000

1 = Interrupt will trigger on a level low (falling edge) event, 0 = level low triggering is disabled

Bits	Name	Description
0 : 0	GPIO_IN_LOW	No description provided for this bit field. 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

GPIO06_INT_LATCHED_CLR | **0x0000C514**

Type: Write

Reset: 0x00000000

writing a 1 to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits

Bits	Name	Description
0 : 0	GPIO_IN_LATCHED_CLR	No description provided for this bit field.

GPIO06_INT_EN_SET | 0x0000C515

Type: Read/Write

Reset: 0x00000000

writing 0 to this register has no effect. Writing a 1 will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	GPIO_IN_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

GPIO06_INT_EN_CLR | 0x0000C516

Type: Read/Write

Reset: 0x00000000

writing 0 to this register has no effect. Writing a 1 will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	GPIO_IN_EN_CLR	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

GPIO06_INT_LATCHED_STS | 0x0000C518

Type: Read

Reset: 0x00000000

Debug: Latched (Sticky) Interrupt. 1 indiecates that the interrutp has triggered. Once the latched bit is set it can only be cleared by writing the clear bit.

Bits	Name	Description
0 : 0	GPIO_IN_LATCHED_STS	No description provided for this bit field. 0: NO_INT_LATCHED 1: INTERRUPT_LATCHED

GPIO06_INT_PENDING_STS | 0x0000C519

Type: Read

Reset: 0x00000000

Debug: Pending is set if interrupt has been sent but not cleared.

Bits	Name	Description
0 : 0	GPIO_IN_PENDING_STS	No description provided for this bit field. 0: NO_INT_PENDING 1: INTERRUPT_PENDING

GPIO06_INT_MID_SEL | 0x0000C51A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: MID0 1: MID1 2: MID2 3: MID3

GPIO06_INT_PRIORITY | 0x0000C51B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: SR 1: A

GPIO06_MODE_CTL | 0x0000C540

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 0	MODE	GPIO Mode: 0: Digital Input 1: Digital Output 2: Digital Input and Output 3: Analog Pass Through 0: DIG_IN 1: DIG_OUT 2: DIG_INOUT 3: ANA_PASS_THRU

GPIO06_DIG_VIN_CTL | 0x0000C541

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
2 : 0	VOLTAGE_SEL	Select Voltage source: 000 = VIN0 (refer to the objective spec.) 001 = VIN1 (refer to the objective spec.) 010 = Reserved 011 = Reserved 100 = Reserved 101 = Reserved 110 = Reserved 111 = Reserved (refer to the objective spec for the definition of VINx) (Voltage supply is for GPIO_MV only, GPIO_LV only has one voltage supply) 0: VIN0 1: VIN1 2: RESERVED2 3: RESERVED3 4: RESERVED4 5: RESERVED5 6: RESERVED6 7: RESERVED7

GPIO06_DIG_PULL_CTL | 0x0000C542

Type: Read/Write

Reset: 0x00000004

No description provided for this register.

Bits	Name	Description
2 : 0	PULLUP_SEL	Current source pulls: - 000 = pull-up 30uA constant - 001 = pull-up 1.5uA - 010 = pull-up 31.5uA constant - 011 = pull-up 1.5uA + 30uA boost - 100 = pull-down 10uA - 101 = no pull - 110 = Reserved - 111 = Reserved Pull up or pull down can only be enabled when master PU/PD is enabled. Pull is disabled when master enable is 0 regardless of this register's setting (see GPIO MDOS for more information of pull enable control) 0: PULLUP_30UA 1: PULLUP_1P5UA 2: PULLUP_31P5UA 3: PULLUP_1P5UA_30UA_BOOST 4: PULLDOWN_10UA 5: NO_PULL 6: RESERVED6 7: RESERVED7

GPIO06_DIG_OUT_DRV_CTL | 0x0000C545

Type: Read/Write

Reset: 0x00000001

No description provided for this register.

Bits	Name	Description
1 : 0	OUTPUT_DRV_SEL	Digital output buffer drive strength select: 00 = Reserved 01 = Low strength 10 = Medium strength 11 = High strength 0: RESERVED 1: LOW_STR 2: MED_STR 3: HIGH_STR

Bits	Name	Description
5 : 4	OUTPUT_TYPE	Digital output buffer type configuration 00= CMOS (drive high and low) 01= Open drain NMOS (only drive low, i.e. I2C, PMOS disabled) 10= Open drain PMOS (only drive high, NMOS disabled) 11= No Drive. Both PMOS and NMOS are disabled 0: CMOS 1: OPEN_DRAIN_NMOS 2: OPEN_DRAIN_PMOS 3: NO_DRIVE

GPIO06_EN_CTL | 0x0000C546

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
7 : 7	PERPH_EN	GPIO Master Enable 0 = puts GPIO_PAD at high Z and disables the block 1 = GPIO is enabled 0: GPIO_DISABLED 1: GPIO_ENABLED

Module Name: GPIO07

Base Address: 0x0000C600

Register Quick Reference:

Register	Offset	Type
GPIO07_REVISION1	0x0000C600	Read
GPIO07_REVISION2	0x0000C601	Read
GPIO07_REVISION3	0x0000C602	Read
GPIO07_REVISION4	0x0000C603	Read
GPIO07_PERPH_TYPE	0x0000C604	Read
GPIO07_PERPH_SUBTYPE	0x0000C605	Read
GPIO07_STATUS1	0x0000C608	Read
GPIO07_INT_RT_STS	0x0000C610	Read
GPIO07_INT_SET_TYPE	0x0000C611	Read/Write
GPIO07_INT_POLARITY_HIGH	0x0000C612	Read/Write

Register	Offset	Type
GPIO07_INT_POLARITY_LOW	0x0000C613	Read/Write
GPIO07_INT_LATCHED_CLR	0x0000C614	Write
GPIO07_INT_EN_SET	0x0000C615	Read/Write
GPIO07_INT_EN_CLR	0x0000C616	Read/Write
GPIO07_INT_LATCHED_STS	0x0000C618	Read
GPIO07_INT_PENDING_STS	0x0000C619	Read
GPIO07_INT_MID_SEL	0x0000C61A	Read/Write
GPIO07_INT_PRIORITY	0x0000C61B	Read/Write
GPIO07_MODE_CTL	0x0000C640	Read/Write
GPIO07_DIG_VIN_CTL	0x0000C641	Read/Write
GPIO07_DIG_PULL_CTL	0x0000C642	Read/Write
GPIO07_DIG_OUT_DRV_CTL	0x0000C645	Read/Write
GPIO07_EN_CTL	0x0000C646	Read/Write

GPIO07_REVISION1 | 0x0000C600

Type: Read

Reset: 0x00000002

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

GPIO07_REVISION2 | 0x0000C601

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

GPIO07_REVISION3 | 0x0000C602

Type: Read

Reset: 0x00000000

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

GPIO07_REVISION4 | 0x0000C603

Type: Read

Reset: 0x00000002

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

GPIO07_PERPH_TYPE | 0x0000C604

Type: Read

Reset: 0x00000010

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	No description provided for this bit field. 16: GPIO

GPIO07_PERPH_SUBTYPE | 0x0000C605

Type: Read

Reset: 0x00000011

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	No description provided for this bit field. 16: GPIO_LV 17: GPIO_MV

GPIO07_STATUS1 | 0x0000C608

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
0 : 0	GPIO_VAL	Value read by the input buffer, if enabled 0: GPIO_INPUT_LOW 1: GPIO_INPUT_HIGH
7 : 7	GPIO_OK	No description provided for this bit field. 0: GPIO_DISABLED 1: GPIO_ENABLED

GPIO07_INT_RT_STS | 0x0000C610

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	GPIO_IN_STS	No description provided for this bit field. 0: INT_RT_STATUS_LOW 1: INT_RT_STATUS_HIGH

GPIO07_INT_SET_TYPE | 0x0000C611

Type: Read/Write

Reset: 0x00000000

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	GPIO_IN_TYPE	No description provided for this bit field. 0: LEVEL 1: EDGE

GPIO07_INT_POLARITY_HIGH | 0x0000C612

Type: Read/Write

Reset: 0x00000000

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	GPIO_IN_HIGH	No description provided for this bit field. 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

GPIO07_INT_POLARITY_LOW | 0x0000C613

Type: Read/Write

Reset: 0x00000000

1 = Interrupt will trigger on a level low (falling edge) event, 0 = level low triggering is disabled

Bits	Name	Description
0 : 0	GPIO_IN_LOW	No description provided for this bit field. 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

GPIO07_INT_LATCHED_CLR | 0x0000C614

Type: Write

Reset: 0x00000000

writing a 1 to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits

Bits	Name	Description
0 : 0	GPIO_IN_LATCHED_CLR	No description provided for this bit field.

GPIO07_INT_EN_SET | 0x0000C615

Type: Read/Write

Reset: 0x00000000

writing 0 to this register has no effect. Writing a 1 will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	GPIO_IN_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

GPIO07_INT_EN_CLR | 0x0000C616

Type: Read/Write

Reset: 0x00000000

writing 0 to this register has no effect. Writing a 1 will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	GPIO_IN_EN_CLR	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

GPIO07_INT_LATCHED_STS | 0x0000C618

Type: Read

Reset: 0x00000000

Debug: Latched (Sticky) Interrupt. 1 indiecates that the interrutp has triggered. Once the latched bit is set it can only be cleared by writing the clear bit.

Bits	Name	Description
0 : 0	GPIO_IN_LATCHED_STS	No description provided for this bit field. 0: NO_INT_LATCHED 1: INTERRUPT_LATCHED

GPIO07_INT_PENDING_STS | 0x0000C619

Type: Read

Reset: 0x00000000

Debug: Pending is set if interrupt has been sent but not cleared.

Bits	Name	Description
0 : 0	GPIO_IN_PENDING_STS	No description provided for this bit field. 0: NO_INT_PENDING 1: INTERRUPT_PENDING

GPIO07_INT_MID_SEL | 0x0000C61A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: MID0 1: MID1 2: MID2 3: MID3

GPIO07_INT_PRIORITY | 0x0000C61B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: SR 1: A

GPIO07_MODE_CTL | 0x0000C640

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 0	MODE	GPIO Mode: 0: Digital Input 1: Digital Output 2: Digital Input and Output 3: Analog Pass Through 0: DIG_IN 1: DIG_OUT 2: DIG_INOUT 3: ANA_PASS_THRU

GPIO07_DIG_VIN_CTL | 0x0000C641

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
2 : 0	VOLTAGE_SEL	Select Voltage source: 000 = VIN0 (refer to the objective spec.) 001 = VIN1 (refer to the objective spec.) 010 = Reserved 011 = Reserved 100 = Reserved 101 = Reserved 110 = Reserved 111 = Reserved (refer to the objective spec for the definition of VINx) (Voltage supply is for GPIO_MV only, GPIO_LV only has one voltage supply) 0: VIN0 1: VIN1 2: RESERVED2 3: RESERVED3 4: RESERVED4 5: RESERVED5 6: RESERVED6 7: RESERVED7

GPIO07_DIG_PULL_CTL | 0x0000C642

Type: Read/Write

Reset: 0x00000004

No description provided for this register.

Bits	Name	Description
2 : 0	PULLUP_SEL	Current source pulls: - 000 = pull-up 30uA constant - 001 = pull-up 1.5uA - 010 = pull-up 31.5uA constant - 011 = pull-up 1.5uA + 30uA boost - 100 = pull-down 10uA - 101 = no pull - 110 = Reserved - 111 = Reserved Pull up or pull down can only be enabled when master PU/PD is enabled. Pull is disabled when master enable is 0 regardless of this register's setting (see GPIO MDOS for more information of pull enable control) 0: PULLUP_30UA 1: PULLUP_1P5UA 2: PULLUP_31P5UA 3: PULLUP_1P5UA_30UA_BOOST 4: PULLDOWN_10UA 5: NO_PULL 6: RESERVED6 7: RESERVED7

GPIO07_DIG_OUT_DRV_CTL | 0x0000C645

Type: Read/Write

Reset: 0x00000001

No description provided for this register.

Bits	Name	Description
1 : 0	OUTPUT_DRV_SEL	Digital output buffer drive strength select: 00 = Reserved 01 = Low strength 10 = Medium strength 11 = High strength 0: RESERVED 1: LOW_STR 2: MED_STR 3: HIGH_STR

Bits	Name	Description
5 : 4	OUTPUT_TYPE	Digital output buffer type configuration 00= CMOS (drive high and low) 01= Open drain NMOS (only drive low, i.e. I2C, PMOS disabled) 10= Open drain PMOS (only drive high, NMOS disabled) 11= No Drive. Both PMOS and NMOS are disabled 0: CMOS 1: OPEN_DRAIN_NMOS 2: OPEN_DRAIN_PMOS 3: NO_DRIVE

GPIO07_EN_CTL | 0x0000C646

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
7 : 7	PERPH_EN	GPIO Master Enable 0 = puts GPIO_PAD at high Z and disables the block 1 = GPIO is enabled 0: GPIO_DISABLED 1: GPIO_ENABLED

Module Name: GPIO08

Base Address: 0x0000C700

Register Quick Reference:

Register	Offset	Type
GPIO08_REVISION1	0x0000C700	Read
GPIO08_REVISION2	0x0000C701	Read
GPIO08_REVISION3	0x0000C702	Read
GPIO08_REVISION4	0x0000C703	Read
GPIO08_PERPH_TYPE	0x0000C704	Read
GPIO08_PERPH_SUBTYPE	0x0000C705	Read
GPIO08_STATUS1	0x0000C708	Read
GPIO08_INT_RT_STS	0x0000C710	Read
GPIO08_INT_SET_TYPE	0x0000C711	Read/Write
GPIO08_INT_POLARITY_HIGH	0x0000C712	Read/Write

Register	Offset	Type
GPIO08_INT_POLARITY_LOW	0x0000C713	Read/Write
GPIO08_INT_LATCHED_CLR	0x0000C714	Write
GPIO08_INT_EN_SET	0x0000C715	Read/Write
GPIO08_INT_EN_CLR	0x0000C716	Read/Write
GPIO08_INT_LATCHED_STS	0x0000C718	Read
GPIO08_INT_PENDING_STS	0x0000C719	Read
GPIO08_INT_MID_SEL	0x0000C71A	Read/Write
GPIO08_INT_PRIORITY	0x0000C71B	Read/Write
GPIO08_MODE_CTL	0x0000C740	Read/Write
GPIO08_DIG_VIN_CTL	0x0000C741	Read/Write
GPIO08_DIG_PULL_CTL	0x0000C742	Read/Write
GPIO08_DIG_OUT_DRV_CTL	0x0000C745	Read/Write
GPIO08_EN_CTL	0x0000C746	Read/Write

GPIO08_REVISION1 | 0x0000C700

Type: Read

Reset: 0x00000002

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

GPIO08_REVISION2 | 0x0000C701

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

GPIO08_REVISION3 | 0x0000C702

Type: Read

Reset: 0x00000000

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

GPIO08_REVISION4 | 0x0000C703

Type: Read

Reset: 0x00000002

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

GPIO08_PERPH_TYPE | 0x0000C704

Type: Read

Reset: 0x00000010

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	No description provided for this bit field. 16: GPIO

GPIO08_PERPH_SUBTYPE | 0x0000C705

Type: Read

Reset: 0x00000011

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	No description provided for this bit field. 16: GPIO_LV 17: GPIO_MV

GPIO08_STATUS1 | 0x0000C708

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
0 : 0	GPIO_VAL	Value read by the input buffer, if enabled 0: GPIO_INPUT_LOW 1: GPIO_INPUT_HIGH
7 : 7	GPIO_OK	No description provided for this bit field. 0: GPIO_DISABLED 1: GPIO_ENABLED

GPIO08_INT_RT_STS | 0x0000C710

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	GPIO_IN_STS	No description provided for this bit field. 0: INT_RT_STATUS_LOW 1: INT_RT_STATUS_HIGH

GPIO08_INT_SET_TYPE | 0x0000C711

Type: Read/Write

Reset: 0x00000000

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	GPIO_IN_TYPE	No description provided for this bit field. 0: LEVEL 1: EDGE

GPIO08_INT_POLARITY_HIGH | 0x0000C712

Type: Read/Write

Reset: 0x00000000

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	GPIO_IN_HIGH	No description provided for this bit field. 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

GPIO08_INT_POLARITY_LOW | 0x0000C713

Type: Read/Write

Reset: 0x00000000

1 = Interrupt will trigger on a level low (falling edge) event, 0 = level low triggering is disabled

Bits	Name	Description
0 : 0	GPIO_IN_LOW	No description provided for this bit field. 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

GPIO08_INT_LATCHED_CLR | **0x0000C714**

Type: Write

Reset: 0x00000000

writing a 1 to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits

Bits	Name	Description
0 : 0	GPIO_IN_LATCHED_CLR	No description provided for this bit field.

GPIO08_INT_EN_SET | 0x0000C715

Type: Read/Write

Reset: 0x00000000

writing 0 to this register has no effect. Writing a 1 will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	GPIO_IN_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

GPIO08_INT_EN_CLR | 0x0000C716

Type: Read/Write

Reset: 0x00000000

writing 0 to this register has no effect. Writing a 1 will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	GPIO_IN_EN_CLR	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

GPIO08_INT_LATCHED_STS | 0x0000C718

Type: Read

Reset: 0x00000000

Debug: Latched (Sticky) Interrupt. 1 indiecates that the interrutp has triggered. Once the latched bit is set it can only be cleared by writing the clear bit.

Bits	Name	Description
0 : 0	GPIO_IN_LATCHED_STS	No description provided for this bit field. 0: NO_INT_LATCHED 1: INTERRUPT_LATCHED

GPIO08_INT_PENDING_STS | 0x0000C719

Type: Read

Reset: 0x00000000

Debug: Pending is set if interrupt has been sent but not cleared.

Bits	Name	Description
0 : 0	GPIO_IN_PENDING_STS	No description provided for this bit field. 0: NO_INT_PENDING 1: INTERRUPT_PENDING

GPIO08_INT_MID_SEL | 0x0000C71A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: MID0 1: MID1 2: MID2 3: MID3

GPIO08_INT_PRIORITY | 0x0000C71B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: SR 1: A

GPIO08_MODE_CTL | 0x0000C740

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
1 : 0	MODE	GPIO Mode: 0: Digital Input 1: Digital Output 2: Digital Input and Output 3: Analog Pass Through 0: DIG_IN 1: DIG_OUT 2: DIG_INOUT 3: ANA_PASS_THRU

GPIO08_DIG_VIN_CTL | 0x0000C741

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
2 : 0	VOLTAGE_SEL	Select Voltage source: 000 = VIN0 (refer to the objective spec.) 001 = VIN1 (refer to the objective spec.) 010 = Reserved 011 = Reserved 100 = Reserved 101 = Reserved 110 = Reserved 111 = Reserved (refer to the objective spec for the definition of VINx) (Voltage supply is for GPIO_MV only, GPIO_LV only has one voltage supply) 0: VIN0 1: VIN1 2: RESERVED2 3: RESERVED3 4: RESERVED4 5: RESERVED5 6: RESERVED6 7: RESERVED7

GPIO08_DIG_PULL_CTL | 0x0000C742

Type: Read/Write

Reset: 0x00000004

No description provided for this register.

Bits	Name	Description
2 : 0	PULLUP_SEL	Current source pulls: - 000 = pull-up 30uA constant - 001 = pull-up 1.5uA - 010 = pull-up 31.5uA constant - 011 = pull-up 1.5uA + 30uA boost - 100 = pull-down 10uA - 101 = no pull - 110 = Reserved - 111 = Reserved Pull up or pull down can only be enabled when master PU/PD is enabled. Pull is disabled when master enable is 0 regardless of this register's setting (see GPIO MDOS for more information of pull enable control) 0: PULLUP_30UA 1: PULLUP_1P5UA 2: PULLUP_31P5UA 3: PULLUP_1P5UA_30UA_BOOST 4: PULLDOWN_10UA 5: NO_PULL 6: RESERVED6 7: RESERVED7

GPIO08_DIG_OUT_DRV_CTL | 0x0000C745

Type: Read/Write

Reset: 0x00000001

No description provided for this register.

Bits	Name	Description
1 : 0	OUTPUT_DRV_SEL	Digital output buffer drive strength select: 00 = Reserved 01 = Low strength 10 = Medium strength 11 = High strength 0: RESERVED 1: LOW_STR 2: MED_STR 3: HIGH_STR

Bits	Name	Description
5 : 4	OUTPUT_TYPE	Digital output buffer type configuration 00= CMOS (drive high and low) 01= Open drain NMOS (only drive low, i.e. I2C, PMOS disabled) 10= Open drain PMOS (only drive high, NMOS disabled) 11= No Drive. Both PMOS and NMOS are disabled 0: CMOS 1: OPEN_DRAIN_NMOS 2: OPEN_DRAIN_PMOS 3: NO_DRIVE

GPIO08_EN_CTL | 0x0000C746

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
7 : 7	PERPH_EN	GPIO Master Enable 0 = puts GPIO_PAD at high Z and disables the block 1 = GPIO is enabled 0: GPIO_DISABLED 1: GPIO_ENABLED

Module Name: S1_CTRL

Base Address: 0x00011400

Register Quick Reference:

Register	Offset	Type
S1_CTRL_REVISION1	0x00011400	Read
S1_CTRL_REVISION2	0x00011401	Read
S1_CTRL_REVISION3	0x00011402	Read
S1_CTRL_REVISION4	0x00011403	Read
S1_CTRL_PERPH_TYPE	0x00011404	Read
S1_CTRL_PERPH_SUBTYPE	0x00011405	Read
S1_CTRL_STATUS1	0x00011408	Read
S1_CTRL_INT_RT_STS	0x00011410	Read
S1_CTRL_INT_SET_TYPE	0x00011411	Read
S1_CTRL_INT_POLARITY_HIGH	0x00011412	Read

Register	Offset	Type
S1_CTRL_INT_POLARITY_LOW	0x00011413	Read/Write
S1_CTRL_INT_LATCHED_CLR	0x00011414	Write
S1_CTRL_INT_EN_SET	0x00011415	Read/Write
S1_CTRL_INT_EN_CLR	0x00011416	Read/Write
S1_CTRL_INT_MID_SEL	0x0001141A	Read/Write
S1_CTRL_INT_PRIORITY	0x0001141B	Read/Write
S1_CTRL_INT_SELF_CLEARING	0x0001141D	Read
S1_CTRL_ULS_VSET_LB	0x00011439	Read/Write
S1_CTRL_ULS_VSET_UB	0x0001143A	Read/Write
S1_CTRL_VSET_LB	0x00011440	Read/Write
S1_CTRL_VSET_UB	0x00011441	Read/Write
S1_CTRL_VSET_VALID_LB	0x00011442	Read
S1_CTRL_VSET_VALID_UB	0x00011443	Read
S1_CTRL_MODE_CTL2	0x00011444	Read/Write
S1_CTRL_MODE_CTL1	0x00011445	Read/Write
S1_CTRL_EN_CTL	0x00011446	Read/Write
S1_CTRL_FOLLOW_HWEN	0x00011447	Read/Write
S1_CTRL_FREQ_CTL	0x00011450	Read/Write
S1_CTRL_OCP	0x00011488	Read/Write
S1_CTRL_STATUS_DEB_CTL	0x00011490	Read/Write
S1_CTRL_PHASE_ID	0x00011495	Read/Write
S1_CTRL_PHASE_CNT_MAX	0x00011496	Read/Write
S1_CTRL_PD_CTL	0x000114A0	Read/Write
S1_CTRL_SI_THRES_VSET_LB_SHADOW	0x000114BC	Read/Write
S1_CTRL_SI_THRES_VSET_UB_SHADOW	0x000114BD	Read/Write
S1_CTRL_GANG_CTL1	0x000114C0	Read/Write
S1_CTRL_GANG_CTL2	0x000114C1	Read/Write

S1_CTRL_REVISION1 | 0x00011400

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	Indicates a change in the HW which is not intended to impact SW compatibility.

S1_CTRL_REVISION2 | 0x00011401

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	Indicates a change in the HW which will impact SW compatibility.

S1_CTRL_REVISION3 | 0x00011402

Type: Read

Reset: 0x00000003

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	Indicates expanded functionality. Minor version adds functionality while being backward compatible for existing features.

S1_CTRL_REVISION4 | 0x00011403

Type: Read

Reset: 0x00000000

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	Indicates different interface version. Major version changes are not backward compatible. There will be a new Programming Guide document per major revision

S1_CTRL_PERPH_TYPE | 0x00011404

Type: Read

Reset: 0x0000001C

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	Peripheral Type

S1_CTRL_PERPH_SUBTYPE | 0x00011405

Type: Read

Reset: 0x0000000B

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	Peripheral SubType 10: Voltage_Mode_PWM_Controller

S1_CTRL_STATUS1 | 0x00011408

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates regulator mode state. 0x0: UNUSED 0x1: STATE==RM_STATE && RETENTION MODE (QUALIFIED)==TRUE 0x2: UNUSED 0x3: STATE==PWM STATE 0: UNUSED_is_0 1: RETENTION_is_1 2: UNUSED_is_2 3: PWM_is_3

Bits	Name	Description
2 : 2	STEPPER_DONE	Indicates voltage stepper is done or not 0: STEPPER_DONE_false 1: STEPPER_DONE_true
3 : 3	PS_TRUE	Indicates whether buck is pulse skipping or not 0: PS_TRUE_false 1: PS_TRUE_true
4 : 4	RETENTION_QUAL	Indicates that retention mode is qualified versus the load current. 0x0: false 0x1: true 0: RM_QUAL_false 1: RM_QUAL_true
5 : 5	VREG_OCP	Indicates if over current or another fault has been detected/latched. 0: NO_FAULT 1: FAULT
6 : 6	VREG_ERROR	Indicates that regulator is not stepping and has fallen below VREG_OK comparator threshold 0 = Regulator is okay; no error condition 0: VREG_ERR_false 1: VREG_ERR_true
7 : 7	VREG_READY	Indicates if VREG is ready or not due to stepping in progress or below analog debounced VREG_OK comp 0: VREG_READY_false 1: VREG_READY_true

S1_CTRL_INT_RT_STS | 0x00011410

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

S1_CTRL_INT_SET_TYPE | 0x00011411

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

S1_CTRL_INT_POLARITY_HIGH | 0x00011412

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

S1_CTRL_INT_POLARITY_LOW | 0x00011413

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

S1_CTRL_INT_LATCHED_CLR | 0x00011414

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

S1_CTRL_INT_EN_SET | 0x00011415

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

S1_CTRL_INT_EN_CLR | 0x00011416

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

S1_CTRL_INT_MID_SEL | 0x0001141A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

S1_CTRL_INT_PRIORITY | 0x0001141B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

S1_CTRL_INT_SELF_CLEARING | 0x0001141D

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

S1_CTRL_ULS_VSET_LB | 0x00011439

Type: Read/Write

Reset: 0x00000074

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	Vreg upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x68 and 0x69). Vuls_vset = 2-byte word (dec) x 1mV

S1_CTRL_ULS_VSET_UB | 0x0001143A

Type: Read/Write

Reset: 0x00000005

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	Vreg upper limit stop-voltage set point bits<3:0>. 2-Byte Word (0x68 and 0x69). Vuls_vset = 2-byte word (dec) x 1mV. Maximum combined supported value is 0x0574 which translates to 1.396mV. Beyond that value, it will be clamped to 0x0574. The default value is already set to the maximum allowed value.

S1_CTRL_VSET_LB | 0x00011440

Type: Read/Write

Reset: 0x00000068

PMIC_LATCHED_WRITE=VSET_UB, PMIC_GANGED

Bits	Name	Description
7 : 0	VSET_LB	Regulator voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S1_CTRL_VSET_UB | 0x00011441

Type: Read/Write

Reset: 0x00000003

PMIC_GANGED

Bits	Name	Description
3 : 0	VSET_UB	Regulator voltage set point bit <11:8>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S1_CTRL_VSET_VALID_LB | 0x00011442

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	VSET_VALID_LB	Lower byte readback for the valid output voltage setpoint value. This register shows the value round up effect of the set voltage since buck doesn't support 1mV resolution.,,,'

S1_CTRL_VSET_VALID_UB | 0x00011443

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Upper byte readback for the valid output voltage setpoint value. This register shows the value round up effect of the set voltage since buck doesn't support 1mV resolution.,,,'

S1_CTRL_MODE_CTL2 | 0x00011444

Type: Read/Write

Reset: 0x00000007

Define Buck Mode Transitions PMIC_GANGED, PMIC_SYNC=d_clk_19p2:d_perph_rb

Bits	Name	Description
2 : 0	MODE_SECONDARY	Controls the mode that buck operates: FORCED_NPM_7 = 7, (DEFAULT) RESERVED,(PRODUCES FORCED_NPM for FTS510)=6, RM_2 =5, RESERVED (PRODUCES FORCED_NPM for FTS510)= 4, RM_1=3, FORCED_NPM_2=2, FORCED_NPM_1=1, FORCED_NPM_0=0 0: FORCED_NPM_0 1: FORCED_NPM_1 2: FORCED_NPM_2 3: RM_1 4: RESERVED_1 5: RM_2 6: RESERVED_2 7: FORCED_NPM_7

S1_CTRL_MODE_CTL1 | 0x00011445

Type: Read/Write

Reset: 0x00000007

Define Buck Mode Transitions PMIC_GANGED, PMIC_SYNC=d_clk_19p2:d_perph_rb

Bits	Name	Description
2 : 0	MODE_PRIMARY	Controls the mode that buck operates: FORCED_NPM_7 = 7,(DEFAULT) RESERVED,(PRODUCES FORCED_NPM for FTS510)=6, RM_2 =5, RESERVED (PRODUCES FORCED_NPM for FTS510)= 4, RM_1=3, FORCED_NPM_2=2, FORCED_NPM_1=1, FORCED_NPM_0=0 0: FORCED_NPM_0 1: FORCED_NPM_1 2: FORCED_NPM_2 3: RM_1 4: RESERVED_1 5: RM_2 6: RESERVED_2 7: FORCED_NPM_7

S1_CTRL_EN_CTL | 0x00011446

Type: Read/Write

Reset: 0x00000000

PMIC_GANGED

Bits	Name	Description
7 : 7	PERPH_EN	Control signal to enable the regulator 0: BUCK_ENABLE_false 1: BUCK_ENABLE_true

S1_CTRL_FOLLOW_HWEN | 0x00011447

Type: Read/Write

Reset: 0x00000000

PMIC_GANGED

Bits	Name	Description
0 : 0	EN_FOLLOW_HWEN0	Enables buck when HWEN0 is asserted 0: EN_FOLLOW_HWEN0_false 1: EN_FOLLOW_HWEN0_true

Bits	Name	Description
1 : 1	EN_FOLLOW_HWEN1	Enables buck when HWEN1 is asserted 0: EN_FOLLOW_HWEN1_false 1: EN_FOLLOW_HWEN1_true
2 : 2	EN_FOLLOW_HWEN2	Enables buck when HWEN2 is asserted 0: EN_FOLLOW_HWEN2_false 1: EN_FOLLOW_HWEN2_true
3 : 3	EN_FOLLOW_PMIC_AWAKE	Enables buck when PMIC_AWAKE is asserted 0: EN_FOLLOW_PMIC_AWAKE_false 1: EN_FOLLOW_PMIC_AWAKE_true
4 : 4	FOLLOW_HWEN0_CFG	If this bit is set low then HWEN0 does NOT control Buck mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN0_CFG_false 1: FOLLOW_HWEN0_CFG_true
5 : 5	FOLLOW_HWEN1_CFG	If this bit is set low then HWEN1 does NOT control Buck mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN1_CFG_false 1: FOLLOW_HWEN1_CFG_true
6 : 6	FOLLOW_HWEN2_CFG	If this bit is set low then HWEN2 does NOT control Buck mode. If this bit is set high then: When HWEN2=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN2_CFG_false 1: FOLLOW_HWEN2_CFG_true
7 : 7	FOLLOW_AWAKE_CFG	If this bit is set low then PMIC_AWAKE does NOT control Buck mode. If this bit is set high then: When PMIC_AWAKE=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_AWAKE_CFG_false 1: FOLLOW_AWAKE_CFG_true

S1_CTRL_FREQ_CTL | 0x00011450

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED

Bits	Name	Description
3 : 0	FREQ_CTL	Clock frequency = 19.2MHz / (CLK_DIV + 1) 2: Fs_6p4MHz 3: Fs_4p8MHz 4: Fs_3p84MHz 5: Fs_3p2MHz 6: Fs_2p74MHz 7: Fs_2p4MHz 8: Fs_2p13MHz 9: Fs_1p92MHz 10: Fs_1p74MHz 11: Fs_1p6MHz

S1_CTRL_OCP | 0x00011488

Type: Read/Write

Reset: 0x00000068

PMIC_GANGED, PMIC_SYNC=d_clk_lfrc:d_perph_rb

Bits	Name	Description
1 : 1	OCP_STATUS_CLEAR	Toggle 1->0 to clear the latched OCP status bit. 0: OCP_STATUS_CLEAR_false 1: OCP_STATUS_CLEAR_true
2 : 2	OCP_GLOBAL_BROADCAST_EN	Determines if the Buck notifies the PON module that an OCP event has occurred. PON module may use this info to shutdown the entire PMIC. 1 = BUCK notifies PON when OCP event occurs 0 = BUCK does not notify PON when OCP even occurs (DEFAULT) 0: OCP_GLOBAL_BROADCAST_EN_false 1: OCP_GLOBAL_BROADCAST_EN_true
4 : 3	INZERO_WAIT_OCP_TIMER	Controls wait time for inzero trigger during current foldback. If inzero wait timer expires then an inzero is forced. OCP timer controls the time at which buck shuts down if OCP_LATCH_EN=1 0: INZERO_WAIT_40us_OCP_TIMER_320us 1: INZERO_WAIT_80us_OCP_TIMER_640us 2: INZERO_WAIT_160us_OCP_TIMER_1280_us 3: INZERO_WAIT_160us_OCP_TIMER_1280us
5 : 5	VREG_FOLDBACK_EN	This bit is used to enable propagating foldback flag to PS. 0: FORCE_FOLDBACK_FALSE 1: FORCE_FOLDBACK_TRUE

Bits	Name	Description
6 : 6	OCP_LATCH_EN	Controls whether OCP event causes buck to shut down when OCP timer expires 0x0: ocp latching is not enabled. Buck will not shut off due to the OCP event. 0x1: ocp latching is enabled. Buck will SHUT OFF due to the OCP event. (DEFAULT) 0: OCP_LATCH_EN_false 1: OCP_LATCH_EN_true

S1_CTRL_STATUS_DEB_CTL | 0x00011490

Type: Read/Write

Reset: 0x00000000

PMIC_GANGED, PMIC_SYNC=d_clk_4p8m:d_perph_rb

Bits	Name	Description
1 : 0	VREG_ERROR_DEB	Configures the debounce on falling edge of VREG_OK comparator output for generating VREG_ERROR. 00 = 1.4us -1.6us 01 = 3.0us -3.2us 10= 6.5us -6.7us 11= 9.8us - 10us 0: ERROR_DEB_0 1: ERROR_DEB_1 2: ERROR_DEB_2 3: ERROR_DEB_3
5 : 4	VREG_READY_DEB	Configures the debounce on rising edge of VREG_OK comparator output for generating VREG_READY. 00 = 1.4us -1.6us 01 = 3.0us -3.2us 10= 6.5us -6.7us 11= 9.8us - 10us 0: READY_DEB_0 1: READY_DEB_1 2: READY_DEB_2 3: READY_DEB_3

S1_CTRL_PHASE_ID | 0x00011495

Type: Read/Write

Reset: 0x00000000

PMIC_LOCKEDBYBIT=LOCKBIT_D1, PMIC_SYNC=d_clk_19p2:d_perph_rb

Bits	Name	Description
3 : 0	PHASE_ID	Unique phase identifier. If less than or equal to SLAVE_COUNT, phase is active. If greater than SLAVE_COUNT, phase is inactive. 0: Phase_Number_1 1: Phase_Number_2 2: Phase_Number_3 3: Phase_Number_4

S1_CTRL_PHASE_CNT_MAX | 0x00011496

Type: Read/Write

Reset: 0x00000000

PMIC_GANGED, PMIC_SYNC=d_clk_19p2:d_perph_rb

Bits	Name	Description
3 : 0	PHASE_CNT_MAX	Sets the maximum number of slave phases we can use.

S1_CTRL_PD_CTL | 0x000114A0

Type: Read/Write

Reset: 0x00000088

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Controls state of lead pull down. (Does not depend on the fsm state). 0: LEAK_PD_EN_false 1: LEAK_PD_EN_true
6 : 6	WEAK_PD_EN	Controls state of weak pull down. (See glue logic for mode fsm state dependency go/fts510). 0: WEAK_PD_ENABLE_false 1: WEAK_PD_ENABLE_true
7 : 7	STRONG_PD_EN	Controls state of strong pull down. (See glue logic for mode fsm state dependency go/fts510). 0: STRONG_PD_ENABLE_false 1: STRONG_PD_ENABLE_true

S1_CTRL_SI_THRES_VSET_LB_SHADOW | 0x000114BC

Type: Read/Write

Reset: 0x00000048

PMIC_LATCHED_WRITE=SI_THRES_VSET_UB_SHADOW, PMIC_GANGED, PMIC_DOC_ONLY

Bits	Name	Description
7 : 0	SI_THRES_VSET_LB	Skip integration threshold for low vset operation: voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV Default = 1.352V

S1_CTRL_SI_THRES_VSET_UB_SHADOW | 0x000114BD

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED, PMIC_DOC_ONLY

Bits	Name	Description
3 : 0	SI_THRES_VSET_UB	Skip integration threshold for low vset operation: voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV Default = 1.352V

S1_CTRL_GANG_CTL1 | 0x000114C0

Type: Read/Write

Reset: 0x00000014

PMIC_LOCKEDBYBIT=LOCKBIT_D1

Bits	Name	Description
7 : 0	GANG_LEADER_PID	When GANG_EN (GANG_CTL2[7]) is set, this peripheral will write the same data that is written to the gang leader Peripheral ID. Reads to the gang leader Peripheral ID are ignored by this peripheral. Ganged peripherals must reside within the same Slave ID

S1_CTRL_GANG_CTL2 | 0x000114C1

Type: Read/Write

Reset: 0x00000000

PMIC_LOCKEDBYBIT=LOCKBIT_D1

Bits	Name	Description
7 : 7	GANG_EN	0 = disable 1 = enable When enabled, this peripheral will write the same data that is written to the gang leader PID. Reads to the gang leader PID are ignored by this peripheral 0: GANG_EN_false 1: GANG_EN_true

Module Name: S1_PS

Base Address: 0x00011500

Register Quick Reference:

Register	Offset	Type
S1_PS_REVISION1	0x00011500	Read
S1_PS_REVISION2	0x00011501	Read
S1_PS_REVISION3	0x00011502	Read
S1_PS_REVISION4	0x00011503	Read
S1_PS_PERPH_TYPE	0x00011504	Read
S1_PS_PERPH_SUBTYPE	0x00011505	Read
S1_PS_VSET_LB	0x00011540	Read/Write
S1_PS_VSET_UB	0x00011541	Read/Write
S1_PS_SI_THRES_VSET_LB	0x000115BC	Read/Write
S1_PS_SI_THRES_VSET_UB	0x000115BD	Read/Write
S1_PS_GANG_CTL1	0x000115C0	Read/Write
S1_PS_GANG_CTL2	0x000115C1	Read/Write

S1_PS_REVISION1 | 0x00011500

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

S1_PS_REVISION2 | 0x00011501

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

S1_PS_REVISION3 | 0x00011502

Type: Read

Reset: 0x00000002

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

S1_PS_REVISION4 | 0x00011503

Type: Read

Reset: 0x00000000

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

S1_PS_PERPH_TYPE | 0x00011504

Type: Read

Reset: 0x0000001C

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	http://projects/sites/PM8941/Shared%20Documents/Systems/Registers/Type%20and%20SubType%20Enumeration%20Table.xlsx

S1_PS_PERPH_SUBTYPE | 0x00011505

Type: Read

Reset: 0x0000000C

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	http://projects/sites/PM8941/Shared%20Documents/Systems/Registers/Type%20and%20SubType%20Enumeration%20Table.xlsx

S1_PS_VSET_LB | 0x00011540

Type: Read/Write

Reset: 0x00000068

PMIC_LATCHED_WRITE=VSET_UB, PMIC_GANGED

Bits	Name	Description
7 : 0	VSET_LB	Regulator voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S1_PS_VSET_UB | 0x00011541

Type: Read/Write

Reset: 0x00000003

PMIC_GANGED

Bits	Name	Description
3 : 0	VSET_UB	Regulator voltage set point bit <11:8>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S1_PS_SI_THRES_VSET_LB | 0x000115BC

Type: Read/Write

Reset: 0x00000048

PMIC_LATCHED_WRITE=SI_THRES_VSET_UB, PMIC_GANGED

Bits	Name	Description
7 : 0	SI_THRES_VSET_LB	Skip integration threshold for low vset operation: voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV Default = 1.352V

S1_PS_SI_THRES_VSET_UB | 0x000115BD

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED

Bits	Name	Description
3 : 0	SI_THRES_VSET_UB	Skip integration threshold for low vset operation: voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV Default = 1.352V

S1_PS_GANG_CTL1 | 0x000115C0

Type: Read/Write

Reset: 0x00000014

PMIC_LOCKEDBYBIT=LOCKBIT_D1

Bits	Name	Description
7 : 0	GANG_LEADER_PID	When GANG_EN (GANG_CTL2[7]) is set, this peripheral will write the same data that is written to the gang leader Peripheral ID. Reads to the gang leader Peripheral ID are ignored by this peripheral. Ganged peripherals must reside within the same Slave ID

S1_PS_GANG_CTL2 | 0x000115C1

Type: Read/Write

Reset: 0x00000080

PMIC_LOCKEDBYBIT=LOCKBIT_D1

Bits	Name	Description
7 : 7	GANG_EN	0 = disable 1 = enable When enabled, this peripheral will write the same data that is written to the gang leader PID. Reads to the gang leader PID are ignored by this peripheral 0: GANG_EN_false 1: GANG_EN_true

Module Name: S1_FREQ

Base Address: 0x00011600

Register Quick Reference:

Register	Offset	Type
S1_FREQ_CLK_ENABLE	0x00011646	Read/Write
S1_FREQ_CLK_DIV	0x00011650	Read/Write
S1_FREQ_CLK_PHASE	0x00011651	Read/Write
S1_FREQ_GANG_CTL1	0x000116C0	Read/Write
S1_FREQ_GANG_CTL2	0x000116C1	Read/Write

S1_FREQ_CLK_ENABLE | 0x00011646

Type: Read/Write

Reset: 0x00000001

No description provided for this register.

Bits	Name	Description
0 : 0	FOLLOW_CLK_SX_REQ	0 = ignore smps_clk_req<X> 1 = clock is enabled when the clocks request is high smps_clk_req<X>='1' 0: FALLOW_CLK_REQ_DISABLED 1: FALLOW_CLK_REQ_ENABLED
7 : 7	EN_CLK_INT	0 = do not force the clock on 1 = enable the clock 0: FORCE_EN_DISABLED 1: FORCE_EN_ENABLED

S1_FREQ_CLK_DIV | 0x00011650

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED

Bits	Name	Description
3 : 0	CLK_DIV	clock_ frequency = 19.2MHz / (CLK_DIV + 1) FTS2 Buck supports 3.2, 4.8, 6.4 and 9.6 MHz HF2 Buck supports 1.6, 2.4, 2.74, 3.2, 3.8, 4.8, and 6.4 MHz CLK_DIV = 0 is not supported, it will generate 9.6 MHz 0: FREQ_9M6HZ_0 1: FREQ_9M6HZ 2: FREQ_6M4HZ 3: FREQ_4M8HZ 4: FREQ_3M8HZ 5: FREQ_3M2HZ 6: FREQ_2M7HZ 7: FREQ_2M4HZ 8: FREQ_2M1HZ 9: FREQ_1M9HZ 10: FREQ_1M7HZ 11: FREQ_1M6HZ 12: FREQ_1M5HZ 13: FREQ_1M4HZ 14: FREQ_1M3HZ 15: FREQ_1M2HZ

S1_FREQ_CLK_PHASE | 0x00011651

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	CLK_PHASE	Distributed clock phase select: clock phase delay = clock period * (CLK_PHASE / 16)

S1_FREQ_GANG_CTL1 | 0x000116C0

Type: Read/Write

Reset: 0x00000014

No description provided for this register.

Bits	Name	Description
7 : 0	GANG_LEADER_PID	When GANG_EN (GANG_CTL2[7]) is set, this peripheral will write the same data that is written to the gang leader Peripheral ID. Reads to the gang leader Peripheral ID are ignored by this peripheral. Ganged peripherals must reside within the same Slave ID

S1_FREQ_GANG_CTL2 | 0x000116C1

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
7 : 7	GANG_EN	0 = disable 1 = enable When enabled, this peripheral will write the same data that is written to the gang leader PID. Reads to the gang leader PID are ignored by this peripheral 0: GANGING_DISABLED 1: GANGING_ENABLED

Module Name: S2_CTRL

Base Address: 0x00011700

Register Quick Reference:

Register	Offset	Type
S2_CTRL_REVISION1	0x00011700	Read
S2_CTRL_REVISION2	0x00011701	Read
S2_CTRL_REVISION3	0x00011702	Read
S2_CTRL_REVISION4	0x00011703	Read
S2_CTRL_PERPH_TYPE	0x00011704	Read
S2_CTRL_PERPH_SUBTYPE	0x00011705	Read
S2_CTRL_STATUS1	0x00011708	Read
S2_CTRL_INT_RT_STS	0x00011710	Read
S2_CTRL_INT_SET_TYPE	0x00011711	Read
S2_CTRL_INT_POLARITY_HIGH	0x00011712	Read
S2_CTRL_INT_POLARITY_LOW	0x00011713	Read/Write
S2_CTRL_INT_LATCHED_CLR	0x00011714	Write
S2_CTRL_INT_EN_SET	0x00011715	Read/Write
S2_CTRL_INT_EN_CLR	0x00011716	Read/Write
S2_CTRL_INT_MID_SEL	0x0001171A	Read/Write
S2_CTRL_INT_PRIORITY	0x0001171B	Read/Write
S2_CTRL_INT_SELF_CLEARING	0x0001171D	Read
S2_CTRL_ULS_VSET_LB	0x00011739	Read/Write
S2_CTRL_ULS_VSET_UB	0x0001173A	Read/Write
S2_CTRL_VSET_LB	0x00011740	Read/Write
S2_CTRL_VSET_UB	0x00011741	Read/Write
S2_CTRL_VSET_VALID_LB	0x00011742	Read
S2_CTRL_VSET_VALID_UB	0x00011743	Read
S2_CTRL_MODE_CTL2	0x00011744	Read/Write
S2_CTRL_MODE_CTL1	0x00011745	Read/Write
S2_CTRL_EN_CTL	0x00011746	Read/Write
S2_CTRL_FOLLOW_HWEN	0x00011747	Read/Write
S2_CTRL_FREQ_CTL	0x00011750	Read/Write
S2_CTRL_OCP	0x00011788	Read/Write

Register	Offset	Type
S2_CTRL_STATUS_DEB_CTL	0x00011790	Read/Write
S2_CTRL_PHASE_ID	0x00011795	Read/Write
S2_CTRL_PHASE_CNT_MAX	0x00011796	Read/Write
S2_CTRL_PD_CTL	0x000117A0	Read/Write
S2_CTRL_SI_THRES_VSET_LB_SHADOW	0x000117BC	Read/Write
S2_CTRL_SI_THRES_VSET_UB_SHADOW	0x000117BD	Read/Write
S2_CTRL_GANG_CTL1	0x000117C0	Read/Write
S2_CTRL_GANG_CTL2	0x000117C1	Read/Write

S2_CTRL_REVISION1 | 0x00011700

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	Indicates a change in the HW which is not intended to impact SW compatibility.

S2_CTRL_REVISION2 | 0x00011701

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	Indicates a change in the HW which will impact SW compatibility.

S2_CTRL_REVISION3 | 0x00011702

Type: Read

Reset: 0x00000003

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	Indicates expanded functionality. Minor version adds functionality while being backward compatible for existing features.

S2_CTRL_REVISION4 | 0x00011703

Type: Read

Reset: 0x00000000

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	Indicates different interface version. Major version changes are not backward compatible. There will be a new Programming Guide document per major revision

S2_CTRL_PERPH_TYPE | 0x00011704

Type: Read

Reset: 0x0000001C

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	Peripheral Type

S2_CTRL_PERPH_SUBTYPE | 0x00011705

Type: Read

Reset: 0x0000000B

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	Peripheral SubType 10: Voltage_Mode_PWM_Controller

S2_CTRL_STATUS1 | 0x00011708

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates regulator mode state. 0x0: UNUSED 0x1: STATE==RM_STATE && RETENTION MODE (QUALIFIED)==TRUE 0x2: UNUSED 0x3: STATE==PWM STATE 0: UNUSED_is_0 1: RETENTION_is_1 2: UNUSED_is_2 3: PWM_is_3
2 : 2	STEPPER_DONE	Indicates voltage stepper is done or not 0: STEPPER_DONE_false 1: STEPPER_DONE_true
3 : 3	PS_TRUE	Indicates whether buck is pulse skipping or not 0: PS_TRUE_false 1: PS_TRUE_true
4 : 4	RETENTION_QUAL	Indicates that retention mode is qualified versus the load current. 0x0: false 0x1: true 0: RM_QUAL_false 1: RM_QUAL_true
5 : 5	VREG_OCP	Indicates if over current or another fault has been detected/latched. 0: NO_FAULT 1: FAULT
6 : 6	VREG_ERROR	Indicates that regulator is not stepping and has fallen below VREG_OK comparator threshold 0 = Regulator is okay; no error condition 0: VREG_ERR_false 1: VREG_ERR_true
7 : 7	VREG_READY	Indicates if VREG is ready or not due to stepping in progress or below analog debounced VREG_OK comp 0: VREG_READY_false 1: VREG_READY_true

S2_CTRL_INT_RT_STS | 0x00011710

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

S2_CTRL_INT_SET_TYPE | 0x00011711

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

S2_CTRL_INT_POLARITY_HIGH | 0x00011712

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

S2_CTRL_INT_POLARITY_LOW | 0x00011713

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

S2_CTRL_INT_LATCHED_CLR | 0x00011714

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

S2_CTRL_INT_EN_SET | 0x00011715

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

S2_CTRL_INT_EN_CLR | 0x00011716

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

S2_CTRL_INT_MID_SEL | 0x0001171A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

S2_CTRL_INT_PRIORITY | **0x0001171B**

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

S2_CTRL_INT_SELF_CLEARING | 0x0001171D

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

S2_CTRL_ULS_VSET_LB | 0x00011739

Type: Read/Write

Reset: 0x00000074

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	Vreg upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x68 and 0x69). Vuls_vset = 2-byte word (dec) x 1mV

S2_CTRL_ULS_VSET_UB | 0x0001173A

Type: Read/Write

Reset: 0x00000005

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	Vreg upper limit stop-voltage set point bits<3:0>. 2-Byte Word (0x68 and 0x69). Vuls_vset = 2-byte word (dec) x 1mV. Maximum combined supported value is 0x0574 which translates to 1.396mV. Beyond that value, it will be clamped to 0x0574. The default value is already set to the maximum allowed value.

S2_CTRL_VSET_LB | 0x00011740

Type: Read/Write

Reset: 0x00000068

PMIC_LATCHED_WRITE=VSET_UB, PMIC_GANGED

Bits	Name	Description
7 : 0	VSET_LB	Regulator voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S2_CTRL_VSET_UB | 0x00011741

Type: Read/Write

Reset: 0x00000003

PMIC_GANGED

Bits	Name	Description
3 : 0	VSET_UB	Regulator voltage set point bit <11:8>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S2_CTRL_VSET_VALID_LB | 0x00011742

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	VSET_VALID_LB	Lower byte readback for the valid output voltage setpoint value. This register shows the value round up effect of the set voltage since buck doesn't support 1mV resolution.,,,'

S2_CTRL_VSET_VALID_UB | 0x00011743

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Upper byte readback for the valid output voltage setpoint value. This register shows the value round up effect of the set voltage since buck doesn't support 1mV resolution.,,,'

S2_CTRL_MODE_CTL2 | 0x00011744

Type: Read/Write

Reset: 0x00000007

Define Buck Mode Transitions PMIC_GANGED, PMIC_SYNC=d_clk_19p2:d_perph_rb

Bits	Name	Description
2 : 0	MODE_SECONDARY	Controls the mode that buck operates: FORCED_NPM_7 = 7, (DEFAULT) RESERVED,(PRODUCES FORCED_NPM for FTS510)=6, RM_2 =5, RESERVED (PRODUCES FORCED_NPM for FTS510)= 4, RM_1=3, FORCED_NPM_2=2, FORCED_NPM_1=1, FORCED_NPM_0=0 0: FORCED_NPM_0 1: FORCED_NPM_1 2: FORCED_NPM_2 3: RM_1 4: RESERVED_1 5: RM_2 6: RESERVED_2 7: FORCED_NPM_7

S2_CTRL_MODE_CTL1 | 0x00011745

Type: Read/Write

Reset: 0x00000007

Define Buck Mode Transitions PMIC_GANGED, PMIC_SYNC=d_clk_19p2:d_perph_rb

Bits	Name	Description
2 : 0	MODE_PRIMARY	Controls the mode that buck operates: FORCED_NPM_7 = 7,(DEFAULT) RESERVED,(PRODUCES FORCED_NPM for FTS510)=6, RM_2 =5, RESERVED (PRODUCES FORCED_NPM for FTS510)= 4, RM_1=3, FORCED_NPM_2=2, FORCED_NPM_1=1, FORCED_NPM_0=0 0: FORCED_NPM_0 1: FORCED_NPM_1 2: FORCED_NPM_2 3: RM_1 4: RESERVED_1 5: RM_2 6: RESERVED_2 7: FORCED_NPM_7

S2_CTRL_EN_CTL | 0x00011746

Type: Read/Write

Reset: 0x00000000

PMIC_GANGED

Bits	Name	Description
7 : 7	PERPH_EN	Control signal to enable the regulator 0: BUCK_ENABLE_false 1: BUCK_ENABLE_true

S2_CTRL_FOLLOW_HWEN | 0x00011747

Type: Read/Write

Reset: 0x00000000

PMIC_GANGED

Bits	Name	Description
0 : 0	EN_FOLLOW_HWEN0	Enables buck when HWEN0 is asserted 0: EN_FOLLOW_HWEN0_false 1: EN_FOLLOW_HWEN0_true
1 : 1	EN_FOLLOW_HWEN1	Enables buck when HWEN1 is asserted 0: EN_FOLLOW_HWEN1_false 1: EN_FOLLOW_HWEN1_true
2 : 2	EN_FOLLOW_HWEN2	Enables buck when HWEN2 is asserted 0: EN_FOLLOW_HWEN2_false 1: EN_FOLLOW_HWEN2_true
3 : 3	EN_FOLLOW_PMIC_AWAKE	Enables buck when PMIC_AWAKE is asserted 0: EN_FOLLOW_PMIC_AWAKE_false 1: EN_FOLLOW_PMIC_AWAKE_true
4 : 4	FOLLOW_HWEN0_CFG	If this bit is set low then HWEN0 does NOT control Buck mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN0_CFG_false 1: FOLLOW_HWEN0_CFG_true

Bits	Name	Description
5 : 5	FOLLOW_HWEN1_CFG	If this bit is set low then HWEN1 does NOT control Buck mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN1_CFG_false 1: FOLLOW_HWEN1_CFG_true
6 : 6	FOLLOW_HWEN2_CFG	If this bit is set low then HWEN2 does NOT control Buck mode. If this bit is set high then: When HWEN2=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN2_CFG_false 1: FOLLOW_HWEN2_CFG_true
7 : 7	FOLLOW_AWAKE_CFG	If this bit is set low then PMIC_AWAKE does NOT control Buck mode. If this bit is set high then: When PMIC_AWAKE=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_AWAKE_CFG_false 1: FOLLOW_AWAKE_CFG_true

S2_CTRL_FREQ_CTL | 0x00011750

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED

Bits	Name	Description
3 : 0	FREQ_CTL	Clock frequency = 19.2MHz / (CLK_DIV + 1) 2: Fs_6p4MHz 3: Fs_4p8MHz 4: Fs_3p84MHz 5: Fs_3p2MHz 6: Fs_2p74MHz 7: Fs_2p4MHz 8: Fs_2p13MHz 9: Fs_1p92MHz 10: Fs_1p74MHz 11: Fs_1p6MHz

S2_CTRL_OCP | 0x00011788

Type: Read/Write

Reset: 0x00000068

PMIC_GANGED, PMIC_SYNC=d_clk_lfrc:d_perph_rb

Bits	Name	Description
1 : 1	OCP_STATUS_CLEAR	Toggle 1->0 to clear the latched OCP status bit. 0: OCP_STATUS_CLEAR_false 1: OCP_STATUS_CLEAR_true
2 : 2	OCP_GLOBAL_BROADCAST_EN	Determines if the Buck notifies the PON module that an OCP event has occurred. PON module may use this info to shutdown the entire PMIC. 1 = BUCK notifies PON when OCP event occurs 0 = BUCK does not notify PON when OCP even occurs (DEFAULT) 0: OCP_GLOBAL_BROADCAST_EN_false 1: OCP_GLOBAL_BROADCAST_EN_true
4 : 3	INZERO_WAIT_OCP_TIMER	Controls wait time for inzero trigger during current foldback. If inzero wait timer expires then an inzero is forced. OCP timer controls the time at which buck shuts down if OCP_LATCH_EN=1 0: INZERO_WAIT_40us_OCP_TIMER_320us 1: INZERO_WAIT_80us_OCP_TIMER_640us 2: INZERO_WAIT_160us_OCP_TIMER_1280_us 3: INZERO_WAIT_160us_OCP_TIMER_1280us
5 : 5	VREG_FOLDBACK_EN	This bit is used to enable propagating foldback flag to PS. 0: FORCE_FOLDBACK_FALSE 1: FORCE_FOLDBACK_TRUE
6 : 6	OCP_LATCH_EN	Controls whether OCP event causes buck to shut down when OCP timer expires 0x0: ocp latching is not enabled. Buck will not shut off due to the OCP event. 0x1: ocp latching is enabled. Buck will SHUT OFF due to the OCP event. (DEFAULT) 0: OCP_LATCH_EN_false 1: OCP_LATCH_EN_true

S2_CTRL_STATUS_DEB_CTL | 0x00011790

Type: Read/Write

Reset: 0x00000000

PMIC_GANGED, PMIC_SYNC=d_clk_4p8m:d_perph_rb

Bits	Name	Description
1 : 0	VREG_ERROR_DEB	Configures the debounce on falling edge of VREG_OK comparator output for generating VREG_ERROR. 00 = 1.4us -1.6us 01 = 3.0us -3.2us 10= 6.5us -6.7us 11= 9.8us - 10us 0: ERROR_DEB_0 1: ERROR_DEB_1 2: ERROR_DEB_2 3: ERROR_DEB_3
5 : 4	VREG_READY_DEB	Configures the debounce on rising edge of VREG_OK comparator output for generating VREG_READY. 00 = 1.4us -1.6us 01 = 3.0us -3.2us 10= 6.5us -6.7us 11= 9.8us - 10us 0: READY_DEB_0 1: READY_DEB_1 2: READY_DEB_2 3: READY_DEB_3

S2_CTRL_PHASE_ID | 0x00011795

Type: Read/Write

Reset: 0x00000000

PMIC_LOCKEDBYBIT=LOCKBIT_D1, PMIC_SYNC=d_clk_19p2:d_perph_rb

Bits	Name	Description
3 : 0	PHASE_ID	Unique phase identifier. If less than or equal to SLAVE_COUNT, phase is active. If greater than SLAVE_COUNT, phase is inactive. 0: Phase_Number_1 1: Phase_Number_2 2: Phase_Number_3 3: Phase_Number_4

S2_CTRL_PHASE_CNT_MAX | 0x00011796

Type: Read/Write

Reset: 0x00000000

PMIC_GANGED, PMIC_SYNC=d_clk_19p2:d_perph_rb

Bits	Name	Description
3 : 0	PHASE_CNT_MAX	Sets the maximum number of slave phases we can use.

S2_CTRL_PD_CTL | 0x000117A0

Type: Read/Write

Reset: 0x00000088

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Controls state of lead pull down. (Does not depend on the fsm state). 0: LEAK_PD_EN_false 1: LEAK_PD_EN_true
6 : 6	WEAK_PD_EN	Controls state of weak pull down. (See glue logic for mode fsm state dependency go/fts510). 0: WEAK_PD_ENABLE_false 1: WEAK_PD_ENABLE_true
7 : 7	STRONG_PD_EN	Controls state of strong pull down. (See glue logic for mode fsm state dependency go/fts510). 0: STRONG_PD_ENABLE_false 1: STRONG_PD_ENABLE_true

S2_CTRL_SI_THRES_VSET_LB_SHADOW | 0x000117BC

Type: Read/Write

Reset: 0x00000048

PMIC_LATCHED_WRITE=SI_THRES_VSET_UB_SHADOW, PMIC_GANGED, PMIC_DOC_ONLY

Bits	Name	Description
7 : 0	SI_THRES_VSET_LB	Skip integration threshold for low vset operation: voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV Default = 1.352V

S2_CTRL_SI_THRES_VSET_UB_SHADOW | 0x000117BD

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED, PMIC_DOC_ONLY

Bits	Name	Description
3 : 0	SI_THRES_VSET_UB	Skip integration threshold for low vset operation: voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV Default = 1.352V

S2_CTRL_GANG_CTL1 | 0x000117C0

Type: Read/Write

Reset: 0x00000017

PMIC_LOCKEDBYBIT=LOCKBIT_D1

Bits	Name	Description
7 : 0	GANG_LEADER_PID	When GANG_EN (GANG_CTL2[7]) is set, this peripheral will write the same data that is written to the gang leader Peripheral ID. Reads to the gang leader Peripheral ID are ignored by this peripheral. Ganged peripherals must reside within the same Slave ID

S2_CTRL_GANG_CTL2 | 0x000117C1

Type: Read/Write

Reset: 0x00000000

PMIC_LOCKEDBYBIT=LOCKBIT_D1

Bits	Name	Description
7 : 7	GANG_EN	0 = disable 1 = enable When enabled, this peripheral will write the same data that is written to the gang leader PID. Reads to the gang leader PID are ignored by this peripheral 0: GANG_EN_false 1: GANG_EN_true

Module Name: S2_PS
Base Address: 0x00011800
Register Quick Reference:

Register	Offset	Type
S2_PS_REVISION1	0x00011800	Read
S2_PS_REVISION2	0x00011801	Read
S2_PS_REVISION3	0x00011802	Read
S2_PS_REVISION4	0x00011803	Read
S2_PS_PERPH_TYPE	0x00011804	Read
S2_PS_PERPH_SUBTYPE	0x00011805	Read
S2_PS_VSET_LB	0x00011840	Read/Write
S2_PS_VSET_UB	0x00011841	Read/Write
S2_PS_SI_THRES_VSET_LB	0x000118BC	Read/Write
S2_PS_SI_THRES_VSET_UB	0x000118BD	Read/Write
S2_PS_GANG_CTL1	0x000118C0	Read/Write
S2_PS_GANG_CTL2	0x000118C1	Read/Write

S2_PS_REVISION1 | 0x00011800

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

S2_PS_REVISION2 | 0x00011801

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

S2_PS_REVISION3 | 0x00011802

Type: Read

Reset: 0x00000002

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

S2_PS_REVISION4 | 0x00011803

Type: Read

Reset: 0x00000000

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

S2_PS_PERPH_TYPE | 0x00011804

Type: Read

Reset: 0x0000001C

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	http://projects/sites/PM8941/Shared%20Documents/Systems/Registers/Type%20and%20SubType%20Enumeration%20Table.xlsx

S2_PS_PERPH_SUBTYPE | 0x00011805

Type: Read

Reset: 0x0000000C

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	http://projects/sites/PM8941/Shared%20Documents/Systems/Registers/Type%20and%20SubType%20Enumeration%20Table.xlsx

S2_PS_VSET_LB | 0x00011840

Type: Read/Write

Reset: 0x00000068

PMIC_LATCHED_WRITE=VSET_UB, PMIC_GANGED

Bits	Name	Description
7 : 0	VSET_LB	Regulator voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S2_PS_VSET_UB | 0x00011841

Type: Read/Write

Reset: 0x00000003

PMIC_GANGED

Bits	Name	Description
3 : 0	VSET_UB	Regulator voltage set point bit <11:8>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S2_PS_SI_THRES_VSET_LB | 0x000118BC

Type: Read/Write

Reset: 0x00000048

PMIC_LATCHED_WRITE=SI_THRES_VSET_UB, PMIC_GANGED

Bits	Name	Description
7 : 0	SI_THRES_VSET_LB	Skip integration threshold for low vset operation: voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV Default = 1.352V

S2_PS_SI_THRES_VSET_UB | 0x000118BD

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED

Bits	Name	Description
3 : 0	SI_THRES_VSET_UB	Skip integration threshold for low vset operation: voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV Default = 1.352V

S2_PS_GANG_CTL1 | 0x000118C0

Type: Read/Write

Reset: 0x00000017

PMIC_LOCKEDBYBIT=LOCKBIT_D1

Bits	Name	Description
7 : 0	GANG_LEADER_PID	When GANG_EN (GANG_CTL2[7]) is set, this peripheral will write the same data that is written to the gang leader Peripheral ID. Reads to the gang leader Peripheral ID are ignored by this peripheral. Ganged peripherals must reside within the same Slave ID

S2_PS_GANG_CTL2 | 0x000118C1

Type: Read/Write

Reset: 0x00000080

PMIC_LOCKEDBYBIT=LOCKBIT_D1

Bits	Name	Description
7 : 7	GANG_EN	0 = disable 1 = enable When enabled, this peripheral will write the same data that is written to the gang leader PID. Reads to the gang leader PID are ignored by this peripheral 0: GANG_EN_false 1: GANG_EN_true

Module Name: S2_FREQ

Base Address: 0x00011900

Register Quick Reference:

Register	Offset	Type
S2_FREQ_CLK_ENABLE	0x00011946	Read/Write
S2_FREQ_CLK_DIV	0x00011950	Read/Write
S2_FREQ_CLK_PHASE	0x00011951	Read/Write
S2_FREQ_GANG_CTL1	0x000119C0	Read/Write
S2_FREQ_GANG_CTL2	0x000119C1	Read/Write

S2_FREQ_CLK_ENABLE | 0x00011946

Type: Read/Write

Reset: 0x00000001

No description provided for this register.

Bits	Name	Description
0 : 0	FOLLOW_CLK_SX_REQ	0 = ignore smps_clk_req<X> 1 = clock is enabled when the clocks request is high smps_clk_req<X>='1' 0: FALLOW_CLK_REQ_DISABLED 1: FALLOW_CLK_REQ_ENABLED
7 : 7	EN_CLK_INT	0 = do not force the clock on 1 = enable the clock 0: FORCE_EN_DISABLED 1: FORCE_EN_ENABLED

S2_FREQ_CLK_DIV | 0x00011950

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED

Bits	Name	Description
3 : 0	CLK_DIV	clock_ frequency = 19.2MHz / (CLK_DIV + 1) FTS2 Buck supports 3.2, 4.8, 6.4 and 9.6 MHz HF2 Buck supports 1.6, 2.4, 2.74, 3.2, 3.8, 4.8, and 6.4 MHz CLK_DIV = 0 is not supported, it will generate 9.6 MHz 0: FREQ_9M6HZ_0 1: FREQ_9M6HZ 2: FREQ_6M4HZ 3: FREQ_4M8HZ 4: FREQ_3M8HZ 5: FREQ_3M2HZ 6: FREQ_2M7HZ 7: FREQ_2M4HZ 8: FREQ_2M1HZ 9: FREQ_1M9HZ 10: FREQ_1M7HZ 11: FREQ_1M6HZ 12: FREQ_1M5HZ 13: FREQ_1M4HZ 14: FREQ_1M3HZ 15: FREQ_1M2HZ

S2_FREQ_CLK_PHASE | 0x00011951

Type: Read/Write

Reset: 0x00000008

No description provided for this register.

Bits	Name	Description
3 : 0	CLK_PHASE	Distributed clock phase select: clock phase delay = clock period * (CLK_PHASE / 16)

S2_FREQ_GANG_CTL1 | 0x000119C0

Type: Read/Write

Reset: 0x00000017

No description provided for this register.

Bits	Name	Description
7 : 0	GANG_LEADER_PID	When GANG_EN (GANG_CTL2[7]) is set, this peripheral will write the same data that is written to the gang leader Peripheral ID. Reads to the gang leader Peripheral ID are ignored by this peripheral. Ganged peripherals must reside within the same Slave ID

S2_FREQ_GANG_CTL2 | 0x000119C1

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
7 : 7	GANG_EN	0 = disable 1 = enable When enabled, this peripheral will write the same data that is written to the gang leader PID. Reads to the gang leader PID are ignored by this peripheral 0: GANGING_DISABLED 1: GANGING_ENABLED

Module Name: S3_CTRL

Base Address: 0x00011A00

Register Quick Reference:

Register	Offset	Type
S3_CTRL_REVISION1	0x00011A00	Read
S3_CTRL_REVISION2	0x00011A01	Read
S3_CTRL_REVISION3	0x00011A02	Read
S3_CTRL_REVISION4	0x00011A03	Read
S3_CTRL_PERPH_TYPE	0x00011A04	Read
S3_CTRL_PERPH_SUBTYPE	0x00011A05	Read
S3_CTRL_STATUS1	0x00011A08	Read
S3_CTRL_INT_RT_STS	0x00011A10	Read
S3_CTRL_INT_SET_TYPE	0x00011A11	Read
S3_CTRL_INT_POLARITY_HIGH	0x00011A12	Read

Register	Offset	Type
S3_CTRL_INT_POLARITY_LOW	0x00011A13	Read/Write
S3_CTRL_INT_LATCHED_CLR	0x00011A14	Write
S3_CTRL_INT_EN_SET	0x00011A15	Read/Write
S3_CTRL_INT_EN_CLR	0x00011A16	Read/Write
S3_CTRL_INT_MID_SEL	0x00011A1A	Read/Write
S3_CTRL_INT_PRIORITY	0x00011A1B	Read/Write
S3_CTRL_INT_SELF_CLEARING	0x00011A1D	Read
S3_CTRL_ULS_VSET_LB	0x00011A39	Read/Write
S3_CTRL_ULS_VSET_UB	0x00011A3A	Read/Write
S3_CTRL_VSET_LB	0x00011A40	Read/Write
S3_CTRL_VSET_UB	0x00011A41	Read/Write
S3_CTRL_VSET_VALID_LB	0x00011A42	Read
S3_CTRL_VSET_VALID_UB	0x00011A43	Read
S3_CTRL_MODE_CTL2	0x00011A44	Read/Write
S3_CTRL_MODE_CTL1	0x00011A45	Read/Write
S3_CTRL_EN_CTL	0x00011A46	Read/Write
S3_CTRL_FOLLOW_HWEN	0x00011A47	Read/Write
S3_CTRL_FREQ_CTL	0x00011A50	Read/Write
S3_CTRL_OCP	0x00011A88	Read/Write
S3_CTRL_STATUS_DEB_CTL	0x00011A90	Read/Write
S3_CTRL_PHASE_ID	0x00011A95	Read/Write
S3_CTRL_PHASE_CNT_MAX	0x00011A96	Read/Write
S3_CTRL_PD_CTL	0x00011AA0	Read/Write
S3_CTRL_SI_THRES_VSET_LB_SHADOW	0x00011ABC	Read/Write
S3_CTRL_SI_THRES_VSET_UB_SHADOW	0x00011ABD	Read/Write
S3_CTRL_GANG_CTL1	0x00011AC0	Read/Write
S3_CTRL_GANG_CTL2	0x00011AC1	Read/Write

S3_CTRL_REVISION1 | 0x00011A00

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	Indicates a change in the HW which is not intended to impact SW compatibility.

S3_CTRL_REVISION2 | 0x00011A01

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	Indicates a change in the HW which will impact SW compatibility.

S3_CTRL_REVISION3 | 0x00011A02

Type: Read

Reset: 0x00000003

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	Indicates expanded functionality. Minor version adds functionality while being backward compatible for existing features.

S3_CTRL_REVISION4 | 0x00011A03

Type: Read

Reset: 0x00000000

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	Indicates different interface version. Major version changes are not backward compatible. There will be a new Programming Guide document per major revision

S3_CTRL_PERPH_TYPE | 0x00011A04

Type: Read

Reset: 0x0000001C

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	Peripheral Type

S3_CTRL_PERPH_SUBTYPE | 0x00011A05

Type: Read

Reset: 0x0000000B

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	Peripheral SubType 10: Voltage_Mode_PWM_Controller

S3_CTRL_STATUS1 | 0x00011A08

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates regulator mode state. 0x0: UNUSED 0x1: STATE==RM_STATE && RETENTION MODE (QUALIFIED)==TRUE 0x2: UNUSED 0x3: STATE==PWM STATE 0: UNUSED_is_0 1: RETENTION_is_1 2: UNUSED_is_2 3: PWM_is_3

Bits	Name	Description
2 : 2	STEPPER_DONE	Indicates voltage stepper is done or not 0: STEPPER_DONE_false 1: STEPPER_DONE_true
3 : 3	PS_TRUE	Indicates whether buck is pulse skipping or not 0: PS_TRUE_false 1: PS_TRUE_true
4 : 4	RETENTION_QUAL	Indicates that retention mode is qualified versus the load current. 0x0: false 0x1: true 0: RM_QUAL_false 1: RM_QUAL_true
5 : 5	VREG_OCP	Indicates if over current or another fault has been detected/latched. 0: NO_FAULT 1: FAULT
6 : 6	VREG_ERROR	Indicates that regulator is not stepping and has fallen below VREG_OK comparator threshold 0 = Regulator is okay; no error condition 0: VREG_ERR_false 1: VREG_ERR_true
7 : 7	VREG_READY	Indicates if VREG is ready or not due to stepping in progress or below analog debounced VREG_OK comp 0: VREG_READY_false 1: VREG_READY_true

S3_CTRL_INT_RT_STS | 0x00011A10

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

S3_CTRL_INT_SET_TYPE | 0x00011A11

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

S3_CTRL_INT_POLARITY_HIGH | 0x00011A12

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

S3_CTRL_INT_POLARITY_LOW | 0x00011A13

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

S3_CTRL_INT_LATCHED_CLR | 0x00011A14

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

S3_CTRL_INT_EN_SET | 0x00011A15

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

S3_CTRL_INT_EN_CLR | 0x00011A16

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

S3_CTRL_INT_MID_SEL | 0x00011A1A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

S3_CTRL_INT_PRIORITY | 0x00011A1B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

S3_CTRL_INT_SELF_CLEARING | 0x00011A1D

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

S3_CTRL_ULS_VSET_LB | 0x00011A39

Type: Read/Write

Reset: 0x00000074

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	Vreg upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x68 and 0x69). Vuls_vset = 2-byte word (dec) x 1mV

S3_CTRL_ULS_VSET_UB | 0x00011A3A

Type: Read/Write

Reset: 0x00000005

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	Vreg upper limit stop-voltage set point bits<3:0>. 2-Byte Word (0x68 and 0x69). Vuls_vset = 2-byte word (dec) x 1mV. Maximum combined supported value is 0x0574 which translates to 1.396mV. Beyond that value, it will be clamped to 0x0574. The default value is already set to the maximum allowed value.

S3_CTRL_VSET_LB | 0x00011A40

Type: Read/Write

Reset: 0x00000068

PMIC_LATCHED_WRITE=VSET_UB, PMIC_GANGED

Bits	Name	Description
7 : 0	VSET_LB	Regulator voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S3_CTRL_VSET_UB | 0x00011A41

Type: Read/Write

Reset: 0x00000003

PMIC_GANGED

Bits	Name	Description
3 : 0	VSET_UB	Regulator voltage set point bit <11:8>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S3_CTRL_VSET_VALID_LB | 0x00011A42

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	VSET_VALID_LB	Lower byte readback for the valid output voltage setpoint value. This register shows the value round up effect of the set voltage since buck doesn't support 1mV resolution.,,,'

S3_CTRL_VSET_VALID_UB | 0x00011A43

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Upper byte readback for the valid output voltage setpoint value. This register shows the value round up effect of the set voltage since buck doesn't support 1mV resolution.,,,'

S3_CTRL_MODE_CTL2 | 0x00011A44

Type: Read/Write

Reset: 0x00000007

Define Buck Mode Transitions PMIC_GANGED, PMIC_SYNC=d_clk_19p2:d_perph_rb

Bits	Name	Description
2 : 0	MODE_SECONDARY	Controls the mode that buck operates: FORCED_NPM_7 = 7, (DEFAULT) RESERVED,(PRODUCES FORCED_NPM for FTS510)=6, RM_2 =5, RESERVED (PRODUCES FORCED_NPM for FTS510)= 4, RM_1=3, FORCED_NPM_2=2, FORCED_NPM_1=1, FORCED_NPM_0=0 0: FORCED_NPM_0 1: FORCED_NPM_1 2: FORCED_NPM_2 3: RM_1 4: RESERVED_1 5: RM_2 6: RESERVED_2 7: FORCED_NPM_7

S3_CTRL_MODE_CTL1 | 0x00011A45

Type: Read/Write

Reset: 0x00000007

Define Buck Mode Transitions PMIC_GANGED, PMIC_SYNC=d_clk_19p2:d_perph_rb

Bits	Name	Description
2 : 0	MODE_PRIMARY	Controls the mode that buck operates: FORCED_NPM_7 = 7,(DEFAULT) RESERVED,(PRODUCES FORCED_NPM for FTS510)=6, RM_2 =5, RESERVED (PRODUCES FORCED_NPM for FTS510)= 4, RM_1=3, FORCED_NPM_2=2, FORCED_NPM_1=1, FORCED_NPM_0=0 0: FORCED_NPM_0 1: FORCED_NPM_1 2: FORCED_NPM_2 3: RM_1 4: RESERVED_1 5: RM_2 6: RESERVED_2 7: FORCED_NPM_7

S3_CTRL_EN_CTL | 0x00011A46

Type: Read/Write

Reset: 0x00000000

PMIC_GANGED

Bits	Name	Description
7 : 7	PERPH_EN	Control signal to enable the regulator 0: BUCK_ENABLE_false 1: BUCK_ENABLE_true

S3_CTRL_FOLLOW_HWEN | 0x00011A47

Type: Read/Write

Reset: 0x00000000

PMIC_GANGED

Bits	Name	Description
0 : 0	EN_FOLLOW_HWEN0	Enables buck when HWEN0 is asserted 0: EN_FOLLOW_HWEN0_false 1: EN_FOLLOW_HWEN0_true

Bits	Name	Description
1 : 1	EN_FOLLOW_HWEN1	Enables buck when HWEN1 is asserted 0: EN_FOLLOW_HWEN1_false 1: EN_FOLLOW_HWEN1_true
2 : 2	EN_FOLLOW_HWEN2	Enables buck when HWEN2 is asserted 0: EN_FOLLOW_HWEN2_false 1: EN_FOLLOW_HWEN2_true
3 : 3	EN_FOLLOW_PMIC_AWAKE	Enables buck when PMIC_AWAKE is asserted 0: EN_FOLLOW_PMIC_AWAKE_false 1: EN_FOLLOW_PMIC_AWAKE_true
4 : 4	FOLLOW_HWEN0_CFG	If this bit is set low then HWEN0 does NOT control Buck mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN0_CFG_false 1: FOLLOW_HWEN0_CFG_true
5 : 5	FOLLOW_HWEN1_CFG	If this bit is set low then HWEN1 does NOT control Buck mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN1_CFG_false 1: FOLLOW_HWEN1_CFG_true
6 : 6	FOLLOW_HWEN2_CFG	If this bit is set low then HWEN2 does NOT control Buck mode. If this bit is set high then: When HWEN2=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN2_CFG_false 1: FOLLOW_HWEN2_CFG_true
7 : 7	FOLLOW_AWAKE_CFG	If this bit is set low then PMIC_AWAKE does NOT control Buck mode. If this bit is set high then: When PMIC_AWAKE=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_AWAKE_CFG_false 1: FOLLOW_AWAKE_CFG_true

S3_CTRL_FREQ_CTL | 0x00011A50

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED

Bits	Name	Description
3 : 0	FREQ_CTL	Clock frequency = 19.2MHz / (CLK_DIV + 1) 2: Fs_6p4MHz 3: Fs_4p8MHz 4: Fs_3p84MHz 5: Fs_3p2MHz 6: Fs_2p74MHz 7: Fs_2p4MHz 8: Fs_2p13MHz 9: Fs_1p92MHz 10: Fs_1p74MHz 11: Fs_1p6MHz

S3_CTRL_OCP | 0x00011A88

Type: Read/Write

Reset: 0x00000068

PMIC_GANGED, PMIC_SYNC=d_clk_lfrc:d_perph_rb

Bits	Name	Description
1 : 1	OCP_STATUS_CLEAR	Toggle 1->0 to clear the latched OCP status bit. 0: OCP_STATUS_CLEAR_false 1: OCP_STATUS_CLEAR_true
2 : 2	OCP_GLOBAL_BROADCAST_EN	Determines if the Buck notifies the PON module that an OCP event has occurred. PON module may use this info to shutdown the entire PMIC. 1 = BUCK notifies PON when OCP event occurs 0 = BUCK does not notify PON when OCP even occurs (DEFAULT) 0: OCP_GLOBAL_BROADCAST_EN_false 1: OCP_GLOBAL_BROADCAST_EN_true
4 : 3	INZERO_WAIT_OCP_TIMER	Controls wait time for inzero trigger during current foldback. If inzero wait timer expires then an inzero is forced. OCP timer controls the time at which buck shuts down if OCP_LATCH_EN=1 0: INZERO_WAIT_40us_OCP_TIMER_320us 1: INZERO_WAIT_80us_OCP_TIMER_640us 2: INZERO_WAIT_160us_OCP_TIMER_1280_us 3: INZERO_WAIT_160us_OCP_TIMER_1280us
5 : 5	VREG_FOLDBACK_EN	This bit is used to enable propagating foldback flag to PS. 0: FORCE_FOLDBACK_FALSE 1: FORCE_FOLDBACK_TRUE

Bits	Name	Description
6 : 6	OCP_LATCH_EN	Controls whether OCP event causes buck to shut down when OCP timer expires 0x0: ocp latching is not enabled. Buck will not shut off due to the OCP event. 0x1: ocp latching is enabled. Buck will SHUT OFF due to the OCP event. (DEFAULT) 0: OCP_LATCH_EN_false 1: OCP_LATCH_EN_true

S3_CTRL_STATUS_DEB_CTL | 0x00011A90

Type: Read/Write

Reset: 0x00000000

PMIC_GANGED, PMIC_SYNC=d_clk_4p8m:d_perph_rb

Bits	Name	Description
1 : 0	VREG_ERROR_DEB	Configures the debounce on falling edge of VREG_OK comparator output for generating VREG_ERROR. 00 = 1.4us -1.6us 01 = 3.0us -3.2us 10= 6.5us -6.7us 11= 9.8us - 10us 0: ERROR_DEB_0 1: ERROR_DEB_1 2: ERROR_DEB_2 3: ERROR_DEB_3
5 : 4	VREG_READY_DEB	Configures the debounce on rising edge of VREG_OK comparator output for generating VREG_READY. 00 = 1.4us -1.6us 01 = 3.0us -3.2us 10= 6.5us -6.7us 11= 9.8us - 10us 0: READY_DEB_0 1: READY_DEB_1 2: READY_DEB_2 3: READY_DEB_3

S3_CTRL_PHASE_ID | 0x00011A95

Type: Read/Write

Reset: 0x00000000

PMIC_LOCKEDBYBIT=LOCKBIT_D1, PMIC_SYNC=d_clk_19p2:d_perph_rb

Bits	Name	Description
3 : 0	PHASE_ID	Unique phase identifier. If less than or equal to SLAVE_COUNT, phase is active. If greater than SLAVE_COUNT, phase is inactive. 0: Phase_Number_1 1: Phase_Number_2 2: Phase_Number_3 3: Phase_Number_4

S3_CTRL_PHASE_CNT_MAX | 0x00011A96

Type: Read/Write

Reset: 0x00000000

PMIC_GANGED, PMIC_SYNC=d_clk_19p2:d_perph_rb

Bits	Name	Description
3 : 0	PHASE_CNT_MAX	Sets the maximum number of slave phases we can use.

S3_CTRL_PD_CTL | 0x00011AA0

Type: Read/Write

Reset: 0x00000088

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Controls state of lead pull down. (Does not depend on the fsm state). 0: LEAK_PD_EN_false 1: LEAK_PD_EN_true
6 : 6	WEAK_PD_EN	Controls state of weak pull down. (See glue logic for mode fsm state dependency go/fts510). 0: WEAK_PD_ENABLE_false 1: WEAK_PD_ENABLE_true
7 : 7	STRONG_PD_EN	Controls state of strong pull down. (See glue logic for mode fsm state dependency go/fts510). 0: STRONG_PD_ENABLE_false 1: STRONG_PD_ENABLE_true

S3_CTRL_SI_THRES_VSET_LB_SHADOW | 0x00011ABC

Type: Read/Write

Reset: 0x00000048

PMIC_LATCHED_WRITE=SI_THRES_VSET_UB_SHADOW, PMIC_GANGED, PMIC_DOC_ONLY

Bits	Name	Description
7 : 0	SI_THRES_VSET_LB	Skip integration threshold for low vset operation: voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV Default = 1.352V

S3_CTRL_SI_THRES_VSET_UB_SHADOW | 0x00011ABD

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED, PMIC_DOC_ONLY

Bits	Name	Description
3 : 0	SI_THRES_VSET_UB	Skip integration threshold for low vset operation: voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV Default = 1.352V

S3_CTRL_GANG_CTL1 | 0x00011AC0

Type: Read/Write

Reset: 0x0000001A

PMIC_LOCKEDBYBIT=LOCKBIT_D1

Bits	Name	Description
7 : 0	GANG_LEADER_PID	When GANG_EN (GANG_CTL2[7]) is set, this peripheral will write the same data that is written to the gang leader Peripheral ID. Reads to the gang leader Peripheral ID are ignored by this peripheral. Ganged peripherals must reside within the same Slave ID

S3_CTRL_GANG_CTL2 | 0x00011AC1

Type: Read/Write

Reset: 0x00000000

PMIC_LOCKEDBYBIT=LOCKBIT_D1

Bits	Name	Description
7 : 7	GANG_EN	0 = disable 1 = enable When enabled, this peripheral will write the same data that is written to the gang leader PID. Reads to the gang leader PID are ignored by this peripheral 0: GANG_EN_false 1: GANG_EN_true

Module Name: S3_PS

Base Address: 0x00011B00

Register Quick Reference:

Register	Offset	Type
S3_PS_REVISION1	0x00011B00	Read
S3_PS_REVISION2	0x00011B01	Read
S3_PS_REVISION3	0x00011B02	Read
S3_PS_REVISION4	0x00011B03	Read
S3_PS_PERPH_TYPE	0x00011B04	Read
S3_PS_PERPH_SUBTYPE	0x00011B05	Read
S3_PS_VSET_LB	0x00011B40	Read/Write
S3_PS_VSET_UB	0x00011B41	Read/Write
S3_PS_SI_THRES_VSET_LB	0x00011BBC	Read/Write
S3_PS_SI_THRES_VSET_UB	0x00011BBD	Read/Write
S3_PS_GANG_CTL1	0x00011BC0	Read/Write
S3_PS_GANG_CTL2	0x00011BC1	Read/Write

S3_PS_REVISION1 | 0x00011B00

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

S3_PS_REVISION2 | 0x00011B01

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

S3_PS_REVISION3 | 0x00011B02

Type: Read

Reset: 0x00000002

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

S3_PS_REVISION4 | 0x00011B03

Type: Read

Reset: 0x00000000

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

S3_PS_PERPH_TYPE | 0x00011B04

Type: Read

Reset: 0x0000001C

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	http://projects/sites/PM8941/Shared%20Documents/Systems/Registers/Type%20and%20SubType%20Enumeration%20Table.xlsx

S3_PS_PERPH_SUBTYPE | 0x00011B05

Type: Read

Reset: 0x0000000C

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	http://projects/sites/PM8941/Shared%20Documents/Systems/Registers/Type%20and%20SubType%20Enumeration%20Table.xlsx

S3_PS_VSET_LB | 0x00011B40

Type: Read/Write

Reset: 0x00000068

PMIC_LATCHED_WRITE=VSET_UB, PMIC_GANGED

Bits	Name	Description
7 : 0	VSET_LB	Regulator voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S3_PS_VSET_UB | 0x00011B41

Type: Read/Write

Reset: 0x00000003

PMIC_GANGED

Bits	Name	Description
3 : 0	VSET_UB	Regulator voltage set point bit <11:8>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S3_PS_SI_THRES_VSET_LB | 0x00011BBC

Type: Read/Write

Reset: 0x00000048

PMIC_LATCHED_WRITE=SI_THRES_VSET_UB, PMIC_GANGED

Bits	Name	Description
7 : 0	SI_THRES_VSET_LB	Skip integration threshold for low vset operation: voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV Default = 1.352V

S3_PS_SI_THRES_VSET_UB | 0x00011BBD

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED

Bits	Name	Description
3 : 0	SI_THRES_VSET_UB	Skip integration threshold for low vset operation: voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV Default = 1.352V

S3_PS_GANG_CTL1 | 0x00011BC0

Type: Read/Write

Reset: 0x0000001A

PMIC_LOCKEDBYBIT=LOCKBIT_D1

Bits	Name	Description
7 : 0	GANG_LEADER_PID	When GANG_EN (GANG_CTL2[7]) is set, this peripheral will write the same data that is written to the gang leader Peripheral ID. Reads to the gang leader Peripheral ID are ignored by this peripheral. Ganged peripherals must reside within the same Slave ID

S3_PS_GANG_CTL2 | 0x00011BC1

Type: Read/Write

Reset: 0x00000080

PMIC_LOCKEDBYBIT=LOCKBIT_D1

Bits	Name	Description
7 : 7	GANG_EN	0 = disable 1 = enable When enabled, this peripheral will write the same data that is written to the gang leader PID. Reads to the gang leader PID are ignored by this peripheral 0: GANG_EN_false 1: GANG_EN_true

Module Name: S3_FREQ

Base Address: 0x00011C00

Register Quick Reference:

Register	Offset	Type
S3_FREQ_CLK_ENABLE	0x00011C46	Read/Write
S3_FREQ_CLK_DIV	0x00011C50	Read/Write
S3_FREQ_CLK_PHASE	0x00011C51	Read/Write
S3_FREQ_GANG_CTL1	0x00011CC0	Read/Write
S3_FREQ_GANG_CTL2	0x00011CC1	Read/Write

S3_FREQ_CLK_ENABLE | 0x00011C46

Type: Read/Write

Reset: 0x00000001

No description provided for this register.

Bits	Name	Description
0 : 0	FOLLOW_CLK_SX_REQ	0 = ignore smps_clk_req<X> 1 = clock is enabled when the clocks request is high smps_clk_req<X>='1' 0: FALLOW_CLK_REQ_DISABLED 1: FALLOW_CLK_REQ_ENABLED
7 : 7	EN_CLK_INT	0 = do not force the clock on 1 = enable the clock 0: FORCE_EN_DISABLED 1: FORCE_EN_ENABLED

S3_FREQ_CLK_DIV | 0x00011C50

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED

Bits	Name	Description
3 : 0	CLK_DIV	clock_ frequency = 19.2MHz / (CLK_DIV + 1) FTS2 Buck supports 3.2, 4.8, 6.4 and 9.6 MHz HF2 Buck supports 1.6, 2.4, 2.74, 3.2, 3.8, 4.8, and 6.4 MHz CLK_DIV = 0 is not supported, it will generate 9.6 MHz 0: FREQ_9M6HZ_0 1: FREQ_9M6HZ 2: FREQ_6M4HZ 3: FREQ_4M8HZ 4: FREQ_3M8HZ 5: FREQ_3M2HZ 6: FREQ_2M7HZ 7: FREQ_2M4HZ 8: FREQ_2M1HZ 9: FREQ_1M9HZ 10: FREQ_1M7HZ 11: FREQ_1M6HZ 12: FREQ_1M5HZ 13: FREQ_1M4HZ 14: FREQ_1M3HZ 15: FREQ_1M2HZ

S3_FREQ_CLK_PHASE | 0x00011C51

Type: Read/Write

Reset: 0x00000004

No description provided for this register.

Bits	Name	Description
3 : 0	CLK_PHASE	Distributed clock phase select: clock phase delay = clock period * (CLK_PHASE / 16)

S3_FREQ_GANG_CTL1 | 0x00011CC0

Type: Read/Write

Reset: 0x0000001A

No description provided for this register.

Bits	Name	Description
7 : 0	GANG_LEADER_PID	When GANG_EN (GANG_CTL2[7]) is set, this peripheral will write the same data that is written to the gang leader Peripheral ID. Reads to the gang leader Peripheral ID are ignored by this peripheral. Ganged peripherals must reside within the same Slave ID

S3_FREQ_GANG_CTL2 | 0x00011CC1

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
7 : 7	GANG_EN	0 = disable 1 = enable When enabled, this peripheral will write the same data that is written to the gang leader PID. Reads to the gang leader PID are ignored by this peripheral 0: GANGING_DISABLED 1: GANGING_ENABLED

Module Name: S4_CTRL

Base Address: 0x00011D00

Register Quick Reference:

Register	Offset	Type
S4_CTRL_REVISION1	0x00011D00	Read
S4_CTRL_REVISION2	0x00011D01	Read
S4_CTRL_REVISION3	0x00011D02	Read
S4_CTRL_REVISION4	0x00011D03	Read
S4_CTRL_PERPH_TYPE	0x00011D04	Read
S4_CTRL_PERPH_SUBTYPE	0x00011D05	Read
S4_CTRL_STATUS1	0x00011D08	Read
S4_CTRL_INT_RT_STS	0x00011D10	Read
S4_CTRL_INT_SET_TYPE	0x00011D11	Read
S4_CTRL_INT_POLARITY_HIGH	0x00011D12	Read
S4_CTRL_INT_POLARITY_LOW	0x00011D13	Read/Write
S4_CTRL_INT_LATCHED_CLR	0x00011D14	Write
S4_CTRL_INT_EN_SET	0x00011D15	Read/Write
S4_CTRL_INT_EN_CLR	0x00011D16	Read/Write
S4_CTRL_INT_MID_SEL	0x00011D1A	Read/Write
S4_CTRL_INT_PRIORITY	0x00011D1B	Read/Write
S4_CTRL_INT_SELF_CLEARING	0x00011D1D	Read
S4_CTRL_ULS_VSET_LB	0x00011D39	Read/Write
S4_CTRL_ULS_VSET_UB	0x00011D3A	Read/Write
S4_CTRL_VSET_LB	0x00011D40	Read/Write
S4_CTRL_VSET_UB	0x00011D41	Read/Write
S4_CTRL_VSET_VALID_LB	0x00011D42	Read
S4_CTRL_VSET_VALID_UB	0x00011D43	Read
S4_CTRL_MODE_CTL2	0x00011D44	Read/Write
S4_CTRL_MODE_CTL1	0x00011D45	Read/Write
S4_CTRL_EN_CTL	0x00011D46	Read/Write
S4_CTRL_FOLLOW_HWEN	0x00011D47	Read/Write
S4_CTRL_FREQ_CTL	0x00011D50	Read/Write
S4_CTRL_OCP	0x00011D88	Read/Write

Register	Offset	Type
S4_CTRL_STATUS_DEB_CTL	0x00011D90	Read/Write
S4_CTRL_PHASE_ID	0x00011D95	Read/Write
S4_CTRL_PHASE_CNT_MAX	0x00011D96	Read/Write
S4_CTRL_PD_CTL	0x00011DA0	Read/Write
S4_CTRL_SI_THRES_VSET_LB_SHADOW	0x00011DBC	Read/Write
S4_CTRL_SI_THRES_VSET_UB_SHADOW	0x00011DBD	Read/Write
S4_CTRL_GANG_CTL1	0x00011DC0	Read/Write
S4_CTRL_GANG_CTL2	0x00011DC1	Read/Write

S4_CTRL_REVISION1 | 0x00011D00

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	Indicates a change in the HW which is not intended to impact SW compatibility.

S4_CTRL_REVISION2 | 0x00011D01

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	Indicates a change in the HW which will impact SW compatibility.

S4_CTRL_REVISION3 | 0x00011D02

Type: Read

Reset: 0x00000003

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	Indicates expanded functionality. Minor version adds functionality while being backward compatible for existing features.

S4_CTRL_REVISION4 | 0x00011D03

Type: Read

Reset: 0x00000000

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	Indicates different interface version. Major version changes are not backward compatible. There will be a new Programming Guide document per major revision

S4_CTRL_PERPH_TYPE | 0x00011D04

Type: Read

Reset: 0x0000001C

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	Peripheral Type

S4_CTRL_PERPH_SUBTYPE | 0x00011D05

Type: Read

Reset: 0x0000000B

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	Peripheral SubType 10: Voltage_Mode_PWM_Controller

S4_CTRL_STATUS1 | 0x00011D08

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates regulator mode state. 0x0: UNUSED 0x1: STATE==RM_STATE && RETENTION MODE (QUALIFIED)==TRUE 0x2: UNUSED 0x3: STATE==PWM STATE 0: UNUSED_is_0 1: RETENTION_is_1 2: UNUSED_is_2 3: PWM_is_3
2 : 2	STEPPER_DONE	Indicates voltage stepper is done or not 0: STEPPER_DONE_false 1: STEPPER_DONE_true
3 : 3	PS_TRUE	Indicates whether buck is pulse skipping or not 0: PS_TRUE_false 1: PS_TRUE_true
4 : 4	RETENTION_QUAL	Indicates that retention mode is qualified versus the load current. 0x0: false 0x1: true 0: RM_QUAL_false 1: RM_QUAL_true
5 : 5	VREG_OCP	Indicates if over current or another fault has been detected/latched. 0: NO_FAULT 1: FAULT
6 : 6	VREG_ERROR	Indicates that regulator is not stepping and has fallen below VREG_OK comparator threshold 0 = Regulator is okay; no error condition 0: VREG_ERR_false 1: VREG_ERR_true
7 : 7	VREG_READY	Indicates if VREG is ready or not due to stepping in progress or below analog debounced VREG_OK comp 0: VREG_READY_false 1: VREG_READY_true

S4_CTRL_INT_RT_STS | **0x00011D10**

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

S4_CTRL_INT_SET_TYPE | 0x00011D11

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

S4_CTRL_INT_POLARITY_HIGH | 0x00011D12

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

S4_CTRL_INT_POLARITY_LOW | 0x00011D13

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

S4_CTRL_INT_LATCHED_CLR | 0x00011D14

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

S4_CTRL_INT_EN_SET | 0x00011D15

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

S4_CTRL_INT_EN_CLR | 0x00011D16

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

S4_CTRL_INT_MID_SEL | 0x00011D1A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

S4_CTRL_INT_PRIORITY **| 0x00011D1B**

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

S4_CTRL_INT_SELF_CLEARING | 0x00011D1D

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

S4_CTRL_ULS_VSET_LB | 0x00011D39

Type: Read/Write

Reset: 0x00000074

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	Vreg upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x68 and 0x69). Vuls_vset = 2-byte word (dec) x 1mV

S4_CTRL_ULS_VSET_UB | 0x00011D3A

Type: Read/Write

Reset: 0x00000005

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	Vreg upper limit stop-voltage set point bits<3:0>. 2-Byte Word (0x68 and 0x69). Vuls_vset = 2-byte word (dec) x 1mV. Maximum combined supported value is 0x0574 which translates to 1.396mV. Beyond that value, it will be clamped to 0x0574. The default value is already set to the maximum allowed value.

S4_CTRL_VSET_LB | 0x00011D40

Type: Read/Write

Reset: 0x00000068

PMIC_LATCHED_WRITE=VSET_UB, PMIC_GANGED

Bits	Name	Description
7 : 0	VSET_LB	Regulator voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S4_CTRL_VSET_UB | 0x00011D41

Type: Read/Write

Reset: 0x00000003

PMIC_GANGED

Bits	Name	Description
3 : 0	VSET_UB	Regulator voltage set point bit <11:8>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S4_CTRL_VSET_VALID_LB | 0x00011D42

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	VSET_VALID_LB	Lower byte readback for the valid output voltage setpoint value. This register shows the value round up effect of the set voltage since buck doesn't support 1mV resolution.,,,'

S4_CTRL_VSET_VALID_UB | 0x00011D43

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Upper byte readback for the valid output voltage setpoint value. This register shows the value round up effect of the set voltage since buck doesn't support 1mV resolution.,,,'

S4_CTRL_MODE_CTL2 | 0x00011D44

Type: Read/Write

Reset: 0x00000007

Define Buck Mode Transitions PMIC_GANGED, PMIC_SYNC=d_clk_19p2:d_perph_rb

Bits	Name	Description
2 : 0	MODE_SECONDARY	Controls the mode that buck operates: FORCED_NPM_7 = 7, (DEFAULT) RESERVED,(PRODUCES FORCED_NPM for FTS510)=6, RM_2 =5, RESERVED (PRODUCES FORCED_NPM for FTS510)= 4, RM_1=3, FORCED_NPM_2=2, FORCED_NPM_1=1, FORCED_NPM_0=0 0: FORCED_NPM_0 1: FORCED_NPM_1 2: FORCED_NPM_2 3: RM_1 4: RESERVED_1 5: RM_2 6: RESERVED_2 7: FORCED_NPM_7

S4_CTRL_MODE_CTL1 | 0x00011D45

Type: Read/Write

Reset: 0x00000007

Define Buck Mode Transitions PMIC_GANGED, PMIC_SYNC=d_clk_19p2:d_perph_rb

Bits	Name	Description
2 : 0	MODE_PRIMARY	Controls the mode that buck operates: FORCED_NPM_7 = 7,(DEFAULT) RESERVED,(PRODUCES FORCED_NPM for FTS510)=6, RM_2 =5, RESERVED (PRODUCES FORCED_NPM for FTS510)= 4, RM_1=3, FORCED_NPM_2=2, FORCED_NPM_1=1, FORCED_NPM_0=0 0: FORCED_NPM_0 1: FORCED_NPM_1 2: FORCED_NPM_2 3: RM_1 4: RESERVED_1 5: RM_2 6: RESERVED_2 7: FORCED_NPM_7

S4_CTRL_EN_CTL | 0x00011D46

Type: Read/Write

Reset: 0x00000000

PMIC_GANGED

Bits	Name	Description
7 : 7	PERPH_EN	Control signal to enable the regulator 0: BUCK_ENABLE_false 1: BUCK_ENABLE_true

S4_CTRL_FOLLOW_HWEN | 0x00011D47

Type: Read/Write

Reset: 0x00000000

PMIC_GANGED

Bits	Name	Description
0 : 0	EN_FOLLOW_HWEN0	Enables buck when HWEN0 is asserted 0: EN_FOLLOW_HWEN0_false 1: EN_FOLLOW_HWEN0_true
1 : 1	EN_FOLLOW_HWEN1	Enables buck when HWEN1 is asserted 0: EN_FOLLOW_HWEN1_false 1: EN_FOLLOW_HWEN1_true
2 : 2	EN_FOLLOW_HWEN2	Enables buck when HWEN2 is asserted 0: EN_FOLLOW_HWEN2_false 1: EN_FOLLOW_HWEN2_true
3 : 3	EN_FOLLOW_PMIC_AWAKE	Enables buck when PMIC_AWAKE is asserted 0: EN_FOLLOW_PMIC_AWAKE_false 1: EN_FOLLOW_PMIC_AWAKE_true
4 : 4	FOLLOW_HWEN0_CFG	If this bit is set low then HWEN0 does NOT control Buck mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN0_CFG_false 1: FOLLOW_HWEN0_CFG_true

Bits	Name	Description
5 : 5	FOLLOW_HWEN1_CFG	If this bit is set low then HWEN1 does NOT control Buck mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN1_CFG_false 1: FOLLOW_HWEN1_CFG_true
6 : 6	FOLLOW_HWEN2_CFG	If this bit is set low then HWEN2 does NOT control Buck mode. If this bit is set high then: When HWEN2=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN2_CFG_false 1: FOLLOW_HWEN2_CFG_true
7 : 7	FOLLOW_AWAKE_CFG	If this bit is set low then PMIC_AWAKE does NOT control Buck mode. If this bit is set high then: When PMIC_AWAKE=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_AWAKE_CFG_false 1: FOLLOW_AWAKE_CFG_true

S4_CTRL_FREQ_CTL | 0x00011D50

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED

Bits	Name	Description
3 : 0	FREQ_CTL	Clock frequency = 19.2MHz / (CLK_DIV + 1) 2: Fs_6p4MHz 3: Fs_4p8MHz 4: Fs_3p84MHz 5: Fs_3p2MHz 6: Fs_2p74MHz 7: Fs_2p4MHz 8: Fs_2p13MHz 9: Fs_1p92MHz 10: Fs_1p74MHz 11: Fs_1p6MHz

S4_CTRL_OCP | 0x00011D88

Type: Read/Write

Reset: 0x00000068

PMIC_GANGED, PMIC_SYNC=d_clk_lfrc:d_perph_rb

Bits	Name	Description
1 : 1	OCP_STATUS_CLEAR	Toggle 1->0 to clear the latched OCP status bit. 0: OCP_STATUS_CLEAR_false 1: OCP_STATUS_CLEAR_true
2 : 2	OCP_GLOBAL_BROADCAST_EN	Determines if the Buck notifies the PON module that an OCP event has occurred. PON module may use this info to shutdown the entire PMIC. 1 = BUCK notifies PON when OCP event occurs 0 = BUCK does not notify PON when OCP even occurs (DEFAULT) 0: OCP_GLOBAL_BROADCAST_EN_false 1: OCP_GLOBAL_BROADCAST_EN_true
4 : 3	INZERO_WAIT_OCP_TIMER	Controls wait time for inzero trigger during current foldback. If inzero wait timer expires then an inzero is forced. OCP timer controls the time at which buck shuts down if OCP_LATCH_EN=1 0: INZERO_WAIT_40us_OCP_TIMER_320us 1: INZERO_WAIT_80us_OCP_TIMER_640us 2: INZERO_WAIT_160us_OCP_TIMER_1280_us 3: INZERO_WAIT_160us_OCP_TIMER_1280us
5 : 5	VREG_FOLDBACK_EN	This bit is used to enable propagating foldback flag to PS. 0: FORCE_FOLDBACK_FALSE 1: FORCE_FOLDBACK_TRUE
6 : 6	OCP_LATCH_EN	Controls whether OCP event causes buck to shut down when OCP timer expires 0x0: ocp latching is not enabled. Buck will not shut off due to the OCP event. 0x1: ocp latching is enabled. Buck will SHUT OFF due to the OCP event. (DEFAULT) 0: OCP_LATCH_EN_false 1: OCP_LATCH_EN_true

S4_CTRL_STATUS_DEB_CTL | 0x00011D90

Type: Read/Write

Reset: 0x00000000

PMIC_GANGED, PMIC_SYNC=d_clk_4p8m:d_perph_rb

Bits	Name	Description
1 : 0	VREG_ERROR_DEB	Configures the debounce on falling edge of VREG_OK comparator output for generating VREG_ERROR. 00 = 1.4us -1.6us 01 = 3.0us -3.2us 10= 6.5us -6.7us 11= 9.8us - 10us 0: ERROR_DEB_0 1: ERROR_DEB_1 2: ERROR_DEB_2 3: ERROR_DEB_3
5 : 4	VREG_READY_DEB	Configures the debounce on rising edge of VREG_OK comparator output for generating VREG_READY. 00 = 1.4us -1.6us 01 = 3.0us -3.2us 10= 6.5us -6.7us 11= 9.8us - 10us 0: READY_DEB_0 1: READY_DEB_1 2: READY_DEB_2 3: READY_DEB_3

S4_CTRL_PHASE_ID | 0x00011D95

Type: Read/Write

Reset: 0x00000000

PMIC_LOCKEDBYBIT=LOCKBIT_D1, PMIC_SYNC=d_clk_19p2:d_perph_rb

Bits	Name	Description
3 : 0	PHASE_ID	Unique phase identifier. If less than or equal to SLAVE_COUNT, phase is active. If greater than SLAVE_COUNT, phase is inactive. 0: Phase_Number_1 1: Phase_Number_2 2: Phase_Number_3 3: Phase_Number_4

S4_CTRL_PHASE_CNT_MAX | 0x00011D96

Type: Read/Write

Reset: 0x00000000

PMIC_GANGED, PMIC_SYNC=d_clk_19p2:d_perph_rb

Bits	Name	Description
3 : 0	PHASE_CNT_MAX	Sets the maximum number of slave phases we can use.

S4_CTRL_PD_CTL | 0x00011DA0

Type: Read/Write

Reset: 0x00000088

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Controls state of lead pull down. (Does not depend on the fsm state). 0: LEAK_PD_EN_false 1: LEAK_PD_EN_true
6 : 6	WEAK_PD_EN	Controls state of weak pull down. (See glue logic for mode fsm state dependency go/fts510). 0: WEAK_PD_ENABLE_false 1: WEAK_PD_ENABLE_true
7 : 7	STRONG_PD_EN	Controls state of strong pull down. (See glue logic for mode fsm state dependency go/fts510). 0: STRONG_PD_ENABLE_false 1: STRONG_PD_ENABLE_true

S4_CTRL_SI_THRES_VSET_LB_SHADOW | 0x00011DBC

Type: Read/Write

Reset: 0x00000048

PMIC_LATCHED_WRITE=SI_THRES_VSET_UB_SHADOW, PMIC_GANGED, PMIC_DOC_ONLY

Bits	Name	Description
7 : 0	SI_THRES_VSET_LB	Skip integration threshold for low vset operation: voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV Default = 1.352V

S4_CTRL_SI_THRES_VSET_UB_SHADOW | 0x00011DBD

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED, PMIC_DOC_ONLY

Bits	Name	Description
3 : 0	SI_THRES_VSET_UB	Skip integration threshold for low vset operation: voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV Default = 1.352V

S4_CTRL_GANG_CTL1 | 0x00011DC0

Type: Read/Write

Reset: 0x0000001D

PMIC_LOCKEDBYBIT=LOCKBIT_D1

Bits	Name	Description
7 : 0	GANG_LEADER_PID	When GANG_EN (GANG_CTL2[7]) is set, this peripheral will write the same data that is written to the gang leader Peripheral ID. Reads to the gang leader Peripheral ID are ignored by this peripheral. Ganged peripherals must reside within the same Slave ID

S4_CTRL_GANG_CTL2 | 0x00011DC1

Type: Read/Write

Reset: 0x00000000

PMIC_LOCKEDBYBIT=LOCKBIT_D1

Bits	Name	Description
7 : 7	GANG_EN	0 = disable 1 = enable When enabled, this peripheral will write the same data that is written to the gang leader PID. Reads to the gang leader PID are ignored by this peripheral 0: GANG_EN_false 1: GANG_EN_true

Module Name: S4_PS

Base Address: 0x00011E00

Register Quick Reference:

Register	Offset	Type
S4_PS_REVISION1	0x00011E00	Read
S4_PS_REVISION2	0x00011E01	Read
S4_PS_REVISION3	0x00011E02	Read
S4_PS_REVISION4	0x00011E03	Read
S4_PS_PERPH_TYPE	0x00011E04	Read
S4_PS_PERPH_SUBTYPE	0x00011E05	Read
S4_PS_VSET_LB	0x00011E40	Read/Write
S4_PS_VSET_UB	0x00011E41	Read/Write
S4_PS_SI_THRES_VSET_LB	0x00011EBC	Read/Write
S4_PS_SI_THRES_VSET_UB	0x00011EBD	Read/Write
S4_PS_GANG_CTL1	0x00011EC0	Read/Write
S4_PS_GANG_CTL2	0x00011EC1	Read/Write

S4_PS_REVISION1 | 0x00011E00

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

S4_PS_REVISION2 | 0x00011E01

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

S4_PS_REVISION3 | 0x00011E02

Type: Read

Reset: 0x00000002

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

S4_PS_REVISION4 | 0x00011E03

Type: Read

Reset: 0x00000000

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

S4_PS_PERPH_TYPE | 0x00011E04

Type: Read

Reset: 0x0000001C

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	http://projects/sites/PM8941/Shared%20Documents/Systems/Registers/Type%20and%20SubType%20Enumeration%20Table.xlsx

S4_PS_PERPH_SUBTYPE | 0x00011E05

Type: Read

Reset: 0x0000000C

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	http://projects/sites/PM8941/Shared%20Documents/Systems/Registers/Type%20and%20SubType%20Enumeration%20Table.xlsx

S4_PS_VSET_LB | 0x00011E40

Type: Read/Write

Reset: 0x00000068

PMIC_LATCHED_WRITE=VSET_UB, PMIC_GANGED

Bits	Name	Description
7 : 0	VSET_LB	Regulator voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S4_PS_VSET_UB | 0x00011E41

Type: Read/Write

Reset: 0x00000003

PMIC_GANGED

Bits	Name	Description
3 : 0	VSET_UB	Regulator voltage set point bit <11:8>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S4_PS_SI_THRES_VSET_LB | 0x00011EBC

Type: Read/Write

Reset: 0x00000048

PMIC_LATCHED_WRITE=SI_THRES_VSET_UB, PMIC_GANGED

Bits	Name	Description
7 : 0	SI_THRES_VSET_LB	Skip integration threshold for low vset operation: voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV Default = 1.352V

S4_PS_SI_THRES_VSET_UB | 0x00011EBD

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED

Bits	Name	Description
3 : 0	SI_THRES_VSET_UB	Skip integration threshold for low vset operation: voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV Default = 1.352V

S4_PS_GANG_CTL1 | 0x00011EC0

Type: Read/Write

Reset: 0x0000001D

PMIC_LOCKEDBYBIT=LOCKBIT_D1

Bits	Name	Description
7 : 0	GANG_LEADER_PID	When GANG_EN (GANG_CTL2[7]) is set, this peripheral will write the same data that is written to the gang leader Peripheral ID. Reads to the gang leader Peripheral ID are ignored by this peripheral. Ganged peripherals must reside within the same Slave ID

S4_PS_GANG_CTL2 | 0x00011EC1

Type: Read/Write

Reset: 0x00000080

PMIC_LOCKEDBYBIT=LOCKBIT_D1

Bits	Name	Description
7 : 7	GANG_EN	0 = disable 1 = enable When enabled, this peripheral will write the same data that is written to the gang leader PID. Reads to the gang leader PID are ignored by this peripheral 0: GANG_EN_false 1: GANG_EN_true

Module Name: S4_FREQ

Base Address: 0x00011F00

Register Quick Reference:

Register	Offset	Type
S4_FREQ_CLK_ENABLE	0x00011F46	Read/Write
S4_FREQ_CLK_DIV	0x00011F50	Read/Write
S4_FREQ_CLK_PHASE	0x00011F51	Read/Write
S4_FREQ_GANG_CTL1	0x00011FC0	Read/Write
S4_FREQ_GANG_CTL2	0x00011FC1	Read/Write

S4_FREQ_CLK_ENABLE | 0x00011F46

Type: Read/Write

Reset: 0x00000001

No description provided for this register.

Bits	Name	Description
0 : 0	FOLLOW_CLK_SX_REQ	0 = ignore smps_clk_req<X> 1 = clock is enabled when the clocks request is high smps_clk_req<X>='1' 0: FALLOW_CLK_REQ_DISABLED 1: FALLOW_CLK_REQ_ENABLED
7 : 7	EN_CLK_INT	0 = do not force the clock on 1 = enable the clock 0: FORCE_EN_DISABLED 1: FORCE_EN_ENABLED

S4_FREQ_CLK_DIV | 0x00011F50

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED

Bits	Name	Description
3 : 0	CLK_DIV	clock_ frequency = 19.2MHz / (CLK_DIV + 1) FTS2 Buck supports 3.2, 4.8, 6.4 and 9.6 MHz HF2 Buck supports 1.6, 2.4, 2.74, 3.2, 3.8, 4.8, and 6.4 MHz CLK_DIV = 0 is not supported, it will generate 9.6 MHz 0: FREQ_9M6HZ_0 1: FREQ_9M6HZ 2: FREQ_6M4HZ 3: FREQ_4M8HZ 4: FREQ_3M8HZ 5: FREQ_3M2HZ 6: FREQ_2M7HZ 7: FREQ_2M4HZ 8: FREQ_2M1HZ 9: FREQ_1M9HZ 10: FREQ_1M7HZ 11: FREQ_1M6HZ 12: FREQ_1M5HZ 13: FREQ_1M4HZ 14: FREQ_1M3HZ 15: FREQ_1M2HZ

S4_FREQ_CLK_PHASE | 0x00011F51

Type: Read/Write

Reset: 0x0000000C

No description provided for this register.

Bits	Name	Description
3 : 0	CLK_PHASE	Distributed clock phase select: clock phase delay = clock period * (CLK_PHASE / 16)

S4_FREQ_GANG_CTL1 | 0x00011FC0

Type: Read/Write

Reset: 0x0000001D

No description provided for this register.

Bits	Name	Description
7 : 0	GANG_LEADER_PID	When GANG_EN (GANG_CTL2[7]) is set, this peripheral will write the same data that is written to the gang leader Peripheral ID. Reads to the gang leader Peripheral ID are ignored by this peripheral. Ganged peripherals must reside within the same Slave ID

S4_FREQ_GANG_CTL2 | 0x00011FC1

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
7 : 7	GANG_EN	0 = disable 1 = enable When enabled, this peripheral will write the same data that is written to the gang leader PID. Reads to the gang leader PID are ignored by this peripheral 0: GANGING_DISABLED 1: GANGING_ENABLED

Module Name: S5_CTRL

Base Address: 0x00012000

Register Quick Reference:

Register	Offset	Type
S5_CTRL_REVISION1	0x00012000	Read
S5_CTRL_REVISION2	0x00012001	Read
S5_CTRL_REVISION3	0x00012002	Read
S5_CTRL_REVISION4	0x00012003	Read
S5_CTRL_PERPH_TYPE	0x00012004	Read
S5_CTRL_PERPH_SUBTYPE	0x00012005	Read
S5_CTRL_STATUS1	0x00012008	Read
S5_CTRL_STATUS2	0x00012009	Read
S5_CTRL_STATUS3	0x0001200A	Read
S5_CTRL_STATUS4	0x0001200B	Read

Register	Offset	Type
S5_CTRL_INT_RT_STS	0x00012010	Read
S5_CTRL_INT_SET_TYPE	0x00012011	Read
S5_CTRL_INT_POLARITY_HIGH	0x00012012	Read
S5_CTRL_INT_POLARITY_LOW	0x00012013	Read/Write
S5_CTRL_INT_LATCHED_CLR	0x00012014	Write
S5_CTRL_INT_EN_SET	0x00012015	Read/Write
S5_CTRL_INT_EN_CLR	0x00012016	Read/Write
S5_CTRL_INT_MID_SEL	0x0001201A	Read/Write
S5_CTRL_INT_PRIORITY	0x0001201B	Read/Write
S5_CTRL_INT_SELF_CLEARING	0x0001201D	Read
S5_CTRL_ULS_VSET_LB	0x00012039	Read/Write
S5_CTRL_ULS_VSET_UB	0x0001203A	Read/Write
S5_CTRL_VSET_LB	0x00012040	Read/Write
S5_CTRL_VSET_UB	0x00012041	Read/Write
S5_CTRL_VSET_VALID_LB	0x00012042	Read
S5_CTRL_VSET_VALID_UB	0x00012043	Read
S5_CTRL_MODE_CTL2	0x00012044	Read/Write
S5_CTRL_MODE_CTL1	0x00012045	Read/Write
S5_CTRL_EN_CTL	0x00012046	Read/Write
S5_CTRL_FOLLOW_HWEN	0x00012047	Read/Write
S5_CTRL_FREQ_CTL	0x00012050	Read/Write
S5_CTRL_OCP	0x00012088	Read/Write
S5_CTRL_PD_CTL	0x000120A0	Read/Write

S5_CTRL_REVISION1 | 0x00012000

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	Indicates a change in the HW which is not intended to impact SW compatibility.

S5_CTRL_REVISION2 | 0x00012001

Type: Read

Reset: 0x00000004

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	Indicates a change in the HW which will impact SW compatibility.

S5_CTRL_REVISION3 | 0x00012002

Type: Read

Reset: 0x00000000

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	Indicates expanded functionality. Minor version adds functionality while being backward compatible for existing features.

S5_CTRL_REVISION4 | 0x00012003

Type: Read

Reset: 0x00000003

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	Indicates different interface version. Major version changes are not backward compatible. There will be a new Programming Guide document per major revision

S5_CTRL_PERPH_TYPE | 0x00012004

Type: Read

Reset: 0x00000003

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	HF BUCK 3: HFS_BUCK

S5_CTRL_PERPH_SUBTYPE | 0x00012005

Type: Read

Reset: 0x0000000A

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	HFS510 (pmic5, 1st generation, first version), VPWM_CTL 10: Voltage_Mode_PWM_Controller

S5_CTRL_STATUS1 | 0x00012008

Type: Read

Reset: 0x00000003

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates the status of the buck 0 = Unused/N/A 1 = Retention Mode 2 = LPM Mode 3 = PWM Mode 0: MODE_STATE_0 1: MODE_STATE_1 2: MODE_STATE_2 3: MODE_STATE_3
2 : 2	STEPPER_DONE	Indicates voltage stepper is done or not 0: STEPPER_DONE_false 1: STEPPER_DONE_true

Bits	Name	Description
3 : 3	PS_TRUE	Indicates whether buck is pulse skipping or not 0: PS_TRUE_false 1: PS_TRUE_true
4 : 4	TIME_SPEND_PWM_READY	Indicates that the status2 register data is ready. 0: TIME_SPEND_PWM_READY_false 1: TIME_SPEND_PWM_READY_true
5 : 5	VREG_OCP	Indicates if over current or another fault has been detected/latched. 0: NO_FAULT 1: FAULT
6 : 6	VREG_ERROR	Indicates that regulator is not stepping and has fallen below VREG_OK comparator threshold 0 = Regulator is okay; no error condition 0: VREG_ERR_false 1: VREG_ERR_true
7 : 7	VREG_READY	Indicates if VREG is ready or not due to stepping in progress or below analog debounced VREG_OK comp 0: VREG_READY_false 1: VREG_READY_true

S5_CTRL_STATUS2 | 0x00012009

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	TIME_SPEND_PWM	Indicates while in any of the operating buck modes how time we spend in PWM mode. Tpwm = auto_mode_time_spend_pwm / 256 * 100%

S5_CTRL_STATUS3 | 0x0001200A

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	CAL_DONE_FLAG	when HI it indicates that the calibration engine has completed autocal/directcal procedure. 0: CTL_CAL_DONE_FLAG_false 1: CTL_CAL_DONE_FLAG_true
5 : 1	CAL_FAULT	when HI it indicates a calibration fault on a particular trim. Calibration value resulted in either then 1st or last code which indicates a lack of calibration range. Bits [5:1] represent faults for following trims: [5]ERR_AMP_TRIM, [4] PWM_THRES_TRIM, [3] PFM_COMP_TRIM, [3] VDIP_COMP_TRIM [1] PWM_RAMP_PEAK_TRIM.
6 : 6	CAL_REQUEST	when HI it indicates that the autocal/direct calibration is in progress. 0: CAL_REQUEST_false 1: CAL_REQUEST_true
7 : 7	CAL_COMP_STATE	when HI it indicates to PBS that the selected trim code has calibrated the selected analog circuit. 0: CAL_COMP_STATE_false 1: CAL_COMP_STATE_true

S5_CTRL_STATUS4 | 0x0001200B

Type: Read

Reset: 0x00000040

No description provided for this register.

Bits	Name	Description
6 : 6	AHC_PRESENT	when HI it indicates to PBS that the hardware for Auto-Headroom Control (AHC) feature is present.
7 : 7	APT_PRESENT	when HI it indicates to PBS that the hardware for APT feature is present.

S5_CTRL_INT_RT_STS | 0x00012010

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

S5_CTRL_INT_SET_TYPE | 0x00012011

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

S5_CTRL_INT_POLARITY_HIGH | 0x00012012

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

S5_CTRL_INT_POLARITY_LOW | 0x00012013

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

S5_CTRL_INT_LATCHED_CLR | 0x00012014

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

S5_CTRL_INT_EN_SET | 0x00012015

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

S5_CTRL_INT_EN_CLR | 0x00012016

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

S5_CTRL_INT_MID_SEL | 0x0001201A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

S5_CTRL_INT_PRIORITY | 0x0001201B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

S5_CTRL_INT_SELF_CLEARING | 0x0001201D

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

S5_CTRL_ULS_VSET_LB | 0x00012039

Type: Read/Write

Reset: 0x000000F8

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	Vreg upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x68 and 0x69). Vuls_vset = 2-byte word (dec) x 1mV

S5_CTRL_ULS_VSET_UB | 0x0001203A

Type: Read/Write

Reset: 0x00000007

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	Vreg upper limit stop-voltage set point bits<3:0>. 2-Byte Word (0x68 and 0x69). Vuls_vset = 2-byte word (dec) x 1mV. Maximum combined supported value is 0x07F8/0x0FF0 which translates to 2040/4080mV for HFS510/HFS515 respectively. Beyond that value, it will be clamped to 0x07F8/0x0FF0 for HFS510/515. The default value is already set to the maximum allowed value.

S5_CTRL_VSET_LB | 0x00012040

Type: Read/Write

Reset: 0x000000A0

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	Regulator voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S5_CTRL_VSET_UB | 0x00012041

Type: Read/Write

Reset: 0x00000003

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	Regulator voltage set point bit <11:8>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S5_CTRL_VSET_VALID_LB | 0x00012042

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	VSET_VALID_LB	Lower byte readback for the valid output voltage setpoint value. This register shows the value round up effect of the set voltage since buck doesn't support 1mV resolution.,,,,,'

S5_CTRL_VSET_VALID_UB | 0x00012043

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Upper byte readback for the valid output voltage setpoint value. This register shows the value round up effect of the set voltage since buck doesn't support 1mV resolution.,,,,,'

S5_CTRL_MODE_CTL2 | 0x00012044

Type: Read/Write

Reset: 0x00000007

Define Buck Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	Controls the mode that buck operates 0: FORCED_NPM_0 1: FORCED_NPM_1 2: FORCED_NPM_2 3: FORCED_RETENTION 4: FORCED_LPM 5: AUTO_RM 6: AUTO_LPM 7: FORCED_NPM_7

S5_CTRL_MODE_CTL1 | 0x00012045

Type: Read/Write

Reset: 0x00000007

Define Buck Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	Controls the mode that buck operates 0: FORCED_NPM_0 1: FORCED_NPM_1 2: FORCED_NPM_2 3: FORCED_RETENTION 4: FORCED_LPM 5: AUTO_RM 6: AUTO_LPM 7: FORCED_NPM_7

S5_CTRL_EN_CTL | 0x00012046

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	PERPH_EN	Control signal to enable the regulator 0: BUCK_ENABLE_false 1: BUCK_ENABLE_true

S5_CTRL_FOLLOW_HWEN | 0x00012047

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HWEN0	Enables buck when HWEN0 is asserted 0: EN_FOLLOW_HWEN0_false 1: EN_FOLLOW_HWEN0_true
1 : 1	EN_FOLLOW_HWEN1	Enables buck when HWEN1 is asserted 0: EN_FOLLOW_HWEN1_false 1: EN_FOLLOW_HWEN1_true

Bits	Name	Description
2 : 2	EN_FOLLOW_HWEN2	Enables buck when HWEN2 is asserted 0: EN_FOLLOW_HWEN2_false 1: EN_FOLLOW_HWEN2_true
3 : 3	EN_FOLLOW_PMIC_AWAKE	Enables buck when PMIC_AWAKE is asserted 0: EN_FOLLOW_PMIC_AWAKE_false 1: EN_FOLLOW_PMIC_AWAKE_true
4 : 4	FOLLOW_HWEN0_CFG	If this bit is set low then HWEN0 does NOT control Buck mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN0_CFG_false 1: FOLLOW_HWEN0_CFG_true
5 : 5	FOLLOW_HWEN1_CFG	If this bit is set low then HWEN1 does NOT control Buck mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN1_CFG_false 1: FOLLOW_HWEN1_CFG_true
6 : 6	FOLLOW_HWEN2_CFG	If this bit is set low then HWEN2 does NOT control Buck mode. If this bit is set high then: When HWEN2=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN2_CFG_false 1: FOLLOW_HWEN2_CFG_true
7 : 7	FOLLOW_AWAKE_CFG	If this bit is set low then PMIC_AWAKE does NOT control Buck mode. If this bit is set high then: When PMIC_AWAKE=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_AWAKE_CFG_false 1: FOLLOW_AWAKE_CFG_true

S5_CTRL_FREQ_CTL | 0x00012050

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED

Bits	Name	Description
3 : 0	FREQ_CTL	Clock frequency = 19.2MHz / (CLK_DIV + 1) 2: Fs_6p4MHz 3: Fs_4p8MHz 4: Fs_3p84MHz 5: Fs_3p2MHz 6: Fs_2p74MHz 7: Fs_2p4MHz 8: Fs_2p13MHz 9: Fs_1p92MHz 10: Fs_1p74MHz 11: Fs_1p6MHz

S5_CTRL_OCP | 0x00012088

Type: Read/Write

Reset: 0x00000068

No description provided for this register.

Bits	Name	Description
1 : 1	OCP_STATUS_CLEAR	Clears latched OCP_STATUS when set to 1. User must set this bit back to 0 after clearing OCP_STATUS. 0: OCP_STATUS_CLEAR_false 1: OCP_STATUS_CLEAR_true
2 : 2	OCP_GLOBAL_BROADCAST_EN	Determines if the Buck notifies the PON module that an OCP event has occurred. PON module may use this info to shutdown the entire PMIC. 1 = BUCK notifies PON when OCP event occurs 0 = BUCK does not notify PON when OCP even occurs 0: OCP_GLOBAL_BROADCAST_EN_false 1: OCP_GLOBAL_BROADCAST_EN_true
4 : 3	INZERO_WAIT_OCP_TIMER	Controls wait time for inzero trigger during current foldback. If inzero wait timer expires then an inzero is forced. OCP timer controls the time at which buck shuts down if OCP_LATCH_EN=1 0: INZERO_WAIT_40us_OCP_TIMER_320us 1: INZERO_WAIT_80us_OCP_TIMER_640us 2: INZERO_WAIT_160us_OCP_TIMER_1280_us 3: INZERO_WAIT_160us_OCP_TIMER_1280us
5 : 5	FOLDBACK_EN	This bit is used to enable/disable the foldback operation. 0: FOLDBACK_EN_false 1: FOLDBACK_EN_true

Bits	Name	Description
6 : 6	OCP_LATCH_EN	Controls whether OCP event causes buck to shut down when OCP timer expires 0: OCP_LATCH_EN_false 1: OCP_LATCH_EN_true
7 : 7	VREGOK_BYP_FOLDBACK_EN	Bit to force foldback (Current to reach zero) mode during a current limit event in PWM mode, and fsm_lat_dis=1 (0x75 PS). 0: FORCE_FOLDBACK_FALSE 1: FORCE_FOLDBACK_TRUE

S5_CTRL_PD_CTL | 0x000120A0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Controls state of lead pull down 0: LEAK_PD_EN_false 1: LEAK_PD_EN_true
4 : 4	WEAK_PD_PWM	Controls state of weak pull down in PWM mode while buck is enabled 0: WEAK_PD_PWM_false 1: WEAK_PD_PWM_true
5 : 5	WEAK_PD_PFM	Controls state of weak pull down in PFM mode while buck is enabled 0: WEAK_PD_PFM_false 1: WEAK_PD_PFM_true
6 : 6	WEAK_PD_EN	Controls state of weak pull down 0: WEAK_PD_ENABLE_false 1: WEAK_PD_ENABLE_true
7 : 7	STRONG_PD_EN	Controls state of strong pull down 0: STRONG_PD_ENABLE_false 1: STRONG_PD_ENABLE_true

Module Name: S5_PS

Base Address: 0x00012100

Register Quick Reference:

Register	Offset	Type
S5_PS_REVISION1	0x00012100	Read
S5_PS_REVISION2	0x00012101	Read

Register	Offset	Type
S5_PS_REVISION3	0x00012102	Read
S5_PS_REVISION4	0x00012103	Read
S5_PS_PERPH_TYPE	0x00012104	Read
S5_PS_PERPH_SUBTYPE	0x00012105	Read

S5_PS_REVISION1 | 0x00012100

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	Indicates a change in the HW which is not intended to impact SW compatibility.

S5_PS_REVISION2 | 0x00012101

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	Indicates a change in the HW which is intended to impact SW compatibility.

S5_PS_REVISION3 | 0x00012102

Type: Read

Reset: 0x00000000

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	Indicates expanded functionality. Minor version adds functionality while being backward compatibility for existing features. 0x01-for HFS510 0x00-for HFS515

S5_PS_REVISION4 | 0x00012103

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	Indicates different interface version. Major version changes are not backward compatible. There will be a new Programming Guide document per major revision. 0x01 -for HFS510 LV 6x 0x00 -for HFS515 HV 6x

S5_PS_PERPH_TYPE | 0x00012104

Type: Read

Reset: 0x00000003

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	SMPS 3: HFS_BUCK

S5_PS_PERPH_SUBTYPE | 0x00012105

Type: Read

Reset: 0x00000017

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	Specifies the current rating for the power stage. 0x15-for HFS510 LV 6x 0x31 -for HFS515 HV 6x 19: PSLV_P4N4_2P0A 20: PSLV_P5N5_2P5A 21: PSLV_P6N6_3P0A 23: PSLV_P8N8_4P0A 33: PSHV_P6N5_1P0A 35: PSHV_P12N12 37: PSHV_P16N16

Module Name: S5_FREQ

Base Address: 0x00012200

Register Quick Reference:

Register	Offset	Type
S5_FREQ_CLK_ENABLE	0x00012246	Read/Write
S5_FREQ_CLK_DIV	0x00012250	Read/Write
S5_FREQ_CLK_PHASE	0x00012251	Read/Write
S5_FREQ_GANG_CTL1	0x000122C0	Read/Write
S5_FREQ_GANG_CTL2	0x000122C1	Read/Write

S5_FREQ_CLK_ENABLE | 0x00012246

Type: Read/Write

Reset: 0x00000001

No description provided for this register.

Bits	Name	Description
0 : 0	FOLLOW_CLK_SX_REQ	0 = ignore smps_clk_req<X> 1 = clock is enabled when the clocks request is high smps_clk_req<X>='1' 0: FALLOW_CLK_REQ_DISABLED 1: FALLOW_CLK_REQ_ENABLED
7 : 7	EN_CLK_INT	0 = do not force the clock on 1 = enable the clock 0: FORCE_EN_DISABLED 1: FORCE_EN_ENABLED

S5_FREQ_CLK_DIV | 0x00012250

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED

Bits	Name	Description
3 : 0	CLK_DIV	clock_ frequency = 19.2MHz / (CLK_DIV + 1) FTS2 Buck supports 3.2, 4.8, 6.4 and 9.6 MHz HF2 Buck supports 1.6, 2.4, 2.74, 3.2, 3.8, 4.8, and 6.4 MHz CLK_DIV = 0 is not supported, it will generate 9.6 MHz 0: FREQ_9M6HZ_0 1: FREQ_9M6HZ 2: FREQ_6M4HZ 3: FREQ_4M8HZ 4: FREQ_3M8HZ 5: FREQ_3M2HZ 6: FREQ_2M7HZ 7: FREQ_2M4HZ 8: FREQ_2M1HZ 9: FREQ_1M9HZ 10: FREQ_1M7HZ 11: FREQ_1M6HZ 12: FREQ_1M5HZ 13: FREQ_1M4HZ 14: FREQ_1M3HZ 15: FREQ_1M2HZ

S5_FREQ_CLK_PHASE | 0x00012251

Type: Read/Write

Reset: 0x00000006

No description provided for this register.

Bits	Name	Description
3 : 0	CLK_PHASE	Distributed clock phase select: clock phase delay = clock period * (CLK_PHASE / 16)

S5_FREQ_GANG_CTL1 | 0x000122C0

Type: Read/Write

Reset: 0x00000020

No description provided for this register.

Bits	Name	Description
7 : 0	GANG_LEADER_PID	When GANG_EN (GANG_CTL2[7]) is set, this peripheral will write the same data that is written to the gang leader Peripheral ID. Reads to the gang leader Peripheral ID are ignored by this peripheral. Ganged peripherals must reside within the same Slave ID

S5_FREQ_GANG_CTL2 | 0x000122C1

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
7 : 7	GANG_EN	0 = disable 1 = enable When enabled, this peripheral will write the same data that is written to the gang leader PID. Reads to the gang leader PID are ignored by this peripheral 0: GANGING_DISABLED 1: GANGING_ENABLED

Module Name: S6_CTRL

Base Address: 0x00012300

Register Quick Reference:

Register	Offset	Type
S6_CTRL_REVISION1	0x00012300	Read
S6_CTRL_REVISION2	0x00012301	Read
S6_CTRL_REVISION3	0x00012302	Read
S6_CTRL_REVISION4	0x00012303	Read
S6_CTRL_PERPH_TYPE	0x00012304	Read
S6_CTRL_PERPH_SUBTYPE	0x00012305	Read
S6_CTRL_STATUS1	0x00012308	Read

Register	Offset	Type
S6_CTRL_STATUS2	0x00012309	Read
S6_CTRL_STATUS3	0x0001230A	Read
S6_CTRL_STATUS4	0x0001230B	Read
S6_CTRL_INT_RT_STS	0x00012310	Read
S6_CTRL_INT_SET_TYPE	0x00012311	Read
S6_CTRL_INT_POLARITY_HIGH	0x00012312	Read
S6_CTRL_INT_POLARITY_LOW	0x00012313	Read/Write
S6_CTRL_INT_LATCHED_CLR	0x00012314	Write
S6_CTRL_INT_EN_SET	0x00012315	Read/Write
S6_CTRL_INT_EN_CLR	0x00012316	Read/Write
S6_CTRL_INT_MID_SEL	0x0001231A	Read/Write
S6_CTRL_INT_PRIORITY	0x0001231B	Read/Write
S6_CTRL_INT_SELF_CLEARING	0x0001231D	Read
S6_CTRL_ULS_VSET_LB	0x00012339	Read/Write
S6_CTRL_ULS_VSET_UB	0x0001233A	Read/Write
S6_CTRL_VSET_LB	0x00012340	Read/Write
S6_CTRL_VSET_UB	0x00012341	Read/Write
S6_CTRL_VSET_VALID_LB	0x00012342	Read
S6_CTRL_VSET_VALID_UB	0x00012343	Read
S6_CTRL_MODE_CTL2	0x00012344	Read/Write
S6_CTRL_MODE_CTL1	0x00012345	Read/Write
S6_CTRL_EN_CTL	0x00012346	Read/Write
S6_CTRL_FOLLOW_HWEN	0x00012347	Read/Write
S6_CTRL_FREQ_CTL	0x00012350	Read/Write
S6_CTRL_OCP	0x00012388	Read/Write
S6_CTRL_PD_CTL	0x000123A0	Read/Write

S6_CTRL_REVISION1 | 0x00012300

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	Indicates a change in the HW which is not intended to impact SW compatibility.

S6_CTRL_REVISION2 | 0x00012301

Type: Read

Reset: 0x00000004

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	Indicates a change in the HW which will impact SW compatibility.

S6_CTRL_REVISION3 | 0x00012302

Type: Read

Reset: 0x00000000

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	Indicates expanded functionality. Minor version adds functionality while being backward compatible for existing features.

S6_CTRL_REVISION4 | 0x00012303

Type: Read

Reset: 0x00000003

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	Indicates different interface version. Major version changes are not backward compatible. There will be a new Programming Guide document per major revision

S6_CTRL_PERPH_TYPE | 0x00012304

Type: Read

Reset: 0x00000003

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	HF BUCK 3: HFS_BUCK

S6_CTRL_PERPH_SUBTYPE | 0x00012305

Type: Read

Reset: 0x0000000A

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	HFS510 (pmic5, 1st generation, first version), VPWM_CTL 10: Voltage_Mode_PWM_Controller

S6_CTRL_STATUS1 | 0x00012308

Type: Read

Reset: 0x00000003

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates the status of the buck 0 = Unused/N/A 1 = Retention Mode 2 = LPM Mode 3 = PWM Mode 0: MODE_STATE_0 1: MODE_STATE_1 2: MODE_STATE_2 3: MODE_STATE_3
2 : 2	STEPPER_DONE	Indicates voltage stepper is done or not 0: STEPPER_DONE_false 1: STEPPER_DONE_true
3 : 3	PS_TRUE	Indicates whether buck is pulse skipping or not 0: PS_TRUE_false 1: PS_TRUE_true
4 : 4	TIME_SPEND_PWM_READY	Indicates that the status2 register data is ready. 0: TIME_SPEND_PWM_READY_false 1: TIME_SPEND_PWM_READY_true
5 : 5	VREG_OCP	Indicates if over current or another fault has been detected/latched. 0: NO_FAULT 1: FAULT
6 : 6	VREG_ERROR	Indicates that regulator is not stepping and has fallen below VREG_OK comparator threshold 0 = Regulator is okay; no error condition 0: VREG_ERR_false 1: VREG_ERR_true
7 : 7	VREG_READY	Indicates if VREG is ready or not due to stepping in progress or below analog debounced VREG_OK comp 0: VREG_READY_false 1: VREG_READY_true

S6_CTRL_STATUS2 | 0x00012309

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	TIME_SPEND_PWM	Indicates while in any of the operating buck modes how time we spend in PWM mode. Tpwm = auto_mode_time_spend_pwm / 256 * 100%

S6_CTRL_STATUS3 | 0x0001230A

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	CAL_DONE_FLAG	when HI it indicates that the calibration engine has completed autocal/directcal procedure. 0: CTL_CAL_DONE_FLAG_false 1: CTL_CAL_DONE_FLAG_true
5 : 1	CAL_FAULT	when HI it indicates a calibration fault on a particular trim. Calibration value resulted in either then 1st or last code which indicates a lack of calibration range. Bits [5:1] represent faults for following trims: [5]ERR_AMP_TRIM, [4] PWM_THRES_TRIM, [3] PFM_COMP_TRIM, [3] VDIP_COMP_TRIM [1] PWM_RAMP_PEAK_TRIM.
6 : 6	CAL_REQUEST	when HI it indicates that the autocal/direct calibration is in progress. 0: CAL_REQUEST_false 1: CAL_REQUEST_true
7 : 7	CAL_COMP_STATE	when HI it indicates to PBS that the selected trim code has calibrated the selected analog circuit. 0: CAL_COMP_STATE_false 1: CAL_COMP_STATE_true

S6_CTRL_STATUS4 | 0x0001230B

Type: Read

Reset: 0x00000040

No description provided for this register.

Bits	Name	Description
6 : 6	AHC_PRESENT	when HI it indicates to PBS that the hardware for Auto-Headroom Control (AHC) feature is present.
7 : 7	APT_PRESENT	when HI it indicates to PBS that the hardware for APT feature is present.

S6_CTRL_INT_RT_STS | 0x00012310

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

S6_CTRL_INT_SET_TYPE | 0x00012311

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

S6_CTRL_INT_POLARITY_HIGH | 0x00012312

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

S6_CTRL_INT_POLARITY_LOW | 0x00012313

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

S6_CTRL_INT_LATCHED_CLR | 0x00012314

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

S6_CTRL_INT_EN_SET | 0x00012315

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

S6_CTRL_INT_EN_CLR | 0x00012316

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

S6_CTRL_INT_MID_SEL | 0x0001231A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

S6_CTRL_INT_PRIORITY | 0x0001231B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

S6_CTRL_INT_SELF_CLEARING | 0x0001231D

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

S6_CTRL_ULS_VSET_LB | 0x00012339

Type: Read/Write

Reset: 0x000000F8

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	Vreg upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x68 and 0x69). Vuls_vset = 2-byte word (dec) x 1mV

S6_CTRL_ULS_VSET_UB | 0x0001233A

Type: Read/Write

Reset: 0x00000007

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	Vreg upper limit stop-voltage set point bits<3:0>. 2-Byte Word (0x68 and 0x69). Vuls_vset = 2-byte word (dec) x 1mV. Maximum combined supported value is 0x07F8/0x0FF0 which translates to 2040/4080mV for HFS510/HFS515 respectively. Beyond that value, it will be clamped to 0x07F8/0x0FF0 for HFS510/515. The default value is already set to the maximum allowed value.

S6_CTRL_VSET_LB | 0x00012340

Type: Read/Write

Reset: 0x00000048

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	Regulator voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S6_CTRL_VSET_UB | 0x00012341

Type: Read/Write

Reset: 0x00000005

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	Regulator voltage set point bit <11:8>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S6_CTRL_VSET_VALID_LB | 0x00012342

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	VSET_VALID_LB	Lower byte readback for the valid output voltage setpoint value. This register shows the value round up effect of the set voltage since buck doesn't support 1mV resolution.,,,,,'

S6_CTRL_VSET_VALID_UB | 0x00012343

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Upper byte readback for the valid output voltage setpoint value. This register shows the value round up effect of the set voltage since buck doesn't support 1mV resolution.,,,,,'

S6_CTRL_MODE_CTL2 | 0x00012344

Type: Read/Write

Reset: 0x00000007

Define Buck Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	Controls the mode that buck operates 0: FORCED_NPM_0 1: FORCED_NPM_1 2: FORCED_NPM_2 3: FORCED_RETENTION 4: FORCED_LPM 5: AUTO_RM 6: AUTO_LPM 7: FORCED_NPM_7

S6_CTRL_MODE_CTL1 | 0x00012345

Type: Read/Write

Reset: 0x00000007

Define Buck Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	Controls the mode that buck operates 0: FORCED_NPM_0 1: FORCED_NPM_1 2: FORCED_NPM_2 3: FORCED_RETENTION 4: FORCED_LPM 5: AUTO_RM 6: AUTO_LPM 7: FORCED_NPM_7

S6_CTRL_EN_CTL | 0x00012346

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	PERPH_EN	Control signal to enable the regulator 0: BUCK_ENABLE_false 1: BUCK_ENABLE_true

S6_CTRL_FOLLOW_HWEN | 0x00012347

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HWEN0	Enables buck when HWEN0 is asserted 0: EN_FOLLOW_HWEN0_false 1: EN_FOLLOW_HWEN0_true

Bits	Name	Description
1 : 1	EN_FOLLOW_HWEN1	Enables buck when HWEN1 is asserted 0: EN_FOLLOW_HWEN1_false 1: EN_FOLLOW_HWEN1_true
2 : 2	EN_FOLLOW_HWEN2	Enables buck when HWEN2 is asserted 0: EN_FOLLOW_HWEN2_false 1: EN_FOLLOW_HWEN2_true
3 : 3	EN_FOLLOW_PMIC_AWAKE	Enables buck when PMIC_AWAKE is asserted 0: EN_FOLLOW_PMIC_AWAKE_false 1: EN_FOLLOW_PMIC_AWAKE_true
4 : 4	FOLLOW_HWEN0_CFG	If this bit is set low then HWEN0 does NOT control Buck mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN0_CFG_false 1: FOLLOW_HWEN0_CFG_true
5 : 5	FOLLOW_HWEN1_CFG	If this bit is set low then HWEN1 does NOT control Buck mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN1_CFG_false 1: FOLLOW_HWEN1_CFG_true
6 : 6	FOLLOW_HWEN2_CFG	If this bit is set low then HWEN2 does NOT control Buck mode. If this bit is set high then: When HWEN2=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN2_CFG_false 1: FOLLOW_HWEN2_CFG_true
7 : 7	FOLLOW_AWAKE_CFG	If this bit is set low then PMIC_AWAKE does NOT control Buck mode. If this bit is set high then: When PMIC_AWAKE=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_AWAKE_CFG_false 1: FOLLOW_AWAKE_CFG_true

S6_CTRL_FREQ_CTL | 0x00012350

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED

Bits	Name	Description
3 : 0	FREQ_CTL	Clock frequency = 19.2MHz / (CLK_DIV + 1) 2: Fs_6p4MHz 3: Fs_4p8MHz 4: Fs_3p84MHz 5: Fs_3p2MHz 6: Fs_2p74MHz 7: Fs_2p4MHz 8: Fs_2p13MHz 9: Fs_1p92MHz 10: Fs_1p74MHz 11: Fs_1p6MHz

S6_CTRL_OCP | 0x00012388

Type: Read/Write

Reset: 0x00000068

No description provided for this register.

Bits	Name	Description
1 : 1	OCP_STATUS_CLEAR	Clears latched OCP_STATUS when set to 1. User must set this bit back to 0 after clearing OCP_STATUS. 0: OCP_STATUS_CLEAR_false 1: OCP_STATUS_CLEAR_true
2 : 2	OCP_GLOBAL_BROADCAST_EN	Determines if the Buck notifies the PON module that an OCP event has occurred. PON module may use this info to shutdown the entire PMIC. 1 = BUCK notifies PON when OCP event occurs 0 = BUCK does not notify PON when OCP even occurs 0: OCP_GLOBAL_BROADCAST_EN_false 1: OCP_GLOBAL_BROADCAST_EN_true
4 : 3	INZERO_WAIT_OCP_TIMER	Controls wait time for inzero trigger during current foldback. If inzero wait timer expires then an inzero is forced. OCP timer controls the time at which buck shuts down if OCP_LATCH_EN=1 0: INZERO_WAIT_40us_OCP_TIMER_320us 1: INZERO_WAIT_80us_OCP_TIMER_640us 2: INZERO_WAIT_160us_OCP_TIMER_1280_us 3: INZERO_WAIT_160us_OCP_TIMER_1280us
5 : 5	FOLDBACK_EN	This bit is used to enable/disable the foldback operation. 0: FOLDBACK_EN_false 1: FOLDBACK_EN_true

Bits	Name	Description
6 : 6	OCP_LATCH_EN	Controls whether OCP event causes buck to shut down when OCP timer expires 0: OCP_LATCH_EN_false 1: OCP_LATCH_EN_true
7 : 7	VREGOK_BYP_FOLDBACK_EN	Bit to force foldback (Current to reach zero) mode during a current limit event in PWM mode, and fsm_lat_dis=1 (0x75 PS). 0: FORCE_FOLDBACK_FALSE 1: FORCE_FOLDBACK_TRUE

S6_CTRL_PD_CTL | 0x000123A0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Controls state of lead pull down 0: LEAK_PD_EN_false 1: LEAK_PD_EN_true
4 : 4	WEAK_PD_PWM	Controls state of weak pull down in PWM mode while buck is enabled 0: WEAK_PD_PWM_false 1: WEAK_PD_PWM_true
5 : 5	WEAK_PD_PFM	Controls state of weak pull down in PFM mode while buck is enabled 0: WEAK_PD_PFM_false 1: WEAK_PD_PFM_true
6 : 6	WEAK_PD_EN	Controls state of weak pull down 0: WEAK_PD_ENABLE_false 1: WEAK_PD_ENABLE_true
7 : 7	STRONG_PD_EN	Controls state of strong pull down 0: STRONG_PD_ENABLE_false 1: STRONG_PD_ENABLE_true

Module Name: S6_PS

Base Address: 0x00012400

Register Quick Reference:

Register	Offset	Type
S6_PS_REVISION1	0x00012400	Read
S6_PS_REVISION2	0x00012401	Read

Register	Offset	Type
S6_PS_REVISION3	0x00012402	Read
S6_PS_REVISION4	0x00012403	Read
S6_PS_PERPH_TYPE	0x00012404	Read
S6_PS_PERPH_SUBTYPE	0x00012405	Read

S6_PS_REVISION1 | 0x00012400

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	Indicates a change in the HW which is not intended to impact SW compatibility.

S6_PS_REVISION2 | 0x00012401

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	Indicates a change in the HW which is intended to impact SW compatibility.

S6_PS_REVISION3 | 0x00012402

Type: Read

Reset: 0x00000000

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	Indicates expanded functionality. Minor version adds functionality while being backward compatibility for existing features. 0x01-for HFS510 0x00-for HFS515

S6_PS_REVISION4 | 0x00012403

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	Indicates different interface version. Major version changes are not backward compatible. There will be a new Programming Guide document per major revision. 0x01 -for HFS510 LV 6x 0x00 -for HFS515 HV 6x

S6_PS_PERPH_TYPE | 0x00012404

Type: Read

Reset: 0x00000003

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	SMPS 3: HFS_BUCK

S6_PS_PERPH_SUBTYPE | 0x00012405

Type: Read

Reset: 0x00000013

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	Specifies the current rating for the power stage. 0x15-for HFS510 LV 6x 0x31 -for HFS515 HV 6x 19: PSLV_P4N4_2P0A 20: PSLV_P5N5_2P5A 21: PSLV_P6N6_3P0A 23: PSLV_P8N8_4P0A 33: PSHV_P6N5_1P0A 35: PSHV_P12N12 37: PSHV_P16N16

Module Name: S6_FREQ

Base Address: 0x00012500

Register Quick Reference:

Register	Offset	Type
S6_FREQ_CLK_ENABLE	0x00012546	Read/Write
S6_FREQ_CLK_DIV	0x00012550	Read/Write
S6_FREQ_CLK_PHASE	0x00012551	Read/Write
S6_FREQ_GANG_CTL1	0x000125C0	Read/Write
S6_FREQ_GANG_CTL2	0x000125C1	Read/Write

S6_FREQ_CLK_ENABLE | 0x00012546

Type: Read/Write

Reset: 0x00000001

No description provided for this register.

Bits	Name	Description
0 : 0	FOLLOW_CLK_SX_REQ	0 = ignore smps_clk_req<X> 1 = clock is enabled when the clocks request is high smps_clk_req<X>='1' 0: FALLOW_CLK_REQ_DISABLED 1: FALLOW_CLK_REQ_ENABLED
7 : 7	EN_CLK_INT	0 = do not force the clock on 1 = enable the clock 0: FORCE_EN_DISABLED 1: FORCE_EN_ENABLED

S6_FREQ_CLK_DIV | 0x00012550

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED

Bits	Name	Description
3 : 0	CLK_DIV	clock_ frequency = 19.2MHz / (CLK_DIV + 1) FTS2 Buck supports 3.2, 4.8, 6.4 and 9.6 MHz HF2 Buck supports 1.6, 2.4, 2.74, 3.2, 3.8, 4.8, and 6.4 MHz CLK_DIV = 0 is not supported, it will generate 9.6 MHz 0: FREQ_9M6HZ_0 1: FREQ_9M6HZ 2: FREQ_6M4HZ 3: FREQ_4M8HZ 4: FREQ_3M8HZ 5: FREQ_3M2HZ 6: FREQ_2M7HZ 7: FREQ_2M4HZ 8: FREQ_2M1HZ 9: FREQ_1M9HZ 10: FREQ_1M7HZ 11: FREQ_1M6HZ 12: FREQ_1M5HZ 13: FREQ_1M4HZ 14: FREQ_1M3HZ 15: FREQ_1M2HZ

S6_FREQ_CLK_PHASE | 0x00012551

Type: Read/Write

Reset: 0x00000002

No description provided for this register.

Bits	Name	Description
3 : 0	CLK_PHASE	Distributed clock phase select: clock phase delay = clock period * (CLK_PHASE / 16)

S6_FREQ_GANG_CTL1 | 0x000125C0

Type: Read/Write

Reset: 0x00000023

No description provided for this register.

Bits	Name	Description
7 : 0	GANG_LEADER_PID	When GANG_EN (GANG_CTL2[7]) is set, this peripheral will write the same data that is written to the gang leader Peripheral ID. Reads to the gang leader Peripheral ID are ignored by this peripheral. Ganged peripherals must reside within the same Slave ID

S6_FREQ_GANG_CTL2 | 0x000125C1

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
7 : 7	GANG_EN	0 = disable 1 = enable When enabled, this peripheral will write the same data that is written to the gang leader PID. Reads to the gang leader PID are ignored by this peripheral 0: GANGING_DISABLED 1: GANGING_ENABLED

Module Name: S7_CTRL

Base Address: 0x00012600

Register Quick Reference:

Register	Offset	Type
S7_CTRL_REVISION1	0x00012600	Read
S7_CTRL_REVISION2	0x00012601	Read
S7_CTRL_REVISION3	0x00012602	Read
S7_CTRL_REVISION4	0x00012603	Read
S7_CTRL_PERPH_TYPE	0x00012604	Read
S7_CTRL_PERPH_SUBTYPE	0x00012605	Read
S7_CTRL_STATUS1	0x00012608	Read

Register	Offset	Type
S7_CTRL_STATUS2	0x00012609	Read
S7_CTRL_STATUS3	0x0001260A	Read
S7_CTRL_STATUS4	0x0001260B	Read
S7_CTRL_INT_RT_STS	0x00012610	Read
S7_CTRL_INT_SET_TYPE	0x00012611	Read
S7_CTRL_INT_POLARITY_HIGH	0x00012612	Read
S7_CTRL_INT_POLARITY_LOW	0x00012613	Read/Write
S7_CTRL_INT_LATCHED_CLR	0x00012614	Write
S7_CTRL_INT_EN_SET	0x00012615	Read/Write
S7_CTRL_INT_EN_CLR	0x00012616	Read/Write
S7_CTRL_INT_MID_SEL	0x0001261A	Read/Write
S7_CTRL_INT_PRIORITY	0x0001261B	Read/Write
S7_CTRL_INT_SELF_CLEARING	0x0001261D	Read
S7_CTRL_ULS_VSET_LB	0x00012639	Read/Write
S7_CTRL_ULS_VSET_UB	0x0001263A	Read/Write
S7_CTRL_VSET_LB	0x00012640	Read/Write
S7_CTRL_VSET_UB	0x00012641	Read/Write
S7_CTRL_VSET_VALID_LB	0x00012642	Read
S7_CTRL_VSET_VALID_UB	0x00012643	Read
S7_CTRL_MODE_CTL2	0x00012644	Read/Write
S7_CTRL_MODE_CTL1	0x00012645	Read/Write
S7_CTRL_EN_CTL	0x00012646	Read/Write
S7_CTRL_FOLLOW_HWEN	0x00012647	Read/Write
S7_CTRL_FREQ_CTL	0x00012650	Read/Write
S7_CTRL_OCP	0x00012688	Read/Write
S7_CTRL_PD_CTL	0x000126A0	Read/Write

S7_CTRL_REVISION1 | 0x00012600

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	Indicates a change in the HW which is not intended to impact SW compatibility.

S7_CTRL_REVISION2 | 0x00012601

Type: Read

Reset: 0x00000004

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	Indicates a change in the HW which will impact SW compatibility.

S7_CTRL_REVISION3 | 0x00012602

Type: Read

Reset: 0x00000000

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	Indicates expanded functionality. Minor version adds functionality while being backward compatible for existing features.

S7_CTRL_REVISION4 | 0x00012603

Type: Read

Reset: 0x00000003

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	Indicates different interface version. Major version changes are not backward compatible. There will be a new Programming Guide document per major revision

S7_CTRL_PERPH_TYPE | 0x00012604

Type: Read

Reset: 0x00000003

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	HF BUCK 3: HFS_BUCK

S7_CTRL_PERPH_SUBTYPE | 0x00012605

Type: Read

Reset: 0x0000000A

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	HFS510 (pmic5, 1st generation, first version), VPWM_CTL 10: Voltage_Mode_PWM_Controller

S7_CTRL_STATUS1 | 0x00012608

Type: Read

Reset: 0x00000003

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates the status of the buck 0 = Unused/N/A 1 = Retention Mode 2 = LPM Mode 3 = PWM Mode 0: MODE_STATE_0 1: MODE_STATE_1 2: MODE_STATE_2 3: MODE_STATE_3
2 : 2	STEPPER_DONE	Indicates voltage stepper is done or not 0: STEPPER_DONE_false 1: STEPPER_DONE_true
3 : 3	PS_TRUE	Indicates whether buck is pulse skipping or not 0: PS_TRUE_false 1: PS_TRUE_true
4 : 4	TIME_SPEND_PWM_READY	Indicates that the status2 register data is ready. 0: TIME_SPEND_PWM_READY_false 1: TIME_SPEND_PWM_READY_true
5 : 5	VREG_OCP	Indicates if over current or another fault has been detected/latched. 0: NO_FAULT 1: FAULT
6 : 6	VREG_ERROR	Indicates that regulator is not stepping and has fallen below VREG_OK comparator threshold 0 = Regulator is okay; no error condition 0: VREG_ERR_false 1: VREG_ERR_true
7 : 7	VREG_READY	Indicates if VREG is ready or not due to stepping in progress or below analog debounced VREG_OK comp 0: VREG_READY_false 1: VREG_READY_true

S7_CTRL_STATUS2 | 0x00012609

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	TIME_SPEND_PWM	Indicates while in any of the operating buck modes how time we spend in PWM mode. Tpwm = auto_mode_time_spend_pwm / 256 * 100%

S7_CTRL_STATUS3 | 0x0001260A

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	CAL_DONE_FLAG	when HI it indicates that the calibration engine has completed autocal/directcal procedure. 0: CTL_CAL_DONE_FLAG_false 1: CTL_CAL_DONE_FLAG_true
5 : 1	CAL_FAULT	when HI it indicates a calibration fault on a particular trim. Calibration value resulted in either then 1st or last code which indicates a lack of calibration range. Bits [5:1] represent faults for following trims: [5]ERR_AMP_TRIM, [4] PWM_THRES_TRIM, [3] PFM_COMP_TRIM, [3] VDIP_COMP_TRIM [1] PWM_RAMP_PEAK_TRIM.
6 : 6	CAL_REQUEST	when HI it indicates that the autocal/direct calibration is in progress. 0: CAL_REQUEST_false 1: CAL_REQUEST_true
7 : 7	CAL_COMP_STATE	when HI it indicates to PBS that the selected trim code has calibrated the selected analog circuit. 0: CAL_COMP_STATE_false 1: CAL_COMP_STATE_true

S7_CTRL_STATUS4 | 0x0001260B

Type: Read

Reset: 0x00000040

No description provided for this register.

Bits	Name	Description
6 : 6	AHC_PRESENT	when HI it indicates to PBS that the hardware for Auto-Headroom Control (AHC) feature is present.
7 : 7	APT_PRESENT	when HI it indicates to PBS that the hardware for APT feature is present.

S7_CTRL_INT_RT_STS | 0x00012610

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

S7_CTRL_INT_SET_TYPE | 0x00012611

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

S7_CTRL_INT_POLARITY_HIGH | 0x00012612

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

S7_CTRL_INT_POLARITY_LOW | 0x00012613

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

S7_CTRL_INT_LATCHED_CLR | 0x00012614

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

S7_CTRL_INT_EN_SET | 0x00012615

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

S7_CTRL_INT_EN_CLR | 0x00012616

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

S7_CTRL_INT_MID_SEL | 0x0001261A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

S7_CTRL_INT_PRIORITY | **0x0001261B**

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

S7_CTRL_INT_SELF_CLEARING | 0x0001261D

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

S7_CTRL_ULS_VSET_LB | 0x00012639

Type: Read/Write

Reset: 0x000000F8

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	Vreg upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x68 and 0x69). Vuls_vset = 2-byte word (dec) x 1mV

S7_CTRL_ULS_VSET_UB | 0x0001263A

Type: Read/Write

Reset: 0x00000007

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	Vreg upper limit stop-voltage set point bits<3:0>. 2-Byte Word (0x68 and 0x69). Vuls_vset = 2-byte word (dec) x 1mV. Maximum combined supported value is 0x07F8/0x0FF0 which translates to 2040/4080mV for HFS510/HFS515 respectively. Beyond that value, it will be clamped to 0x07F8/0x0FF0 for HFS510/515. The default value is already set to the maximum allowed value.

S7_CTRL_VSET_LB | 0x00012640

Type: Read/Write

Reset: 0x000000F8

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	Regulator voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S7_CTRL_VSET_UB | 0x00012641

Type: Read/Write

Reset: 0x00000007

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	Regulator voltage set point bit <11:8>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S7_CTRL_VSET_VALID_LB | 0x00012642

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	VSET_VALID_LB	Lower byte readback for the valid output voltage setpoint value. This register shows the value round up effect of the set voltage since buck doesn't support 1mV resolution.,,,,,'

S7_CTRL_VSET_VALID_UB | 0x00012643

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Upper byte readback for the valid output voltage setpoint value. This register shows the value round up effect of the set voltage since buck doesn't support 1mV resolution.,,,,,'

S7_CTRL_MODE_CTL2 | 0x00012644

Type: Read/Write

Reset: 0x00000007

Define Buck Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	Controls the mode that buck operates 0: FORCED_NPM_0 1: FORCED_NPM_1 2: FORCED_NPM_2 3: FORCED_RETENTION 4: FORCED_LPM 5: AUTO_RM 6: AUTO_LPM 7: FORCED_NPM_7

S7_CTRL_MODE_CTL1 | 0x00012645

Type: Read/Write

Reset: 0x00000007

Define Buck Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	Controls the mode that buck operates 0: FORCED_NPM_0 1: FORCED_NPM_1 2: FORCED_NPM_2 3: FORCED_RETENTION 4: FORCED_LPM 5: AUTO_RM 6: AUTO_LPM 7: FORCED_NPM_7

S7_CTRL_EN_CTL | 0x00012646

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	PERPH_EN	Control signal to enable the regulator 0: BUCK_ENABLE_false 1: BUCK_ENABLE_true

S7_CTRL_FOLLOW_HWEN | 0x00012647

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HWEN0	Enables buck when HWEN0 is asserted 0: EN_FOLLOW_HWEN0_false 1: EN_FOLLOW_HWEN0_true

Bits	Name	Description
1 : 1	EN_FOLLOW_HWEN1	Enables buck when HWEN1 is asserted 0: EN_FOLLOW_HWEN1_false 1: EN_FOLLOW_HWEN1_true
2 : 2	EN_FOLLOW_HWEN2	Enables buck when HWEN2 is asserted 0: EN_FOLLOW_HWEN2_false 1: EN_FOLLOW_HWEN2_true
3 : 3	EN_FOLLOW_PMIC_AWAKE	Enables buck when PMIC_AWAKE is asserted 0: EN_FOLLOW_PMIC_AWAKE_false 1: EN_FOLLOW_PMIC_AWAKE_true
4 : 4	FOLLOW_HWEN0_CFG	If this bit is set low then HWEN0 does NOT control Buck mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN0_CFG_false 1: FOLLOW_HWEN0_CFG_true
5 : 5	FOLLOW_HWEN1_CFG	If this bit is set low then HWEN1 does NOT control Buck mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN1_CFG_false 1: FOLLOW_HWEN1_CFG_true
6 : 6	FOLLOW_HWEN2_CFG	If this bit is set low then HWEN2 does NOT control Buck mode. If this bit is set high then: When HWEN2=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN2_CFG_false 1: FOLLOW_HWEN2_CFG_true
7 : 7	FOLLOW_AWAKE_CFG	If this bit is set low then PMIC_AWAKE does NOT control Buck mode. If this bit is set high then: When PMIC_AWAKE=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_AWAKE_CFG_false 1: FOLLOW_AWAKE_CFG_true

S7_CTRL_FREQ_CTL | 0x00012650

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED

Bits	Name	Description
3 : 0	FREQ_CTL	Clock frequency = 19.2MHz / (CLK_DIV + 1) 2: Fs_6p4MHz 3: Fs_4p8MHz 4: Fs_3p84MHz 5: Fs_3p2MHz 6: Fs_2p74MHz 7: Fs_2p4MHz 8: Fs_2p13MHz 9: Fs_1p92MHz 10: Fs_1p74MHz 11: Fs_1p6MHz

S7_CTRL_OCP | 0x00012688

Type: Read/Write

Reset: 0x00000068

No description provided for this register.

Bits	Name	Description
1 : 1	OCP_STATUS_CLEAR	Clears latched OCP_STATUS when set to 1. User must set this bit back to 0 after clearing OCP_STATUS. 0: OCP_STATUS_CLEAR_false 1: OCP_STATUS_CLEAR_true
2 : 2	OCP_GLOBAL_BROADCAST_EN	Determines if the Buck notifies the PON module that an OCP event has occurred. PON module may use this info to shutdown the entire PMIC. 1 = BUCK notifies PON when OCP event occurs 0 = BUCK does not notify PON when OCP even occurs 0: OCP_GLOBAL_BROADCAST_EN_false 1: OCP_GLOBAL_BROADCAST_EN_true
4 : 3	INZERO_WAIT_OCP_TIMER	Controls wait time for inzero trigger during current foldback. If inzero wait timer expires then an inzero is forced. OCP timer controls the time at which buck shuts down if OCP_LATCH_EN=1 0: INZERO_WAIT_40us_OCP_TIMER_320us 1: INZERO_WAIT_80us_OCP_TIMER_640us 2: INZERO_WAIT_160us_OCP_TIMER_1280_us 3: INZERO_WAIT_160us_OCP_TIMER_1280us
5 : 5	FOLDBACK_EN	This bit is used to enable/disable the foldback operation. 0: FOLDBACK_EN_false 1: FOLDBACK_EN_true

Bits	Name	Description
6 : 6	OCP_LATCH_EN	Controls whether OCP event causes buck to shut down when OCP timer expires 0: OCP_LATCH_EN_false 1: OCP_LATCH_EN_true
7 : 7	VREGOK_BYP_FOLDBACK_EN	Bit to force foldback (Current to reach zero) mode during a current limit event in PWM mode, and fsm_lat_dis=1 (0x75 PS). 0: FORCE_FOLDBACK_FALSE 1: FORCE_FOLDBACK_TRUE

S7_CTRL_PD_CTL | 0x000126A0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Controls state of lead pull down 0: LEAK_PD_EN_false 1: LEAK_PD_EN_true
4 : 4	WEAK_PD_PWM	Controls state of weak pull down in PWM mode while buck is enabled 0: WEAK_PD_PWM_false 1: WEAK_PD_PWM_true
5 : 5	WEAK_PD_PFM	Controls state of weak pull down in PFM mode while buck is enabled 0: WEAK_PD_PFM_false 1: WEAK_PD_PFM_true
6 : 6	WEAK_PD_EN	Controls state of weak pull down 0: WEAK_PD_ENABLE_false 1: WEAK_PD_ENABLE_true
7 : 7	STRONG_PD_EN	Controls state of strong pull down 0: STRONG_PD_ENABLE_false 1: STRONG_PD_ENABLE_true

Module Name: S7_PS

Base Address: 0x00012700

Register Quick Reference:

Register	Offset	Type
S7_PS_REVISION1	0x00012700	Read
S7_PS_REVISION2	0x00012701	Read

Register	Offset	Type
S7_PS_REVISION3	0x00012702	Read
S7_PS_REVISION4	0x00012703	Read
S7_PS_PERPH_TYPE	0x00012704	Read
S7_PS_PERPH_SUBTYPE	0x00012705	Read

S7_PS_REVISION1 | 0x00012700

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	Indicates a change in the HW which is not intended to impact SW compatibility.

S7_PS_REVISION2 | 0x00012701

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	Indicates a change in the HW which is intended to impact SW compatibility.

S7_PS_REVISION3 | 0x00012702

Type: Read

Reset: 0x00000000

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	Indicates expanded functionality. Minor version adds functionality while being backward compatibility for existing features. 0x01-for HFS510 0x00-for HFS515

S7_PS_REVISION4 | 0x00012703

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	Indicates different interface version. Major version changes are not backward compatible. There will be a new Programming Guide document per major revision. 0x01 -for HFS510 LV 6x 0x00 -for HFS515 HV 6x

S7_PS_PERPH_TYPE | 0x00012704

Type: Read

Reset: 0x00000003

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	SMPS 3: HFS_BUCK

S7_PS_PERPH_SUBTYPE | 0x00012705

Type: Read

Reset: 0x00000023

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	Specifies the current rating for the power stage. 0x15-for HFS510 LV 6x 0x31 -for HFS515 HV 6x 19: PSLV_P4N4_2P0A 20: PSLV_P5N5_2P5A 21: PSLV_P6N6_3P0A 23: PSLV_P8N8_4P0A 33: PSHV_P6N5_1P0A 35: PSHV_P12N12 37: PSHV_P16N16

Module Name: S7_FREQ

Base Address: 0x00012800

Register Quick Reference:

Register	Offset	Type
S7_FREQ_CLK_ENABLE	0x00012846	Read/Write
S7_FREQ_CLK_DIV	0x00012850	Read/Write
S7_FREQ_CLK_PHASE	0x00012851	Read/Write
S7_FREQ_GANG_CTL1	0x000128C0	Read/Write
S7_FREQ_GANG_CTL2	0x000128C1	Read/Write

S7_FREQ_CLK_ENABLE | 0x00012846

Type: Read/Write

Reset: 0x00000001

No description provided for this register.

Bits	Name	Description
0 : 0	FOLLOW_CLK_SX_REQ	0 = ignore smps_clk_req<X> 1 = clock is enabled when the clocks request is high smps_clk_req<X>='1' 0: FALLOW_CLK_REQ_DISABLED 1: FALLOW_CLK_REQ_ENABLED
7 : 7	EN_CLK_INT	0 = do not force the clock on 1 = enable the clock 0: FORCE_EN_DISABLED 1: FORCE_EN_ENABLED

S7_FREQ_CLK_DIV | 0x00012850

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED

Bits	Name	Description
3 : 0	CLK_DIV	clock_ frequency = 19.2MHz / (CLK_DIV + 1) FTS2 Buck supports 3.2, 4.8, 6.4 and 9.6 MHz HF2 Buck supports 1.6, 2.4, 2.74, 3.2, 3.8, 4.8, and 6.4 MHz CLK_DIV = 0 is not supported, it will generate 9.6 MHz 0: FREQ_9M6HZ_0 1: FREQ_9M6HZ 2: FREQ_6M4HZ 3: FREQ_4M8HZ 4: FREQ_3M8HZ 5: FREQ_3M2HZ 6: FREQ_2M7HZ 7: FREQ_2M4HZ 8: FREQ_2M1HZ 9: FREQ_1M9HZ 10: FREQ_1M7HZ 11: FREQ_1M6HZ 12: FREQ_1M5HZ 13: FREQ_1M4HZ 14: FREQ_1M3HZ 15: FREQ_1M2HZ

S7_FREQ_CLK_PHASE | 0x00012851

Type: Read/Write

Reset: 0x0000000A

No description provided for this register.

Bits	Name	Description
3 : 0	CLK_PHASE	Distributed clock phase select: clock phase delay = clock period * (CLK_PHASE / 16)

S7_FREQ_GANG_CTL1 | 0x000128C0

Type: Read/Write

Reset: 0x00000026

No description provided for this register.

Bits	Name	Description
7 : 0	GANG_LEADER_PID	When GANG_EN (GANG_CTL2[7]) is set, this peripheral will write the same data that is written to the gang leader Peripheral ID. Reads to the gang leader Peripheral ID are ignored by this peripheral. Ganged peripherals must reside within the same Slave ID

S7_FREQ_GANG_CTL2 | 0x000128C1

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
7 : 7	GANG_EN	0 = disable 1 = enable When enabled, this peripheral will write the same data that is written to the gang leader PID. Reads to the gang leader PID are ignored by this peripheral 0: GANGING_DISABLED 1: GANGING_ENABLED

Module Name: S8_CTRL

Base Address: 0x00012900

Register Quick Reference:

Register	Offset	Type
S8_CTRL_REVISION1	0x00012900	Read
S8_CTRL_REVISION2	0x00012901	Read
S8_CTRL_REVISION3	0x00012902	Read
S8_CTRL_REVISION4	0x00012903	Read
S8_CTRL_PERPH_TYPE	0x00012904	Read
S8_CTRL_PERPH_SUBTYPE	0x00012905	Read
S8_CTRL_STATUS1	0x00012908	Read

Register	Offset	Type
S8_CTRL_INT_RT_STS	0x00012910	Read
S8_CTRL_INT_SET_TYPE	0x00012911	Read
S8_CTRL_INT_POLARITY_HIGH	0x00012912	Read
S8_CTRL_INT_POLARITY_LOW	0x00012913	Read/Write
S8_CTRL_INT_LATCHED_CLR	0x00012914	Write
S8_CTRL_INT_EN_SET	0x00012915	Read/Write
S8_CTRL_INT_EN_CLR	0x00012916	Read/Write
S8_CTRL_INT_MID_SEL	0x0001291A	Read/Write
S8_CTRL_INT_PRIORITY	0x0001291B	Read/Write
S8_CTRL_INT_SELF_CLEARING	0x0001291D	Read
S8_CTRL_ULS_VSET_LB	0x00012939	Read/Write
S8_CTRL_ULS_VSET_UB	0x0001293A	Read/Write
S8_CTRL_VSET_LB	0x00012940	Read/Write
S8_CTRL_VSET_UB	0x00012941	Read/Write
S8_CTRL_VSET_VALID_LB	0x00012942	Read
S8_CTRL_VSET_VALID_UB	0x00012943	Read
S8_CTRL_MODE_CTL2	0x00012944	Read/Write
S8_CTRL_MODE_CTL1	0x00012945	Read/Write
S8_CTRL_EN_CTL	0x00012946	Read/Write
S8_CTRL_FOLLOW_HWEN	0x00012947	Read/Write
S8_CTRL_FREQ_CTL	0x00012950	Read/Write
S8_CTRL_OCP	0x00012988	Read/Write
S8_CTRL_STATUS_DEB_CTL	0x00012990	Read/Write
S8_CTRL_PHASE_ID	0x00012995	Read/Write
S8_CTRL_PHASE_CNT_MAX	0x00012996	Read/Write
S8_CTRL_PD_CTL	0x000129A0	Read/Write
S8_CTRL_SI_THRES_VSET_LB_SHADOW	0x000129BC	Read/Write
S8_CTRL_SI_THRES_VSET_UB_SHADOW	0x000129BD	Read/Write
S8_CTRL_GANG_CTL1	0x000129C0	Read/Write
S8_CTRL_GANG_CTL2	0x000129C1	Read/Write

S8_CTRL_REVISION1 | **0x00012900**

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	Indicates a change in the HW which is not intended to impact SW compatibility.

S8_CTRL_REVISION2 | 0x00012901

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	Indicates a change in the HW which will impact SW compatibility.

S8_CTRL_REVISION3 | 0x00012902

Type: Read

Reset: 0x00000003

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	Indicates expanded functionality. Minor version adds functionality while being backward compatible for existing features.

S8_CTRL_REVISION4 | 0x00012903

Type: Read

Reset: 0x00000000

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	Indicates different interface version. Major version changes are not backward compatible. There will be a new Programming Guide document per major revision

S8_CTRL_PERPH_TYPE | 0x00012904

Type: Read

Reset: 0x0000001C

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	Peripheral Type

S8_CTRL_PERPH_SUBTYPE | 0x00012905

Type: Read

Reset: 0x0000000B

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	Peripheral SubType 10: Voltage_Mode_PWM_Controller

S8_CTRL_STATUS1 | 0x00012908

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates regulator mode state. 0x0: UNUSED 0x1: STATE==RM_STATE && RETENTION MODE (QUALIFIED)==TRUE 0x2: UNUSED 0x3: STATE==PWM STATE 0: UNUSED_is_0 1: RETENTION_is_1 2: UNUSED_is_2 3: PWM_is_3
2 : 2	STEPPER_DONE	Indicates voltage stepper is done or not 0: STEPPER_DONE_false 1: STEPPER_DONE_true
3 : 3	PS_TRUE	Indicates whether buck is pulse skipping or not 0: PS_TRUE_false 1: PS_TRUE_true
4 : 4	RETENTION_QUAL	Indicates that retention mode is qualified versus the load current. 0x0: false 0x1: true 0: RM_QUAL_false 1: RM_QUAL_true
5 : 5	VREG_OCP	Indicates if over current or another fault has been detected/latched. 0: NO_FAULT 1: FAULT
6 : 6	VREG_ERROR	Indicates that regulator is not stepping and has fallen below VREG_OK comparator threshold 0 = Regulator is okay; no error condition 0: VREG_ERR_false 1: VREG_ERR_true
7 : 7	VREG_READY	Indicates if VREG is ready or not due to stepping in progress or below analog debounced VREG_OK comp 0: VREG_READY_false 1: VREG_READY_true

S8_CTRL_INT_RT_STS | 0x00012910

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

S8_CTRL_INT_SET_TYPE | 0x00012911

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

S8_CTRL_INT_POLARITY_HIGH | 0x00012912

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

S8_CTRL_INT_POLARITY_LOW | 0x00012913

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

S8_CTRL_INT_LATCHED_CLR | 0x00012914

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

S8_CTRL_INT_EN_SET | 0x00012915

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

S8_CTRL_INT_EN_CLR | 0x00012916

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

S8_CTRL_INT_MID_SEL | 0x0001291A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

S8_CTRL_INT_PRIORITY | 0x0001291B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

S8_CTRL_INT_SELF_CLEARING | 0x0001291D

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

S8_CTRL_ULS_VSET_LB | 0x00012939

Type: Read/Write

Reset: 0x00000074

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	Vreg upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x68 and 0x69). Vuls_vset = 2-byte word (dec) x 1mV

S8_CTRL_ULS_VSET_UB | 0x0001293A

Type: Read/Write

Reset: 0x00000005

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	Vreg upper limit stop-voltage set point bits<3:0>. 2-Byte Word (0x68 and 0x69). Vuls_vset = 2-byte word (dec) x 1mV. Maximum combined supported value is 0x0574 which translates to 1.396mV. Beyond that value, it will be clamped to 0x0574. The default value is already set to the maximum allowed value.

S8_CTRL_VSET_LB | 0x00012940

Type: Read/Write

Reset: 0x00000068

PMIC_LATCHED_WRITE=VSET_UB, PMIC_GANGED

Bits	Name	Description
7 : 0	VSET_LB	Regulator voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S8_CTRL_VSET_UB | 0x00012941

Type: Read/Write

Reset: 0x00000004

PMIC_GANGED

Bits	Name	Description
3 : 0	VSET_UB	Regulator voltage set point bit <11:8>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S8_CTRL_VSET_VALID_LB | 0x00012942

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 0	VSET_VALID_LB	Lower byte readback for the valid output voltage setpoint value. This register shows the value round up effect of the set voltage since buck doesn't support 1mV resolution.,,,'

S8_CTRL_VSET_VALID_UB | 0x00012943

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Upper byte readback for the valid output voltage setpoint value. This register shows the value round up effect of the set voltage since buck doesn't support 1mV resolution.,,,'

S8_CTRL_MODE_CTL2 | 0x00012944

Type: Read/Write

Reset: 0x00000007

Define Buck Mode Transitions PMIC_GANGED, PMIC_SYNC=d_clk_19p2:d_perph_rb

Bits	Name	Description
2 : 0	MODE_SECONDARY	Controls the mode that buck operates: FORCED_NPM_7 = 7, (DEFAULT) RESERVED,(PRODUCES FORCED_NPM for FTS510)=6, RM_2 =5, RESERVED (PRODUCES FORCED_NPM for FTS510)= 4, RM_1=3, FORCED_NPM_2=2, FORCED_NPM_1=1, FORCED_NPM_0=0 0: FORCED_NPM_0 1: FORCED_NPM_1 2: FORCED_NPM_2 3: RM_1 4: RESERVED_1 5: RM_2 6: RESERVED_2 7: FORCED_NPM_7

S8_CTRL_MODE_CTL1 | 0x00012945

Type: Read/Write

Reset: 0x00000007

Define Buck Mode Transitions PMIC_GANGED, PMIC_SYNC=d_clk_19p2:d_perph_rb

Bits	Name	Description
2 : 0	MODE_PRIMARY	Controls the mode that buck operates: FORCED_NPM_7 = 7,(DEFAULT) RESERVED,(PRODUCES FORCED_NPM for FTS510)=6, RM_2 =5, RESERVED (PRODUCES FORCED_NPM for FTS510)= 4, RM_1=3, FORCED_NPM_2=2, FORCED_NPM_1=1, FORCED_NPM_0=0 0: FORCED_NPM_0 1: FORCED_NPM_1 2: FORCED_NPM_2 3: RM_1 4: RESERVED_1 5: RM_2 6: RESERVED_2 7: FORCED_NPM_7

S8_CTRL_EN_CTL | 0x00012946

Type: Read/Write

Reset: 0x00000000

PMIC_GANGED

Bits	Name	Description
7 : 7	PERPH_EN	Control signal to enable the regulator 0: BUCK_ENABLE_false 1: BUCK_ENABLE_true

S8_CTRL_FOLLOW_HWEN | 0x00012947

Type: Read/Write

Reset: 0x00000000

PMIC_GANGED

Bits	Name	Description
0 : 0	EN_FOLLOW_HWEN0	Enables buck when HWEN0 is asserted 0: EN_FOLLOW_HWEN0_false 1: EN_FOLLOW_HWEN0_true
1 : 1	EN_FOLLOW_HWEN1	Enables buck when HWEN1 is asserted 0: EN_FOLLOW_HWEN1_false 1: EN_FOLLOW_HWEN1_true
2 : 2	EN_FOLLOW_HWEN2	Enables buck when HWEN2 is asserted 0: EN_FOLLOW_HWEN2_false 1: EN_FOLLOW_HWEN2_true
3 : 3	EN_FOLLOW_PMIC_AWAKE	Enables buck when PMIC_AWAKE is asserted 0: EN_FOLLOW_PMIC_AWAKE_false 1: EN_FOLLOW_PMIC_AWAKE_true
4 : 4	FOLLOW_HWEN0_CFG	If this bit is set low then HWEN0 does NOT control Buck mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN0_CFG_false 1: FOLLOW_HWEN0_CFG_true
5 : 5	FOLLOW_HWEN1_CFG	If this bit is set low then HWEN1 does NOT control Buck mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN1_CFG_false 1: FOLLOW_HWEN1_CFG_true
6 : 6	FOLLOW_HWEN2_CFG	If this bit is set low then HWEN2 does NOT control Buck mode. If this bit is set high then: When HWEN2=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_HWEN2_CFG_false 1: FOLLOW_HWEN2_CFG_true
7 : 7	FOLLOW_AWAKE_CFG	If this bit is set low then PMIC_AWAKE does NOT control Buck mode. If this bit is set high then: When PMIC_AWAKE=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: FOLLOW_AWAKE_CFG_false 1: FOLLOW_AWAKE_CFG_true

S8_CTRL_FREQ_CTL | **0x00012950**

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED

Bits	Name	Description
3 : 0	FREQ_CTL	Clock frequency = 19.2MHz / (CLK_DIV + 1) 2: Fs_6p4MHz 3: Fs_4p8MHz 4: Fs_3p84MHz 5: Fs_3p2MHz 6: Fs_2p74MHz 7: Fs_2p4MHz 8: Fs_2p13MHz 9: Fs_1p92MHz 10: Fs_1p74MHz 11: Fs_1p6MHz

S8_CTRL_OCP | 0x00012988

Type: Read/Write

Reset: 0x00000068

PMIC_GANGED, PMIC_SYNC=d_clk_lfrc:d_perph_rb

Bits	Name	Description
1 : 1	OCP_STATUS_CLEAR	Toggle 1->0 to clear the latched OCP status bit. 0: OCP_STATUS_CLEAR_false 1: OCP_STATUS_CLEAR_true
2 : 2	OCP_GLOBAL_BROADCAST_EN	Determines if the Buck notifies the PON module that an OCP event has occurred. PON module may use this info to shutdown the entire PMIC. 1 = BUCK notifies PON when OCP event occurs 0 = BUCK does not notify PON when OCP even occurs (DEFAULT) 0: OCP_GLOBAL_BROADCAST_EN_false 1: OCP_GLOBAL_BROADCAST_EN_true

Bits	Name	Description
4 : 3	INZERO_WAIT_OCP_TIMER	Controls wait time for inzero trigger during current foldback. If inzero wait timer expires then an inzero is forced. OCP timer controls the time at which buck shuts down if OCP_LATCH_EN=1 0: INZERO_WAIT_40us_OCP_TIMER_320us 1: INZERO_WAIT_80us_OCP_TIMER_640us 2: INZERO_WAIT_160us_OCP_TIMER_1280_us 3: INZERO_WAIT_160us_OCP_TIMER_1280us
5 : 5	VREG_FOLDBACK_EN	This bit is used to enable propagating foldback flag to PS. 0: FORCE_FOLDBACK_FALSE 1: FORCE_FOLDBACK_TRUE
6 : 6	OCP_LATCH_EN	Controls whether OCP event causes buck to shut down when OCP timer expires 0x0: ocp latching is not enabled. Buck will not shut off due to the OCP event. 0x1: ocp latching is enabled. Buck will SHUT OFF due to the OCP event. (DEFAULT) 0: OCP_LATCH_EN_false 1: OCP_LATCH_EN_true

S8_CTRL_STATUS_DEB_CTL | 0x00012990

Type: Read/Write

Reset: 0x00000000

PMIC_GANGED, PMIC_SYNC=d_clk_4p8m:d_perph_rb

Bits	Name	Description
1 : 0	VREG_ERROR_DEB	Configures the debounce on falling edge of VREG_OK comparator output for generating VREG_ERROR. 00 = 1.4us -1.6us 01 = 3.0us -3.2us 10= 6.5us -6.7us 11= 9.8us - 10us 0: ERROR_DEB_0 1: ERROR_DEB_1 2: ERROR_DEB_2 3: ERROR_DEB_3

Bits	Name	Description
5 : 4	VREG_READY_DEB	Configures the debounce on rising edge of VREG_OK comparator output for generating VREG_READY. 00 = 1.4us -1.6us 01 = 3.0us -3.2us 10= 6.5us -6.7us 11= 9.8us - 10us 0: READY_DEB_0 1: READY_DEB_1 2: READY_DEB_2 3: READY_DEB_3

S8_CTRL_PHASE_ID | 0x00012995

Type: Read/Write

Reset: 0x00000000

PMIC_LOCKEDBYBIT=LOCKBIT_D1, PMIC_SYNC=d_clk_19p2:d_perph_rb

Bits	Name	Description
3 : 0	PHASE_ID	Unique phase identifier. If less than or equal to SLAVE_COUNT, phase is active. If greater than SLAVE_COUNT, phase is inactive. 0: Phase_Number_1 1: Phase_Number_2 2: Phase_Number_3 3: Phase_Number_4

S8_CTRL_PHASE_CNT_MAX | 0x00012996

Type: Read/Write

Reset: 0x00000000

PMIC_GANGED, PMIC_SYNC=d_clk_19p2:d_perph_rb

Bits	Name	Description
3 : 0	PHASE_CNT_MAX	Sets the maximum number of slave phases we can use.

S8_CTRL_PD_CTL | 0x000129A0

Type: Read/Write

Reset: 0x00000088

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Controls state of lead pull down. (Does not depend on the fsm state). 0: LEAK_PD_EN_false 1: LEAK_PD_EN_true
6 : 6	WEAK_PD_EN	Controls state of weak pull down. (See glue logic for mode fsm state dependency go/fts510). 0: WEAK_PD_ENABLE_false 1: WEAK_PD_ENABLE_true
7 : 7	STRONG_PD_EN	Controls state of strong pull down. (See glue logic for mode fsm state dependency go/fts510). 0: STRONG_PD_ENABLE_false 1: STRONG_PD_ENABLE_true

S8_CTRL_SI_THRES_VSET_LB_SHADOW | 0x000129BC

Type: Read/Write

Reset: 0x00000048

PMIC_LATCHED_WRITE=SI_THRES_VSET_UB_SHADOW, PMIC_GANGED, PMIC_DOC_ONLY

Bits	Name	Description
7 : 0	SI_THRES_VSET_LB	Skip integration threshold for low vset operation: voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV Default = 1.352V

S8_CTRL_SI_THRES_VSET_UB_SHADOW | 0x000129BD

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED, PMIC_DOC_ONLY

Bits	Name	Description
3 : 0	SI_THRES_VSET_UB	Skip integration threshold for low vset operation: voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV Default = 1.352V

S8_CTRL_GANG_CTL1 | 0x000129C0

Type: Read/Write

Reset: 0x00000029

PMIC_LOCKEDBYBIT=LOCKBIT_D1

Bits	Name	Description
7 : 0	GANG_LEADER_PID	When GANG_EN (GANG_CTL2[7]) is set, this peripheral will write the same data that is written to the gang leader Peripheral ID. Reads to the gang leader Peripheral ID are ignored by this peripheral. Ganged peripherals must reside within the same Slave ID

S8_CTRL_GANG_CTL2 | 0x000129C1

Type: Read/Write

Reset: 0x00000000

PMIC_LOCKEDBYBIT=LOCKBIT_D1

Bits	Name	Description
7 : 7	GANG_EN	0 = disable 1 = enable When enabled, this peripheral will write the same data that is written to the gang leader PID. Reads to the gang leader PID are ignored by this peripheral 0: GANG_EN_false 1: GANG_EN_true

Module Name: S8_PS

Base Address: 0x00012A00

Register Quick Reference:

Register	Offset	Type
S8_PS_REVISION1	0x00012A00	Read
S8_PS_REVISION2	0x00012A01	Read

Register	Offset	Type
S8_PS_REVISION3	0x00012A02	Read
S8_PS_REVISION4	0x00012A03	Read
S8_PS_PERPH_TYPE	0x00012A04	Read
S8_PS_PERPH_SUBTYPE	0x00012A05	Read
S8_PS_VSET_LB	0x00012A40	Read/Write
S8_PS_VSET_UB	0x00012A41	Read/Write
S8_PS_SI_THRES_VSET_LB	0x00012ABC	Read/Write
S8_PS_SI_THRES_VSET_UB	0x00012ABD	Read/Write
S8_PS_GANG_CTL1	0x00012AC0	Read/Write
S8_PS_GANG_CTL2	0x00012AC1	Read/Write

S8_PS_REVISION1 | 0x00012A00

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

S8_PS_REVISION2 | 0x00012A01

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
7 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

S8_PS_REVISION3 | 0x00012A02

Type: Read

Reset: 0x00000002

HW Version Regster [23:16]

Bits	Name	Description
7 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

S8_PS_REVISION4 | 0x00012A03

Type: Read

Reset: 0x00000000

HW Version Regster [31:24]

Bits	Name	Description
7 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

S8_PS_PERPH_TYPE | 0x00012A04

Type: Read

Reset: 0x0000001C

Peripheral Type PMIC_CONSTANT

Bits	Name	Description
7 : 0	TYPE	http://projects/sites/PM8941/Shared%20Documents/Systems/Registers/Type%20and%20SubType%20Enumeration%20Table.xlsx

S8_PS_PERPH_SUBTYPE | 0x00012A05

Type: Read

Reset: 0x0000000C

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	http://projects/sites/PM8941/Shared%20Documents/Systems/Registers/Type%20and%20SubType%20Enumeration%20Table.xlsx

S8_PS_VSET_LB | 0x00012A40

Type: Read/Write

Reset: 0x00000068

PMIC_LATCHED_WRITE=VSET_UB, PMIC_GANGED

Bits	Name	Description
7 : 0	VSET_LB	Regulator voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S8_PS_VSET_UB | 0x00012A41

Type: Read/Write

Reset: 0x00000004

PMIC_GANGED

Bits	Name	Description
3 : 0	VSET_UB	Regulator voltage set point bit <11:8>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

S8_PS_SI_THRES_VSET_LB | 0x00012ABC

Type: Read/Write

Reset: 0x00000048

PMIC_LATCHED_WRITE=SI_THRES_VSET_UB, PMIC_GANGED

Bits	Name	Description
7 : 0	SI_THRES_VSET_LB	Skip integration threshold for low vset operation: voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV Default = 1.352V

S8_PS_SI_THRES_VSET_UB | 0x00012ABD

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED

Bits	Name	Description
3 : 0	SI_THRES_VSET_UB	Skip integration threshold for low vset operation: voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV Default = 1.352V

S8_PS_GANG_CTL1 | 0x00012AC0

Type: Read/Write

Reset: 0x00000029

PMIC_LOCKEDBYBIT=LOCKBIT_D1

Bits	Name	Description
7 : 0	GANG_LEADER_PID	When GANG_EN (GANG_CTL2[7]) is set, this peripheral will write the same data that is written to the gang leader Peripheral ID. Reads to the gang leader Peripheral ID are ignored by this peripheral. Ganged peripherals must reside within the same Slave ID

S8_PS_GANG_CTL2 | 0x00012AC1

Type: Read/Write

Reset: 0x00000080

PMIC_LOCKEDBYBIT=LOCKBIT_D1

Bits	Name	Description
7 : 7	GANG_EN	0 = disable 1 = enable When enabled, this peripheral will write the same data that is written to the gang leader PID. Reads to the gang leader PID are ignored by this peripheral 0: GANG_EN_false 1: GANG_EN_true

Module Name: S8_FREQ

Base Address: 0x00012B00

Register Quick Reference:

Register	Offset	Type
S8_FREQ_CLK_ENABLE	0x00012B46	Read/Write
S8_FREQ_CLK_DIV	0x00012B50	Read/Write
S8_FREQ_CLK_PHASE	0x00012B51	Read/Write
S8_FREQ_GANG_CTL1	0x00012BC0	Read/Write
S8_FREQ_GANG_CTL2	0x00012BC1	Read/Write

S8_FREQ_CLK_ENABLE | 0x00012B46

Type: Read/Write

Reset: 0x00000001

No description provided for this register.

Bits	Name	Description
0 : 0	FOLLOW_CLK_SX_REQ	0 = ignore smps_clk_req<X> 1 = clock is enabled when the clocks request is high smps_clk_req<X>='1' 0: FALLOW_CLK_REQ_DISABLED 1: FALLOW_CLK_REQ_ENABLED
7 : 7	EN_CLK_INT	0 = do not force the clock on 1 = enable the clock 0: FORCE_EN_DISABLED 1: FORCE_EN_ENABLED

S8_FREQ_CLK_DIV | 0x00012B50

Type: Read/Write

Reset: 0x00000005

PMIC_GANGED

Bits	Name	Description
3 : 0	CLK_DIV	clock_ frequency = 19.2MHz / (CLK_DIV + 1) FTS2 Buck supports 3.2, 4.8, 6.4 and 9.6 MHz HF2 Buck supports 1.6, 2.4, 2.74, 3.2, 3.8, 4.8, and 6.4 MHz CLK_DIV = 0 is not supported, it will generate 9.6 MHz 0: FREQ_9M6HZ_0 1: FREQ_9M6HZ 2: FREQ_6M4HZ 3: FREQ_4M8HZ 4: FREQ_3M8HZ 5: FREQ_3M2HZ 6: FREQ_2M7HZ 7: FREQ_2M4HZ 8: FREQ_2M1HZ 9: FREQ_1M9HZ 10: FREQ_1M7HZ 11: FREQ_1M6HZ 12: FREQ_1M5HZ 13: FREQ_1M4HZ 14: FREQ_1M3HZ 15: FREQ_1M2HZ

S8_FREQ_CLK_PHASE | 0x00012B51

Type: Read/Write

Reset: 0x0000000B

No description provided for this register.

Bits	Name	Description
3 : 0	CLK_PHASE	Distributed clock phase select: clock phase delay = clock period * (CLK_PHASE / 16)

S8_FREQ_GANG_CTL1 | 0x00012BC0

Type: Read/Write

Reset: 0x00000029

No description provided for this register.

Bits	Name	Description
7 : 0	GANG_LEADER_PID	When GANG_EN (GANG_CTL2[7]) is set, this peripheral will write the same data that is written to the gang leader Peripheral ID. Reads to the gang leader Peripheral ID are ignored by this peripheral. Ganged peripherals must reside within the same Slave ID

S8_FREQ_GANG_CTL2 | 0x00012BC1

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
7 : 7	GANG_EN	0 = disable 1 = enable When enabled, this peripheral will write the same data that is written to the gang leader PID. Reads to the gang leader PID are ignored by this peripheral 0: GANGING_DISABLED 1: GANGING_ENABLED

Module Name: LDO01

Base Address: 0x00014000

Register Quick Reference:

Register	Offset	Type
LDO01_REVISION1	0x00014000	Read
LDO01_REVISION2	0x00014001	Read
LDO01_REVISION3	0x00014002	Read
LDO01_REVISION4	0x00014003	Read
LDO01_PERPH_TYPE	0x00014004	Read
LDO01_PERPH_SUBTYPE	0x00014005	Read
LDO01_STATUS1	0x00014008	Read
LDO01_INT_RT_STS	0x00014010	Read
LDO01_INT_SET_TYPE	0x00014011	Read
LDO01_INT_POLARITY_HIGH	0x00014012	Read
LDO01_INT_POLARITY_LOW	0x00014013	Read/Write
LDO01_INT_LATCHED_CLR	0x00014014	Write

Register	Offset	Type
LDO01_INT_EN_SET	0x00014015	Read/Write
LDO01_INT_EN_CLR	0x00014016	Read/Write
LDO01_INT_MID_SEL	0x0001401A	Read/Write
LDO01_INT_PRIORITY	0x0001401B	Read/Write
LDO01_ULS_VSET_LB	0x00014039	Read/Write
LDO01_ULS_VSET_UB	0x0001403A	Read/Write
LDO01_STEPPER_CTL	0x0001403B	Read/Write
LDO01_VSET_LB	0x00014040	Read/Write
LDO01_VSET_UB	0x00014041	Read/Write
LDO01_VSET_VALID_LB	0x00014042	Read
LDO01_VSET_VALID_UB	0x00014043	Read
LDO01_MODE_CTL2	0x00014044	Read/Write
LDO01_MODE_CTL1	0x00014045	Read/Write
LDO01_EN_CTL	0x00014046	Read/Write
LDO01_FOLLOW_HWEN	0x00014047	Read/Write
LDO01_PBS_VOTE_CTL	0x00014049	Read/Write
LDO01_PD_CTL	0x000140A0	Read/Write

LDO01_REVISION1 | 0x00014000

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO01_REVISION2 | 0x00014001

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

LDO01_REVISION3 | 0x00014002

Type: Read

Reset: 0x00000001

HW Version Regster [23:16]

Bits	Name	Description
2 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO01_REVISION4 | 0x00014003

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
2 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

LDO01_PERPH_TYPE | 0x00014004

Type: Read

Reset: 0x00000004

Peripheral Type

Bits	Name	Description
7 : 0	TYPE	LDO

LDO01_PERPH_SUBTYPE | 0x00014005

Type: Read

Reset: 0x0000006B

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	P50 for example

LDO01_STATUS1 | 0x00014008

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Retention Mode if STATUS3__RETENTION_PRESENT == 1. Otherwise unused. 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_RM_OR_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	STEPPER_DONE	Indicated stepper is done when LDO is enabled or Vset is changed. 1 = Stepper done 0 = Stepper is stepping 0: NOT_DONE_STEPPING 1: DONE_STEPPING

Bits	Name	Description
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass/Retention 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass/Retention 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO01_INT_RT_STS | 0x00014010

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

LDO01_INT_SET_TYPE | 0x00014011

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

LDO01_INT_POLARITY_HIGH | 0x00014012

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

LDO01_INT_POLARITY_LOW | 0x00014013

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

LDO01_INT_LATCHED_CLR | 0x00014014

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

LDO01_INT_EN_SET | 0x00014015

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

LDO01_INT_EN_CLR | 0x00014016

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

LDO01_INT_MID_SEL | 0x0001401A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

LDO01_INT_PRIORITY | 0x0001401B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

LDO01_ULS_VSET_LB | 0x00014039

Type: Read/Write

Reset: 0x00000030

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	LDO upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

LDO01_ULS_VSET_UB | 0x0001403A

Type: Read/Write

Reset: 0x00000005

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	LDO upper limit stopvoltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV. Max combined value is 1.328mv (set as a default for both regs). Write will update ULS_VSET[15:0]. User must program ULS in LSB (8mV) increments to avoid undesired behavior

LDO01_STEPPER_CTL | 0x0001403B

Type: Read/Write

Reset: 0x00000083

No description provided for this register.

Bits	Name	Description
1 : 0	STEP_RATE	Voltage Stepper Rate (voltage always increases 1 LSB per step) 0 = 38.4 mV/us 1 = 19.2 mV/us 2 = 9.6 mV/us 3 = 4.8 mV/us (DEFAULT) 0: STEP_RATE_38p4_mV_us 1: STEP_RATE_19p2_mV_us 2: STEP_RATE_9p6_mV_us 3: STEP_RATE_4p8_mV_us
7 : 7	VOLTAGE_STEPPER_EN	Enables the voltage stepper. Stepper should never be disabled. 0: VOLTAGE_STEPPER_DIABLED 1: VOLTAGE_STEPPER_ENABLED

LDO01_VSET_LB | 0x00014040

Type: Read/Write

Reset: 0x000000B0

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO01_VSET_UB | 0x00014041

Type: Read/Write

Reset: 0x00000004

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO01_VSET_VALID_LB | 0x00014042

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 3	VSET_VALID_LB	Readback the valid output voltage setpoint Lower Byte (Bits 7:0). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO01_VSET_VALID_UB | 0x00014043

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Readback the valid output voltage setpoint Upper Byte (Bits 15:8). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO01_MODE_CTL2 | 0x00014044

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. This setting only matters when 0x47 FOLLOW_HWEN or 0x49 PBS_VOTE_CTL is programmed. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Retention if STATUS3__RETENTION_PRESENT == 1. Otherwise, NPM. 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: RM_OR_NPM 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO01_MODE_CTL1 | 0x00014045

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Retention if STATUS3__RETENTION_PRESENT == 1. Otherwise, NPM. 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: RM_OR_NPM 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO01_EN_CTL | 0x00014046

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

LDO01_FOLLOW_HWEN | 0x00014047

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO state. If this bit is set high then: When HWEN0=1 --> LDO if forced on (regardless of reg 0x46) When HWEN0=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN0_FALSE 1: EN_FOLLOW_HW_EN0_TRUE
1 : 1	EN_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO state. If this bit is set high then: When HWEN1=1 --> LDO if forced on (regardless of reg 0x46) When HWEN1=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN1_FALSE 1: EN_FOLLOW_HW_EN1_TRUE
2 : 2	EN_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO state. If this bit is set high then: When HWEN2=1 --> LDO if forced on (regardless of reg 0x46) When HWEN2=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN2_FALSE 1: EN_FOLLOW_HW_EN2_TRUE

Bits	Name	Description
3 : 3	EN_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE state does NOT control LDO state. If this bit is set high then: When PMIC_AWAKE=1 --> LDO if forced on (regardless of reg 0x46) When PMIC_AWAKE=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_PMIC_AWAKE_FALSE 1: EN_FOLLOW_PMIC_AWAKE_TRUE
4 : 4	MODE_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN0_FALSE 1: MODE_FOLLOW_HW_EN0_TRUE
5 : 5	MODE_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN1_FALSE 1: MODE_FOLLOW_HW_EN1_TRUE
6 : 6	MODE_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO mode. If this bit is set high then: When HWEN2=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN2_FALSE 1: MODE_FOLLOW_HW_EN2_TRUE
7 : 7	MODE_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE (sleep_b) state does NOT control LDO mode. If this bit is set high then: When PMIC_AWAKE=0--> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_PMIC_AWAKE_FALSE 1: MODE_FOLLOW_PMIC_AWAKE_TRUE

LDO01_PBS_VOTE_CTL | 0x00014049

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
6 : 6	PBS_MODE	PBS Mode vote 0 = Mode is set by MODE_CTL1 reg (DEFAULT) 1 = Mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: PBS_MODE_false 1: PBS_MODE_true
7 : 7	PBS_EN	Enables the LDO when asserted to HI. 0 = LDO will NOT be enabled (DEFAULT) 1= ENABLE LDO 0: PBS_EN_false 1: PBS_EN_true

LDO01_PD_CTL | 0x000140A0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Intended for keeping Vout from rising due to passFET leakage at high temperature. 1 = Leakage pulldown is always enabled 0 = Leakage pulldown is always disabled (DEFAULT) 0: LEAKAGE_PD_ALWAYS_DISABLED 1: LEAKAGE_PD_ALWAYS_ENABLED
6 : 6	WEAK_PD_EN	Weak PD is never active while LDO is on (VREG_READY=1). 1 = Weak pulldown is enabled when the regulator is disabled 0 = Weak pulldown is always inactive (DEFAULT) 0: WEAK_PD_DISABLED 1: WEAK_PD_WHEN_LDO_IS_OFF
7 : 7	STRONG_PD_EN	Strong PD is never active while LDO is on (VREG_READY=1). Strong pull-down FET Rdson will vary with Vout. 1 = Strong pulldown is enabled when the regulator is disabled (DEFAULT) 0 = Strong pulldown is always inactive. 0: STRONG_PD_ALWAYS_OFF 1: STRONG_PD_WHEN_LDO_IS_OFF

Module Name: LDO02

Base Address: 0x00014100

Register Quick Reference:

Register	Offset	Type
LDO02_REVISION1	0x00014100	Read
LDO02_REVISION2	0x00014101	Read
LDO02_REVISION3	0x00014102	Read
LDO02_REVISION4	0x00014103	Read
LDO02_PERPH_TYPE	0x00014104	Read
LDO02_PERPH_SUBTYPE	0x00014105	Read
LDO02_STATUS1	0x00014108	Read
LDO02_STATUS3	0x0001410A	Read
LDO02_INT_RT_STS	0x00014110	Read
LDO02_INT_SET_TYPE	0x00014111	Read
LDO02_INT_POLARITY_HIGH	0x00014112	Read
LDO02_INT_POLARITY_LOW	0x00014113	Read/Write
LDO02_INT_LATCHED_CLR	0x00014114	Write
LDO02_INT_EN_SET	0x00014115	Read/Write
LDO02_INT_EN_CLR	0x00014116	Read/Write
LDO02_INT_MID_SEL	0x0001411A	Read/Write
LDO02_INT_PRIORITY	0x0001411B	Read/Write
LDO02_ULS_VSET_LB	0x00014139	Read/Write
LDO02_ULS_VSET_UB	0x0001413A	Read/Write
LDO02_STEPPER_CTL	0x0001413B	Read/Write
LDO02_VSET_LB	0x00014140	Read/Write
LDO02_VSET_UB	0x00014141	Read/Write
LDO02_VSET_VALID_LB	0x00014142	Read
LDO02_VSET_VALID_UB	0x00014143	Read
LDO02_MODE_CTL2	0x00014144	Read/Write
LDO02_MODE_CTL1	0x00014145	Read/Write
LDO02_EN_CTL	0x00014146	Read/Write
LDO02_FOLLOW_HWEN	0x00014147	Read/Write
LDO02_PD_CTL	0x000141A0	Read/Write

LDO02_REVISION1 | 0x00014100

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO02_REVISION2 | 0x00014101

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

LDO02_REVISION3 | 0x00014102

Type: Read

Reset: 0x00000001

HW Version Regster [23:16]

Bits	Name	Description
2 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO02_REVISION4 | 0x00014103

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
2 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

LDO02_PERPH_TYPE | 0x00014104

Type: Read

Reset: 0x00000004

Peripheral Type

Bits	Name	Description
7 : 0	TYPE	LDO

LDO02_PERPH_SUBTYPE | 0x00014105

Type: Read

Reset: 0x0000006A

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	P50 for example

LDO02_STATUS1 | 0x00014108

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Retention Mode if STATUS3__RETENTION_PRESENT == 1. Otherwise unused. 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_RM_OR_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	STEPPER_DONE	Indicated stepper is done when LDO is enabled or Vset is changed. 1 = Stepper done 0 = Stepper is stepping 0: NOT_DONE_STEPPING 1: DONE_STEPPING
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass/Retention 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass/Retention 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO02_STATUS3 | 0x0001410A

Type: Read

Reset: 0x00000003

No description provided for this register.

Bits	Name	Description
0 : 0	AHC_PRESENT	Indicates to SW/user that this regulator supports AHC (automatic headroom control) 0: NO_AHC 1: AHC
1 : 1	LLC_PRESENT	Indicates to SW/user that this regulator supports LLC (load-line compensation) 0: NO_LLC 1: LLC
2 : 2	RETENTION_PRESENT	Indicates to SW/user that this regulator supports retention 0: NO_RETENTION 1: RETENTION

LDO02_INT_RT_STS | 0x00014110

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

LDO02_INT_SET_TYPE | 0x00014111

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

Bits	Name	Description
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

LDO02_INT_POLARITY_HIGH | 0x00014112

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

LDO02_INT_POLARITY_LOW | 0x00014113

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

LDO02_INT_LATCHED_CLR | 0x00014114

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

LDO02_INT_EN_SET | 0x00014115

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

LDO02_INT_EN_CLR | 0x00014116

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

LDO02_INT_MID_SEL | 0x0001411A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

LDO02_INT_PRIORITY | 0x0001411B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

LDO02_ULS_VSET_LB | 0x00014139

Type: Read/Write

Reset: 0x00000030

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	LDO upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

LDO02_ULS_VSET_UB | 0x0001413A

Type: Read/Write

Reset: 0x00000005

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	LDO upper limit stopvoltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV. Max combined value is 1.328mv (set as a default for both regs). Write will update ULS_VSET[15:0]. User must program ULS in LSB (8mV) increments to avoid undesired behavior

LDO02_STEPPER_CTL | 0x0001413B

Type: Read/Write

Reset: 0x00000083

No description provided for this register.

Bits	Name	Description
1 : 0	STEP_RATE	Voltage Stepper Rate (voltage always increases 1 LSB per step) 0 = 38.4 mV/us 1 = 19.2 mV/us 2 = 9.6 mV/us 3 = 4.8 mV/us (DEFAULT) 0: STEP_RATE_38p4_mV_us 1: STEP_RATE_19p2_mV_us 2: STEP_RATE_9p6_mV_us 3: STEP_RATE_4p8_mV_us
7 : 7	VOLTAGE_STEPPER_EN	Enables the voltage stepper. Stepper should never be disabled. 0: VOLTAGE_STEPPER_DIABLED 1: VOLTAGE_STEPPER_ENABLED

LDO02_VSET_LB | 0x00014140

Type: Read/Write

Reset: 0x000000E8

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO02_VSET_UB | 0x00014141

Type: Read/Write

Reset: 0x00000003

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO02_VSET_VALID_LB | 0x00014142

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 3	VSET_VALID_LB	Readback the valid output voltage setpoint Lower Byte (Bits 7:0). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO02_VSET_VALID_UB | 0x00014143

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Readback the valid output voltage setpoint Upper Byte (Bits 15:8). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO02_MODE_CTL2 | 0x00014144

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. This setting only matters when 0x47 FOLLOW_HWEN or 0x49 PBS_VOTE_CTL is programmed. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Retention if STATUS3__RETENTION_PRESENT == 1. Otherwise, NPM. 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: RM_OR_NPM 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO02_MODE_CTL1 | 0x00014145

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Retention if STATUS3__RETENTION_PRESENT == 1. Otherwise, NPM. 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: RM_OR_NPM 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO02_EN_CTL | 0x00014146

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

LDO02_FOLLOW_HWEN | 0x00014147

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO state. If this bit is set high then: When HWEN0=1 --> LDO if forced on (regardless of reg 0x46) When HWEN0=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN0_FALSE 1: EN_FOLLOW_HW_EN0_TRUE
1 : 1	EN_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO state. If this bit is set high then: When HWEN1=1 --> LDO if forced on (regardless of reg 0x46) When HWEN1=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN1_FALSE 1: EN_FOLLOW_HW_EN1_TRUE
2 : 2	EN_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO state. If this bit is set high then: When HWEN2=1 --> LDO if forced on (regardless of reg 0x46) When HWEN2=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN2_FALSE 1: EN_FOLLOW_HW_EN2_TRUE
3 : 3	EN_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE state does NOT control LDO state. If this bit is set high then: When PMIC_AWAKE=1 --> LDO if forced on (regardless of reg 0x46) When PMIC_AWAKE=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_PMIC_AWAKE_FALSE 1: EN_FOLLOW_PMIC_AWAKE_TRUE
4 : 4	MODE_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN0_FALSE 1: MODE_FOLLOW_HW_EN0_TRUE
5 : 5	MODE_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN1_FALSE 1: MODE_FOLLOW_HW_EN1_TRUE
6 : 6	MODE_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO mode. If this bit is set high then: When HWEN2=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN2_FALSE 1: MODE_FOLLOW_HW_EN2_TRUE

Bits	Name	Description
7 : 7	MODE_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE (sleep_b) state does NOT control LDO mode. If this bit is set high then: When PMIC_AWAKE=0--> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_PMIC_AWAKE_FALSE 1: MODE_FOLLOW_PMIC_AWAKE_TRUE

LDO02_PD_CTL | 0x000141A0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Intended for keeping Vout from rising due to passFET leakage at high temperature. 1 = Leakage pulldown is always enabled 0 = Leakage pulldown is always disabled (DEFAULT) 0: LEAKAGE_PD_ALWAYS_DISABLED 1: LEAKAGE_PD_ALWAYS_ENABLED
6 : 6	WEAK_PD_EN	Weak PD is never active while LDO is on (VREG_READY=1). 1 = Weak pulldown is enabled when the regulator is disabled 0 = Weak pulldown is always inactive (DEFAULT) 0: WEAK_PD_DISABLED 1: WEAK_PD_WHEN_LDO_IS_OFF
7 : 7	STRONG_PD_EN	Strong PD is never active while LDO is on (VREG_READY=1). Strong pull-down FET Rdson will vary with Vout. 1 = Strong pulldown is enabled when the regulator is disabled (DEFAULT) 0 = Strong pulldown is always inactive. 0: STRONG_PD_ALWAYS_OFF 1: STRONG_PD_WHEN_LDO_IS_OFF

Module Name: LDO03

Base Address: 0x00014200

Register Quick Reference:

Register	Offset	Type
LDO03_REVISION1	0x00014200	Read

Register	Offset	Type
LDO03_REVISION2	0x00014201	Read
LDO03_REVISION3	0x00014202	Read
LDO03_REVISION4	0x00014203	Read
LDO03_PERPH_TYPE	0x00014204	Read
LDO03_PERPH_SUBTYPE	0x00014205	Read
LDO03_STATUS1	0x00014208	Read
LDO03_INT_RT_STS	0x00014210	Read
LDO03_INT_SET_TYPE	0x00014211	Read
LDO03_INT_POLARITY_HIGH	0x00014212	Read
LDO03_INT_POLARITY_LOW	0x00014213	Read/Write
LDO03_INT_LATCHED_CLR	0x00014214	Write
LDO03_INT_EN_SET	0x00014215	Read/Write
LDO03_INT_EN_CLR	0x00014216	Read/Write
LDO03_INT_MID_SEL	0x0001421A	Read/Write
LDO03_INT_PRIORITY	0x0001421B	Read/Write
LDO03_ULS_VSET_LB	0x00014239	Read/Write
LDO03_ULS_VSET_UB	0x0001423A	Read/Write
LDO03_STEPPER_CTL	0x0001423B	Read/Write
LDO03_VSET_LB	0x00014240	Read/Write
LDO03_VSET_UB	0x00014241	Read/Write
LDO03_VSET_VALID_LB	0x00014242	Read
LDO03_VSET_VALID_UB	0x00014243	Read
LDO03_MODE_CTL2	0x00014244	Read/Write
LDO03_MODE_CTL1	0x00014245	Read/Write
LDO03_EN_CTL	0x00014246	Read/Write
LDO03_FOLLOW_HWEN	0x00014247	Read/Write
LDO03_PBS_VOTE_CTL	0x00014249	Read/Write
LDO03_PD_CTL	0x000142A0	Read/Write

LDO03_REVISION1 | 0x00014200

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO03_REVISION2 | 0x00014201

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

LDO03_REVISION3 | 0x00014202

Type: Read

Reset: 0x00000001

HW Version Regster [23:16]

Bits	Name	Description
2 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO03_REVISION4 | 0x00014203

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
2 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

LDO03_PERPH_TYPE | 0x00014204

Type: Read

Reset: 0x00000004

Peripheral Type

Bits	Name	Description
7 : 0	TYPE	LDO

LDO03_PERPH_SUBTYPE | 0x00014205

Type: Read

Reset: 0x0000006B

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	P50 for example

LDO03_STATUS1 | 0x00014208

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Retention Mode if STATUS3__RETENTION_PRESENT == 1. Otherwise unused. 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_RM_OR_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	STEPPER_DONE	Indicated stepper is done when LDO is enabled or Vset is changed. 1 = Stepper done 0 = Stepper is stepping 0: NOT_DONE_STEPPING 1: DONE_STEPPING
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass/Retention 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass/Retention 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO03_INT_RT_STS | 0x00014210

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT

Bits	Name	Description
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

LDO03_INT_SET_TYPE | 0x00014211

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

LDO03_INT_POLARITY_HIGH | 0x00014212

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

LDO03_INT_POLARITY_LOW | 0x00014213

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

LDO03_INT_LATCHED_CLR | 0x00014214

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

LDO03_INT_EN_SET | 0x00014215

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

LDO03_INT_EN_CLR | 0x00014216

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

LDO03_INT_MID_SEL | 0x0001421A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

LDO03_INT_PRIORITY | 0x0001421B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

LDO03_ULS_VSET_LB | **0x00014239**

Type: Read/Write

Reset: 0x00000030

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	LDO upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

LDO03_ULS_VSET_UB | 0x0001423A

Type: Read/Write

Reset: 0x00000005

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	LDO upper limit stopvoltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV. Max combined value is 1.328mv (set as a default for both regs). Write will update ULS_VSET[15:0]. User must program ULS in LSB (8mV) increments to avoid undesired behavior

LDO03_STEPPER_CTL | 0x0001423B

Type: Read/Write

Reset: 0x00000083

No description provided for this register.

Bits	Name	Description
1 : 0	STEP_RATE	Voltage Stepper Rate (voltage always increases 1 LSB per step) 0 = 38.4 mV/us 1 = 19.2 mV/us 2 = 9.6 mV/us 3 = 4.8 mV/us (DEFAULT) 0: STEP_RATE_38p4_mV_us 1: STEP_RATE_19p2_mV_us 2: STEP_RATE_9p6_mV_us 3: STEP_RATE_4p8_mV_us
7 : 7	VOLTAGE_STEPPER_EN	Enables the voltage stepper. Stepper should never be disabled. 0: VOLTAGE_STEPPER_DIABLED 1: VOLTAGE_STEPPER_ENABLED

LDO03_VSET_LB | 0x00014240

Type: Read/Write

Reset: 0x000000E8

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO03_VSET_UB | 0x00014241

Type: Read/Write

Reset: 0x00000003

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO03_VSET_VALID_LB | 0x00014242

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 3	VSET_VALID_LB	Readback the valid output voltage setpoint Lower Byte (Bits 7:0). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO03_VSET_VALID_UB | 0x00014243

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Readback the valid output voltage setpoint Upper Byte (Bits 15:8). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO03_MODE_CTL2 | 0x00014244

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. This setting only matters when 0x47 FOLLOW_HWEN or 0x49 PBS_VOTE_CTL is programmed. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Retention if STATUS3__RETENTION_PRESENT == 1. Otherwise, NPM. 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: RM_OR_NPM 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO03_MODE_CTL1 | 0x00014245

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Retention if STATUS3__RETENTION_PRESENT == 1. Otherwise, NPM. 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: RM_OR_NPM 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO03_EN_CTL | 0x00014246

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

LDO03_FOLLOW_HWEN | 0x00014247

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO state. If this bit is set high then: When HWEN0=1 --> LDO if forced on (regardless of reg 0x46) When HWEN0=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN0_FALSE 1: EN_FOLLOW_HW_EN0_TRUE
1 : 1	EN_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO state. If this bit is set high then: When HWEN1=1 --> LDO if forced on (regardless of reg 0x46) When HWEN1=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN1_FALSE 1: EN_FOLLOW_HW_EN1_TRUE
2 : 2	EN_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO state. If this bit is set high then: When HWEN2=1 --> LDO if forced on (regardless of reg 0x46) When HWEN2=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN2_FALSE 1: EN_FOLLOW_HW_EN2_TRUE
3 : 3	EN_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE state does NOT control LDO state. If this bit is set high then: When PMIC_AWAKE=1 --> LDO if forced on (regardless of reg 0x46) When PMIC_AWAKE=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_PMIC_AWAKE_FALSE 1: EN_FOLLOW_PMIC_AWAKE_TRUE
4 : 4	MODE_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN0_FALSE 1: MODE_FOLLOW_HW_EN0_TRUE
5 : 5	MODE_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN1_FALSE 1: MODE_FOLLOW_HW_EN1_TRUE
6 : 6	MODE_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO mode. If this bit is set high then: When HWEN2=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN2_FALSE 1: MODE_FOLLOW_HW_EN2_TRUE

Bits	Name	Description
7 : 7	MODE_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE (sleep_b) state does NOT control LDO mode. If this bit is set high then: When PMIC_AWAKE=0--> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_PMIC_AWAKE_FALSE 1: MODE_FOLLOW_PMIC_AWAKE_TRUE

LDO03_PBS_VOTE_CTL | 0x00014249

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
6 : 6	PBS_MODE	PBS Mode vote 0 = Mode is set by MODE_CTL1 reg (DEFAULT) 1 = Mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: PBS_MODE_false 1: PBS_MODE_true
7 : 7	PBS_EN	Enables the LDO when asserted to HI. 0 = LDO will NOT be enabled (DEFAULT) 1= ENABLE LDO 0: PBS_EN_false 1: PBS_EN_true

LDO03_PD_CTL | 0x000142A0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Intended for keeping Vout from rising due to passFET leakage at high temperature. 1 = Leakage pulldown is always enabled 0 = Leakage pulldown is always disabled (DEFAULT) 0: LEAKAGE_PD_ALWAYS_DISABLED 1: LEAKAGE_PD_ALWAYS_ENABLED
6 : 6	WEAK_PD_EN	Weak PD is never active while LDO is on (VREG_READY=1). 1 = Weak pulldown is enabled when the regulator is disabled 0 = Weak pulldown is always inactive (DEFAULT) 0: WEAK_PD_DISABLED 1: WEAK_PD_WHEN_LDO_IS_OFF
7 : 7	STRONG_PD_EN	Strong PD is never active while LDO is on (VREG_READY=1). Strong pull-down FET Rdson will vary with Vout. 1 = Strong pulldown is enabled when the regulator is disabled (DEFAULT) 0 = Strong pulldown is always inactive. 0: STRONG_PD_ALWAYS_OFF 1: STRONG_PD_WHEN_LDO_IS_OFF

Module Name: LDO04

Base Address: 0x00014300

Register Quick Reference:

Register	Offset	Type
LDO04_STATUS1	0x00014308	Read
LDO04_STEPPER_CTL	0x0001433B	Read/Write
LDO04_VSET_LB	0x00014340	Read/Write
LDO04_VSET_UB	0x00014341	Read/Write
LDO04_EN_CTL	0x00014346	Read/Write

LDO04_STATUS1 | 0x00014308

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Retention Mode if STATUS3__RETENTION_PRESENT == 1. Otherwise unused. 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_RM_OR_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	STEPPER_DONE	Indicated stepper is done when LDO is enabled or Vset is changed. 1 = Stepper done 0 = Stepper is stepping 0: NOT_DONE_STEPPING 1: DONE_STEPPING
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass/Retention 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass/Retention 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO04_STEPPER_CTL | 0x0001433B

Type: Read/Write

Reset: 0x00000083

No description provided for this register.

Bits	Name	Description
1 : 0	STEP_RATE	Voltage Stepper Rate (voltage always increases 1 LSB per step) 0 = 38.4 mV/us 1 = 19.2 mV/us 2 = 9.6 mV/us 3 = 4.8 mV/us (DEFAULT) 0: STEP_RATE_38p4_mV_us 1: STEP_RATE_19p2_mV_us 2: STEP_RATE_9p6_mV_us 3: STEP_RATE_4p8_mV_us
7 : 7	VOLTAGE_STEPPER_EN	Enables the voltage stepper. Stepper should never be disabled. 0: VOLTAGE_STEPPER_DIABLED 1: VOLTAGE_STEPPER_ENABLED

LDO04_VSET_LB | 0x00014340

Type: Read/Write

Reset: 0x000000A0

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO04_VSET_UB | 0x00014341

Type: Read/Write

Reset: 0x00000003

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO04_EN_CTL | 0x00014346

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

Module Name: LDO05

Base Address: 0x00014400

Register Quick Reference:

Register	Offset	Type
LDO05_REVISION1	0x00014400	Read
LDO05_REVISION2	0x00014401	Read
LDO05_REVISION3	0x00014402	Read
LDO05_REVISION4	0x00014403	Read
LDO05_PERPH_TYPE	0x00014404	Read
LDO05_PERPH_SUBTYPE	0x00014405	Read
LDO05_STATUS1	0x00014408	Read
LDO05_STATUS3	0x0001440A	Read
LDO05_INT_RT_STS	0x00014410	Read
LDO05_INT_SET_TYPE	0x00014411	Read
LDO05_INT_POLARITY_HIGH	0x00014412	Read
LDO05_INT_POLARITY_LOW	0x00014413	Read/Write
LDO05_INT_LATCHED_CLR	0x00014414	Write

Register	Offset	Type
LDO05_INT_EN_SET	0x00014415	Read/Write
LDO05_INT_EN_CLR	0x00014416	Read/Write
LDO05_INT_MID_SEL	0x0001441A	Read/Write
LDO05_INT_PRIORITY	0x0001441B	Read/Write
LDO05_ULS_VSET_LB	0x00014439	Read/Write
LDO05_ULS_VSET_UB	0x0001443A	Read/Write
LDO05_VSET_LB	0x00014440	Read/Write
LDO05_VSET_UB	0x00014441	Read/Write
LDO05_VSET_VALID_LB	0x00014442	Read
LDO05_VSET_VALID_UB	0x00014443	Read
LDO05_MODE_CTL2	0x00014444	Read/Write
LDO05_MODE_CTL1	0x00014445	Read/Write
LDO05_EN_CTL	0x00014446	Read/Write
LDO05_FOLLOW_HWEN	0x00014447	Read/Write
LDO05_PD_CTL	0x000144A0	Read/Write

LDO05_REVISION1 | 0x00014400

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO05_REVISION2 | 0x00014401

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

LDO05_REVISION3 | 0x00014402

Type: Read

Reset: 0x00000000

HW Version Regster [23:16]

Bits	Name	Description
2 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO05_REVISION4 | 0x00014403

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
2 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

LDO05_PERPH_TYPE | 0x00014404

Type: Read

Reset: 0x00000004

Peripheral Type

Bits	Name	Description
7 : 0	TYPE	LDO

LDO05_PERPH_SUBTYPE | 0x00014405

Type: Read

Reset: 0x0000007A

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	P50 for example

LDO05_STATUS1 | 0x00014408

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Unused 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	SOFT_START_DONE	Indicates soft start is done when LDO is enabled. 1 = Softstart done 0 = Softstart in progress 0: NOT_DONE_SOFT_START 1: DONE_SOFT_START
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP

Bits	Name	Description
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO05_STATUS3 | 0x0001440A

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	AHC_PRESENT	Indicates to SW/user that this regulator supports AHC (automatic headroom control) 0: NO_AHC 1: AHC
1 : 1	LLC_PRESENT	Indicates to SW/user that this regulator supports LLC (load-line compensation) 0: NO_LLC 1: LLC

LDO05_INT_RT_STS | 0x00014410

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT

Bits	Name	Description
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

LDO05_INT_SET_TYPE | 0x00014411

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

LDO05_INT_POLARITY_HIGH | 0x00014412

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

LDO05_INT_POLARITY_LOW | 0x00014413

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

LDO05_INT_LATCHED_CLR | 0x00014414

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

LDO05_INT_EN_SET | 0x00014415

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

LDO05_INT_EN_CLR | 0x00014416

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

LDO05_INT_MID_SEL | 0x0001441A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

LDO05_INT_PRIORITY | 0x0001441B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

LDO05_ULS_VSET_LB | **0x00014439**

Type: Read/Write

Reset: 0x000000D8

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	LDO upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

LDO05_ULS_VSET_UB | 0x0001443A

Type: Read/Write

Reset: 0x0000000D

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	LDO upper limit stopvoltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV. Max combined value is 1.328mv (set as a default for both regs). Write will update ULS_VSET[15:0]. User must program ULS in LSB (8mV) increments to avoid undesired behavior

LDO05_VSET_LB | 0x00014440

Type: Read/Write

Reset: 0x00000090

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO05_VSET_UB | 0x00014441

Type: Read/Write

Reset: 0x0000000B

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO05_VSET_VALID_LB | 0x00014442

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 3	VSET_VALID_LB	Readback the valid output voltage setpoint Lower Byte (Bits 7:0). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO05_VSET_VALID_UB | 0x00014443

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Readback the valid output voltage setpoint Upper Byte (Bits 15:8). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO05_MODE_CTL2 | 0x00014444

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. This setting only matters when 0x47 FOLLOW_HWEN or 0x49 PBS_VOTE_CTL is programmed. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO05_MODE_CTL1 | 0x00014445

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO05_EN_CTL | 0x00014446

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

LDO05_FOLLOW_HWEN | 0x00014447

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO state. If this bit is set high then: When HWEN0=1 --> LDO if forced on (regardless of reg 0x46) When HWEN0=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN0_FALSE 1: EN_FOLLOW_HW_EN0_TRUE
1 : 1	EN_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO state. If this bit is set high then: When HWEN1=1 --> LDO if forced on (regardless of reg 0x46) When HWEN1=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN1_FALSE 1: EN_FOLLOW_HW_EN1_TRUE
2 : 2	EN_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO state. If this bit is set high then: When HWEN2=1 --> LDO if forced on (regardless of reg 0x46) When HWEN2=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN2_FALSE 1: EN_FOLLOW_HW_EN2_TRUE
3 : 3	EN_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE state does NOT control LDO state. If this bit is set high then: When PMIC_AWAKE=1 --> LDO if forced on (regardless of reg 0x46) When PMIC_AWAKE=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_PMIC_AWAKE_FALSE 1: EN_FOLLOW_PMIC_AWAKE_TRUE
4 : 4	MODE_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN0_FALSE 1: MODE_FOLLOW_HW_EN0_TRUE
5 : 5	MODE_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN1_FALSE 1: MODE_FOLLOW_HW_EN1_TRUE
6 : 6	MODE_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO mode. If this bit is set high then: When HWEN2=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN2_FALSE 1: MODE_FOLLOW_HW_EN2_TRUE

Bits	Name	Description
7 : 7	MODE_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE (sleep_b) state does NOT control LDO mode. If this bit is set high then: When PMIC_AWAKE=0--> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_PMIC_AWAKE_FALSE 1: MODE_FOLLOW_PMIC_AWAKE_TRUE

LDO05_PD_CTL | 0x000144A0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Intended for keeping Vout from rising due to passFET leakage at high temperature. 1 = Leakage pulldown is always enabled 0 = Leakage pulldown is always disabled (DEFAULT) 0: LEAKAGE_PD_ALWAYS_DISABLED 1: LEAKAGE_PD_ALWAYS_ENABLED
6 : 6	WEAK_PD_EN	Weak PD is never active while LDO is on (VREG_READY=1). 1 = Weak pulldown is enabled when the regulator is disabled 0 = Weak pulldown is always inactive (DEFAULT) 0: WEAK_PD_DISABLED 1: WEAK_PD_WHEN_LDO_IS_OFF
7 : 7	STRONG_PD_EN	Strong PD is never active while LDO is on (VREG_READY=1). Strong pull-down FET Rdson will vary with Vout. 1 = Strong pulldown is enabled when the regulator is disabled (DEFAULT) 0 = Strong pulldown is always inactive. 0: STRONG_PD_ALWAYS_OFF 1: STRONG_PD_WHEN_LDO_IS_OFF

Module Name: LDO06

Base Address: 0x00014500

Register Quick Reference:

Register	Offset	Type
LDO06_REVISION1	0x00014500	Read

Register	Offset	Type
LDO06_REVISION2	0x00014501	Read
LDO06_REVISION3	0x00014502	Read
LDO06_REVISION4	0x00014503	Read
LDO06_PERPH_TYPE	0x00014504	Read
LDO06_PERPH_SUBTYPE	0x00014505	Read
LDO06_STATUS1	0x00014508	Read
LDO06_STATUS3	0x0001450A	Read
LDO06_INT_RT_STS	0x00014510	Read
LDO06_INT_SET_TYPE	0x00014511	Read
LDO06_INT_POLARITY_HIGH	0x00014512	Read
LDO06_INT_POLARITY_LOW	0x00014513	Read/Write
LDO06_INT_LATCHED_CLR	0x00014514	Write
LDO06_INT_EN_SET	0x00014515	Read/Write
LDO06_INT_EN_CLR	0x00014516	Read/Write
LDO06_INT_MID_SEL	0x0001451A	Read/Write
LDO06_INT_PRIORITY	0x0001451B	Read/Write
LDO06_ULS_VSET_LB	0x00014539	Read/Write
LDO06_ULS_VSET_UB	0x0001453A	Read/Write
LDO06_STEPPER_CTL	0x0001453B	Read/Write
LDO06_VSET_LB	0x00014540	Read/Write
LDO06_VSET_UB	0x00014541	Read/Write
LDO06_VSET_VALID_LB	0x00014542	Read
LDO06_VSET_VALID_UB	0x00014543	Read
LDO06_MODE_CTL2	0x00014544	Read/Write
LDO06_MODE_CTL1	0x00014545	Read/Write
LDO06_EN_CTL	0x00014546	Read/Write
LDO06_FOLLOW_HWEN	0x00014547	Read/Write
LDO06_PD_CTL	0x000145A0	Read/Write

LDO06_REVISION1 | 0x00014500

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO06_REVISION2 | 0x00014501

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

LDO06_REVISION3 | 0x00014502

Type: Read

Reset: 0x00000001

HW Version Regster [23:16]

Bits	Name	Description
2 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO06_REVISION4 | 0x00014503

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
2 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

LDO06_PERPH_TYPE | 0x00014504

Type: Read

Reset: 0x00000004

Peripheral Type

Bits	Name	Description
7 : 0	TYPE	LDO

LDO06_PERPH_SUBTYPE | 0x00014505

Type: Read

Reset: 0x0000006C

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	P50 for example

LDO06_STATUS1 | 0x00014508

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Retention Mode if STATUS3__RETENTION_PRESENT == 1. Otherwise unused. 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_RM_OR_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	STEPPER_DONE	Indicated stepper is done when LDO is enabled or Vset is changed. 1 = Stepper done 0 = Stepper is stepping 0: NOT_DONE_STEPPING 1: DONE_STEPPING
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass/Retention 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass/Retention 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO06_STATUS3 | 0x0001450A

Type: Read

Reset: 0x00000003

No description provided for this register.

Bits	Name	Description
0 : 0	AHC_PRESENT	Indicates to SW/user that this regulator supports AHC (automatic headroom control) 0: NO_AHC 1: AHC
1 : 1	LLC_PRESENT	Indicates to SW/user that this regulator supports LLC (load-line compensation) 0: NO_LLC 1: LLC
2 : 2	RETENTION_PRESENT	Indicates to SW/user that this regulator supports retention 0: NO_RETENTION 1: RETENTION

LDO06_INT_RT_STS | 0x00014510

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

LDO06_INT_SET_TYPE | 0x00014511

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

Bits	Name	Description
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

LDO06_INT_POLARITY_HIGH | 0x00014512

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

LDO06_INT_POLARITY_LOW | 0x00014513

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

LDO06_INT_LATCHED_CLR | 0x00014514

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

LDO06_INT_EN_SET | 0x00014515

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

LDO06_INT_EN_CLR | 0x00014516

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

LDO06_INT_MID_SEL | 0x0001451A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

LDO06_INT_PRIORITY | 0x0001451B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

LDO06_ULS_VSET_LB | 0x00014539

Type: Read/Write

Reset: 0x00000030

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	LDO upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

LDO06_ULS_VSET_UB | 0x0001453A

Type: Read/Write

Reset: 0x00000005

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	LDO upper limit stopvoltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV. Max combined value is 1.328mv (set as a default for both regs). Write will update ULS_VSET[15:0]. User must program ULS in LSB (8mV) increments to avoid undesired behavior

LDO06_STEPPER_CTL | 0x0001453B

Type: Read/Write

Reset: 0x00000083

No description provided for this register.

Bits	Name	Description
1 : 0	STEP_RATE	Voltage Stepper Rate (voltage always increases 1 LSB per step) 0 = 38.4 mV/us 1 = 19.2 mV/us 2 = 9.6 mV/us 3 = 4.8 mV/us (DEFAULT) 0: STEP_RATE_38p4_mV_us 1: STEP_RATE_19p2_mV_us 2: STEP_RATE_9p6_mV_us 3: STEP_RATE_4p8_mV_us
7 : 7	VOLTAGE_STEPPER_EN	Enables the voltage stepper. Stepper should never be disabled. 0: VOLTAGE_STEPPER_DIABLED 1: VOLTAGE_STEPPER_ENABLED

LDO06_VSET_LB | 0x00014540

Type: Read/Write

Reset: 0x00000058

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO06_VSET_UB | 0x00014541

Type: Read/Write

Reset: 0x00000002

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO06_VSET_VALID_LB | 0x00014542

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 3	VSET_VALID_LB	Readback the valid output voltage setpoint Lower Byte (Bits 7:0). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO06_VSET_VALID_UB | 0x00014543

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Readback the valid output voltage setpoint Upper Byte (Bits 15:8). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO06_MODE_CTL2 | 0x00014544

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. This setting only matters when 0x47 FOLLOW_HWEN or 0x49 PBS_VOTE_CTL is programmed. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Retention if STATUS3__RETENTION_PRESENT == 1. Otherwise, NPM. 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: RM_OR_NPM 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO06_MODE_CTL1 | 0x00014545

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Retention if STATUS3__RETENTION_PRESENT == 1. Otherwise, NPM. 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: RM_OR_NPM 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO06_EN_CTL | 0x00014546

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

LDO06_FOLLOW_HWEN | 0x00014547

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO state. If this bit is set high then: When HWEN0=1 --> LDO if forced on (regardless of reg 0x46) When HWEN0=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN0_FALSE 1: EN_FOLLOW_HW_EN0_TRUE
1 : 1	EN_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO state. If this bit is set high then: When HWEN1=1 --> LDO if forced on (regardless of reg 0x46) When HWEN1=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN1_FALSE 1: EN_FOLLOW_HW_EN1_TRUE
2 : 2	EN_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO state. If this bit is set high then: When HWEN2=1 --> LDO if forced on (regardless of reg 0x46) When HWEN2=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN2_FALSE 1: EN_FOLLOW_HW_EN2_TRUE
3 : 3	EN_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE state does NOT control LDO state. If this bit is set high then: When PMIC_AWAKE=1 --> LDO if forced on (regardless of reg 0x46) When PMIC_AWAKE=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_PMIC_AWAKE_FALSE 1: EN_FOLLOW_PMIC_AWAKE_TRUE
4 : 4	MODE_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN0_FALSE 1: MODE_FOLLOW_HW_EN0_TRUE
5 : 5	MODE_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN1_FALSE 1: MODE_FOLLOW_HW_EN1_TRUE
6 : 6	MODE_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO mode. If this bit is set high then: When HWEN2=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN2_FALSE 1: MODE_FOLLOW_HW_EN2_TRUE

Bits	Name	Description
7 : 7	MODE_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE (sleep_b) state does NOT control LDO mode. If this bit is set high then: When PMIC_AWAKE=0--> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_PMIC_AWAKE_FALSE 1: MODE_FOLLOW_PMIC_AWAKE_TRUE

LDO06_PD_CTL | 0x000145A0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Intended for keeping Vout from rising due to passFET leakage at high temperature. 1 = Leakage pulldown is always enabled 0 = Leakage pulldown is always disabled (DEFAULT) 0: LEAKAGE_PD_ALWAYS_DISABLED 1: LEAKAGE_PD_ALWAYS_ENABLED
6 : 6	WEAK_PD_EN	Weak PD is never active while LDO is on (VREG_READY=1). 1 = Weak pulldown is enabled when the regulator is disabled 0 = Weak pulldown is always inactive (DEFAULT) 0: WEAK_PD_DISABLED 1: WEAK_PD_WHEN_LDO_IS_OFF
7 : 7	STRONG_PD_EN	Strong PD is never active while LDO is on (VREG_READY=1). Strong pull-down FET Rdson will vary with Vout. 1 = Strong pulldown is enabled when the regulator is disabled (DEFAULT) 0 = Strong pulldown is always inactive. 0: STRONG_PD_ALWAYS_OFF 1: STRONG_PD_WHEN_LDO_IS_OFF

Module Name: LDO07

Base Address: 0x00014600

Register Quick Reference:

Register	Offset	Type
LDO07_REVISION1	0x00014600	Read

Register	Offset	Type
LDO07_REVISION2	0x00014601	Read
LDO07_REVISION3	0x00014602	Read
LDO07_REVISION4	0x00014603	Read
LDO07_PERPH_TYPE	0x00014604	Read
LDO07_PERPH_SUBTYPE	0x00014605	Read
LDO07_STATUS1	0x00014608	Read
LDO07_STATUS3	0x0001460A	Read
LDO07_INT_RT_STS	0x00014610	Read
LDO07_INT_SET_TYPE	0x00014611	Read
LDO07_INT_POLARITY_HIGH	0x00014612	Read
LDO07_INT_POLARITY_LOW	0x00014613	Read/Write
LDO07_INT_LATCHED_CLR	0x00014614	Write
LDO07_INT_EN_SET	0x00014615	Read/Write
LDO07_INT_EN_CLR	0x00014616	Read/Write
LDO07_INT_MID_SEL	0x0001461A	Read/Write
LDO07_INT_PRIORITY	0x0001461B	Read/Write
LDO07_ULS_VSET_LB	0x00014639	Read/Write
LDO07_ULS_VSET_UB	0x0001463A	Read/Write
LDO07_STEPPER_CTL	0x0001463B	Read/Write
LDO07_VSET_LB	0x00014640	Read/Write
LDO07_VSET_UB	0x00014641	Read/Write
LDO07_VSET_VALID_LB	0x00014642	Read
LDO07_VSET_VALID_UB	0x00014643	Read
LDO07_MODE_CTL2	0x00014644	Read/Write
LDO07_MODE_CTL1	0x00014645	Read/Write
LDO07_EN_CTL	0x00014646	Read/Write
LDO07_FOLLOW_HWEN	0x00014647	Read/Write
LDO07_PD_CTL	0x000146A0	Read/Write

LDO07_REVISION1 | 0x00014600

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO07_REVISION2 | 0x00014601

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

LDO07_REVISION3 | 0x00014602

Type: Read

Reset: 0x00000001

HW Version Regster [23:16]

Bits	Name	Description
2 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO07_REVISION4 | 0x00014603

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
2 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

LDO07_PERPH_TYPE | 0x00014604

Type: Read

Reset: 0x00000004

Peripheral Type

Bits	Name	Description
7 : 0	TYPE	LDO

LDO07_PERPH_SUBTYPE | 0x00014605

Type: Read

Reset: 0x0000006B

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	P50 for example

LDO07_STATUS1 | 0x00014608

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Retention Mode if STATUS3__RETENTION_PRESENT == 1. Otherwise unused. 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_RM_OR_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	STEPPER_DONE	Indicated stepper is done when LDO is enabled or Vset is changed. 1 = Stepper done 0 = Stepper is stepping 0: NOT_DONE_STEPPING 1: DONE_STEPPING
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass/Retention 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass/Retention 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO07_STATUS3 | 0x0001460A

Type: Read

Reset: 0x00000003

No description provided for this register.

Bits	Name	Description
0 : 0	AHC_PRESENT	Indicates to SW/user that this regulator supports AHC (automatic headroom control) 0: NO_AHC 1: AHC
1 : 1	LLC_PRESENT	Indicates to SW/user that this regulator supports LLC (load-line compensation) 0: NO_LLC 1: LLC
2 : 2	RETENTION_PRESENT	Indicates to SW/user that this regulator supports retention 0: NO_RETENTION 1: RETENTION

LDO07_INT_RT_STS | 0x00014610

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

LDO07_INT_SET_TYPE | 0x00014611

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

Bits	Name	Description
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

LDO07_INT_POLARITY_HIGH | 0x00014612

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

LDO07_INT_POLARITY_LOW | 0x00014613

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

LDO07_INT_LATCHED_CLR | 0x00014614

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

LDO07_INT_EN_SET | 0x00014615

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

LDO07_INT_EN_CLR | 0x00014616

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

LDO07_INT_MID_SEL | 0x0001461A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

LDO07_INT_PRIORITY | 0x0001461B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

LDO07_ULS_VSET_LB | 0x00014639

Type: Read/Write

Reset: 0x00000030

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	LDO upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

LDO07_ULS_VSET_UB | 0x0001463A

Type: Read/Write

Reset: 0x00000005

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	LDO upper limit stopvoltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV. Max combined value is 1.328mv (set as a default for both regs). Write will update ULS_VSET[15:0]. User must program ULS in LSB (8mV) increments to avoid undesired behavior

LDO07_STEPPER_CTL | 0x0001463B

Type: Read/Write

Reset: 0x00000083

No description provided for this register.

Bits	Name	Description
1 : 0	STEP_RATE	Voltage Stepper Rate (voltage always increases 1 LSB per step) 0 = 38.4 mV/us 1 = 19.2 mV/us 2 = 9.6 mV/us 3 = 4.8 mV/us (DEFAULT) 0: STEP_RATE_38p4_mV_us 1: STEP_RATE_19p2_mV_us 2: STEP_RATE_9p6_mV_us 3: STEP_RATE_4p8_mV_us
7 : 7	VOLTAGE_STEPPER_EN	Enables the voltage stepper. Stepper should never be disabled. 0: VOLTAGE_STEPPER_DIABLED 1: VOLTAGE_STEPPER_ENABLED

LDO07_VSET_LB | 0x00014640

Type: Read/Write

Reset: 0x000000B0

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO07_VSET_UB | 0x00014641

Type: Read/Write

Reset: 0x00000004

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO07_VSET_VALID_LB | 0x00014642

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 3	VSET_VALID_LB	Readback the valid output voltage setpoint Lower Byte (Bits 7:0). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO07_VSET_VALID_UB | 0x00014643

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Readback the valid output voltage setpoint Upper Byte (Bits 15:8). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO07_MODE_CTL2 | 0x00014644

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. This setting only matters when 0x47 FOLLOW_HWEN or 0x49 PBS_VOTE_CTL is programmed. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Retention if STATUS3__RETENTION_PRESENT == 1. Otherwise, NPM. 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: RM_OR_NPM 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO07_MODE_CTL1 | 0x00014645

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Retention if STATUS3__RETENTION_PRESENT == 1. Otherwise, NPM. 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: RM_OR_NPM 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO07_EN_CTL | 0x00014646

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

LDO07_FOLLOW_HWEN | 0x00014647

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO state. If this bit is set high then: When HWEN0=1 --> LDO if forced on (regardless of reg 0x46) When HWEN0=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN0_FALSE 1: EN_FOLLOW_HW_EN0_TRUE
1 : 1	EN_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO state. If this bit is set high then: When HWEN1=1 --> LDO if forced on (regardless of reg 0x46) When HWEN1=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN1_FALSE 1: EN_FOLLOW_HW_EN1_TRUE
2 : 2	EN_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO state. If this bit is set high then: When HWEN2=1 --> LDO if forced on (regardless of reg 0x46) When HWEN2=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN2_FALSE 1: EN_FOLLOW_HW_EN2_TRUE
3 : 3	EN_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE state does NOT control LDO state. If this bit is set high then: When PMIC_AWAKE=1 --> LDO if forced on (regardless of reg 0x46) When PMIC_AWAKE=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_PMIC_AWAKE_FALSE 1: EN_FOLLOW_PMIC_AWAKE_TRUE
4 : 4	MODE_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN0_FALSE 1: MODE_FOLLOW_HW_EN0_TRUE
5 : 5	MODE_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN1_FALSE 1: MODE_FOLLOW_HW_EN1_TRUE
6 : 6	MODE_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO mode. If this bit is set high then: When HWEN2=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN2_FALSE 1: MODE_FOLLOW_HW_EN2_TRUE

Bits	Name	Description
7 : 7	MODE_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE (sleep_b) state does NOT control LDO mode. If this bit is set high then: When PMIC_AWAKE=0--> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_PMIC_AWAKE_FALSE 1: MODE_FOLLOW_PMIC_AWAKE_TRUE

LDO07_PD_CTL | 0x000146A0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Intended for keeping Vout from rising due to passFET leakage at high temperature. 1 = Leakage pulldown is always enabled 0 = Leakage pulldown is always disabled (DEFAULT) 0: LEAKAGE_PD_ALWAYS_DISABLED 1: LEAKAGE_PD_ALWAYS_ENABLED
6 : 6	WEAK_PD_EN	Weak PD is never active while LDO is on (VREG_READY=1). 1 = Weak pulldown is enabled when the regulator is disabled 0 = Weak pulldown is always inactive (DEFAULT) 0: WEAK_PD_DISABLED 1: WEAK_PD_WHEN_LDO_IS_OFF
7 : 7	STRONG_PD_EN	Strong PD is never active while LDO is on (VREG_READY=1). Strong pull-down FET Rdson will vary with Vout. 1 = Strong pulldown is enabled when the regulator is disabled (DEFAULT) 0 = Strong pulldown is always inactive. 0: STRONG_PD_ALWAYS_OFF 1: STRONG_PD_WHEN_LDO_IS_OFF

Module Name: LDO08

Base Address: 0x00014700

Register Quick Reference:

Register	Offset	Type
LDO08_REVISION1	0x00014700	Read

Register	Offset	Type
LDO08_REVISION2	0x00014701	Read
LDO08_REVISION3	0x00014702	Read
LDO08_REVISION4	0x00014703	Read
LDO08_PERPH_TYPE	0x00014704	Read
LDO08_PERPH_SUBTYPE	0x00014705	Read
LDO08_STATUS1	0x00014708	Read
LDO08_STATUS3	0x0001470A	Read
LDO08_INT_RT_STS	0x00014710	Read
LDO08_INT_SET_TYPE	0x00014711	Read
LDO08_INT_POLARITY_HIGH	0x00014712	Read
LDO08_INT_POLARITY_LOW	0x00014713	Read/Write
LDO08_INT_LATCHED_CLR	0x00014714	Write
LDO08_INT_EN_SET	0x00014715	Read/Write
LDO08_INT_EN_CLR	0x00014716	Read/Write
LDO08_INT_MID_SEL	0x0001471A	Read/Write
LDO08_INT_PRIORITY	0x0001471B	Read/Write
LDO08_INT_SELF_CLEARING	0x0001471D	Read
LDO08_ULS_VSET_LB	0x00014739	Read/Write
LDO08_ULS_VSET_UB	0x0001473A	Read/Write
LDO08_STEPPER_CTL	0x0001473B	Read/Write
LDO08_VSET_LB	0x00014740	Read/Write
LDO08_VSET_UB	0x00014741	Read/Write
LDO08_VSET_VALID_LB	0x00014742	Read
LDO08_VSET_VALID_UB	0x00014743	Read
LDO08_MODE_CTL2	0x00014744	Read/Write
LDO08_MODE_CTL1	0x00014745	Read/Write
LDO08_EN_CTL	0x00014746	Read/Write
LDO08_FOLLOW_HWEN	0x00014747	Read/Write
LDO08_PBS_VOTE_CTL	0x00014749	Read/Write
LDO08_MISC_CTL1	0x00014789	Read/Write
LDO08_PD_CTL	0x000147A0	Read/Write

LDO08_REVISION1 | 0x00014700

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO08_REVISION2 | 0x00014701

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

LDO08_REVISION3 | 0x00014702

Type: Read

Reset: 0x00000001

HW Version Regster [23:16]

Bits	Name	Description
2 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO08_REVISION4 | 0x00014703

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
2 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

LDO08_PERPH_TYPE | 0x00014704

Type: Read

Reset: 0x00000004

Peripheral Type

Bits	Name	Description
7 : 0	TYPE	LDO

LDO08_PERPH_SUBTYPE | 0x00014705

Type: Read

Reset: 0x0000006A

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	P50 for example

LDO08_STATUS1 | 0x00014708

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Retention Mode if STATUS3__RETENTION_PRESENT == 1. Otherwise unused. 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_RM_OR_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	STEPPER_DONE	Indicated stepper is done when LDO is enabled or Vset is changed. 1 = Stepper done 0 = Stepper is stepping 0: NOT_DONE_STEPPING 1: DONE_STEPPING
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass/Retention 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass/Retention 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO08_STATUS3 | 0x0001470A

Type: Read

Reset: 0x00000003

No description provided for this register.

Bits	Name	Description
0 : 0	AHC_PRESENT	Indicates to SW/user that this regulator supports AHC (automatic headroom control) 0: NO_AHC 1: AHC
1 : 1	LLC_PRESENT	Indicates to SW/user that this regulator supports LLC (load-line compensation) 0: NO_LLC 1: LLC
2 : 2	RETENTION_PRESENT	Indicates to SW/user that this regulator supports retention 0: NO_RETENTION 1: RETENTION

LDO08_INT_RT_STS | 0x00014710

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

LDO08_INT_SET_TYPE | 0x00014711

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

Bits	Name	Description
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

LDO08_INT_POLARITY_HIGH | 0x00014712

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

LDO08_INT_POLARITY_LOW | 0x00014713

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

LDO08_INT_LATCHED_CLR | 0x00014714

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

LDO08_INT_EN_SET | 0x00014715

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

LDO08_INT_EN_CLR | 0x00014716

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

LDO08_INT_MID_SEL | 0x0001471A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

LDO08_INT_PRIORITY | 0x0001471B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

LDO08_INT_SELF_CLEARING | 0x0001471D

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

LDO08_ULS_VSET_LB | 0x00014739

Type: Read/Write

Reset: 0x00000030

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	LDO upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

LDO08_ULS_VSET_UB | 0x0001473A

Type: Read/Write

Reset: 0x00000005

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	LDO upper limit stopvoltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV. Max combined value is 1.328mv (set as a default for both regs). Write will update ULS_VSET[15:0]. User must program ULS in LSB (8mV) increments to avoid undesired behavior

LDO08_STEPPER_CTL | 0x0001473B

Type: Read/Write

Reset: 0x00000083

No description provided for this register.

Bits	Name	Description
1 : 0	STEP_RATE	Voltage Stepper Rate (voltage always increases 1 LSB per step) 0 = 38.4 mV/us 1 = 19.2 mV/us 2 = 9.6 mV/us 3 = 4.8 mV/us (DEFAULT) 0: STEP_RATE_38p4_mV_us 1: STEP_RATE_19p2_mV_us 2: STEP_RATE_9p6_mV_us 3: STEP_RATE_4p8_mV_us
7 : 7	VOLTAGE_STEPPER_EN	Enables the voltage stepper. Stepper should never be disabled. 0: VOLTAGE_STEPPER_DIABLED 1: VOLTAGE_STEPPER_ENABLED

LDO08_VSET_LB | 0x00014740

Type: Read/Write

Reset: 0x000000D8

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO08_VSET_UB | 0x00014741

Type: Read/Write

Reset: 0x00000002

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO08_VSET_VALID_LB | 0x00014742

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 3	VSET_VALID_LB	Readback the valid output voltage setpoint Lower Byte (Bits 7:0). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO08_VSET_VALID_UB | 0x00014743

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Readback the valid output voltage setpoint Upper Byte (Bits 15:8). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO08_MODE_CTL2 | 0x00014744

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. This setting only matters when 0x47 FOLLOW_HWEN or 0x49 PBS_VOTE_CTL is programmed. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Retention if STATUS3__RETENTION_PRESENT == 1. Otherwise, NPM. 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: RM_OR_NPM 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO08_MODE_CTL1 | 0x00014745

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Retention if STATUS3__RETENTION_PRESENT == 1. Otherwise, NPM. 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: RM_OR_NPM 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO08_EN_CTL | 0x00014746

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

LDO08_FOLLOW_HWEN | 0x00014747

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO state. If this bit is set high then: When HWEN0=1 --> LDO if forced on (regardless of reg 0x46) When HWEN0=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN0_FALSE 1: EN_FOLLOW_HW_EN0_TRUE
1 : 1	EN_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO state. If this bit is set high then: When HWEN1=1 --> LDO if forced on (regardless of reg 0x46) When HWEN1=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN1_FALSE 1: EN_FOLLOW_HW_EN1_TRUE
2 : 2	EN_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO state. If this bit is set high then: When HWEN2=1 --> LDO if forced on (regardless of reg 0x46) When HWEN2=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN2_FALSE 1: EN_FOLLOW_HW_EN2_TRUE

Bits	Name	Description
3 : 3	EN_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE state does NOT control LDO state. If this bit is set high then: When PMIC_AWAKE=1 --> LDO if forced on (regardless of reg 0x46) When PMIC_AWAKE=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_PMIC_AWAKE_FALSE 1: EN_FOLLOW_PMIC_AWAKE_TRUE
4 : 4	MODE_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN0_FALSE 1: MODE_FOLLOW_HW_EN0_TRUE
5 : 5	MODE_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN1_FALSE 1: MODE_FOLLOW_HW_EN1_TRUE
6 : 6	MODE_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO mode. If this bit is set high then: When HWEN2=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN2_FALSE 1: MODE_FOLLOW_HW_EN2_TRUE
7 : 7	MODE_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE (sleep_b) state does NOT control LDO mode. If this bit is set high then: When PMIC_AWAKE=0--> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_PMIC_AWAKE_FALSE 1: MODE_FOLLOW_PMIC_AWAKE_TRUE

LDO08_PBS_VOTE_CTL | 0x00014749

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
6 : 6	PBS_MODE	PBS Mode vote 0 = Mode is set by MODE_CTL1 reg (DEFAULT) 1 = Mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: PBS_MODE_false 1: PBS_MODE_true
7 : 7	PBS_EN	Enables the LDO when asserted to HI. 0 = LDO will NOT be enabled (DEFAULT) 1= ENABLE LDO 0: PBS_EN_false 1: PBS_EN_true

LDO08_MISC_CTL1 | 0x00014789

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	READY_STEPDOWN_QUAL	Enables Vout qualification when stepping down Vset. 0 = Disable (default). VREG_READY stays high during downward stepping becaue VOUT > Vset 1 = Enable. VREG_READY stays low during downard stepping until Vout settles below Vset + Margin 0: STEPDOWN_QUAL_DISABLED 1: STEPDOWN_QUAL_ENABLED

LDO08_PD_CTL | 0x000147A0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Intended for keeping Vout from rising due to passFET leakage at high temperature. 1 = Leakage pulldown is always enabled 0 = Leakage pulldown is always disabled (DEFAULT) 0: LEAKAGE_PD_ALWAYS_DISABLED 1: LEAKAGE_PD_ALWAYS_ENABLED
6 : 6	WEAK_PD_EN	Weak PD is never active while LDO is on (VREG_READY=1). 1 = Weak pulldown is enabled when the regulator is disabled 0 = Weak pulldown is always inactive (DEFAULT) 0: WEAK_PD_DISABLED 1: WEAK_PD_WHEN_LDO_IS_OFF
7 : 7	STRONG_PD_EN	Strong PD is never active while LDO is on (VREG_READY=1). Strong pull-down FET Rdson will vary with Vout. 1 = Strong pulldown is enabled when the regulator is disabled (DEFAULT) 0 = Strong pulldown is always inactive. 0: STRONG_PD_ALWAYS_OFF 1: STRONG_PD_WHEN_LDO_IS_OFF

Module Name: LDO09

Base Address: 0x00014800

Register Quick Reference:

Register	Offset	Type
LDO09_REVISION1	0x00014800	Read
LDO09_REVISION2	0x00014801	Read
LDO09_REVISION3	0x00014802	Read
LDO09_REVISION4	0x00014803	Read
LDO09_PERPH_TYPE	0x00014804	Read
LDO09_PERPH_SUBTYPE	0x00014805	Read
LDO09_STATUS1	0x00014808	Read
LDO09_STATUS3	0x0001480A	Read
LDO09_INT_RT_STS	0x00014810	Read
LDO09_INT_SET_TYPE	0x00014811	Read
LDO09_INT_POLARITY_HIGH	0x00014812	Read
LDO09_INT_POLARITY_LOW	0x00014813	Read/Write
LDO09_INT_LATCHED_CLR	0x00014814	Write
LDO09_INT_EN_SET	0x00014815	Read/Write
LDO09_INT_EN_CLR	0x00014816	Read/Write

Register	Offset	Type
LDO09_INT_MID_SEL	0x0001481A	Read/Write
LDO09_INT_PRIORITY	0x0001481B	Read/Write
LDO09_INT_SELF_CLEARING	0x0001481D	Read
LDO09_ULS_VSET_LB	0x00014839	Read/Write
LDO09_ULS_VSET_UB	0x0001483A	Read/Write
LDO09_VSET_LB	0x00014840	Read/Write
LDO09_VSET_UB	0x00014841	Read/Write
LDO09_VSET_VALID_LB	0x00014842	Read
LDO09_VSET_VALID_UB	0x00014843	Read
LDO09_MODE_CTL2	0x00014844	Read/Write
LDO09_MODE_CTL1	0x00014845	Read/Write
LDO09_EN_CTL	0x00014846	Read/Write
LDO09_FOLLOW_HWEN	0x00014847	Read/Write
LDO09_PBS_VOTE_CTL	0x00014849	Read/Write
LDO09_PD_CTL	0x000148A0	Read/Write

LDO09_REVISION1 | 0x00014800

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO09_REVISION2 | 0x00014801

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

LDO09_REVISION3 | 0x00014802

Type: Read

Reset: 0x00000001

HW Version Regster [23:16]

Bits	Name	Description
2 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO09_REVISION4 | 0x00014803

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
2 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

LDO09_PERPH_TYPE | 0x00014804

Type: Read

Reset: 0x00000004

Peripheral Type

Bits	Name	Description
7 : 0	TYPE	LDO

LDO09_PERPH_SUBTYPE | 0x00014805

Type: Read

Reset: 0x00000073

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	P50 for example

LDO09_STATUS1 | 0x00014808

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Unused 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	SOFT_START_DONE	Indicates soft start is done when LDO is enabled. 1 = Softstart done 0 = Softstart in progress 0: NOT_DONE_SOFT_START 1: DONE_SOFT_START
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP

Bits	Name	Description
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO09_STATUS3 | 0x0001480A

Type: Read

Reset: 0x00000003

No description provided for this register.

Bits	Name	Description
0 : 0	AHC_PRESENT	Indicates to SW/user that this regulator supports AHC (automatic headroom control) 0: NO_AHC 1: AHC
1 : 1	LLC_PRESENT	Indicates to SW/user that this regulator supports LLC (load-line compensation) 0: NO_LLC 1: LLC

LDO09_INT_RT_STS | 0x00014810

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT

Bits	Name	Description
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

LDO09_INT_SET_TYPE | 0x00014811

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

LDO09_INT_POLARITY_HIGH | 0x00014812

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

LDO09_INT_POLARITY_LOW | 0x00014813

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

LDO09_INT_LATCHED_CLR | 0x00014814

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

LDO09_INT_EN_SET | 0x00014815

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

LDO09_INT_EN_CLR | 0x00014816

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

LDO09_INT_MID_SEL | 0x0001481A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

LDO09_INT_PRIORITY | 0x0001481B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

LDO09_INT_SELF_CLEARING | 0x0001481D

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

LDO09_ULS_VSET_LB | 0x00014839

Type: Read/Write

Reset: 0x000000D8

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	LDO upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

LDO09_ULS_VSET_UB | 0x0001483A

Type: Read/Write

Reset: 0x0000000D

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	LDO upper limit stopvoltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV. Max combined value is 1.328mv (set as a default for both regs). Write will update ULS_VSET[15:0]. User must program ULS in LSB (8mV) increments to avoid undesired behavior

LDO09_VSET_LB | 0x00014840

Type: Read/Write

Reset: 0x00000008

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO09_VSET_UB | 0x00014841

Type: Read/Write

Reset: 0x00000007

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO09_VSET_VALID_LB | 0x00014842

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 3	VSET_VALID_LB	Readback the valid output voltage setpoint Lower Byte (Bits 7:0). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO09_VSET_VALID_UB | 0x00014843

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Readback the valid output voltage setpoint Upper Byte (Bits 15:8). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO09_MODE_CTL2 | 0x00014844

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. This setting only matters when 0x47 FOLLOW_HWEN or 0x49 PBS_VOTE_CTL is programmed. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO09_MODE_CTL1 | 0x00014845

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO09_EN_CTL | 0x00014846

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

LDO09_FOLLOW_HWEN | 0x00014847

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO state. If this bit is set high then: When HWEN0=1 --> LDO if forced on (regardless of reg 0x46) When HWEN0=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN0_FALSE 1: EN_FOLLOW_HW_EN0_TRUE
1 : 1	EN_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO state. If this bit is set high then: When HWEN1=1 --> LDO if forced on (regardless of reg 0x46) When HWEN1=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN1_FALSE 1: EN_FOLLOW_HW_EN1_TRUE
2 : 2	EN_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO state. If this bit is set high then: When HWEN2=1 --> LDO if forced on (regardless of reg 0x46) When HWEN2=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN2_FALSE 1: EN_FOLLOW_HW_EN2_TRUE

Bits	Name	Description
3 : 3	EN_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE state does NOT control LDO state. If this bit is set high then: When PMIC_AWAKE=1 --> LDO if forced on (regardless of reg 0x46) When PMIC_AWAKE=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_PMIC_AWAKE_FALSE 1: EN_FOLLOW_PMIC_AWAKE_TRUE
4 : 4	MODE_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN0_FALSE 1: MODE_FOLLOW_HW_EN0_TRUE
5 : 5	MODE_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN1_FALSE 1: MODE_FOLLOW_HW_EN1_TRUE
6 : 6	MODE_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO mode. If this bit is set high then: When HWEN2=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN2_FALSE 1: MODE_FOLLOW_HW_EN2_TRUE
7 : 7	MODE_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE (sleep_b) state does NOT control LDO mode. If this bit is set high then: When PMIC_AWAKE=0--> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_PMIC_AWAKE_FALSE 1: MODE_FOLLOW_PMIC_AWAKE_TRUE

LDO09_PBS_VOTE_CTL | 0x00014849

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
6 : 6	PBS_MODE	PBS Mode vote 0 = Mode is set by MODE_CTL1 reg (DEFAULT) 1 = Mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: PBS_MODE_false 1: PBS_MODE_true
7 : 7	PBS_EN	Enables the LDO when asserted to HI. 0 = LDO will NOT be enabled (DEFAULT) 1= ENABLE LDO 0: PBS_EN_false 1: PBS_EN_true

LDO09_PD_CTL | 0x000148A0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Intended for keeping Vout from rising due to passFET leakage at high temperature. 1 = Leakage pulldown is always enabled 0 = Leakage pulldown is always disabled (DEFAULT) 0: LEAKAGE_PD_ALWAYS_DISABLED 1: LEAKAGE_PD_ALWAYS_ENABLED
6 : 6	WEAK_PD_EN	Weak PD is never active while LDO is on (VREG_READY=1). 1 = Weak pulldown is enabled when the regulator is disabled 0 = Weak pulldown is always inactive (DEFAULT) 0: WEAK_PD_DISABLED 1: WEAK_PD_WHEN_LDO_IS_OFF
7 : 7	STRONG_PD_EN	Strong PD is never active while LDO is on (VREG_READY=1). Strong pull-down FET Rdson will vary with Vout. 1 = Strong pulldown is enabled when the regulator is disabled (DEFAULT) 0 = Strong pulldown is always inactive. 0: STRONG_PD_ALWAYS_OFF 1: STRONG_PD_WHEN_LDO_IS_OFF

Module Name: LDO10

Base Address: 0x00014900

Register Quick Reference:

Register	Offset	Type
LDO10_REVISION1	0x00014900	Read
LDO10_REVISION2	0x00014901	Read
LDO10_REVISION3	0x00014902	Read
LDO10_REVISION4	0x00014903	Read
LDO10_PERPH_TYPE	0x00014904	Read
LDO10_PERPH_SUBTYPE	0x00014905	Read
LDO10_STATUS1	0x00014908	Read
LDO10_STATUS3	0x0001490A	Read
LDO10_INT_RT_STS	0x00014910	Read
LDO10_INT_SET_TYPE	0x00014911	Read
LDO10_INT_POLARITY_HIGH	0x00014912	Read
LDO10_INT_POLARITY_LOW	0x00014913	Read/Write
LDO10_INT_LATCHED_CLR	0x00014914	Write
LDO10_INT_EN_SET	0x00014915	Read/Write
LDO10_INT_EN_CLR	0x00014916	Read/Write
LDO10_INT_MID_SEL	0x0001491A	Read/Write
LDO10_INT_PRIORITY	0x0001491B	Read/Write
LDO10_INT_SELF_CLEARING	0x0001491D	Read
LDO10_ULS_VSET_LB	0x00014939	Read/Write
LDO10_ULS_VSET_UB	0x0001493A	Read/Write
LDO10_VSET_LB	0x00014940	Read/Write
LDO10_VSET_UB	0x00014941	Read/Write
LDO10_VSET_VALID_LB	0x00014942	Read
LDO10_VSET_VALID_UB	0x00014943	Read
LDO10_MODE_CTL2	0x00014944	Read/Write
LDO10_MODE_CTL1	0x00014945	Read/Write
LDO10_EN_CTL	0x00014946	Read/Write
LDO10_FOLLOW_HWEN	0x00014947	Read/Write
LDO10_PBS_VOTE_CTL	0x00014949	Read/Write
LDO10_PD_CTL	0x000149A0	Read/Write

LDO10_REVISION1 | 0x00014900

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO10_REVISION2 | 0x00014901

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

LDO10_REVISION3 | 0x00014902

Type: Read

Reset: 0x00000001

HW Version Regster [23:16]

Bits	Name	Description
2 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO10_REVISION4 | 0x00014903

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
2 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

LDO10_PERPH_TYPE | 0x00014904

Type: Read

Reset: 0x00000004

Peripheral Type

Bits	Name	Description
7 : 0	TYPE	LDO

LDO10_PERPH_SUBTYPE | 0x00014905

Type: Read

Reset: 0x00000071

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	P50 for example

LDO10_STATUS1 | 0x00014908

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Unused 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	SOFT_START_DONE	Indicates soft start is done when LDO is enabled. 1 = Softstart done 0 = Softstart in progress 0: NOT_DONE_SOFT_START 1: DONE_SOFT_START
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO10_STATUS3 | 0x0001490A

Type: Read

Reset: 0x00000003

No description provided for this register.

Bits	Name	Description
0 : 0	AHC_PRESENT	Indicates to SW/user that this regulator supports AHC (automatic headroom control) 0: NO_AHC 1: AHC
1 : 1	LLC_PRESENT	Indicates to SW/user that this regulator supports LLC (load-line compensation) 0: NO_LLC 1: LLC

LDO10_INT_RT_STS | 0x00014910

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

LDO10_INT_SET_TYPE | 0x00014911

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

LDO10_INT_POLARITY_HIGH | 0x00014912

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

LDO10_INT_POLARITY_LOW | 0x00014913

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

LDO10_INT_LATCHED_CLR | 0x00014914

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

LDO10_INT_EN_SET | 0x00014915

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

LDO10_INT_EN_CLR | 0x00014916

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

LDO10_INT_MID_SEL | 0x0001491A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

LDO10_INT_PRIORITY | 0x0001491B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

LDO10_INT_SELF_CLEARING | 0x0001491D

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

LDO10_ULS_VSET_LB | 0x00014939

Type: Read/Write

Reset: 0x000000D8

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	LDO upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

LDO10_ULS_VSET_UB | 0x0001493A

Type: Read/Write

Reset: 0x0000000D

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	LDO upper limit stopvoltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV. Max combined value is 1.328mv (set as a default for both regs). Write will update ULS_VSET[15:0]. User must program ULS in LSB (8mV) increments to avoid undesired behavior

LDO10_VSET_LB | 0x00014940

Type: Read/Write

Reset: 0x00000008

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO10_VSET_UB | 0x00014941

Type: Read/Write

Reset: 0x00000007

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO10_VSET_VALID_LB | 0x00014942

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 3	VSET_VALID_LB	Readback the valid output voltage setpoint Lower Byte (Bits 7:0). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO10_VSET_VALID_UB | 0x00014943

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Readback the valid output voltage setpoint Upper Byte (Bits 15:8). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO10_MODE_CTL2 | 0x00014944

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. This setting only matters when 0x47 FOLLOW_HWEN or 0x49 PBS_VOTE_CTL is programmed. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO10_MODE_CTL1 | 0x00014945

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO10_EN_CTL | 0x00014946

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

LDO10_FOLLOW_HWEN | 0x00014947

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO state. If this bit is set high then: When HWEN0=1 --> LDO if forced on (regardless of reg 0x46) When HWEN0=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN0_FALSE 1: EN_FOLLOW_HW_EN0_TRUE
1 : 1	EN_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO state. If this bit is set high then: When HWEN1=1 --> LDO if forced on (regardless of reg 0x46) When HWEN1=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN1_FALSE 1: EN_FOLLOW_HW_EN1_TRUE
2 : 2	EN_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO state. If this bit is set high then: When HWEN2=1 --> LDO if forced on (regardless of reg 0x46) When HWEN2=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN2_FALSE 1: EN_FOLLOW_HW_EN2_TRUE
3 : 3	EN_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE state does NOT control LDO state. If this bit is set high then: When PMIC_AWAKE=1 --> LDO if forced on (regardless of reg 0x46) When PMIC_AWAKE=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_PMIC_AWAKE_FALSE 1: EN_FOLLOW_PMIC_AWAKE_TRUE
4 : 4	MODE_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN0_FALSE 1: MODE_FOLLOW_HW_EN0_TRUE
5 : 5	MODE_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN1_FALSE 1: MODE_FOLLOW_HW_EN1_TRUE
6 : 6	MODE_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO mode. If this bit is set high then: When HWEN2=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN2_FALSE 1: MODE_FOLLOW_HW_EN2_TRUE

Bits	Name	Description
7 : 7	MODE_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE (sleep_b) state does NOT control LDO mode. If this bit is set high then: When PMIC_AWAKE=0--> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_PMIC_AWAKE_FALSE 1: MODE_FOLLOW_PMIC_AWAKE_TRUE

LDO10_PBS_VOTE_CTL | 0x00014949

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
6 : 6	PBS_MODE	PBS Mode vote 0 = Mode is set by MODE_CTL1 reg (DEFAULT) 1 = Mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: PBS_MODE_false 1: PBS_MODE_true
7 : 7	PBS_EN	Enables the LDO when asserted to HI. 0 = LDO will NOT be enabled (DEFAULT) 1= ENABLE LDO 0: PBS_EN_false 1: PBS_EN_true

LDO10_PD_CTL | 0x000149A0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Intended for keeping Vout from rising due to passFET leakage at high temperature. 1 = Leakage pulldown is always enabled 0 = Leakage pulldown is always disabled (DEFAULT) 0: LEAKAGE_PD_ALWAYS_DISABLED 1: LEAKAGE_PD_ALWAYS_ENABLED
6 : 6	WEAK_PD_EN	Weak PD is never active while LDO is on (VREG_READY=1). 1 = Weak pulldown is enabled when the regulator is disabled 0 = Weak pulldown is always inactive (DEFAULT) 0: WEAK_PD_DISABLED 1: WEAK_PD_WHEN_LDO_IS_OFF
7 : 7	STRONG_PD_EN	Strong PD is never active while LDO is on (VREG_READY=1). Strong pull-down FET Rdson will vary with Vout. 1 = Strong pulldown is enabled when the regulator is disabled (DEFAULT) 0 = Strong pulldown is always inactive. 0: STRONG_PD_ALWAYS_OFF 1: STRONG_PD_WHEN_LDO_IS_OFF

Module Name: LDO11

Base Address: 0x00014A00

Register Quick Reference:

Register	Offset	Type
LDO11_REVISION1	0x00014A00	Read
LDO11_REVISION2	0x00014A01	Read
LDO11_REVISION3	0x00014A02	Read
LDO11_REVISION4	0x00014A03	Read
LDO11_PERPH_TYPE	0x00014A04	Read
LDO11_PERPH_SUBTYPE	0x00014A05	Read
LDO11_STATUS1	0x00014A08	Read
LDO11_STATUS3	0x00014A0A	Read
LDO11_INT_RT_STS	0x00014A10	Read
LDO11_INT_SET_TYPE	0x00014A11	Read
LDO11_INT_POLARITY_HIGH	0x00014A12	Read
LDO11_INT_POLARITY_LOW	0x00014A13	Read/Write
LDO11_INT_LATCHED_CLR	0x00014A14	Write
LDO11_INT_EN_SET	0x00014A15	Read/Write
LDO11_INT_EN_CLR	0x00014A16	Read/Write

Register	Offset	Type
LDO11_INT_MID_SEL	0x00014A1A	Read/Write
LDO11_INT_PRIORITY	0x00014A1B	Read/Write
LDO11_INT_SELF_CLEARING	0x00014A1D	Read
LDO11_ULS_VSET_LB	0x00014A39	Read/Write
LDO11_ULS_VSET_UB	0x00014A3A	Read/Write
LDO11_VSET_LB	0x00014A40	Read/Write
LDO11_VSET_UB	0x00014A41	Read/Write
LDO11_VSET_VALID_LB	0x00014A42	Read
LDO11_VSET_VALID_UB	0x00014A43	Read
LDO11_MODE_CTL2	0x00014A44	Read/Write
LDO11_MODE_CTL1	0x00014A45	Read/Write
LDO11_EN_CTL	0x00014A46	Read/Write
LDO11_FOLLOW_HWEN	0x00014A47	Read/Write
LDO11_PBS_VOTE_CTL	0x00014A49	Read/Write
LDO11_PD_CTL	0x00014AA0	Read/Write

LDO11_REVISION1 | 0x00014A00

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO11_REVISION2 | 0x00014A01

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

LDO11_REVISION3 | 0x00014A02

Type: Read

Reset: 0x00000001

HW Version Regster [23:16]

Bits	Name	Description
2 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO11_REVISION4 | 0x00014A03

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
2 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

LDO11_PERPH_TYPE | 0x00014A04

Type: Read

Reset: 0x00000004

Peripheral Type

Bits	Name	Description
7 : 0	TYPE	LDO

LDO11_PERPH_SUBTYPE | 0x00014A05

Type: Read

Reset: 0x00000073

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	P50 for example

LDO11_STATUS1 | 0x00014A08

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Unused 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	SOFT_START_DONE	Indicates soft start is done when LDO is enabled. 1 = Softstart done 0 = Softstart in progress 0: NOT_DONE_SOFT_START 1: DONE_SOFT_START
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP

Bits	Name	Description
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO11_STATUS3 | 0x00014A0A

Type: Read

Reset: 0x00000003

No description provided for this register.

Bits	Name	Description
0 : 0	AHC_PRESENT	Indicates to SW/user that this regulator supports AHC (automatic headroom control) 0: NO_AHC 1: AHC
1 : 1	LLC_PRESENT	Indicates to SW/user that this regulator supports LLC (load-line compensation) 0: NO_LLC 1: LLC

LDO11_INT_RT_STS | 0x00014A10

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT

Bits	Name	Description
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

LDO11_INT_SET_TYPE | 0x00014A11

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

LDO11_INT_POLARITY_HIGH | 0x00014A12

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

LDO11_INT_POLARITY_LOW | 0x00014A13

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

LDO11_INT_LATCHED_CLR | 0x00014A14

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

LDO11_INT_EN_SET | 0x00014A15

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

LDO11_INT_EN_CLR | 0x00014A16

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

LDO11_INT_MID_SEL | 0x00014A1A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

LDO11_INT_PRIORITY | 0x00014A1B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

LDO11_INT_SELF_CLEARING | **0x00014A1D**

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

LDO11_ULS_VSET_LB | 0x00014A39

Type: Read/Write

Reset: 0x000000D8

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	LDO upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

LDO11_ULS_VSET_UB | 0x00014A3A

Type: Read/Write

Reset: 0x0000000D

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	LDO upper limit stopvoltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV. Max combined value is 1.328mv (set as a default for both regs). Write will update ULS_VSET[15:0]. User must program ULS in LSB (8mV) increments to avoid undesired behavior

LDO11_VSET_LB | 0x00014A40

Type: Read/Write

Reset: 0x00000008

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO11_VSET_UB | 0x00014A41

Type: Read/Write

Reset: 0x00000007

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO11_VSET_VALID_LB | 0x00014A42

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 3	VSET_VALID_LB	Readback the valid output voltage setpoint Lower Byte (Bits 7:0). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO11_VSET_VALID_UB | 0x00014A43

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Readback the valid output voltage setpoint Upper Byte (Bits 15:8). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO11_MODE_CTL2 | 0x00014A44

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. This setting only matters when 0x47 FOLLOW_HWEN or 0x49 PBS_VOTE_CTL is programmed. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO11_MODE_CTL1 | 0x00014A45

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO11_EN_CTL | 0x00014A46

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

LDO11_FOLLOW_HWEN | 0x00014A47

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO state. If this bit is set high then: When HWEN0=1 --> LDO if forced on (regardless of reg 0x46) When HWEN0=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN0_FALSE 1: EN_FOLLOW_HW_EN0_TRUE
1 : 1	EN_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO state. If this bit is set high then: When HWEN1=1 --> LDO if forced on (regardless of reg 0x46) When HWEN1=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN1_FALSE 1: EN_FOLLOW_HW_EN1_TRUE
2 : 2	EN_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO state. If this bit is set high then: When HWEN2=1 --> LDO if forced on (regardless of reg 0x46) When HWEN2=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN2_FALSE 1: EN_FOLLOW_HW_EN2_TRUE

Bits	Name	Description
3 : 3	EN_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE state does NOT control LDO state. If this bit is set high then: When PMIC_AWAKE=1 --> LDO if forced on (regardless of reg 0x46) When PMIC_AWAKE=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_PMIC_AWAKE_FALSE 1: EN_FOLLOW_PMIC_AWAKE_TRUE
4 : 4	MODE_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN0_FALSE 1: MODE_FOLLOW_HW_EN0_TRUE
5 : 5	MODE_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN1_FALSE 1: MODE_FOLLOW_HW_EN1_TRUE
6 : 6	MODE_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO mode. If this bit is set high then: When HWEN2=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN2_FALSE 1: MODE_FOLLOW_HW_EN2_TRUE
7 : 7	MODE_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE (sleep_b) state does NOT control LDO mode. If this bit is set high then: When PMIC_AWAKE=0--> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_PMIC_AWAKE_FALSE 1: MODE_FOLLOW_PMIC_AWAKE_TRUE

LDO11_PBS_VOTE_CTL | 0x00014A49

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
6 : 6	PBS_MODE	PBS Mode vote 0 = Mode is set by MODE_CTL1 reg (DEFAULT) 1 = Mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: PBS_MODE_false 1: PBS_MODE_true
7 : 7	PBS_EN	Enables the LDO when asserted to HI. 0 = LDO will NOT be enabled (DEFAULT) 1= ENABLE LDO 0: PBS_EN_false 1: PBS_EN_true

LDO11_PD_CTL | 0x00014AA0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Intended for keeping Vout from rising due to passFET leakage at high temperature. 1 = Leakage pulldown is always enabled 0 = Leakage pulldown is always disabled (DEFAULT) 0: LEAKAGE_PD_ALWAYS_DISABLED 1: LEAKAGE_PD_ALWAYS_ENABLED
6 : 6	WEAK_PD_EN	Weak PD is never active while LDO is on (VREG_READY=1). 1 = Weak pulldown is enabled when the regulator is disabled 0 = Weak pulldown is always inactive (DEFAULT) 0: WEAK_PD_DISABLED 1: WEAK_PD_WHEN_LDO_IS_OFF
7 : 7	STRONG_PD_EN	Strong PD is never active while LDO is on (VREG_READY=1). Strong pull-down FET Rdson will vary with Vout. 1 = Strong pulldown is enabled when the regulator is disabled (DEFAULT) 0 = Strong pulldown is always inactive. 0: STRONG_PD_ALWAYS_OFF 1: STRONG_PD_WHEN_LDO_IS_OFF

Module Name: LDO12

Base Address: 0x00014B00

Register Quick Reference:

Register	Offset	Type
LDO12_REVISION1	0x00014B00	Read
LDO12_REVISION2	0x00014B01	Read
LDO12_REVISION3	0x00014B02	Read
LDO12_REVISION4	0x00014B03	Read
LDO12_PERPH_TYPE	0x00014B04	Read
LDO12_PERPH_SUBTYPE	0x00014B05	Read
LDO12_STATUS1	0x00014B08	Read
LDO12_STATUS3	0x00014B0A	Read
LDO12_INT_RT_STS	0x00014B10	Read
LDO12_INT_SET_TYPE	0x00014B11	Read
LDO12_INT_POLARITY_HIGH	0x00014B12	Read
LDO12_INT_POLARITY_LOW	0x00014B13	Read/Write
LDO12_INT_LATCHED_CLR	0x00014B14	Write
LDO12_INT_EN_SET	0x00014B15	Read/Write
LDO12_INT_EN_CLR	0x00014B16	Read/Write
LDO12_INT_MID_SEL	0x00014B1A	Read/Write
LDO12_INT_PRIORITY	0x00014B1B	Read/Write
LDO12_INT_SELF_CLEARING	0x00014B1D	Read
LDO12_ULS_VSET_LB	0x00014B39	Read/Write
LDO12_ULS_VSET_UB	0x00014B3A	Read/Write
LDO12_VSET_LB	0x00014B40	Read/Write
LDO12_VSET_UB	0x00014B41	Read/Write
LDO12_VSET_VALID_LB	0x00014B42	Read
LDO12_VSET_VALID_UB	0x00014B43	Read
LDO12_MODE_CTL2	0x00014B44	Read/Write
LDO12_MODE_CTL1	0x00014B45	Read/Write
LDO12_EN_CTL	0x00014B46	Read/Write
LDO12_FOLLOW_HWEN	0x00014B47	Read/Write
LDO12_PBS_VOTE_CTL	0x00014B49	Read/Write
LDO12_PD_CTL	0x00014BA0	Read/Write

LDO12_REVISION1 | **0x00014B00**

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO12_REVISION2 | 0x00014B01

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

LDO12_REVISION3 | 0x00014B02

Type: Read

Reset: 0x00000001

HW Version Regster [23:16]

Bits	Name	Description
2 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO12_REVISION4 | 0x00014B03

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
2 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

LDO12_PERPH_TYPE | 0x00014B04

Type: Read

Reset: 0x00000004

Peripheral Type

Bits	Name	Description
7 : 0	TYPE	LDO

LDO12_PERPH_SUBTYPE | 0x00014B05

Type: Read

Reset: 0x00000072

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	P50 for example

LDO12_STATUS1 | 0x00014B08

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Unused 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	SOFT_START_DONE	Indicates soft start is done when LDO is enabled. 1 = Softstart done 0 = Softstart in progress 0: NOT_DONE_SOFT_START 1: DONE_SOFT_START
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO12_STATUS3 | 0x00014B0A

Type: Read

Reset: 0x00000003

No description provided for this register.

Bits	Name	Description
0 : 0	AHC_PRESENT	Indicates to SW/user that this regulator supports AHC (automatic headroom control) 0: NO_AHC 1: AHC
1 : 1	LLC_PRESENT	Indicates to SW/user that this regulator supports LLC (load-line compensation) 0: NO_LLC 1: LLC

LDO12_INT_RT_STS | 0x00014B10

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

LDO12_INT_SET_TYPE | 0x00014B11

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

LDO12_INT_POLARITY_HIGH | 0x00014B12

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

LDO12_INT_POLARITY_LOW | 0x00014B13

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

LDO12_INT_LATCHED_CLR | 0x00014B14

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

LDO12_INT_EN_SET | 0x00014B15

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

LDO12_INT_EN_CLR | 0x00014B16

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

LDO12_INT_MID_SEL | 0x00014B1A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

LDO12_INT_PRIORITY | 0x00014B1B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

LDO12_INT_SELF_CLEARING | 0x00014B1D

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

LDO12_ULS_VSET_LB | 0x00014B39

Type: Read/Write

Reset: 0x000000D8

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	LDO upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

LDO12_ULS_VSET_UB | 0x00014B3A

Type: Read/Write

Reset: 0x0000000D

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	LDO upper limit stopvoltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV. Max combined value is 1.328mv (set as a default for both regs). Write will update ULS_VSET[15:0]. User must program ULS in LSB (8mV) increments to avoid undesired behavior

LDO12_VSET_LB | 0x00014B40

Type: Read/Write

Reset: 0x00000008

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO12_VSET_UB | 0x00014B41

Type: Read/Write

Reset: 0x00000007

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO12_VSET_VALID_LB | 0x00014B42

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 3	VSET_VALID_LB	Readback the valid output voltage setpoint Lower Byte (Bits 7:0). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO12_VSET_VALID_UB | 0x00014B43

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Readback the valid output voltage setpoint Upper Byte (Bits 15:8). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO12_MODE_CTL2 | 0x00014B44

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. This setting only matters when 0x47 FOLLOW_HWEN or 0x49 PBS_VOTE_CTL is programmed. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO12_MODE_CTL1 | 0x00014B45

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO12_EN_CTL | 0x00014B46

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

LDO12_FOLLOW_HWEN | 0x00014B47

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO state. If this bit is set high then: When HWEN0=1 --> LDO if forced on (regardless of reg 0x46) When HWEN0=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN0_FALSE 1: EN_FOLLOW_HW_EN0_TRUE
1 : 1	EN_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO state. If this bit is set high then: When HWEN1=1 --> LDO if forced on (regardless of reg 0x46) When HWEN1=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN1_FALSE 1: EN_FOLLOW_HW_EN1_TRUE
2 : 2	EN_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO state. If this bit is set high then: When HWEN2=1 --> LDO if forced on (regardless of reg 0x46) When HWEN2=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN2_FALSE 1: EN_FOLLOW_HW_EN2_TRUE
3 : 3	EN_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE state does NOT control LDO state. If this bit is set high then: When PMIC_AWAKE=1 --> LDO if forced on (regardless of reg 0x46) When PMIC_AWAKE=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_PMIC_AWAKE_FALSE 1: EN_FOLLOW_PMIC_AWAKE_TRUE
4 : 4	MODE_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN0_FALSE 1: MODE_FOLLOW_HW_EN0_TRUE
5 : 5	MODE_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN1_FALSE 1: MODE_FOLLOW_HW_EN1_TRUE
6 : 6	MODE_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO mode. If this bit is set high then: When HWEN2=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN2_FALSE 1: MODE_FOLLOW_HW_EN2_TRUE

Bits	Name	Description
7 : 7	MODE_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE (sleep_b) state does NOT control LDO mode. If this bit is set high then: When PMIC_AWAKE=0--> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_PMIC_AWAKE_FALSE 1: MODE_FOLLOW_PMIC_AWAKE_TRUE

LDO12_PBS_VOTE_CTL | 0x00014B49

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
6 : 6	PBS_MODE	PBS Mode vote 0 = Mode is set by MODE_CTL1 reg (DEFAULT) 1 = Mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: PBS_MODE_false 1: PBS_MODE_true
7 : 7	PBS_EN	Enables the LDO when asserted to HI. 0 = LDO will NOT be enabled (DEFAULT) 1= ENABLE LDO 0: PBS_EN_false 1: PBS_EN_true

LDO12_PD_CTL | 0x00014BA0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Intended for keeping Vout from rising due to passFET leakage at high temperature. 1 = Leakage pulldown is always enabled 0 = Leakage pulldown is always disabled (DEFAULT) 0: LEAKAGE_PD_ALWAYS_DISABLED 1: LEAKAGE_PD_ALWAYS_ENABLED
6 : 6	WEAK_PD_EN	Weak PD is never active while LDO is on (VREG_READY=1). 1 = Weak pulldown is enabled when the regulator is disabled 0 = Weak pulldown is always inactive (DEFAULT) 0: WEAK_PD_DISABLED 1: WEAK_PD_WHEN_LDO_IS_OFF
7 : 7	STRONG_PD_EN	Strong PD is never active while LDO is on (VREG_READY=1). Strong pull-down FET Rdson will vary with Vout. 1 = Strong pulldown is enabled when the regulator is disabled (DEFAULT) 0 = Strong pulldown is always inactive. 0: STRONG_PD_ALWAYS_OFF 1: STRONG_PD_WHEN_LDO_IS_OFF

Module Name: LDO13

Base Address: 0x00014C00

Register Quick Reference:

Register	Offset	Type
LDO13_REVISION1	0x00014C00	Read
LDO13_REVISION2	0x00014C01	Read
LDO13_REVISION3	0x00014C02	Read
LDO13_REVISION4	0x00014C03	Read
LDO13_PERPH_TYPE	0x00014C04	Read
LDO13_PERPH_SUBTYPE	0x00014C05	Read
LDO13_STATUS1	0x00014C08	Read
LDO13_STATUS3	0x00014C0A	Read
LDO13_INT_RT_STS	0x00014C10	Read
LDO13_INT_SET_TYPE	0x00014C11	Read
LDO13_INT_POLARITY_HIGH	0x00014C12	Read
LDO13_INT_POLARITY_LOW	0x00014C13	Read/Write
LDO13_INT_LATCHED_CLR	0x00014C14	Write
LDO13_INT_EN_SET	0x00014C15	Read/Write
LDO13_INT_EN_CLR	0x00014C16	Read/Write

Register	Offset	Type
LDO13_INT_MID_SEL	0x00014C1A	Read/Write
LDO13_INT_PRIORITY	0x00014C1B	Read/Write
LDO13_INT_SELF_CLEARING	0x00014C1D	Read
LDO13_ULS_VSET_LB	0x00014C39	Read/Write
LDO13_ULS_VSET_UB	0x00014C3A	Read/Write
LDO13_VSET_LB	0x00014C40	Read/Write
LDO13_VSET_UB	0x00014C41	Read/Write
LDO13_VSET_VALID_LB	0x00014C42	Read
LDO13_VSET_VALID_UB	0x00014C43	Read
LDO13_MODE_CTL2	0x00014C44	Read/Write
LDO13_MODE_CTL1	0x00014C45	Read/Write
LDO13_EN_CTL	0x00014C46	Read/Write
LDO13_FOLLOW_HWEN	0x00014C47	Read/Write
LDO13_PBS_VOTE_CTL	0x00014C49	Read/Write
LDO13_PD_CTL	0x00014CA0	Read/Write

LDO13_REVISION1 | 0x00014C00

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO13_REVISION2 | 0x00014C01

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

LDO13_REVISION3 | 0x00014C02

Type: Read

Reset: 0x00000001

HW Version Regster [23:16]

Bits	Name	Description
2 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO13_REVISION4 | 0x00014C03

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
2 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

LDO13_PERPH_TYPE | 0x00014C04

Type: Read

Reset: 0x00000004

Peripheral Type

Bits	Name	Description
7 : 0	TYPE	LDO

LDO13_PERPH_SUBTYPE | 0x00014C05

Type: Read

Reset: 0x00000072

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	P50 for example

LDO13_STATUS1 | 0x00014C08

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Unused 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	SOFT_START_DONE	Indicates soft start is done when LDO is enabled. 1 = Softstart done 0 = Softstart in progress 0: NOT_DONE_SOFT_START 1: DONE_SOFT_START
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP

Bits	Name	Description
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO13_STATUS3 | 0x00014C0A

Type: Read

Reset: 0x00000003

No description provided for this register.

Bits	Name	Description
0 : 0	AHC_PRESENT	Indicates to SW/user that this regulator supports AHC (automatic headroom control) 0: NO_AHC 1: AHC
1 : 1	LLC_PRESENT	Indicates to SW/user that this regulator supports LLC (load-line compensation) 0: NO_LLC 1: LLC

LDO13_INT_RT_STS | 0x00014C10

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT

Bits	Name	Description
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

LDO13_INT_SET_TYPE | 0x00014C11

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

LDO13_INT_POLARITY_HIGH | 0x00014C12

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

LDO13_INT_POLARITY_LOW | 0x00014C13

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

LDO13_INT_LATCHED_CLR | 0x00014C14

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

LDO13_INT_EN_SET | 0x00014C15

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

LDO13_INT_EN_CLR | 0x00014C16

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

LDO13_INT_MID_SEL | 0x00014C1A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

LDO13_INT_PRIORITY | 0x00014C1B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

LDO13_INT_SELF_CLEARING **| 0x00014C1D**

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

LDO13_ULS_VSET_LB | 0x00014C39

Type: Read/Write

Reset: 0x000000D8

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	LDO upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

LDO13_ULS_VSET_UB | 0x00014C3A

Type: Read/Write

Reset: 0x0000000D

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	LDO upper limit stopvoltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV. Max combined value is 1.328mv (set as a default for both regs). Write will update ULS_VSET[15:0]. User must program ULS in LSB (8mV) increments to avoid undesired behavior

LDO13_VSET_LB | 0x00014C40

Type: Read/Write

Reset: 0x00000008

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO13_VSET_UB | 0x00014C41

Type: Read/Write

Reset: 0x00000007

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO13_VSET_VALID_LB | 0x00014C42

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 3	VSET_VALID_LB	Readback the valid output voltage setpoint Lower Byte (Bits 7:0). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO13_VSET_VALID_UB | 0x00014C43

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Readback the valid output voltage setpoint Upper Byte (Bits 15:8). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO13_MODE_CTL2 | 0x00014C44

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. This setting only matters when 0x47 FOLLOW_HWEN or 0x49 PBS_VOTE_CTL is programmed. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO13_MODE_CTL1 | 0x00014C45

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO13_EN_CTL | 0x00014C46

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

LDO13_FOLLOW_HWEN | 0x00014C47

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO state. If this bit is set high then: When HWEN0=1 --> LDO if forced on (regardless of reg 0x46) When HWEN0=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN0_FALSE 1: EN_FOLLOW_HW_EN0_TRUE
1 : 1	EN_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO state. If this bit is set high then: When HWEN1=1 --> LDO if forced on (regardless of reg 0x46) When HWEN1=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN1_FALSE 1: EN_FOLLOW_HW_EN1_TRUE
2 : 2	EN_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO state. If this bit is set high then: When HWEN2=1 --> LDO if forced on (regardless of reg 0x46) When HWEN2=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN2_FALSE 1: EN_FOLLOW_HW_EN2_TRUE

Bits	Name	Description
3 : 3	EN_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE state does NOT control LDO state. If this bit is set high then: When PMIC_AWAKE=1 --> LDO if forced on (regardless of reg 0x46) When PMIC_AWAKE=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_PMIC_AWAKE_FALSE 1: EN_FOLLOW_PMIC_AWAKE_TRUE
4 : 4	MODE_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN0_FALSE 1: MODE_FOLLOW_HW_EN0_TRUE
5 : 5	MODE_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN1_FALSE 1: MODE_FOLLOW_HW_EN1_TRUE
6 : 6	MODE_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO mode. If this bit is set high then: When HWEN2=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN2_FALSE 1: MODE_FOLLOW_HW_EN2_TRUE
7 : 7	MODE_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE (sleep_b) state does NOT control LDO mode. If this bit is set high then: When PMIC_AWAKE=0--> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_PMIC_AWAKE_FALSE 1: MODE_FOLLOW_PMIC_AWAKE_TRUE

LDO13_PBS_VOTE_CTL | 0x00014C49

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
6 : 6	PBS_MODE	PBS Mode vote 0 = Mode is set by MODE_CTL1 reg (DEFAULT) 1 = Mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: PBS_MODE_false 1: PBS_MODE_true
7 : 7	PBS_EN	Enables the LDO when asserted to HI. 0 = LDO will NOT be enabled (DEFAULT) 1= ENABLE LDO 0: PBS_EN_false 1: PBS_EN_true

LDO13_PD_CTL | 0x00014CA0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Intended for keeping Vout from rising due to passFET leakage at high temperature. 1 = Leakage pulldown is always enabled 0 = Leakage pulldown is always disabled (DEFAULT) 0: LEAKAGE_PD_ALWAYS_DISABLED 1: LEAKAGE_PD_ALWAYS_ENABLED
6 : 6	WEAK_PD_EN	Weak PD is never active while LDO is on (VREG_READY=1). 1 = Weak pulldown is enabled when the regulator is disabled 0 = Weak pulldown is always inactive (DEFAULT) 0: WEAK_PD_DISABLED 1: WEAK_PD_WHEN_LDO_IS_OFF
7 : 7	STRONG_PD_EN	Strong PD is never active while LDO is on (VREG_READY=1). Strong pull-down FET Rdson will vary with Vout. 1 = Strong pulldown is enabled when the regulator is disabled (DEFAULT) 0 = Strong pulldown is always inactive. 0: STRONG_PD_ALWAYS_OFF 1: STRONG_PD_WHEN_LDO_IS_OFF

Module Name: LDO14

Base Address: 0x00014D00

Register Quick Reference:

Register	Offset	Type
LDO14_REVISION1	0x00014D00	Read
LDO14_REVISION2	0x00014D01	Read
LDO14_REVISION3	0x00014D02	Read
LDO14_REVISION4	0x00014D03	Read
LDO14_PERPH_TYPE	0x00014D04	Read
LDO14_PERPH_SUBTYPE	0x00014D05	Read
LDO14_STATUS1	0x00014D08	Read
LDO14_STATUS3	0x00014D0A	Read
LDO14_INT_RT_STS	0x00014D10	Read
LDO14_INT_SET_TYPE	0x00014D11	Read
LDO14_INT_POLARITY_HIGH	0x00014D12	Read
LDO14_INT_POLARITY_LOW	0x00014D13	Read/Write
LDO14_INT_LATCHED_CLR	0x00014D14	Write
LDO14_INT_EN_SET	0x00014D15	Read/Write
LDO14_INT_EN_CLR	0x00014D16	Read/Write
LDO14_INT_MID_SEL	0x00014D1A	Read/Write
LDO14_INT_PRIORITY	0x00014D1B	Read/Write
LDO14_INT_SELF_CLEARING	0x00014D1D	Read
LDO14_ULS_VSET_LB	0x00014D39	Read/Write
LDO14_ULS_VSET_UB	0x00014D3A	Read/Write
LDO14_VSET_LB	0x00014D40	Read/Write
LDO14_VSET_UB	0x00014D41	Read/Write
LDO14_VSET_VALID_LB	0x00014D42	Read
LDO14_VSET_VALID_UB	0x00014D43	Read
LDO14_MODE_CTL2	0x00014D44	Read/Write
LDO14_MODE_CTL1	0x00014D45	Read/Write
LDO14_EN_CTL	0x00014D46	Read/Write
LDO14_FOLLOW_HWEN	0x00014D47	Read/Write
LDO14_PBS_VOTE_CTL	0x00014D49	Read/Write
LDO14_PD_CTL	0x00014DA0	Read/Write

LDO14_REVISION1 | **0x00014D00**

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO14_REVISION2 | 0x00014D01

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

LDO14_REVISION3 | 0x00014D02

Type: Read

Reset: 0x00000001

HW Version Regster [23:16]

Bits	Name	Description
2 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO14_REVISION4 | 0x00014D03

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
2 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

LDO14_PERPH_TYPE | 0x00014D04

Type: Read

Reset: 0x00000004

Peripheral Type

Bits	Name	Description
7 : 0	TYPE	LDO

LDO14_PERPH_SUBTYPE | 0x00014D05

Type: Read

Reset: 0x00000073

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	P50 for example

LDO14_STATUS1 | 0x00014D08

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Unused 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	SOFT_START_DONE	Indicates soft start is done when LDO is enabled. 1 = Softstart done 0 = Softstart in progress 0: NOT_DONE_SOFT_START 1: DONE_SOFT_START
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO14_STATUS3 | 0x00014D0A

Type: Read

Reset: 0x00000003

No description provided for this register.

Bits	Name	Description
0 : 0	AHC_PRESENT	Indicates to SW/user that this regulator supports AHC (automatic headroom control) 0: NO_AHC 1: AHC
1 : 1	LLC_PRESENT	Indicates to SW/user that this regulator supports LLC (load-line compensation) 0: NO_LLC 1: LLC

LDO14_INT_RT_STS | 0x00014D10

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

LDO14_INT_SET_TYPE | 0x00014D11

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

LDO14_INT_POLARITY_HIGH | 0x00014D12

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

LDO14_INT_POLARITY_LOW | 0x00014D13

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

LDO14_INT_LATCHED_CLR | 0x00014D14

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

LDO14_INT_EN_SET | 0x00014D15

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

LDO14_INT_EN_CLR | 0x00014D16

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

LDO14_INT_MID_SEL | 0x00014D1A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

LDO14_INT_PRIORITY | 0x00014D1B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

LDO14_INT_SELF_CLEARING | 0x00014D1D

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

LDO14_ULS_VSET_LB | 0x00014D39

Type: Read/Write

Reset: 0x000000D8

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	LDO upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

LDO14_ULS_VSET_UB | 0x00014D3A

Type: Read/Write

Reset: 0x0000000D

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	LDO upper limit stopvoltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV. Max combined value is 1.328mv (set as a default for both regs). Write will update ULS_VSET[15:0]. User must program ULS in LSB (8mV) increments to avoid undesired behavior

LDO14_VSET_LB | 0x00014D40

Type: Read/Write

Reset: 0x00000008

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO14_VSET_UB | 0x00014D41

Type: Read/Write

Reset: 0x00000007

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO14_VSET_VALID_LB | 0x00014D42

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 3	VSET_VALID_LB	Readback the valid output voltage setpoint Lower Byte (Bits 7:0). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO14_VSET_VALID_UB | 0x00014D43

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Readback the valid output voltage setpoint Upper Byte (Bits 15:8). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO14_MODE_CTL2 | 0x00014D44

Type: Read/Write

Reset: 0x00000007

Bits	Name	Description
2 : 0	MODE_SECONDARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. This setting only matters when 0x47 FOLLOW_HWEN or 0x49 PBS_VOTE_CTL is programmed. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO14_MODE_CTL1 | 0x00014D45

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO14_EN_CTL | 0x00014D46

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

LDO14_FOLLOW_HWEN | 0x00014D47

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO state. If this bit is set high then: When HWEN0=1 --> LDO if forced on (regardless of reg 0x46) When HWEN0=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN0_FALSE 1: EN_FOLLOW_HW_EN0_TRUE
1 : 1	EN_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO state. If this bit is set high then: When HWEN1=1 --> LDO if forced on (regardless of reg 0x46) When HWEN1=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN1_FALSE 1: EN_FOLLOW_HW_EN1_TRUE
2 : 2	EN_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO state. If this bit is set high then: When HWEN2=1 --> LDO if forced on (regardless of reg 0x46) When HWEN2=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN2_FALSE 1: EN_FOLLOW_HW_EN2_TRUE
3 : 3	EN_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE state does NOT control LDO state. If this bit is set high then: When PMIC_AWAKE=1 --> LDO if forced on (regardless of reg 0x46) When PMIC_AWAKE=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_PMIC_AWAKE_FALSE 1: EN_FOLLOW_PMIC_AWAKE_TRUE
4 : 4	MODE_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN0_FALSE 1: MODE_FOLLOW_HW_EN0_TRUE
5 : 5	MODE_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN1_FALSE 1: MODE_FOLLOW_HW_EN1_TRUE
6 : 6	MODE_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO mode. If this bit is set high then: When HWEN2=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN2_FALSE 1: MODE_FOLLOW_HW_EN2_TRUE

Bits	Name	Description
7 : 7	MODE_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE (sleep_b) state does NOT control LDO mode. If this bit is set high then: When PMIC_AWAKE=0--> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_PMIC_AWAKE_FALSE 1: MODE_FOLLOW_PMIC_AWAKE_TRUE

LDO14_PBS_VOTE_CTL | 0x00014D49

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
6 : 6	PBS_MODE	PBS Mode vote 0 = Mode is set by MODE_CTL1 reg (DEFAULT) 1 = Mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: PBS_MODE_false 1: PBS_MODE_true
7 : 7	PBS_EN	Enables the LDO when asserted to HI. 0 = LDO will NOT be enabled (DEFAULT) 1= ENABLE LDO 0: PBS_EN_false 1: PBS_EN_true

LDO14_PD_CTL | 0x00014DA0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Intended for keeping Vout from rising due to passFET leakage at high temperature. 1 = Leakage pulldown is always enabled 0 = Leakage pulldown is always disabled (DEFAULT) 0: LEAKAGE_PD_ALWAYS_DISABLED 1: LEAKAGE_PD_ALWAYS_ENABLED
6 : 6	WEAK_PD_EN	Weak PD is never active while LDO is on (VREG_READY=1). 1 = Weak pulldown is enabled when the regulator is disabled 0 = Weak pulldown is always inactive (DEFAULT) 0: WEAK_PD_DISABLED 1: WEAK_PD_WHEN_LDO_IS_OFF
7 : 7	STRONG_PD_EN	Strong PD is never active while LDO is on (VREG_READY=1). Strong pull-down FET Rdson will vary with Vout. 1 = Strong pulldown is enabled when the regulator is disabled (DEFAULT) 0 = Strong pulldown is always inactive. 0: STRONG_PD_ALWAYS_OFF 1: STRONG_PD_WHEN_LDO_IS_OFF

Module Name: LDO15

Base Address: 0x00014E00

Register Quick Reference:

Register	Offset	Type
LDO15_REVISION1	0x00014E00	Read
LDO15_REVISION2	0x00014E01	Read
LDO15_REVISION3	0x00014E02	Read
LDO15_REVISION4	0x00014E03	Read
LDO15_PERPH_TYPE	0x00014E04	Read
LDO15_PERPH_SUBTYPE	0x00014E05	Read
LDO15_STATUS1	0x00014E08	Read
LDO15_STATUS3	0x00014E0A	Read
LDO15_INT_RT_STS	0x00014E10	Read
LDO15_INT_SET_TYPE	0x00014E11	Read
LDO15_INT_POLARITY_HIGH	0x00014E12	Read
LDO15_INT_POLARITY_LOW	0x00014E13	Read/Write
LDO15_INT_LATCHED_CLR	0x00014E14	Write
LDO15_INT_EN_SET	0x00014E15	Read/Write
LDO15_INT_EN_CLR	0x00014E16	Read/Write

Register	Offset	Type
LDO15_INT_MID_SEL	0x00014E1A	Read/Write
LDO15_INT_PRIORITY	0x00014E1B	Read/Write
LDO15_INT_SELF_CLEARING	0x00014E1D	Read
LDO15_ULS_VSET_LB	0x00014E39	Read/Write
LDO15_ULS_VSET_UB	0x00014E3A	Read/Write
LDO15_VSET_LB	0x00014E40	Read/Write
LDO15_VSET_UB	0x00014E41	Read/Write
LDO15_VSET_VALID_LB	0x00014E42	Read
LDO15_VSET_VALID_UB	0x00014E43	Read
LDO15_MODE_CTL2	0x00014E44	Read/Write
LDO15_MODE_CTL1	0x00014E45	Read/Write
LDO15_EN_CTL	0x00014E46	Read/Write
LDO15_FOLLOW_HWEN	0x00014E47	Read/Write
LDO15_PBS_VOTE_CTL	0x00014E49	Read/Write
LDO15_PD_CTL	0x00014EA0	Read/Write

LDO15_REVISION1 | 0x00014E00

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO15_REVISION2 | 0x00014E01

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

LDO15_REVISION3 | 0x00014E02

Type: Read

Reset: 0x00000000

HW Version Regster [23:16]

Bits	Name	Description
2 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO15_REVISION4 | 0x00014E03

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
2 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

LDO15_PERPH_TYPE | 0x00014E04

Type: Read

Reset: 0x00000004

Peripheral Type

Bits	Name	Description
7 : 0	TYPE	LDO

LDO15_PERPH_SUBTYPE | 0x00014E05

Type: Read

Reset: 0x0000007B

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	P50 for example

LDO15_STATUS1 | 0x00014E08

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Unused 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	SOFT_START_DONE	Indicates soft start is done when LDO is enabled. 1 = Softstart done 0 = Softstart in progress 0: NOT_DONE_SOFT_START 1: DONE_SOFT_START
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP

Bits	Name	Description
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO15_STATUS3 | 0x00014E0A

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	AHC_PRESENT	Indicates to SW/user that this regulator supports AHC (automatic headroom control) 0: NO_AHC 1: AHC
1 : 1	LLC_PRESENT	Indicates to SW/user that this regulator supports LLC (load-line compensation) 0: NO_LLC 1: LLC

LDO15_INT_RT_STS | 0x00014E10

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT

Bits	Name	Description
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

LDO15_INT_SET_TYPE | 0x00014E11

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

LDO15_INT_POLARITY_HIGH | 0x00014E12

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

LDO15_INT_POLARITY_LOW | 0x00014E13

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

LDO15_INT_LATCHED_CLR | 0x00014E14

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

LDO15_INT_EN_SET | 0x00014E15

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

LDO15_INT_EN_CLR | 0x00014E16

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

LDO15_INT_MID_SEL | 0x00014E1A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

LDO15_INT_PRIORITY | 0x00014E1B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

LDO15_INT_SELF_CLEARING **| 0x00014E1D**

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

LDO15_ULS_VSET_LB | 0x00014E39

Type: Read/Write

Reset: 0x000000D8

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	LDO upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

LDO15_ULS_VSET_UB | 0x00014E3A

Type: Read/Write

Reset: 0x0000000D

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	LDO upper limit stopvoltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV. Max combined value is 1.328mv (set as a default for both regs). Write will update ULS_VSET[15:0]. User must program ULS in LSB (8mV) increments to avoid undesired behavior

LDO15_VSET_LB | 0x00014E40

Type: Read/Write

Reset: 0x00000050

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO15_VSET_UB | 0x00014E41

Type: Read/Write

Reset: 0x0000000C

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO15_VSET_VALID_LB | 0x00014E42

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 3	VSET_VALID_LB	Readback the valid output voltage setpoint Lower Byte (Bits 7:0). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO15_VSET_VALID_UB | 0x00014E43

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Readback the valid output voltage setpoint Upper Byte (Bits 15:8). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO15_MODE_CTL2 | 0x00014E44

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. This setting only matters when 0x47 FOLLOW_HWEN or 0x49 PBS_VOTE_CTL is programmed. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO15_MODE_CTL1 | 0x00014E45

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO15_EN_CTL | 0x00014E46

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

LDO15_FOLLOW_HWEN | 0x00014E47

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO state. If this bit is set high then: When HWEN0=1 --> LDO if forced on (regardless of reg 0x46) When HWEN0=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN0_FALSE 1: EN_FOLLOW_HW_EN0_TRUE
1 : 1	EN_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO state. If this bit is set high then: When HWEN1=1 --> LDO if forced on (regardless of reg 0x46) When HWEN1=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN1_FALSE 1: EN_FOLLOW_HW_EN1_TRUE
2 : 2	EN_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO state. If this bit is set high then: When HWEN2=1 --> LDO if forced on (regardless of reg 0x46) When HWEN2=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN2_FALSE 1: EN_FOLLOW_HW_EN2_TRUE

Bits	Name	Description
3 : 3	EN_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE state does NOT control LDO state. If this bit is set high then: When PMIC_AWAKE=1 --> LDO if forced on (regardless of reg 0x46) When PMIC_AWAKE=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_PMIC_AWAKE_FALSE 1: EN_FOLLOW_PMIC_AWAKE_TRUE
4 : 4	MODE_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN0_FALSE 1: MODE_FOLLOW_HW_EN0_TRUE
5 : 5	MODE_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN1_FALSE 1: MODE_FOLLOW_HW_EN1_TRUE
6 : 6	MODE_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO mode. If this bit is set high then: When HWEN2=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN2_FALSE 1: MODE_FOLLOW_HW_EN2_TRUE
7 : 7	MODE_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE (sleep_b) state does NOT control LDO mode. If this bit is set high then: When PMIC_AWAKE=0--> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_PMIC_AWAKE_FALSE 1: MODE_FOLLOW_PMIC_AWAKE_TRUE

LDO15_PBS_VOTE_CTL | 0x00014E49

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
6 : 6	PBS_MODE	PBS Mode vote 0 = Mode is set by MODE_CTL1 reg (DEFAULT) 1 = Mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: PBS_MODE_false 1: PBS_MODE_true
7 : 7	PBS_EN	Enables the LDO when asserted to HI. 0 = LDO will NOT be enabled (DEFAULT) 1= ENABLE LDO 0: PBS_EN_false 1: PBS_EN_true

LDO15_PD_CTL | 0x00014EA0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Intended for keeping Vout from rising due to passFET leakage at high temperature. 1 = Leakage pulldown is always enabled 0 = Leakage pulldown is always disabled (DEFAULT) 0: LEAKAGE_PD_ALWAYS_DISABLED 1: LEAKAGE_PD_ALWAYS_ENABLED
6 : 6	WEAK_PD_EN	Weak PD is never active while LDO is on (VREG_READY=1). 1 = Weak pulldown is enabled when the regulator is disabled 0 = Weak pulldown is always inactive (DEFAULT) 0: WEAK_PD_DISABLED 1: WEAK_PD_WHEN_LDO_IS_OFF
7 : 7	STRONG_PD_EN	Strong PD is never active while LDO is on (VREG_READY=1). Strong pull-down FET Rdson will vary with Vout. 1 = Strong pulldown is enabled when the regulator is disabled (DEFAULT) 0 = Strong pulldown is always inactive. 0: STRONG_PD_ALWAYS_OFF 1: STRONG_PD_WHEN_LDO_IS_OFF

Module Name: LDO16

Base Address: 0x00014F00

Register Quick Reference:

Register	Offset	Type
LDO16_REVISION1	0x00014F00	Read
LDO16_REVISION2	0x00014F01	Read
LDO16_REVISION3	0x00014F02	Read
LDO16_REVISION4	0x00014F03	Read
LDO16_PERPH_TYPE	0x00014F04	Read
LDO16_PERPH_SUBTYPE	0x00014F05	Read
LDO16_STATUS1	0x00014F08	Read
LDO16_STATUS3	0x00014F0A	Read
LDO16_INT_RT_STS	0x00014F10	Read
LDO16_INT_SET_TYPE	0x00014F11	Read
LDO16_INT_POLARITY_HIGH	0x00014F12	Read
LDO16_INT_POLARITY_LOW	0x00014F13	Read/Write
LDO16_INT_LATCHED_CLR	0x00014F14	Write
LDO16_INT_EN_SET	0x00014F15	Read/Write
LDO16_INT_EN_CLR	0x00014F16	Read/Write
LDO16_INT_MID_SEL	0x00014F1A	Read/Write
LDO16_INT_PRIORITY	0x00014F1B	Read/Write
LDO16_INT_SELF_CLEARING	0x00014F1D	Read
LDO16_ULS_VSET_LB	0x00014F39	Read/Write
LDO16_ULS_VSET_UB	0x00014F3A	Read/Write
LDO16_VSET_LB	0x00014F40	Read/Write
LDO16_VSET_UB	0x00014F41	Read/Write
LDO16_VSET_VALID_LB	0x00014F42	Read
LDO16_VSET_VALID_UB	0x00014F43	Read
LDO16_MODE_CTL2	0x00014F44	Read/Write
LDO16_MODE_CTL1	0x00014F45	Read/Write
LDO16_EN_CTL	0x00014F46	Read/Write
LDO16_FOLLOW_HWEN	0x00014F47	Read/Write
LDO16_PBS_VOTE_CTL	0x00014F49	Read/Write
LDO16_PD_CTL	0x00014FA0	Read/Write

LDO16_REVISION1 | **0x00014F00**

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO16_REVISION2 | 0x00014F01

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

LDO16_REVISION3 | 0x00014F02

Type: Read

Reset: 0x00000001

HW Version Regster [23:16]

Bits	Name	Description
2 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO16_REVISION4 | 0x00014F03

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
2 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

LDO16_PERPH_TYPE | 0x00014F04

Type: Read

Reset: 0x00000004

Peripheral Type

Bits	Name	Description
7 : 0	TYPE	LDO

LDO16_PERPH_SUBTYPE | 0x00014F05

Type: Read

Reset: 0x00000071

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	P50 for example

LDO16_STATUS1 | 0x00014F08

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Unused 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	SOFT_START_DONE	Indicates soft start is done when LDO is enabled. 1 = Softstart done 0 = Softstart in progress 0: NOT_DONE_SOFT_START 1: DONE_SOFT_START
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO16_STATUS3 | 0x00014F0A

Type: Read

Reset: 0x00000003

No description provided for this register.

Bits	Name	Description
0 : 0	AHC_PRESENT	Indicates to SW/user that this regulator supports AHC (automatic headroom control) 0: NO_AHC 1: AHC
1 : 1	LLC_PRESENT	Indicates to SW/user that this regulator supports LLC (load-line compensation) 0: NO_LLC 1: LLC

LDO16_INT_RT_STS | 0x00014F10

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

LDO16_INT_SET_TYPE | 0x00014F11

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

LDO16_INT_POLARITY_HIGH | 0x00014F12

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

LDO16_INT_POLARITY_LOW | 0x00014F13

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

LDO16_INT_LATCHED_CLR | 0x00014F14

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

LDO16_INT_EN_SET | 0x00014F15

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

LDO16_INT_EN_CLR | 0x00014F16

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

LDO16_INT_MID_SEL | 0x00014F1A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

LDO16_INT_PRIORITY | 0x00014F1B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

LDO16_INT_SELF_CLEARING | 0x00014F1D

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

LDO16_ULS_VSET_LB | 0x00014F39

Type: Read/Write

Reset: 0x000000D8

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	LDO upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

LDO16_ULS_VSET_UB | 0x00014F3A

Type: Read/Write

Reset: 0x0000000D

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	LDO upper limit stopvoltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV. Max combined value is 1.328mv (set as a default for both regs). Write will update ULS_VSET[15:0]. User must program ULS in LSB (8mV) increments to avoid undesired behavior

LDO16_VSET_LB | 0x00014F40

Type: Read/Write

Reset: 0x00000008

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO16_VSET_UB | 0x00014F41

Type: Read/Write

Reset: 0x00000007

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO16_VSET_VALID_LB | 0x00014F42

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 3	VSET_VALID_LB	Readback the valid output voltage setpoint Lower Byte (Bits 7:0). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO16_VSET_VALID_UB | 0x00014F43

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Readback the valid output voltage setpoint Upper Byte (Bits 15:8). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO16_MODE_CTL2 | 0x00014F44

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. This setting only matters when 0x47 FOLLOW_HWEN or 0x49 PBS_VOTE_CTL is programmed. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO16_MODE_CTL1 | 0x00014F45

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO16_EN_CTL | 0x00014F46

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

LDO16_FOLLOW_HWEN | 0x00014F47

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO state. If this bit is set high then: When HWEN0=1 --> LDO if forced on (regardless of reg 0x46) When HWEN0=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN0_FALSE 1: EN_FOLLOW_HW_EN0_TRUE
1 : 1	EN_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO state. If this bit is set high then: When HWEN1=1 --> LDO if forced on (regardless of reg 0x46) When HWEN1=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN1_FALSE 1: EN_FOLLOW_HW_EN1_TRUE
2 : 2	EN_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO state. If this bit is set high then: When HWEN2=1 --> LDO if forced on (regardless of reg 0x46) When HWEN2=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN2_FALSE 1: EN_FOLLOW_HW_EN2_TRUE
3 : 3	EN_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE state does NOT control LDO state. If this bit is set high then: When PMIC_AWAKE=1 --> LDO if forced on (regardless of reg 0x46) When PMIC_AWAKE=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_PMIC_AWAKE_FALSE 1: EN_FOLLOW_PMIC_AWAKE_TRUE
4 : 4	MODE_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN0_FALSE 1: MODE_FOLLOW_HW_EN0_TRUE
5 : 5	MODE_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN1_FALSE 1: MODE_FOLLOW_HW_EN1_TRUE
6 : 6	MODE_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO mode. If this bit is set high then: When HWEN2=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN2_FALSE 1: MODE_FOLLOW_HW_EN2_TRUE

Bits	Name	Description
7 : 7	MODE_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE (sleep_b) state does NOT control LDO mode. If this bit is set high then: When PMIC_AWAKE=0--> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_PMIC_AWAKE_FALSE 1: MODE_FOLLOW_PMIC_AWAKE_TRUE

LDO16_PBS_VOTE_CTL | 0x00014F49

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
6 : 6	PBS_MODE	PBS Mode vote 0 = Mode is set by MODE_CTL1 reg (DEFAULT) 1 = Mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: PBS_MODE_false 1: PBS_MODE_true
7 : 7	PBS_EN	Enables the LDO when asserted to HI. 0 = LDO will NOT be enabled (DEFAULT) 1= ENABLE LDO 0: PBS_EN_false 1: PBS_EN_true

LDO16_PD_CTL | 0x00014FA0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Intended for keeping Vout from rising due to passFET leakage at high temperature. 1 = Leakage pulldown is always enabled 0 = Leakage pulldown is always disabled (DEFAULT) 0: LEAKAGE_PD_ALWAYS_DISABLED 1: LEAKAGE_PD_ALWAYS_ENABLED
6 : 6	WEAK_PD_EN	Weak PD is never active while LDO is on (VREG_READY=1). 1 = Weak pulldown is enabled when the regulator is disabled 0 = Weak pulldown is always inactive (DEFAULT) 0: WEAK_PD_DISABLED 1: WEAK_PD_WHEN_LDO_IS_OFF
7 : 7	STRONG_PD_EN	Strong PD is never active while LDO is on (VREG_READY=1). Strong pull-down FET Rdson will vary with Vout. 1 = Strong pulldown is enabled when the regulator is disabled (DEFAULT) 0 = Strong pulldown is always inactive. 0: STRONG_PD_ALWAYS_OFF 1: STRONG_PD_WHEN_LDO_IS_OFF

Module Name: LDO17

Base Address: 0x00015000

Register Quick Reference:

Register	Offset	Type
LDO17_REVISION1	0x00015000	Read
LDO17_REVISION2	0x00015001	Read
LDO17_REVISION3	0x00015002	Read
LDO17_REVISION4	0x00015003	Read
LDO17_PERPH_TYPE	0x00015004	Read
LDO17_PERPH_SUBTYPE	0x00015005	Read
LDO17_STATUS1	0x00015008	Read
LDO17_STATUS3	0x0001500A	Read
LDO17_STATUS4	0x0001500B	Read
LDO17_INT_RT_STS	0x00015010	Read
LDO17_INT_SET_TYPE	0x00015011	Read
LDO17_INT_POLARITY_HIGH	0x00015012	Read
LDO17_INT_POLARITY_LOW	0x00015013	Read/Write
LDO17_INT_LATCHED_CLR	0x00015014	Write
LDO17_INT_EN_SET	0x00015015	Read/Write

Register	Offset	Type
LDO17_INT_EN_CLR	0x00015016	Read/Write
LDO17_INT_MID_SEL	0x0001501A	Read/Write
LDO17_INT_PRIORITY	0x0001501B	Read/Write
LDO17_INT_SELF_CLEARING	0x0001501D	Read
LDO17_ULS_VSET_LB	0x00015039	Read/Write
LDO17_ULS_VSET_UB	0x0001503A	Read/Write
LDO17_STEPPER_CTL	0x0001503B	Read/Write
LDO17_VSET_LB	0x00015040	Read/Write
LDO17_VSET_UB	0x00015041	Read/Write
LDO17_VSET_VALID_LB	0x00015042	Read
LDO17_VSET_VALID_UB	0x00015043	Read
LDO17_MODE_CTL2	0x00015044	Read/Write
LDO17_MODE_CTL1	0x00015045	Read/Write
LDO17_EN_CTL	0x00015046	Read/Write
LDO17_FOLLOW_HWEN	0x00015047	Read/Write
LDO17_PBS_VOTE_CTL	0x00015049	Read/Write
LDO17_MISC_CTL1	0x00015089	Read/Write
LDO17_PD_CTL	0x000150A0	Read/Write

LDO17_REVISION1 | 0x00015000

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO17_REVISION2 | 0x00015001

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

LDO17_REVISION3 | 0x00015002

Type: Read

Reset: 0x00000001

HW Version Regster [23:16]

Bits	Name	Description
2 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO17_REVISION4 | 0x00015003

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
2 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

LDO17_PERPH_TYPE | 0x00015004

Type: Read

Reset: 0x00000004

Peripheral Type

Bits	Name	Description
7 : 0	TYPE	LDO

LDO17_PERPH_SUBTYPE | 0x00015005

Type: Read

Reset: 0x0000006A

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	P50 for example

LDO17_STATUS1 | 0x00015008

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Retention Mode if STATUS3__RETENTION_PRESENT == 1. Otherwise unused. 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_RM_OR_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	STEPPER_DONE	Indicated stepper is done when LDO is enabled or Vset is changed. 1 = Stepper done 0 = Stepper is stepping 0: NOT_DONE_STEPPING 1: DONE_STEPPING

Bits	Name	Description
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass/Retention 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass/Retention 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO17_STATUS3 | 0x0001500A

Type: Read

Reset: 0x00000003

No description provided for this register.

Bits	Name	Description
0 : 0	AHC_PRESENT	Indicates to SW/user that this regulator supports AHC (automatic headroom control) 0: NO_AHC 1: AHC
1 : 1	LLC_PRESENT	Indicates to SW/user that this regulator supports LLC (load-line compensation) 0: NO_LLC 1: LLC
2 : 2	RETENTION_PRESENT	Indicates to SW/user that this regulator supports retention 0: NO_RETENTION 1: RETENTION

LDO17_STATUS4 | 0x0001500B

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 1	ECM_READY	Indicates that ECM has powered up and is ready to operate when 0x1. This register is only used when STATUS4__ECM_PRESENT==1. 0: ECM_READY_false 1: ECM_READY_true
3 : 3	ECM_DATA_READY	Indicates that data is ready for PBS to perform read when 0x1. This register is only used when STATUS4__ECM_PRESENT==1. 0: ECM_DATA_READY_false 1: ECM_DATA_READY_true
4 : 4	ECM_PRESENT	Indicates to SW/user that this regulator supports ECM 0: NO_ECM 1: ECM
7 : 7	ECM_DATA_OVERWRITTEN	ECM measurement data overwritten before read. The bit should clear when both data registers are read. This register is only used when STATUS4__ECM_PRESENT==1. 0 = ECM data not overwritten 1 = ECM data overwritten 0: DATA_NOT_OVERWRITTEN 1: DATA_OVERWRITTEN

LDO17_INT_RT_STS | 0x00015010

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

LDO17_INT_SET_TYPE | 0x00015011

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

LDO17_INT_POLARITY_HIGH | 0x00015012

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

LDO17_INT_POLARITY_LOW | 0x00015013

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

LDO17_INT_LATCHED_CLR | 0x00015014

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

LDO17_INT_EN_SET | 0x00015015

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

LDO17_INT_EN_CLR | 0x00015016

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

LDO17_INT_MID_SEL | 0x0001501A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

LDO17_INT_PRIORITY | 0x0001501B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

LDO17_INT_SELF_CLEARING | 0x0001501D

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

LDO17_ULS_VSET_LB | 0x00015039

Type: Read/Write

Reset: 0x00000030

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	LDO upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

LDO17_ULS_VSET_UB | 0x0001503A

Type: Read/Write

Reset: 0x00000005

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	LDO upper limit stopvoltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV. Max combined value is 1.328mv (set as a default for both regs). Write will update ULS_VSET[15:0]. User must program ULS in LSB (8mV) increments to avoid undesired behavior

LDO17_STEPPER_CTL | 0x0001503B

Type: Read/Write

Reset: 0x00000083

No description provided for this register.

Bits	Name	Description
1 : 0	STEP_RATE	Voltage Stepper Rate (voltage always increases 1 LSB per step) 0 = 38.4 mV/us 1 = 19.2 mV/us 2 = 9.6 mV/us 3 = 4.8 mV/us (DEFAULT) 0: STEP_RATE_38p4_mV_us 1: STEP_RATE_19p2_mV_us 2: STEP_RATE_9p6_mV_us 3: STEP_RATE_4p8_mV_us
7 : 7	VOLTAGE_STEPPER_EN	Enables the voltage stepper. Stepper should never be disabled. 0: VOLTAGE_STEPPER_DIABLED 1: VOLTAGE_STEPPER_ENABLED

LDO17_VSET_LB | 0x00015040

Type: Read/Write

Reset: 0x00000018

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO17_VSET_UB | 0x00015041

Type: Read/Write

Reset: 0x00000005

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO17_VSET_VALID_LB | 0x00015042

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 3	VSET_VALID_LB	Readback the valid output voltage setpoint Lower Byte (Bits 7:0). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO17_VSET_VALID_UB | 0x00015043

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Readback the valid output voltage setpoint Upper Byte (Bits 15:8). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO17_MODE_CTL2 | 0x00015044

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. This setting only matters when 0x47 FOLLOW_HWEN or 0x49 PBS_VOTE_CTL is programmed. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Retention if STATUS3__RETENTION_PRESENT == 1. Otherwise, NPM. 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: RM_OR_NPM 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO17_MODE_CTL1 | 0x00015045

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Retention if STATUS3__RETENTION_PRESENT == 1. Otherwise, NPM. 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: RM_OR_NPM 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO17_EN_CTL | 0x00015046

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

LDO17_FOLLOW_HWEN | 0x00015047

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO state. If this bit is set high then: When HWEN0=1 --> LDO if forced on (regardless of reg 0x46) When HWEN0=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN0_FALSE 1: EN_FOLLOW_HW_EN0_TRUE
1 : 1	EN_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO state. If this bit is set high then: When HWEN1=1 --> LDO if forced on (regardless of reg 0x46) When HWEN1=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN1_FALSE 1: EN_FOLLOW_HW_EN1_TRUE
2 : 2	EN_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO state. If this bit is set high then: When HWEN2=1 --> LDO if forced on (regardless of reg 0x46) When HWEN2=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN2_FALSE 1: EN_FOLLOW_HW_EN2_TRUE
3 : 3	EN_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE state does NOT control LDO state. If this bit is set high then: When PMIC_AWAKE=1 --> LDO if forced on (regardless of reg 0x46) When PMIC_AWAKE=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_PMIC_AWAKE_FALSE 1: EN_FOLLOW_PMIC_AWAKE_TRUE
4 : 4	MODE_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN0_FALSE 1: MODE_FOLLOW_HW_EN0_TRUE
5 : 5	MODE_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN1_FALSE 1: MODE_FOLLOW_HW_EN1_TRUE
6 : 6	MODE_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO mode. If this bit is set high then: When HWEN2=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN2_FALSE 1: MODE_FOLLOW_HW_EN2_TRUE

Bits	Name	Description
7 : 7	MODE_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE (sleep_b) state does NOT control LDO mode. If this bit is set high then: When PMIC_AWAKE=0--> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_PMIC_AWAKE_FALSE 1: MODE_FOLLOW_PMIC_AWAKE_TRUE

LDO17_PBS_VOTE_CTL | 0x00015049

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
6 : 6	PBS_MODE	PBS Mode vote 0 = Mode is set by MODE_CTL1 reg (DEFAULT) 1 = Mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: PBS_MODE_false 1: PBS_MODE_true
7 : 7	PBS_EN	Enables the LDO when asserted to HI. 0 = LDO will NOT be enabled (DEFAULT) 1= ENABLE LDO 0: PBS_EN_false 1: PBS_EN_true

LDO17_MISC_CTL1 | 0x00015089

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	READY_STEPDOWN_QUAL	Enables Vout qualification when stepping down Vset. 0 = Disable (default). VREG_READY stays high during downward stepping becaue VOUT > Vset 1 = Enable. VREG_READY stays low during downard stepping until Vout settles below Vset + Margin 0: STEPDOWN_QUAL_DISABLED 1: STEPDOWN_QUAL_ENABLED

LDO17_PD_CTL | 0x000150A0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Intended for keeping Vout from rising due to passFET leakage at high temperature. 1 = Leakage pulldown is always enabled 0 = Leakage pulldown is always disabled (DEFAULT) 0: LEAKAGE_PD_ALWAYS_DISABLED 1: LEAKAGE_PD_ALWAYS_ENABLED
6 : 6	WEAK_PD_EN	Weak PD is never active while LDO is on (VREG_READY=1). 1 = Weak pulldown is enabled when the regulator is disabled 0 = Weak pulldown is always inactive (DEFAULT) 0: WEAK_PD_DISABLED 1: WEAK_PD_WHEN_LDO_IS_OFF
7 : 7	STRONG_PD_EN	Strong PD is never active while LDO is on (VREG_READY=1). Strong pull-down FET Rdson will vary with Vout. 1 = Strong pulldown is enabled when the regulator is disabled (DEFAULT) 0 = Strong pulldown is always inactive. 0: STRONG_PD_ALWAYS_OFF 1: STRONG_PD_WHEN_LDO_IS_OFF

Module Name: LDO18

Base Address: 0x00015100

Register Quick Reference:

Register	Offset	Type
LDO18_REVISION1	0x00015100	Read
LDO18_REVISION2	0x00015101	Read

Register	Offset	Type
LDO18_REVISION3	0x00015102	Read
LDO18_REVISION4	0x00015103	Read
LDO18_PERPH_TYPE	0x00015104	Read
LDO18_PERPH_SUBTYPE	0x00015105	Read
LDO18_STATUS1	0x00015108	Read
LDO18_STATUS3	0x0001510A	Read
LDO18_STATUS4	0x0001510B	Read
LDO18_INT_RT_STS	0x00015110	Read
LDO18_INT_SET_TYPE	0x00015111	Read
LDO18_INT_POLARITY_HIGH	0x00015112	Read
LDO18_INT_POLARITY_LOW	0x00015113	Read/Write
LDO18_INT_LATCHED_CLR	0x00015114	Write
LDO18_INT_EN_SET	0x00015115	Read/Write
LDO18_INT_EN_CLR	0x00015116	Read/Write
LDO18_INT_MID_SEL	0x0001511A	Read/Write
LDO18_INT_PRIORITY	0x0001511B	Read/Write
LDO18_INT_SELF_CLEARING	0x0001511D	Read
LDO18_ULS_VSET_LB	0x00015139	Read/Write
LDO18_ULS_VSET_UB	0x0001513A	Read/Write
LDO18_STEPPER_CTL	0x0001513B	Read/Write
LDO18_VSET_LB	0x00015140	Read/Write
LDO18_VSET_UB	0x00015141	Read/Write
LDO18_VSET_VALID_LB	0x00015142	Read
LDO18_VSET_VALID_UB	0x00015143	Read
LDO18_MODE_CTL2	0x00015144	Read/Write
LDO18_MODE_CTL1	0x00015145	Read/Write
LDO18_EN_CTL	0x00015146	Read/Write
LDO18_FOLLOW_HWEN	0x00015147	Read/Write
LDO18_PBS_VOTE_CTL	0x00015149	Read/Write
LDO18_MISC_CTL1	0x00015189	Read/Write
LDO18_PD_CTL	0x000151A0	Read/Write

LDO18_REVISION1 | 0x00015100

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO18_REVISION2 | 0x00015101

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

LDO18_REVISION3 | 0x00015102

Type: Read

Reset: 0x00000001

HW Version Regster [23:16]

Bits	Name	Description
2 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO18_REVISION4 | 0x00015103

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
2 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

LDO18_PERPH_TYPE | 0x00015104

Type: Read

Reset: 0x00000004

Peripheral Type

Bits	Name	Description
7 : 0	TYPE	LDO

LDO18_PERPH_SUBTYPE | 0x00015105

Type: Read

Reset: 0x0000006A

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	P50 for example

LDO18_STATUS1 | 0x00015108

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Retention Mode if STATUS3__RETENTION_PRESENT == 1. Otherwise unused. 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_RM_OR_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	STEPPER_DONE	Indicated stepper is done when LDO is enabled or Vset is changed. 1 = Stepper done 0 = Stepper is stepping 0: NOT_DONE_STEPPING 1: DONE_STEPPING
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass/Retention 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass/Retention 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO18_STATUS3 | 0x0001510A

Type: Read

Reset: 0x00000003

No description provided for this register.

Bits	Name	Description
0 : 0	AHC_PRESENT	Indicates to SW/user that this regulator supports AHC (automatic headroom control) 0: NO_AHC 1: AHC
1 : 1	LLC_PRESENT	Indicates to SW/user that this regulator supports LLC (load-line compensation) 0: NO_LLC 1: LLC
2 : 2	RETENTION_PRESENT	Indicates to SW/user that this regulator supports retention 0: NO_RETENTION 1: RETENTION

LDO18_STATUS4 | 0x0001510B

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 1	ECM_READY	Indicates that ECM has powered up and is ready to operate when 0x1. This register is only used when STATUS4__ECM_PRESENT==1. 0: ECM_READY_false 1: ECM_READY_true
3 : 3	ECM_DATA_READY	Indicates that data is ready for PBS to perform read when 0x1. This register is only used when STATUS4__ECM_PRESENT==1. 0: ECM_DATA_READY_false 1: ECM_DATA_READY_true
4 : 4	ECM_PRESENT	Indicates to SW/user that this regulator supports ECM 0: NO_ECM 1: ECM
7 : 7	ECM_DATA_OVERWRITTEN	ECM measurement data overwritten before read. The bit should clear when both data registers are read. This register is only used when STATUS4__ECM_PRESENT==1. 0 = ECM data not overwritten 1 = ECM data overwritten 0: DATA_NOT_OVERWRITTEN 1: DATA_OVERWRITTEN

LDO18_INT_RT_STS | 0x00015110

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

LDO18_INT_SET_TYPE | 0x00015111

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

LDO18_INT_POLARITY_HIGH | 0x00015112

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

LDO18_INT_POLARITY_LOW | 0x00015113

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

LDO18_INT_LATCHED_CLR | 0x00015114

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

LDO18_INT_EN_SET | 0x00015115

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

LDO18_INT_EN_CLR | 0x00015116

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

LDO18_INT_MID_SEL | 0x0001511A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

LDO18_INT_PRIORITY | 0x0001511B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

LDO18_INT_SELF_CLEARING | 0x0001511D

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

LDO18_ULS_VSET_LB | 0x00015139

Type: Read/Write

Reset: 0x00000030

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	LDO upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

LDO18_ULS_VSET_UB | 0x0001513A

Type: Read/Write

Reset: 0x00000005

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	LDO upper limit stopvoltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV. Max combined value is 1.328mv (set as a default for both regs). Write will update ULS_VSET[15:0]. User must program ULS in LSB (8mV) increments to avoid undesired behavior

LDO18_STEPPER_CTL | 0x0001513B

Type: Read/Write

Reset: 0x00000083

No description provided for this register.

Bits	Name	Description
1 : 0	STEP_RATE	Voltage Stepper Rate (voltage always increases 1 LSB per step) 0 = 38.4 mV/us 1 = 19.2 mV/us 2 = 9.6 mV/us 3 = 4.8 mV/us (DEFAULT) 0: STEP_RATE_38p4_mV_us 1: STEP_RATE_19p2_mV_us 2: STEP_RATE_9p6_mV_us 3: STEP_RATE_4p8_mV_us
7 : 7	VOLTAGE_STEPPER_EN	Enables the voltage stepper. Stepper should never be disabled. 0: VOLTAGE_STEPPER_DIABLED 1: VOLTAGE_STEPPER_ENABLED

LDO18_VSET_LB | 0x00015140

Type: Read/Write

Reset: 0x000000D0

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO18_VSET_UB | 0x00015141

Type: Read/Write

Reset: 0x00000004

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO18_VSET_VALID_LB | 0x00015142

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 3	VSET_VALID_LB	Readback the valid output voltage setpoint Lower Byte (Bits 7:0). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO18_VSET_VALID_UB | 0x00015143

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Readback the valid output voltage setpoint Upper Byte (Bits 15:8). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO18_MODE_CTL2 | 0x00015144

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. This setting only matters when 0x47 FOLLOW_HWEN or 0x49 PBS_VOTE_CTL is programmed. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Retention if STATUS3__RETENTION_PRESENT == 1. Otherwise, NPM. 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: RM_OR_NPM 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO18_MODE_CTL1 | 0x00015145

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Retention if STATUS3__RETENTION_PRESENT == 1. Otherwise, NPM. 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: RM_OR_NPM 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO18_EN_CTL | 0x00015146

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

LDO18_FOLLOW_HWEN | 0x00015147

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO state. If this bit is set high then: When HWEN0=1 --> LDO if forced on (regardless of reg 0x46) When HWEN0=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN0_FALSE 1: EN_FOLLOW_HW_EN0_TRUE
1 : 1	EN_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO state. If this bit is set high then: When HWEN1=1 --> LDO if forced on (regardless of reg 0x46) When HWEN1=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN1_FALSE 1: EN_FOLLOW_HW_EN1_TRUE
2 : 2	EN_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO state. If this bit is set high then: When HWEN2=1 --> LDO if forced on (regardless of reg 0x46) When HWEN2=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN2_FALSE 1: EN_FOLLOW_HW_EN2_TRUE
3 : 3	EN_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE state does NOT control LDO state. If this bit is set high then: When PMIC_AWAKE=1 --> LDO if forced on (regardless of reg 0x46) When PMIC_AWAKE=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_PMIC_AWAKE_FALSE 1: EN_FOLLOW_PMIC_AWAKE_TRUE
4 : 4	MODE_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN0_FALSE 1: MODE_FOLLOW_HW_EN0_TRUE
5 : 5	MODE_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN1_FALSE 1: MODE_FOLLOW_HW_EN1_TRUE
6 : 6	MODE_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO mode. If this bit is set high then: When HWEN2=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN2_FALSE 1: MODE_FOLLOW_HW_EN2_TRUE

Bits	Name	Description
7 : 7	MODE_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE (sleep_b) state does NOT control LDO mode. If this bit is set high then: When PMIC_AWAKE=0--> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_PMIC_AWAKE_FALSE 1: MODE_FOLLOW_PMIC_AWAKE_TRUE

LDO18_PBS_VOTE_CTL | 0x00015149

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
6 : 6	PBS_MODE	PBS Mode vote 0 = Mode is set by MODE_CTL1 reg (DEFAULT) 1 = Mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: PBS_MODE_false 1: PBS_MODE_true
7 : 7	PBS_EN	Enables the LDO when asserted to HI. 0 = LDO will NOT be enabled (DEFAULT) 1= ENABLE LDO 0: PBS_EN_false 1: PBS_EN_true

LDO18_MISC_CTL1 | 0x00015189

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	READY_STEPDOWN_QUAL	Enables Vout qualification when stepping down Vset. 0 = Disable (default). VREG_READY stays high during downward stepping becaue VOUT > Vset 1 = Enable. VREG_READY stays low during downard stepping until Vout settles below Vset + Margin 0: STEPDOWN_QUAL_DISABLED 1: STEPDOWN_QUAL_ENABLED

LDO18_PD_CTL | 0x000151A0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Intended for keeping Vout from rising due to passFET leakage at high temperature. 1 = Leakage pulldown is always enabled 0 = Leakage pulldown is always disabled (DEFAULT) 0: LEAKAGE_PD_ALWAYS_DISABLED 1: LEAKAGE_PD_ALWAYS_ENABLED
6 : 6	WEAK_PD_EN	Weak PD is never active while LDO is on (VREG_READY=1). 1 = Weak pulldown is enabled when the regulator is disabled 0 = Weak pulldown is always inactive (DEFAULT) 0: WEAK_PD_DISABLED 1: WEAK_PD_WHEN_LDO_IS_OFF
7 : 7	STRONG_PD_EN	Strong PD is never active while LDO is on (VREG_READY=1). Strong pull-down FET Rdson will vary with Vout. 1 = Strong pulldown is enabled when the regulator is disabled (DEFAULT) 0 = Strong pulldown is always inactive. 0: STRONG_PD_ALWAYS_OFF 1: STRONG_PD_WHEN_LDO_IS_OFF

Module Name: LDO19

Base Address: 0x00015200

Register Quick Reference:

Register	Offset	Type
LDO19_REVISION1	0x00015200	Read
LDO19_REVISION2	0x00015201	Read

Register	Offset	Type
LDO19_REVISION3	0x00015202	Read
LDO19_REVISION4	0x00015203	Read
LDO19_PERPH_TYPE	0x00015204	Read
LDO19_PERPH_SUBTYPE	0x00015205	Read
LDO19_STATUS1	0x00015208	Read
LDO19_STATUS3	0x0001520A	Read
LDO19_STATUS4	0x0001520B	Read
LDO19_INT_RT_STS	0x00015210	Read
LDO19_INT_SET_TYPE	0x00015211	Read
LDO19_INT_POLARITY_HIGH	0x00015212	Read
LDO19_INT_POLARITY_LOW	0x00015213	Read/Write
LDO19_INT_LATCHED_CLR	0x00015214	Write
LDO19_INT_EN_SET	0x00015215	Read/Write
LDO19_INT_EN_CLR	0x00015216	Read/Write
LDO19_INT_MID_SEL	0x0001521A	Read/Write
LDO19_INT_PRIORITY	0x0001521B	Read/Write
LDO19_INT_SELF_CLEARING	0x0001521D	Read
LDO19_ULS_VSET_LB	0x00015239	Read/Write
LDO19_ULS_VSET_UB	0x0001523A	Read/Write
LDO19_VSET_LB	0x00015240	Read/Write
LDO19_VSET_UB	0x00015241	Read/Write
LDO19_VSET_VALID_LB	0x00015242	Read
LDO19_VSET_VALID_UB	0x00015243	Read
LDO19_MODE_CTL2	0x00015244	Read/Write
LDO19_MODE_CTL1	0x00015245	Read/Write
LDO19_EN_CTL	0x00015246	Read/Write
LDO19_FOLLOW_HWEN	0x00015247	Read/Write
LDO19_PBS_VOTE_CTL	0x00015249	Read/Write
LDO19_PD_CTL	0x000152A0	Read/Write

LDO19_REVISION1 | 0x00015200

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO19_REVISION2 | 0x00015201

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

LDO19_REVISION3 | 0x00015202

Type: Read

Reset: 0x00000000

HW Version Regster [23:16]

Bits	Name	Description
2 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO19_REVISION4 | 0x00015203

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
2 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

LDO19_PERPH_TYPE | 0x00015204

Type: Read

Reset: 0x00000004

Peripheral Type

Bits	Name	Description
7 : 0	TYPE	LDO

LDO19_PERPH_SUBTYPE | 0x00015205

Type: Read

Reset: 0x0000007B

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	P50 for example

LDO19_STATUS1 | 0x00015208

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Unused 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	SOFT_START_DONE	Indicates soft start is done when LDO is enabled. 1 = Softstart done 0 = Softstart in progress 0: NOT_DONE_SOFT_START 1: DONE_SOFT_START
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO19_STATUS3 | 0x0001520A

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	AHC_PRESENT	Indicates to SW/user that this regulator supports AHC (automatic headroom control) 0: NO_AHC 1: AHC
1 : 1	LLC_PRESENT	Indicates to SW/user that this regulator supports LLC (load-line compensation) 0: NO_LLC 1: LLC

LDO19_STATUS4 | 0x0001520B

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 1	ECM_READY	Indicates that ECM has powered up and is ready to operate when 0x1. This register is only used when STATUS4__ECM_PRESENT==1. 0: ECM_READY_false 1: ECM_READY_true
3 : 3	ECM_DATA_READY	Indicates that data is ready for PBS to perform read when 0x1. This register is only used when STATUS4__ECM_PRESENT==1. 0: ECM_DATA_READY_false 1: ECM_DATA_READY_true
4 : 4	ECM_PRESENT	Indicates to SW/user that this regulator supports ECM 0: NO_ECM 1: ECM
7 : 7	ECM_DATA_OVERWRITTEN	ECM measurement data overwritten before read. The bit should clear when both data registers are read. This register is only used when STATUS4__ECM_PRESENT==1. 0 = ECM data not overwritten 1 = ECM data overwritten 0: DATA_NOT_OVERWRITTEN 1: DATA_OVERWRITTEN

LDO19_INT_RT_STS | 0x00015210

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

LDO19_INT_SET_TYPE | 0x00015211

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

LDO19_INT_POLARITY_HIGH | 0x00015212

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

LDO19_INT_POLARITY_LOW | 0x00015213

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

LDO19_INT_LATCHED_CLR | 0x00015214

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

LDO19_INT_EN_SET | 0x00015215

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

LDO19_INT_EN_CLR | 0x00015216

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

LDO19_INT_MID_SEL | 0x0001521A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

LDO19_INT_PRIORITY | 0x0001521B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

LDO19_INT_SELF_CLEARING | 0x0001521D

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

LDO19_ULS_VSET_LB | 0x00015239

Type: Read/Write

Reset: 0x000000D8

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	LDO upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

LDO19_ULS_VSET_UB | 0x0001523A

Type: Read/Write

Reset: 0x0000000D

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	LDO upper limit stopvoltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV. Max combined value is 1.328mv (set as a default for both regs). Write will update ULS_VSET[15:0]. User must program ULS in LSB (8mV) increments to avoid undesired behavior

LDO19_VSET_LB | 0x00015240

Type: Read/Write

Reset: 0x00000008

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO19_VSET_UB | 0x00015241

Type: Read/Write

Reset: 0x00000007

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO19_VSET_VALID_LB | 0x00015242

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 3	VSET_VALID_LB	Readback the valid output voltage setpoint Lower Byte (Bits 7:0). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO19_VSET_VALID_UB | 0x00015243

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Readback the valid output voltage setpoint Upper Byte (Bits 15:8). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO19_MODE_CTL2 | 0x00015244

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. This setting only matters when 0x47 FOLLOW_HWEN or 0x49 PBS_VOTE_CTL is programmed. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO19_MODE_CTL1 | 0x00015245

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO19_EN_CTL | 0x00015246

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

LDO19_FOLLOW_HWEN | 0x00015247

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO state. If this bit is set high then: When HWEN0=1 --> LDO if forced on (regardless of reg 0x46) When HWEN0=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN0_FALSE 1: EN_FOLLOW_HW_EN0_TRUE
1 : 1	EN_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO state. If this bit is set high then: When HWEN1=1 --> LDO if forced on (regardless of reg 0x46) When HWEN1=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN1_FALSE 1: EN_FOLLOW_HW_EN1_TRUE
2 : 2	EN_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO state. If this bit is set high then: When HWEN2=1 --> LDO if forced on (regardless of reg 0x46) When HWEN2=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN2_FALSE 1: EN_FOLLOW_HW_EN2_TRUE

Bits	Name	Description
3 : 3	EN_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE state does NOT control LDO state. If this bit is set high then: When PMIC_AWAKE=1 --> LDO if forced on (regardless of reg 0x46) When PMIC_AWAKE=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_PMIC_AWAKE_FALSE 1: EN_FOLLOW_PMIC_AWAKE_TRUE
4 : 4	MODE_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN0_FALSE 1: MODE_FOLLOW_HW_EN0_TRUE
5 : 5	MODE_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN1_FALSE 1: MODE_FOLLOW_HW_EN1_TRUE
6 : 6	MODE_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO mode. If this bit is set high then: When HWEN2=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN2_FALSE 1: MODE_FOLLOW_HW_EN2_TRUE
7 : 7	MODE_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE (sleep_b) state does NOT control LDO mode. If this bit is set high then: When PMIC_AWAKE=0--> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_PMIC_AWAKE_FALSE 1: MODE_FOLLOW_PMIC_AWAKE_TRUE

LDO19_PBS_VOTE_CTL | 0x00015249

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
6 : 6	PBS_MODE	PBS Mode vote 0 = Mode is set by MODE_CTL1 reg (DEFAULT) 1 = Mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: PBS_MODE_false 1: PBS_MODE_true
7 : 7	PBS_EN	Enables the LDO when asserted to HI. 0 = LDO will NOT be enabled (DEFAULT) 1= ENABLE LDO 0: PBS_EN_false 1: PBS_EN_true

LDO19_PD_CTL | 0x000152A0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Intended for keeping Vout from rising due to passFET leakage at high temperature. 1 = Leakage pulldown is always enabled 0 = Leakage pulldown is always disabled (DEFAULT) 0: LEAKAGE_PD_ALWAYS_DISABLED 1: LEAKAGE_PD_ALWAYS_ENABLED
6 : 6	WEAK_PD_EN	Weak PD is never active while LDO is on (VREG_READY=1). 1 = Weak pulldown is enabled when the regulator is disabled 0 = Weak pulldown is always inactive (DEFAULT) 0: WEAK_PD_DISABLED 1: WEAK_PD_WHEN_LDO_IS_OFF
7 : 7	STRONG_PD_EN	Strong PD is never active while LDO is on (VREG_READY=1). Strong pull-down FET Rdson will vary with Vout. 1 = Strong pulldown is enabled when the regulator is disabled (DEFAULT) 0 = Strong pulldown is always inactive. 0: STRONG_PD_ALWAYS_OFF 1: STRONG_PD_WHEN_LDO_IS_OFF

Module Name: LDO20

Base Address: 0x00015300

Register Quick Reference:

Register	Offset	Type
LDO20_REVISION1	0x00015300	Read
LDO20_REVISION2	0x00015301	Read
LDO20_REVISION3	0x00015302	Read
LDO20_REVISION4	0x00015303	Read
LDO20_PERPH_TYPE	0x00015304	Read
LDO20_PERPH_SUBTYPE	0x00015305	Read
LDO20_STATUS1	0x00015308	Read
LDO20_STATUS3	0x0001530A	Read
LDO20_STATUS4	0x0001530B	Read
LDO20_INT_RT_STS	0x00015310	Read
LDO20_INT_SET_TYPE	0x00015311	Read
LDO20_INT_POLARITY_HIGH	0x00015312	Read
LDO20_INT_POLARITY_LOW	0x00015313	Read/Write
LDO20_INT_LATCHED_CLR	0x00015314	Write
LDO20_INT_EN_SET	0x00015315	Read/Write
LDO20_INT_EN_CLR	0x00015316	Read/Write
LDO20_INT_MID_SEL	0x0001531A	Read/Write
LDO20_INT_PRIORITY	0x0001531B	Read/Write
LDO20_INT_SELF_CLEARING	0x0001531D	Read
LDO20_ULS_VSET_LB	0x00015339	Read/Write
LDO20_ULS_VSET_UB	0x0001533A	Read/Write
LDO20_VSET_LB	0x00015340	Read/Write
LDO20_VSET_UB	0x00015341	Read/Write
LDO20_VSET_VALID_LB	0x00015342	Read
LDO20_VSET_VALID_UB	0x00015343	Read
LDO20_MODE_CTL2	0x00015344	Read/Write
LDO20_MODE_CTL1	0x00015345	Read/Write
LDO20_EN_CTL	0x00015346	Read/Write
LDO20_FOLLOW_HWEN	0x00015347	Read/Write
LDO20_PBS_VOTE_CTL	0x00015349	Read/Write
LDO20_OCP_CTL1	0x00015388	Read/Write
LDO20_PD_CTL	0x000153A0	Read/Write

LDO20_REVISION1 | 0x00015300

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO20_REVISION2 | 0x00015301

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

LDO20_REVISION3 | 0x00015302

Type: Read

Reset: 0x00000000

HW Version Regster [23:16]

Bits	Name	Description
2 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO20_REVISION4 | 0x00015303

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
2 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

LDO20_PERPH_TYPE | 0x00015304

Type: Read

Reset: 0x00000004

Peripheral Type

Bits	Name	Description
7 : 0	TYPE	LDO

LDO20_PERPH_SUBTYPE | 0x00015305

Type: Read

Reset: 0x0000007B

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	P50 for example

LDO20_STATUS1 | 0x00015308

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Unused 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	SOFT_START_DONE	Indicates soft start is done when LDO is enabled. 1 = Softstart done 0 = Softstart in progress 0: NOT_DONE_SOFT_START 1: DONE_SOFT_START
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO20_STATUS3 | 0x0001530A

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	AHC_PRESENT	Indicates to SW/user that this regulator supports AHC (automatic headroom control) 0: NO_AHC 1: AHC
1 : 1	LLC_PRESENT	Indicates to SW/user that this regulator supports LLC (load-line compensation) 0: NO_LLC 1: LLC

LDO20_STATUS4 | 0x0001530B

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 1	ECM_READY	Indicates that ECM has powered up and is ready to operate when 0x1. This register is only used when STATUS4__ECM_PRESENT==1. 0: ECM_READY_false 1: ECM_READY_true
3 : 3	ECM_DATA_READY	Indicates that data is ready for PBS to perform read when 0x1. This register is only used when STATUS4__ECM_PRESENT==1. 0: ECM_DATA_READY_false 1: ECM_DATA_READY_true
4 : 4	ECM_PRESENT	Indicates to SW/user that this regulator supports ECM 0: NO_ECM 1: ECM
7 : 7	ECM_DATA_OVERWRITTEN	ECM measurement data overwritten before read. The bit should clear when both data registers are read. This register is only used when STATUS4__ECM_PRESENT==1. 0 = ECM data not overwritten 1 = ECM data overwritten 0: DATA_NOT_OVERWRITTEN 1: DATA_OVERWRITTEN

LDO20_INT_RT_STS | 0x00015310

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

LDO20_INT_SET_TYPE | 0x00015311

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

LDO20_INT_POLARITY_HIGH | 0x00015312

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

LDO20_INT_POLARITY_LOW | 0x00015313

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

LDO20_INT_LATCHED_CLR | 0x00015314

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

LDO20_INT_EN_SET | 0x00015315

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

LDO20_INT_EN_CLR | 0x00015316

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

LDO20_INT_MID_SEL | 0x0001531A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

LDO20_INT_PRIORITY | 0x0001531B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

LDO20_INT_SELF_CLEARING | 0x0001531D

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

LDO20_ULS_VSET_LB | 0x00015339

Type: Read/Write

Reset: 0x000000D8

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	LDO upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

LDO20_ULS_VSET_UB | 0x0001533A

Type: Read/Write

Reset: 0x0000000D

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	LDO upper limit stopvoltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV. Max combined value is 1.328mv (set as a default for both regs). Write will update ULS_VSET[15:0]. User must program ULS in LSB (8mV) increments to avoid undesired behavior

LDO20_VSET_LB | 0x00015340

Type: Read/Write

Reset: 0x00000008

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO20_VSET_UB | 0x00015341

Type: Read/Write

Reset: 0x00000007

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO20_VSET_VALID_LB | 0x00015342

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 3	VSET_VALID_LB	Readback the valid output voltage setpoint Lower Byte (Bits 7:0). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO20_VSET_VALID_UB | 0x00015343

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Readback the valid output voltage setpoint Upper Byte (Bits 15:8). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO20_MODE_CTL2 | 0x00015344

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. This setting only matters when 0x47 FOLLOW_HWEN or 0x49 PBS_VOTE_CTL is programmed. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO20_MODE_CTL1 | 0x00015345

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO20_EN_CTL | 0x00015346

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

LDO20_FOLLOW_HWEN | 0x00015347

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO state. If this bit is set high then: When HWEN0=1 --> LDO if forced on (regardless of reg 0x46) When HWEN0=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN0_FALSE 1: EN_FOLLOW_HW_EN0_TRUE
1 : 1	EN_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO state. If this bit is set high then: When HWEN1=1 --> LDO if forced on (regardless of reg 0x46) When HWEN1=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN1_FALSE 1: EN_FOLLOW_HW_EN1_TRUE
2 : 2	EN_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO state. If this bit is set high then: When HWEN2=1 --> LDO if forced on (regardless of reg 0x46) When HWEN2=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN2_FALSE 1: EN_FOLLOW_HW_EN2_TRUE

Bits	Name	Description
3 : 3	EN_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE state does NOT control LDO state. If this bit is set high then: When PMIC_AWAKE=1 --> LDO if forced on (regardless of reg 0x46) When PMIC_AWAKE=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_PMIC_AWAKE_FALSE 1: EN_FOLLOW_PMIC_AWAKE_TRUE
4 : 4	MODE_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN0_FALSE 1: MODE_FOLLOW_HW_EN0_TRUE
5 : 5	MODE_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN1_FALSE 1: MODE_FOLLOW_HW_EN1_TRUE
6 : 6	MODE_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO mode. If this bit is set high then: When HWEN2=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN2_FALSE 1: MODE_FOLLOW_HW_EN2_TRUE
7 : 7	MODE_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE (sleep_b) state does NOT control LDO mode. If this bit is set high then: When PMIC_AWAKE=0--> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_PMIC_AWAKE_FALSE 1: MODE_FOLLOW_PMIC_AWAKE_TRUE

LDO20_PBS_VOTE_CTL | 0x00015349

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
6 : 6	PBS_MODE	PBS Mode vote 0 = Mode is set by MODE_CTL1 reg (DEFAULT) 1 = Mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: PBS_MODE_false 1: PBS_MODE_true
7 : 7	PBS_EN	Enables the LDO when asserted to HI. 0 = LDO will NOT be enabled (DEFAULT) 1= ENABLE LDO 0: PBS_EN_false 1: PBS_EN_true

LDO20_OCP_CTL1 | 0x00015388

Type: Read/Write

Reset: 0x000000D0

No description provided for this register.

Bits	Name	Description
0 : 0	OCP_TESTMODE_EN	Enable OCP test mode. In test mode the current limit is lowered. 0: OCP_TESTMODE_DISABLED 1: OCP_TESTMODE_ENABLED
1 : 1	OCP_STATUS_CLR	Writing a 1 to this bit clears the VREG_OCP bit in STATUS1. It does not recover the regulator from OCP shutdown. This bit must then be toggled from 1 to 0 to re-arm the status bit so that it can be latched in the event of another OCP event.
2 : 2	OCP_GLOBAL_BROADCAST_EN	Determines if the LDO notifies the PON module that an OCP event has occurred. PON module may use this info to shutdown the entire PMIC. 1 = LDO notifies PON when OCP event occurs 0 = LDO does not notify PON when OCP even occurs (DEFAULT) 0: OCP_GLOBAL_BROADCAST_DISABLED 1: OCP_GLOBAL_BROADCAST_ENABLED
5 : 4	OCP_DEB	Debounce time for OCP detection 0 = ~120us 1 = ~240us (DEFAULT) 2 = ~480us 3 = ~960us 0: DEB_120US 1: DEB_240US 2: DEB_480US 3: DEB_960US

Bits	Name	Description
6 : 6	OCP_SELF_SHUTDOWN_EN	Determines if the LDO performs self-shutdown upon OCP event. 1 = LDO self-shutdowns upon OCP event (DEFAULT) 0 = LDO does not self-shutdown upon OCP event 0: OCP_SELF_SHUTDOWN_DISABLED 1: OCP_SELF_SHUTDOWN_ENABLED
7 : 7	OCP_EN	1 = Enable OCP feature (DEFAULT) 0 = Disable OCP feature 0: OCP_DISABLED 1: OCP_ENABLED

LDO20_PD_CTL | 0x000153A0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Intended for keeping Vout from rising due to passFET leakage at high temperature. 1 = Leakage pulldown is always enabled 0 = Leakage pulldown is always disabled (DEFAULT) 0: LEAKAGE_PD_ALWAYS_DISABLED 1: LEAKAGE_PD_ALWAYS_ENABLED
6 : 6	WEAK_PD_EN	Weak PD is never active while LDO is on (VREG_READY=1). 1 = Weak pulldown is enabled when the regulator is disabled 0 = Weak pulldown is always inactive (DEFAULT) 0: WEAK_PD_DISABLED 1: WEAK_PD_WHEN_LDO_IS_OFF
7 : 7	STRONG_PD_EN	Strong PD is never active while LDO is on (VREG_READY=1). Strong pull-down FET Rdson will vary with Vout. 1 = Strong pulldown is enabled when the regulator is disabled (DEFAULT) 0 = Strong pulldown is always inactive. 0: STRONG_PD_ALWAYS_OFF 1: STRONG_PD_WHEN_LDO_IS_OFF

Module Name: LDO21

Base Address: 0x00015400

Register Quick Reference:

Register	Offset	Type
LDO21_REVISION1	0x00015400	Read

Register	Offset	Type
LDO21_REVISION2	0x00015401	Read
LDO21_REVISION3	0x00015402	Read
LDO21_REVISION4	0x00015403	Read
LDO21_PERPH_TYPE	0x00015404	Read
LDO21_PERPH_SUBTYPE	0x00015405	Read
LDO21_STATUS1	0x00015408	Read
LDO21_STATUS3	0x0001540A	Read
LDO21_STATUS4	0x0001540B	Read
LDO21_INT_RT_STS	0x00015410	Read
LDO21_INT_SET_TYPE	0x00015411	Read
LDO21_INT_POLARITY_HIGH	0x00015412	Read
LDO21_INT_POLARITY_LOW	0x00015413	Read/Write
LDO21_INT_LATCHED_CLR	0x00015414	Write
LDO21_INT_EN_SET	0x00015415	Read/Write
LDO21_INT_EN_CLR	0x00015416	Read/Write
LDO21_INT_MID_SEL	0x0001541A	Read/Write
LDO21_INT_PRIORITY	0x0001541B	Read/Write
LDO21_INT_SELF_CLEARING	0x0001541D	Read
LDO21_ULS_VSET_LB	0x00015439	Read/Write
LDO21_ULS_VSET_UB	0x0001543A	Read/Write
LDO21_VSET_LB	0x00015440	Read/Write
LDO21_VSET_UB	0x00015441	Read/Write
LDO21_VSET_VALID_LB	0x00015442	Read
LDO21_VSET_VALID_UB	0x00015443	Read
LDO21_MODE_CTL2	0x00015444	Read/Write
LDO21_MODE_CTL1	0x00015445	Read/Write
LDO21_EN_CTL	0x00015446	Read/Write
LDO21_FOLLOW_HWEN	0x00015447	Read/Write
LDO21_PBS_VOTE_CTL	0x00015449	Read/Write
LDO21_OCP_CTL1	0x00015488	Read/Write
LDO21_PD_CTL	0x000154A0	Read/Write

LDO21_REVISION1 | 0x00015400

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO21_REVISION2 | 0x00015401

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

LDO21_REVISION3 | 0x00015402

Type: Read

Reset: 0x00000000

HW Version Regster [23:16]

Bits	Name	Description
2 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO21_REVISION4 | 0x00015403

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
2 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

LDO21_PERPH_TYPE | 0x00015404

Type: Read

Reset: 0x00000004

Peripheral Type

Bits	Name	Description
7 : 0	TYPE	LDO

LDO21_PERPH_SUBTYPE | 0x00015405

Type: Read

Reset: 0x0000007D

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	P50 for example

LDO21_STATUS1 | 0x00015408

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Unused 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	SOFT_START_DONE	Indicates soft start is done when LDO is enabled. 1 = Softstart done 0 = Softstart in progress 0: NOT_DONE_SOFT_START 1: DONE_SOFT_START
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO21_STATUS3 | 0x0001540A

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	AHC_PRESENT	Indicates to SW/user that this regulator supports AHC (automatic headroom control) 0: NO_AHC 1: AHC
1 : 1	LLC_PRESENT	Indicates to SW/user that this regulator supports LLC (load-line compensation) 0: NO_LLC 1: LLC

LDO21_STATUS4 | 0x0001540B

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 1	ECM_READY	Indicates that ECM has powered up and is ready to operate when 0x1. This register is only used when STATUS4__ECM_PRESENT==1. 0: ECM_READY_false 1: ECM_READY_true
3 : 3	ECM_DATA_READY	Indicates that data is ready for PBS to perform read when 0x1. This register is only used when STATUS4__ECM_PRESENT==1. 0: ECM_DATA_READY_false 1: ECM_DATA_READY_true
4 : 4	ECM_PRESENT	Indicates to SW/user that this regulator supports ECM 0: NO_ECM 1: ECM
7 : 7	ECM_DATA_OVERWRITTEN	ECM measurement data overwritten before read. The bit should clear when both data registers are read. This register is only used when STATUS4__ECM_PRESENT==1. 0 = ECM data not overwritten 1 = ECM data overwritten 0: DATA_NOT_OVERWRITTEN 1: DATA_OVERWRITTEN

LDO21_INT_RT_STS | 0x00015410

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

LDO21_INT_SET_TYPE | 0x00015411

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

LDO21_INT_POLARITY_HIGH | 0x00015412

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

LDO21_INT_POLARITY_LOW | 0x00015413

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

LDO21_INT_LATCHED_CLR | 0x00015414

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

LDO21_INT_EN_SET | 0x00015415

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

LDO21_INT_EN_CLR | 0x00015416

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

LDO21_INT_MID_SEL | 0x0001541A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

LDO21_INT_PRIORITY | 0x0001541B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

LDO21_INT_SELF_CLEARING | 0x0001541D

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

LDO21_ULS_VSET_LB | 0x00015439

Type: Read/Write

Reset: 0x000000D8

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	LDO upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

LDO21_ULS_VSET_UB | 0x0001543A

Type: Read/Write

Reset: 0x0000000D

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	LDO upper limit stopvoltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV. Max combined value is 1.328mv (set as a default for both regs). Write will update ULS_VSET[15:0]. User must program ULS in LSB (8mV) increments to avoid undesired behavior

LDO21_VSET_LB | 0x00015440

Type: Read/Write

Reset: 0x00000090

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO21_VSET_UB | 0x00015441

Type: Read/Write

Reset: 0x0000000A

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO21_VSET_VALID_LB | 0x00015442

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 3	VSET_VALID_LB	Readback the valid output voltage setpoint Lower Byte (Bits 7:0). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO21_VSET_VALID_UB | 0x00015443

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Readback the valid output voltage setpoint Upper Byte (Bits 15:8). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO21_MODE_CTL2 | 0x00015444

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. This setting only matters when 0x47 FOLLOW_HWEN or 0x49 PBS_VOTE_CTL is programmed. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO21_MODE_CTL1 | 0x00015445

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO21_EN_CTL | 0x00015446

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

LDO21_FOLLOW_HWEN | 0x00015447

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO state. If this bit is set high then: When HWEN0=1 --> LDO if forced on (regardless of reg 0x46) When HWEN0=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN0_FALSE 1: EN_FOLLOW_HW_EN0_TRUE
1 : 1	EN_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO state. If this bit is set high then: When HWEN1=1 --> LDO if forced on (regardless of reg 0x46) When HWEN1=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN1_FALSE 1: EN_FOLLOW_HW_EN1_TRUE
2 : 2	EN_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO state. If this bit is set high then: When HWEN2=1 --> LDO if forced on (regardless of reg 0x46) When HWEN2=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN2_FALSE 1: EN_FOLLOW_HW_EN2_TRUE

Bits	Name	Description
3 : 3	EN_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE state does NOT control LDO state. If this bit is set high then: When PMIC_AWAKE=1 --> LDO if forced on (regardless of reg 0x46) When PMIC_AWAKE=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_PMIC_AWAKE_FALSE 1: EN_FOLLOW_PMIC_AWAKE_TRUE
4 : 4	MODE_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN0_FALSE 1: MODE_FOLLOW_HW_EN0_TRUE
5 : 5	MODE_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN1_FALSE 1: MODE_FOLLOW_HW_EN1_TRUE
6 : 6	MODE_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO mode. If this bit is set high then: When HWEN2=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN2_FALSE 1: MODE_FOLLOW_HW_EN2_TRUE
7 : 7	MODE_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE (sleep_b) state does NOT control LDO mode. If this bit is set high then: When PMIC_AWAKE=0--> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_PMIC_AWAKE_FALSE 1: MODE_FOLLOW_PMIC_AWAKE_TRUE

LDO21_PBS_VOTE_CTL | 0x00015449

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
6 : 6	PBS_MODE	PBS Mode vote 0 = Mode is set by MODE_CTL1 reg (DEFAULT) 1 = Mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: PBS_MODE_false 1: PBS_MODE_true
7 : 7	PBS_EN	Enables the LDO when asserted to HI. 0 = LDO will NOT be enabled (DEFAULT) 1= ENABLE LDO 0: PBS_EN_false 1: PBS_EN_true

LDO21_OCP_CTL1 | 0x00015488

Type: Read/Write

Reset: 0x000000D0

No description provided for this register.

Bits	Name	Description
0 : 0	OCP_TESTMODE_EN	Enable OCP test mode. In test mode the current limit is lowered. 0: OCP_TESTMODE_DISABLED 1: OCP_TESTMODE_ENABLED
1 : 1	OCP_STATUS_CLR	Writing a 1 to this bit clears the VREG_OCP bit in STATUS1. It does not recover the regulator from OCP shutdown. This bit must then be toggled from 1 to 0 to re-arm the status bit so that it can be latched in the event of another OCP event.
2 : 2	OCP_GLOBAL_BROADCAST_EN	Determines if the LDO notifies the PON module that an OCP event has occurred. PON module may use this info to shutdown the entire PMIC. 1 = LDO notifies PON when OCP event occurs 0 = LDO does not notify PON when OCP even occurs (DEFAULT) 0: OCP_GLOBAL_BROADCAST_DISABLED 1: OCP_GLOBAL_BROADCAST_ENABLED
5 : 4	OCP_DEB	Debounce time for OCP detection 0 = ~120us 1 = ~240us (DEFAULT) 2 = ~480us 3 = ~960us 0: DEB_120US 1: DEB_240US 2: DEB_480US 3: DEB_960US

Bits	Name	Description
6 : 6	OCP_SELF_SHUTDOWN_EN	Determines if the LDO performs self-shutdown upon OCP event. 1 = LDO self-shutdowns upon OCP event (DEFAULT) 0 = LDO does not self-shutdown upon OCP event 0: OCP_SELF_SHUTDOWN_DISABLED 1: OCP_SELF_SHUTDOWN_ENABLED
7 : 7	OCP_EN	1 = Enable OCP feature (DEFAULT) 0 = Disable OCP feature 0: OCP_DISABLED 1: OCP_ENABLED

LDO21_PD_CTL | 0x000154A0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Intended for keeping Vout from rising due to passFET leakage at high temperature. 1 = Leakage pulldown is always enabled 0 = Leakage pulldown is always disabled (DEFAULT) 0: LEAKAGE_PD_ALWAYS_DISABLED 1: LEAKAGE_PD_ALWAYS_ENABLED
6 : 6	WEAK_PD_EN	Weak PD is never active while LDO is on (VREG_READY=1). 1 = Weak pulldown is enabled when the regulator is disabled 0 = Weak pulldown is always inactive (DEFAULT) 0: WEAK_PD_DISABLED 1: WEAK_PD_WHEN_LDO_IS_OFF
7 : 7	STRONG_PD_EN	Strong PD is never active while LDO is on (VREG_READY=1). Strong pull-down FET Rdson will vary with Vout. 1 = Strong pulldown is enabled when the regulator is disabled (DEFAULT) 0 = Strong pulldown is always inactive. 0: STRONG_PD_ALWAYS_OFF 1: STRONG_PD_WHEN_LDO_IS_OFF

Module Name: LDO22

Base Address: 0x00015500

Register Quick Reference:

Register	Offset	Type
LDO22_REVISION1	0x00015500	Read

Register	Offset	Type
LDO22_REVISION2	0x00015501	Read
LDO22_REVISION3	0x00015502	Read
LDO22_REVISION4	0x00015503	Read
LDO22_PERPH_TYPE	0x00015504	Read
LDO22_PERPH_SUBTYPE	0x00015505	Read
LDO22_STATUS1	0x00015508	Read
LDO22_STATUS3	0x0001550A	Read
LDO22_STATUS4	0x0001550B	Read
LDO22_INT_RT_STS	0x00015510	Read
LDO22_INT_SET_TYPE	0x00015511	Read
LDO22_INT_POLARITY_HIGH	0x00015512	Read
LDO22_INT_POLARITY_LOW	0x00015513	Read/Write
LDO22_INT_LATCHED_CLR	0x00015514	Write
LDO22_INT_EN_SET	0x00015515	Read/Write
LDO22_INT_EN_CLR	0x00015516	Read/Write
LDO22_INT_MID_SEL	0x0001551A	Read/Write
LDO22_INT_PRIORITY	0x0001551B	Read/Write
LDO22_INT_SELF_CLEARING	0x0001551D	Read
LDO22_ULS_VSET_LB	0x00015539	Read/Write
LDO22_ULS_VSET_UB	0x0001553A	Read/Write
LDO22_VSET_LB	0x00015540	Read/Write
LDO22_VSET_UB	0x00015541	Read/Write
LDO22_VSET_VALID_LB	0x00015542	Read
LDO22_VSET_VALID_UB	0x00015543	Read
LDO22_MODE_CTL2	0x00015544	Read/Write
LDO22_MODE_CTL1	0x00015545	Read/Write
LDO22_EN_CTL	0x00015546	Read/Write
LDO22_FOLLOW_HWEN	0x00015547	Read/Write
LDO22_PBS_VOTE_CTL	0x00015549	Read/Write
LDO22_OCP_CTL1	0x00015588	Read/Write
LDO22_PD_CTL	0x000155A0	Read/Write

LDO22_REVISION1 | 0x00015500

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO22_REVISION2 | 0x00015501

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

LDO22_REVISION3 | 0x00015502

Type: Read

Reset: 0x00000000

HW Version Regster [23:16]

Bits	Name	Description
2 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO22_REVISION4 | 0x00015503

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
2 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

LDO22_PERPH_TYPE | 0x00015504

Type: Read

Reset: 0x00000004

Peripheral Type

Bits	Name	Description
7 : 0	TYPE	LDO

LDO22_PERPH_SUBTYPE | 0x00015505

Type: Read

Reset: 0x0000007D

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	P50 for example

LDO22_STATUS1 | 0x00015508

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Unused 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	SOFT_START_DONE	Indicates soft start is done when LDO is enabled. 1 = Softstart done 0 = Softstart in progress 0: NOT_DONE_SOFT_START 1: DONE_SOFT_START
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO22_STATUS3 | 0x0001550A

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	AHC_PRESENT	Indicates to SW/user that this regulator supports AHC (automatic headroom control) 0: NO_AHC 1: AHC
1 : 1	LLC_PRESENT	Indicates to SW/user that this regulator supports LLC (load-line compensation) 0: NO_LLC 1: LLC

LDO22_STATUS4 | 0x0001550B

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 1	ECM_READY	Indicates that ECM has powered up and is ready to operate when 0x1. This register is only used when STATUS4__ECM_PRESENT==1. 0: ECM_READY_false 1: ECM_READY_true
3 : 3	ECM_DATA_READY	Indicates that data is ready for PBS to perform read when 0x1. This register is only used when STATUS4__ECM_PRESENT==1. 0: ECM_DATA_READY_false 1: ECM_DATA_READY_true
4 : 4	ECM_PRESENT	Indicates to SW/user that this regulator supports ECM 0: NO_ECM 1: ECM
7 : 7	ECM_DATA_OVERWRITTEN	ECM measurement data overwritten before read. The bit should clear when both data registers are read. This register is only used when STATUS4__ECM_PRESENT==1. 0 = ECM data not overwritten 1 = ECM data overwritten 0: DATA_NOT_OVERWRITTEN 1: DATA_OVERWRITTEN

LDO22_INT_RT_STS | 0x00015510

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

LDO22_INT_SET_TYPE | 0x00015511

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

LDO22_INT_POLARITY_HIGH | 0x00015512

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

LDO22_INT_POLARITY_LOW | 0x00015513

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

LDO22_INT_LATCHED_CLR | 0x00015514

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

LDO22_INT_EN_SET | 0x00015515

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

LDO22_INT_EN_CLR | 0x00015516

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

LDO22_INT_MID_SEL | 0x0001551A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

LDO22_INT_PRIORITY | 0x0001551B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

LDO22_INT_SELF_CLEARING | 0x0001551D

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

LDO22_ULS_VSET_LB | 0x00015539

Type: Read/Write

Reset: 0x000000D8

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	LDO upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

LDO22_ULS_VSET_UB | 0x0001553A

Type: Read/Write

Reset: 0x0000000D

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	LDO upper limit stopvoltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV. Max combined value is 1.328mv (set as a default for both regs). Write will update ULS_VSET[15:0]. User must program ULS in LSB (8mV) increments to avoid undesired behavior

LDO22_VSET_LB | 0x00015540

Type: Read/Write

Reset: 0x00000090

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO22_VSET_UB | 0x00015541

Type: Read/Write

Reset: 0x0000000B

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO22_VSET_VALID_LB | 0x00015542

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 3	VSET_VALID_LB	Readback the valid output voltage setpoint Lower Byte (Bits 7:0). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO22_VSET_VALID_UB | 0x00015543

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Readback the valid output voltage setpoint Upper Byte (Bits 15:8). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO22_MODE_CTL2 | 0x00015544

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. This setting only matters when 0x47 FOLLOW_HWEN or 0x49 PBS_VOTE_CTL is programmed. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO22_MODE_CTL1 | 0x00015545

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO22_EN_CTL | 0x00015546

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

LDO22_FOLLOW_HWEN | 0x00015547

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO state. If this bit is set high then: When HWEN0=1 --> LDO if forced on (regardless of reg 0x46) When HWEN0=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN0_FALSE 1: EN_FOLLOW_HW_EN0_TRUE
1 : 1	EN_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO state. If this bit is set high then: When HWEN1=1 --> LDO if forced on (regardless of reg 0x46) When HWEN1=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN1_FALSE 1: EN_FOLLOW_HW_EN1_TRUE
2 : 2	EN_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO state. If this bit is set high then: When HWEN2=1 --> LDO if forced on (regardless of reg 0x46) When HWEN2=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN2_FALSE 1: EN_FOLLOW_HW_EN2_TRUE

Bits	Name	Description
3 : 3	EN_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE state does NOT control LDO state. If this bit is set high then: When PMIC_AWAKE=1 --> LDO if forced on (regardless of reg 0x46) When PMIC_AWAKE=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_PMIC_AWAKE_FALSE 1: EN_FOLLOW_PMIC_AWAKE_TRUE
4 : 4	MODE_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN0_FALSE 1: MODE_FOLLOW_HW_EN0_TRUE
5 : 5	MODE_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN1_FALSE 1: MODE_FOLLOW_HW_EN1_TRUE
6 : 6	MODE_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO mode. If this bit is set high then: When HWEN2=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN2_FALSE 1: MODE_FOLLOW_HW_EN2_TRUE
7 : 7	MODE_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE (sleep_b) state does NOT control LDO mode. If this bit is set high then: When PMIC_AWAKE=0--> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_PMIC_AWAKE_FALSE 1: MODE_FOLLOW_PMIC_AWAKE_TRUE

LDO22_PBS_VOTE_CTL | 0x00015549

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
6 : 6	PBS_MODE	PBS Mode vote 0 = Mode is set by MODE_CTL1 reg (DEFAULT) 1 = Mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: PBS_MODE_false 1: PBS_MODE_true
7 : 7	PBS_EN	Enables the LDO when asserted to HI. 0 = LDO will NOT be enabled (DEFAULT) 1= ENABLE LDO 0: PBS_EN_false 1: PBS_EN_true

LDO22_OCP_CTL1 | 0x00015588

Type: Read/Write

Reset: 0x000000D0

No description provided for this register.

Bits	Name	Description
0 : 0	OCP_TESTMODE_EN	Enable OCP test mode. In test mode the current limit is lowered. 0: OCP_TESTMODE_DISABLED 1: OCP_TESTMODE_ENABLED
1 : 1	OCP_STATUS_CLR	Writing a 1 to this bit clears the VREG_OCP bit in STATUS1. It does not recover the regulator from OCP shutdown. This bit must then be toggled from 1 to 0 to re-arm the status bit so that it can be latched in the event of another OCP event.
2 : 2	OCP_GLOBAL_BROADCAST_EN	Determines if the LDO notifies the PON module that an OCP event has occurred. PON module may use this info to shutdown the entire PMIC. 1 = LDO notifies PON when OCP event occurs 0 = LDO does not notify PON when OCP even occurs (DEFAULT) 0: OCP_GLOBAL_BROADCAST_DISABLED 1: OCP_GLOBAL_BROADCAST_ENABLED
5 : 4	OCP_DEB	Debounce time for OCP detection 0 = ~120us 1 = ~240us (DEFAULT) 2 = ~480us 3 = ~960us 0: DEB_120US 1: DEB_240US 2: DEB_480US 3: DEB_960US

Bits	Name	Description
6 : 6	OCP_SELF_SHUTDOWN_EN	Determines if the LDO performs self-shutdown upon OCP event. 1 = LDO self-shutdowns upon OCP event (DEFAULT) 0 = LDO does not self-shutdown upon OCP event 0: OCP_SELF_SHUTDOWN_DISABLED 1: OCP_SELF_SHUTDOWN_ENABLED
7 : 7	OCP_EN	1 = Enable OCP feature (DEFAULT) 0 = Disable OCP feature 0: OCP_DISABLED 1: OCP_ENABLED

LDO22_PD_CTL | 0x000155A0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Intended for keeping Vout from rising due to passFET leakage at high temperature. 1 = Leakage pulldown is always enabled 0 = Leakage pulldown is always disabled (DEFAULT) 0: LEAKAGE_PD_ALWAYS_DISABLED 1: LEAKAGE_PD_ALWAYS_ENABLED
6 : 6	WEAK_PD_EN	Weak PD is never active while LDO is on (VREG_READY=1). 1 = Weak pulldown is enabled when the regulator is disabled 0 = Weak pulldown is always inactive (DEFAULT) 0: WEAK_PD_DISABLED 1: WEAK_PD_WHEN_LDO_IS_OFF
7 : 7	STRONG_PD_EN	Strong PD is never active while LDO is on (VREG_READY=1). Strong pull-down FET Rdson will vary with Vout. 1 = Strong pulldown is enabled when the regulator is disabled (DEFAULT) 0 = Strong pulldown is always inactive. 0: STRONG_PD_ALWAYS_OFF 1: STRONG_PD_WHEN_LDO_IS_OFF

Module Name: LDO23

Base Address: 0x00015600

Register Quick Reference:

Register	Offset	Type
LDO23_REVISION1	0x00015600	Read

Register	Offset	Type
LDO23_REVISION2	0x00015601	Read
LDO23_REVISION3	0x00015602	Read
LDO23_REVISION4	0x00015603	Read
LDO23_PERPH_TYPE	0x00015604	Read
LDO23_PERPH_SUBTYPE	0x00015605	Read
LDO23_STATUS1	0x00015608	Read
LDO23_STATUS3	0x0001560A	Read
LDO23_STATUS4	0x0001560B	Read
LDO23_INT_RT_STS	0x00015610	Read
LDO23_INT_SET_TYPE	0x00015611	Read
LDO23_INT_POLARITY_HIGH	0x00015612	Read
LDO23_INT_POLARITY_LOW	0x00015613	Read/Write
LDO23_INT_LATCHED_CLR	0x00015614	Write
LDO23_INT_EN_SET	0x00015615	Read/Write
LDO23_INT_EN_CLR	0x00015616	Read/Write
LDO23_INT_MID_SEL	0x0001561A	Read/Write
LDO23_INT_PRIORITY	0x0001561B	Read/Write
LDO23_INT_SELF_CLEARING	0x0001561D	Read
LDO23_ULS_VSET_LB	0x00015639	Read/Write
LDO23_ULS_VSET_UB	0x0001563A	Read/Write
LDO23_VSET_LB	0x00015640	Read/Write
LDO23_VSET_UB	0x00015641	Read/Write
LDO23_VSET_VALID_LB	0x00015642	Read
LDO23_VSET_VALID_UB	0x00015643	Read
LDO23_MODE_CTL2	0x00015644	Read/Write
LDO23_MODE_CTL1	0x00015645	Read/Write
LDO23_EN_CTL	0x00015646	Read/Write
LDO23_FOLLOW_HWEN	0x00015647	Read/Write
LDO23_PBS_VOTE_CTL	0x00015649	Read/Write
LDO23_OCP_CTL1	0x00015688	Read/Write
LDO23_PD_CTL	0x000156A0	Read/Write

LDO23_REVISION1 | 0x00015600

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO23_REVISION2 | 0x00015601

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

LDO23_REVISION3 | 0x00015602

Type: Read

Reset: 0x00000000

HW Version Regster [23:16]

Bits	Name	Description
2 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO23_REVISION4 | 0x00015603

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
2 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

LDO23_PERPH_TYPE | 0x00015604

Type: Read

Reset: 0x00000004

Peripheral Type

Bits	Name	Description
7 : 0	TYPE	LDO

LDO23_PERPH_SUBTYPE | 0x00015605

Type: Read

Reset: 0x0000007D

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	P50 for example

LDO23_STATUS1 | 0x00015608

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Unused 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	SOFT_START_DONE	Indicates soft start is done when LDO is enabled. 1 = Softstart done 0 = Softstart in progress 0: NOT_DONE_SOFT_START 1: DONE_SOFT_START
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO23_STATUS3 | 0x0001560A

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	AHC_PRESENT	Indicates to SW/user that this regulator supports AHC (automatic headroom control) 0: NO_AHC 1: AHC
1 : 1	LLC_PRESENT	Indicates to SW/user that this regulator supports LLC (load-line compensation) 0: NO_LLC 1: LLC

LDO23_STATUS4 | 0x0001560B

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 1	ECM_READY	Indicates that ECM has powered up and is ready to operate when 0x1. This register is only used when STATUS4__ECM_PRESENT==1. 0: ECM_READY_false 1: ECM_READY_true
3 : 3	ECM_DATA_READY	Indicates that data is ready for PBS to perform read when 0x1. This register is only used when STATUS4__ECM_PRESENT==1. 0: ECM_DATA_READY_false 1: ECM_DATA_READY_true
4 : 4	ECM_PRESENT	Indicates to SW/user that this regulator supports ECM 0: NO_ECM 1: ECM
7 : 7	ECM_DATA_OVERWRITTEN	ECM measurement data overwritten before read. The bit should clear when both data registers are read. This register is only used when STATUS4__ECM_PRESENT==1. 0 = ECM data not overwritten 1 = ECM data overwritten 0: DATA_NOT_OVERWRITTEN 1: DATA_OVERWRITTEN

LDO23_INT_RT_STS | 0x00015610

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

LDO23_INT_SET_TYPE | 0x00015611

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

LDO23_INT_POLARITY_HIGH | 0x00015612

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

LDO23_INT_POLARITY_LOW | 0x00015613

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

LDO23_INT_LATCHED_CLR | 0x00015614

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

LDO23_INT_EN_SET | 0x00015615

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

LDO23_INT_EN_CLR | 0x00015616

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

LDO23_INT_MID_SEL | 0x0001561A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

LDO23_INT_PRIORITY | 0x0001561B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

LDO23_INT_SELF_CLEARING | 0x0001561D

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

LDO23_ULS_VSET_LB | 0x00015639

Type: Read/Write

Reset: 0x000000D8

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	LDO upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

LDO23_ULS_VSET_UB | 0x0001563A

Type: Read/Write

Reset: 0x0000000D

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	LDO upper limit stopvoltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV. Max combined value is 1.328mv (set as a default for both regs). Write will update ULS_VSET[15:0]. User must program ULS in LSB (8mV) increments to avoid undesired behavior

LDO23_VSET_LB | 0x00015640

Type: Read/Write

Reset: 0x000000E8

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO23_VSET_UB | 0x00015641

Type: Read/Write

Reset: 0x0000000C

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO23_VSET_VALID_LB | 0x00015642

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 3	VSET_VALID_LB	Readback the valid output voltage setpoint Lower Byte (Bits 7:0). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO23_VSET_VALID_UB | 0x00015643

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Readback the valid output voltage setpoint Upper Byte (Bits 15:8). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO23_MODE_CTL2 | 0x00015644

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. This setting only matters when 0x47 FOLLOW_HWEN or 0x49 PBS_VOTE_CTL is programmed. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO23_MODE_CTL1 | 0x00015645

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO23_EN_CTL | 0x00015646

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

LDO23_FOLLOW_HWEN | 0x00015647

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO state. If this bit is set high then: When HWEN0=1 --> LDO if forced on (regardless of reg 0x46) When HWEN0=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN0_FALSE 1: EN_FOLLOW_HW_EN0_TRUE
1 : 1	EN_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO state. If this bit is set high then: When HWEN1=1 --> LDO if forced on (regardless of reg 0x46) When HWEN1=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN1_FALSE 1: EN_FOLLOW_HW_EN1_TRUE
2 : 2	EN_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO state. If this bit is set high then: When HWEN2=1 --> LDO if forced on (regardless of reg 0x46) When HWEN2=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN2_FALSE 1: EN_FOLLOW_HW_EN2_TRUE

Bits	Name	Description
3 : 3	EN_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE state does NOT control LDO state. If this bit is set high then: When PMIC_AWAKE=1 --> LDO if forced on (regardless of reg 0x46) When PMIC_AWAKE=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_PMIC_AWAKE_FALSE 1: EN_FOLLOW_PMIC_AWAKE_TRUE
4 : 4	MODE_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN0_FALSE 1: MODE_FOLLOW_HW_EN0_TRUE
5 : 5	MODE_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN1_FALSE 1: MODE_FOLLOW_HW_EN1_TRUE
6 : 6	MODE_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO mode. If this bit is set high then: When HWEN2=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN2_FALSE 1: MODE_FOLLOW_HW_EN2_TRUE
7 : 7	MODE_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE (sleep_b) state does NOT control LDO mode. If this bit is set high then: When PMIC_AWAKE=0--> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_PMIC_AWAKE_FALSE 1: MODE_FOLLOW_PMIC_AWAKE_TRUE

LDO23_PBS_VOTE_CTL | 0x00015649

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
6 : 6	PBS_MODE	PBS Mode vote 0 = Mode is set by MODE_CTL1 reg (DEFAULT) 1 = Mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: PBS_MODE_false 1: PBS_MODE_true
7 : 7	PBS_EN	Enables the LDO when asserted to HI. 0 = LDO will NOT be enabled (DEFAULT) 1= ENABLE LDO 0: PBS_EN_false 1: PBS_EN_true

LDO23_OCP_CTL1 | 0x00015688

Type: Read/Write

Reset: 0x000000D0

No description provided for this register.

Bits	Name	Description
0 : 0	OCP_TESTMODE_EN	Enable OCP test mode. In test mode the current limit is lowered. 0: OCP_TESTMODE_DISABLED 1: OCP_TESTMODE_ENABLED
1 : 1	OCP_STATUS_CLR	Writing a 1 to this bit clears the VREG_OCP bit in STATUS1. It does not recover the regulator from OCP shutdown. This bit must then be toggled from 1 to 0 to re-arm the status bit so that it can be latched in the event of another OCP event.
2 : 2	OCP_GLOBAL_BROADCAST_EN	Determines if the LDO notifies the PON module that an OCP event has occurred. PON module may use this info to shutdown the entire PMIC. 1 = LDO notifies PON when OCP event occurs 0 = LDO does not notify PON when OCP even occurs (DEFAULT) 0: OCP_GLOBAL_BROADCAST_DISABLED 1: OCP_GLOBAL_BROADCAST_ENABLED
5 : 4	OCP_DEB	Debounce time for OCP detection 0 = ~120us 1 = ~240us (DEFAULT) 2 = ~480us 3 = ~960us 0: DEB_120US 1: DEB_240US 2: DEB_480US 3: DEB_960US

Bits	Name	Description
6 : 6	OCP_SELF_SHUTDOWN_EN	Determines if the LDO performs self-shutdown upon OCP event. 1 = LDO self-shutdowns upon OCP event (DEFAULT) 0 = LDO does not self-shutdown upon OCP event 0: OCP_SELF_SHUTDOWN_DISABLED 1: OCP_SELF_SHUTDOWN_ENABLED
7 : 7	OCP_EN	1 = Enable OCP feature (DEFAULT) 0 = Disable OCP feature 0: OCP_DISABLED 1: OCP_ENABLED

LDO23_PD_CTL | 0x000156A0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Intended for keeping Vout from rising due to passFET leakage at high temperature. 1 = Leakage pulldown is always enabled 0 = Leakage pulldown is always disabled (DEFAULT) 0: LEAKAGE_PD_ALWAYS_DISABLED 1: LEAKAGE_PD_ALWAYS_ENABLED
6 : 6	WEAK_PD_EN	Weak PD is never active while LDO is on (VREG_READY=1). 1 = Weak pulldown is enabled when the regulator is disabled 0 = Weak pulldown is always inactive (DEFAULT) 0: WEAK_PD_DISABLED 1: WEAK_PD_WHEN_LDO_IS_OFF
7 : 7	STRONG_PD_EN	Strong PD is never active while LDO is on (VREG_READY=1). Strong pull-down FET Rdson will vary with Vout. 1 = Strong pulldown is enabled when the regulator is disabled (DEFAULT) 0 = Strong pulldown is always inactive. 0: STRONG_PD_ALWAYS_OFF 1: STRONG_PD_WHEN_LDO_IS_OFF

Module Name: LDO24

Base Address: 0x00015700

Register Quick Reference:

Register	Offset	Type
LDO24_REVISION1	0x00015700	Read

Register	Offset	Type
LDO24_REVISION2	0x00015701	Read
LDO24_REVISION3	0x00015702	Read
LDO24_REVISION4	0x00015703	Read
LDO24_PERPH_TYPE	0x00015704	Read
LDO24_PERPH_SUBTYPE	0x00015705	Read
LDO24_STATUS1	0x00015708	Read
LDO24_STATUS3	0x0001570A	Read
LDO24_STATUS4	0x0001570B	Read
LDO24_INT_RT_STS	0x00015710	Read
LDO24_INT_SET_TYPE	0x00015711	Read
LDO24_INT_POLARITY_HIGH	0x00015712	Read
LDO24_INT_POLARITY_LOW	0x00015713	Read/Write
LDO24_INT_LATCHED_CLR	0x00015714	Write
LDO24_INT_EN_SET	0x00015715	Read/Write
LDO24_INT_EN_CLR	0x00015716	Read/Write
LDO24_INT_MID_SEL	0x0001571A	Read/Write
LDO24_INT_PRIORITY	0x0001571B	Read/Write
LDO24_INT_SELF_CLEARING	0x0001571D	Read
LDO24_ULS_VSET_LB	0x00015739	Read/Write
LDO24_ULS_VSET_UB	0x0001573A	Read/Write
LDO24_VSET_LB	0x00015740	Read/Write
LDO24_VSET_UB	0x00015741	Read/Write
LDO24_VSET_VALID_LB	0x00015742	Read
LDO24_VSET_VALID_UB	0x00015743	Read
LDO24_MODE_CTL2	0x00015744	Read/Write
LDO24_MODE_CTL1	0x00015745	Read/Write
LDO24_EN_CTL	0x00015746	Read/Write
LDO24_FOLLOW_HWEN	0x00015747	Read/Write
LDO24_PBS_VOTE_CTL	0x00015749	Read/Write
LDO24_OCP_CTL1	0x00015788	Read/Write
LDO24_PD_CTL	0x000157A0	Read/Write

LDO24_REVISION1 | 0x00015700

Type: Read

Reset: 0x00000000

HW Version Regster [7:0] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MINOR	This number is incremented for digital change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO24_REVISION2 | 0x00015701

Type: Read

Reset: 0x00000002

HW Version Regster [15:8] PMIC_CONSTANT

Bits	Name	Description
2 : 0	DIG_MAJOR	This number is incremented when changes are made to the digital HW that are not backwards compatible with existing software.

LDO24_REVISION3 | 0x00015702

Type: Read

Reset: 0x00000000

HW Version Regster [23:16]

Bits	Name	Description
2 : 0	ANA_MINOR	This number is incremented for analog change that is not intended to affect software or any change that adds a new feature but is backwards compatible with old software. Software changes may be required to take advantage of the new features. Minor resets to zero when Major increments.

LDO24_REVISION4 | 0x00015703

Type: Read

Reset: 0x00000001

HW Version Regster [31:24]

Bits	Name	Description
2 : 0	ANA_MAJOR	This number is incremented when changes are made to the analog HW that are not backwards compatible with existing software.

LDO24_PERPH_TYPE | 0x00015704

Type: Read

Reset: 0x00000004

Peripheral Type

Bits	Name	Description
7 : 0	TYPE	LDO

LDO24_PERPH_SUBTYPE | 0x00015705

Type: Read

Reset: 0x0000007D

Peripheral SubType

Bits	Name	Description
7 : 0	SUBTYPE	P50 for example

LDO24_STATUS1 | 0x00015708

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 0	MODE_STATE	Indicates that the status of the LDO. Value is only valid when LDO is enabled. 0 : Bypass Mode 1 : Unused 2 : LPM Mode 3 : NPM Mode 0: MODE_STATE_BYP 1: MODE_STATE_UNUSED 2: MODE_STATE_LPM 3: MODE_STATE_NPM
2 : 2	SOFT_START_DONE	Indicates soft start is done when LDO is enabled. 1 = Softstart done 0 = Softstart in progress 0: NOT_DONE_SOFT_START 1: DONE_SOFT_START
5 : 5	VREG_OCP	The signal driving this bit should be a OCP_LATCHED_STS signal that only resets with dVdd_rb or by write to OCP_STATUS_CLR bit in reg 0x66. 1 = Over Current has been detected/latched. 0 = No fault detected 0: NO_OCP 1: OCP
6 : 6	VREG_ERROR	1 = Regulator is not stepping and has fallen below VREG_OK comparator threshold, or done stepping/softstart and not reached VREG_OK comparator threshold. 0 = Regulator is okay; no error condition or in Bypass 0: LDO_NO_ERR 1: LDO_ERR
7 : 7	VREG_READY	1 = Regulator is settled and ready to use or in Bypass 0 = Regulator is not ready due to stepping or soft-start in progress or below the de-bounced VREG_OK comparator threshold 0: LDO_NOT_READY 1: LDO_READY

LDO24_STATUS3 | 0x0001570A

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	AHC_PRESENT	Indicates to SW/user that this regulator supports AHC (automatic headroom control) 0: NO_AHC 1: AHC
1 : 1	LLC_PRESENT	Indicates to SW/user that this regulator supports LLC (load-line compensation) 0: NO_LLC 1: LLC

LDO24_STATUS4 | 0x0001570B

Type: Read

Reset: 0x00000000

Status Registers

Bits	Name	Description
1 : 1	ECM_READY	Indicates that ECM has powered up and is ready to operate when 0x1. This register is only used when STATUS4__ECM_PRESENT==1. 0: ECM_READY_false 1: ECM_READY_true
3 : 3	ECM_DATA_READY	Indicates that data is ready for PBS to perform read when 0x1. This register is only used when STATUS4__ECM_PRESENT==1. 0: ECM_DATA_READY_false 1: ECM_DATA_READY_true
4 : 4	ECM_PRESENT	Indicates to SW/user that this regulator supports ECM 0: NO_ECM 1: ECM
7 : 7	ECM_DATA_OVERWRITTEN	ECM measurement data overwritten before read. The bit should clear when both data registers are read. This register is only used when STATUS4__ECM_PRESENT==1. 0 = ECM data not overwritten 1 = ECM data overwritten 0: DATA_NOT_OVERWRITTEN 1: DATA_OVERWRITTEN

LDO24_INT_RT_STS | 0x00015710

Type: Read

Reset: 0x00000000

Interrupt Real Time Status Bits

Bits	Name	Description
0 : 0	VREG_FAULT_RT_STS	FAULT condition has occurred 0: VREG_NO_FAULT 1: VREG_FAULT
1 : 1	VREG_ACK_RT_STS	Regulator is enabled and ready to use 0: VREG_NO_ACK 1: VREG_ACK

LDO24_INT_SET_TYPE | 0x00015711

Type: Read

Reset: 0x00000003

0 = use level trigger interrupts, 1 = use edge trigger interrupts

Bits	Name	Description
0 : 0	VREG_FAULT_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED
1 : 1	VREG_ACK_TYPE	Interrupt type, edge or level 0: LEVEL_TRIGGERED 1: EDGE_TRIGGERED

LDO24_INT_POLARITY_HIGH | 0x00015712

Type: Read

Reset: 0x00000003

1= Interrupt will trigger on a level high (rising edge) event, 0 = level high triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED
1 : 1	VREG_ACK_HIGH	Edge type, rising or Level type, high true 0: HIGH_TRIGGER_DISABLED 1: HIGH_TRIGGER_ENABLED

LDO24_INT_POLARITY_LOW | 0x00015713

Type: Read/Write

Reset: 0x00000000

1' = Interrupt will trigger on a level low (falling edge) event, '0' = level low triggering is disabled

Bits	Name	Description
0 : 0	VREG_FAULT_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED
1 : 1	VREG_ACK_LOW	Edge type, falling or Level type, low true 0: LOW_TRIGGER_DISABLED 1: LOW_TRIGGER_ENABLED

LDO24_INT_LATCHED_CLR | 0x00015714

Type: Write

Reset: 0x00000000

Writing a '1' to this interrupt will rearm the interrupt when an interrupt is pending. It clears the internal sticky and sent bits. This register is a back-up just in case the self-clearing interrupt logic does not work.

Bits	Name	Description
0 : 0	VREG_FAULT_LATCHED_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_LATCHED_CLR	No description provided for this bit field.

LDO24_INT_EN_SET | 0x00015715

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will enable the corresponding interrupt. Reading this register will readback enable status PMIC_SET_MASK

Bits	Name	Description
0 : 0	VREG_FAULT_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED
1 : 1	VREG_ACK_EN_SET	No description provided for this bit field. 0: INT_DISABLED 1: INT_ENABLED

LDO24_INT_EN_CLR | 0x00015716

Type: Read/Write

Reset: 0x00000000

writing '0' to this register has no effect. Writing a '1' will disable the corresponding interrupt. Reading this register will readback enable status PMIC_CLR_MASK=INT_EN_SET

Bits	Name	Description
0 : 0	VREG_FAULT_EN_CLR	No description provided for this bit field.
1 : 1	VREG_ACK_EN_CLR	No description provided for this bit field.

LDO24_INT_MID_SEL | 0x0001571A

Type: Read/Write

Reset: 0x00000000

Selects the MID that will receive the interrupt

Bits	Name	Description
1 : 0	INT_MID_SEL	No description provided for this bit field. 0: INT_MID_SEL_0 1: INT_MID_SEL_1

LDO24_INT_PRIORITY | 0x0001571B

Type: Read/Write

Reset: 0x00000000

SR=0 A=1

Bits	Name	Description
0 : 0	INT_PRIORITY	No description provided for this bit field. 0: INT_PRIORITY_SR 1: INT_PRIORITY_A

LDO24_INT_SELF_CLEARING | 0x0001571D

Type: Read

Reset: 0x00000001

Indicate self clearing or regular interrupt 0 = Regular interrupt 1 = Self clearing interrupt

Bits	Name	Description
0 : 0	INT_SELF_CLEARING	No description provided for this bit field. 0: INT_SELF_CLEARING_0 1: INT_SELF_CLEARING_1

LDO24_ULS_VSET_LB | 0x00015739

Type: Read/Write

Reset: 0x000000D8

PMIC_LATCHED_WRITE=ULS_VSET_UB

Bits	Name	Description
7 : 0	ULS_VSET_LB	LDO upper limit stop voltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV

LDO24_ULS_VSET_UB | 0x0001573A

Type: Read/Write

Reset: 0x0000000D

No description provided for this register.

Bits	Name	Description
3 : 0	ULS_VSET_UB	LDO upper limit stopvoltage set point bits<7:0>. 2-Byte Word (0x40 and 0x41). Vout = 2-byte word (dec) x 1mV. Max combined value is 1.328mv (set as a default for both regs). Write will update ULS_VSET[15:0]. User must program ULS in LSB (8mV) increments to avoid undesired behavior

LDO24_VSET_LB | 0x00015740

Type: Read/Write

Reset: 0x00000090

PMIC_LATCHED_WRITE=VSET_UB

Bits	Name	Description
7 : 0	VSET_LB	LDO voltage set point Lower Byte (bits 7:0). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO24_VSET_UB | 0x00015741

Type: Read/Write

Reset: 0x0000000B

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_UB	LDO voltage set point Upper Byte (bits 15:8). Set point is a 2-Byte Word, reg 0x40 and 0x41. Vout = 2-byte word (dec) x 1mV. Although this register can be programmed in 1mV increments, the actual LDO LSB is as follows: NMOS, LV PMOS and MV PMOS LSB = 8mV

LDO24_VSET_VALID_LB | 0x00015742

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 3	VSET_VALID_LB	Readback the valid output voltage setpoint Lower Byte (Bits 7:0). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO24_VSET_VALID_UB | 0x00015743

Type: Read

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
3 : 0	VSET_VALID_UB	Readback the valid output voltage setpoint Upper Byte (Bits 15:8). This value will also reflect any additional VSET offset due to the digital LLC feature (if present and enabled). VSET_VALID is not ENUM'd. This register should read 0x00 when LDO is off.

LDO24_MODE_CTL2 | 0x00015744

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_SECONDARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. This setting only matters when 0x47 FOLLOW_HWEN or 0x49 PBS_VOTE_CTL is programmed. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO24_MODE_CTL1 | 0x00015745

Type: Read/Write

Reset: 0x00000007

Define LDO Mode Transitions

Bits	Name	Description
2 : 0	MODE_PRIMARY	LDO Mode: LDO will run in NPM when programmed to an unused setting. 111 = NPM (DEFAULT) 110 = Unused (-->NPM) 101 = Unused (-->NPM) 100 = LPM 011 = Unused (-->NPM) 010 = FORCED BYPASS 001 = Active Bypass NPM if LDO500. NPM if LDO510. 000 = Active Bypass LPM if LDO500. NPM if LDO510. 0: ACTIVE_BYPASS_LPM_OR_NPM 1: ACTIVE_BYPASS_NPM_OR_NPM 2: FORCED_BYPASS 3: UNUSED_3 4: LPM 5: UNUSED_5 6: UNUSED_6 7: NPM

LDO24_EN_CTL | 0x00015746

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
7 : 7	EN_LDO	1' = Enable the LDO, '0' = do not force LDO on 0: EN_LDO_FALSE 1: EN_LDO_TRUE

LDO24_FOLLOW_HWEN | 0x00015747

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
0 : 0	EN_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO state. If this bit is set high then: When HWEN0=1 --> LDO if forced on (regardless of reg 0x46) When HWEN0=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN0_FALSE 1: EN_FOLLOW_HW_EN0_TRUE
1 : 1	EN_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO state. If this bit is set high then: When HWEN1=1 --> LDO if forced on (regardless of reg 0x46) When HWEN1=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN1_FALSE 1: EN_FOLLOW_HW_EN1_TRUE
2 : 2	EN_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO state. If this bit is set high then: When HWEN2=1 --> LDO if forced on (regardless of reg 0x46) When HWEN2=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_HW_EN2_FALSE 1: EN_FOLLOW_HW_EN2_TRUE

Bits	Name	Description
3 : 3	EN_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE state does NOT control LDO state. If this bit is set high then: When PMIC_AWAKE=1 --> LDO if forced on (regardless of reg 0x46) When PMIC_AWAKE=0 --> LDO state depends on reg 0x46 0: EN_FOLLOW_PMIC_AWAKE_FALSE 1: EN_FOLLOW_PMIC_AWAKE_TRUE
4 : 4	MODE_FOLLOW_HW_EN0	If this bit is set low then HWEN0 state does NOT control LDO mode. If this bit is set high then: When HWEN0=0--> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN0=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN0_FALSE 1: MODE_FOLLOW_HW_EN0_TRUE
5 : 5	MODE_FOLLOW_HW_EN1	If this bit is set low then HWEN1 state does NOT control LDO mode. If this bit is set high then: When HWEN1=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN1=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN1_FALSE 1: MODE_FOLLOW_HW_EN1_TRUE
6 : 6	MODE_FOLLOW_HW_EN2	If this bit is set low then HWEN2 state does NOT control LDO mode. If this bit is set high then: When HWEN2=0 --> mode is set by MODE_CTL1 reg (DEFAULT) When HWEN2=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_HW_EN2_FALSE 1: MODE_FOLLOW_HW_EN2_TRUE
7 : 7	MODE_FOLLOW_PMIC_AWAKE	If this bit is set low then PMIC_AWAKE (sleep_b) state does NOT control LDO mode. If this bit is set high then: When PMIC_AWAKE=0--> mode is set by MODE_CTL1 reg (DEFAULT) When PMIC_AWAKE=1 --> mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: MODE_FOLLOW_PMIC_AWAKE_FALSE 1: MODE_FOLLOW_PMIC_AWAKE_TRUE

LDO24_PBS_VOTE_CTL | 0x00015749

Type: Read/Write

Reset: 0x00000000

No description provided for this register.

Bits	Name	Description
6 : 6	PBS_MODE	PBS Mode vote 0 = Mode is set by MODE_CTL1 reg (DEFAULT) 1 = Mode is set by max(MODE_CTL1,MODE_CTL2) reg 0: PBS_MODE_false 1: PBS_MODE_true
7 : 7	PBS_EN	Enables the LDO when asserted to HI. 0 = LDO will NOT be enabled (DEFAULT) 1= ENABLE LDO 0: PBS_EN_false 1: PBS_EN_true

LDO24_OCP_CTL1 | 0x00015788

Type: Read/Write

Reset: 0x000000D0

No description provided for this register.

Bits	Name	Description
0 : 0	OCP_TESTMODE_EN	Enable OCP test mode. In test mode the current limit is lowered. 0: OCP_TESTMODE_DISABLED 1: OCP_TESTMODE_ENABLED
1 : 1	OCP_STATUS_CLR	Writing a 1 to this bit clears the VREG_OCP bit in STATUS1. It does not recover the regulator from OCP shutdown. This bit must then be toggled from 1 to 0 to re-arm the status bit so that it can be latched in the event of another OCP event.
2 : 2	OCP_GLOBAL_BROADCAST_EN	Determines if the LDO notifies the PON module that an OCP event has occurred. PON module may use this info to shutdown the entire PMIC. 1 = LDO notifies PON when OCP event occurs 0 = LDO does not notify PON when OCP even occurs (DEFAULT) 0: OCP_GLOBAL_BROADCAST_DISABLED 1: OCP_GLOBAL_BROADCAST_ENABLED
5 : 4	OCP_DEB	Debounce time for OCP detection 0 = ~120us 1 = ~240us (DEFAULT) 2 = ~480us 3 = ~960us 0: DEB_120US 1: DEB_240US 2: DEB_480US 3: DEB_960US

Bits	Name	Description
6 : 6	OCP_SELF_SHUTDOWN_EN	Determines if the LDO performs self-shutdown upon OCP event. 1 = LDO self-shutdowns upon OCP event (DEFAULT) 0 = LDO does not self-shutdown upon OCP event 0: OCP_SELF_SHUTDOWN_DISABLED 1: OCP_SELF_SHUTDOWN_ENABLED
7 : 7	OCP_EN	1 = Enable OCP feature (DEFAULT) 0 = Disable OCP feature 0: OCP_DISABLED 1: OCP_ENABLED

LDO24_PD_CTL | 0x000157A0

Type: Read/Write

Reset: 0x00000080

No description provided for this register.

Bits	Name	Description
3 : 3	LEAK_PD_EN	Intended for keeping Vout from rising due to passFET leakage at high temperature. 1 = Leakage pulldown is always enabled 0 = Leakage pulldown is always disabled (DEFAULT) 0: LEAKAGE_PD_ALWAYS_DISABLED 1: LEAKAGE_PD_ALWAYS_ENABLED
6 : 6	WEAK_PD_EN	Weak PD is never active while LDO is on (VREG_READY=1). 1 = Weak pulldown is enabled when the regulator is disabled 0 = Weak pulldown is always inactive (DEFAULT) 0: WEAK_PD_DISABLED 1: WEAK_PD_WHEN_LDO_IS_OFF
7 : 7	STRONG_PD_EN	Strong PD is never active while LDO is on (VREG_READY=1). Strong pull-down FET Rdson will vary with Vout. 1 = Strong pulldown is enabled when the regulator is disabled (DEFAULT) 0 = Strong pulldown is always inactive. 0: STRONG_PD_ALWAYS_OFF 1: STRONG_PD_WHEN_LDO_IS_OFF

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